ELE 717 · POSTGRADUATE DIPLOMA · NIGERIAN DEFENCE ACADEMY

Solid State Electronics

A week-by-week course following the 15-week Lecture 2 plan

With learning objectives, worked examples, practice questions, the 50 PGD examination questions solved, CAT practice tests and laboratory practicals.

Contents

1. Orientation · Course — Start Here
2. Week 1 · Module 1 · Material Physics & Semiconductor Growth — Introduction to Semiconductor Materials
3. Week 2 · Module 1 · Material Physics & Semiconductor Growth — Material Growth Techniques
4. Week 3 · Module 1 · Material Physics & Semiconductor Growth — Impurity Control and Doping
5. Week 4 · Module 2 · Energy Bands & Carrier Statistics — Energy Bands and Carrier Concentrations
6. Week 5 · Module 2 · Energy Bands & Carrier Statistics — Carrier Statistics
7. Week 6 · Module 2 · Energy Bands & Carrier Statistics — Continuous Assessment Test 1 and Review
8. Week 7 · Module 3 · Carrier Transport & Injection Phenomena — Drift and Diffusion Currents
9. Week 8 · Module 3 · Carrier Transport & Injection Phenomena — High-Field Effects and Excess Carriers
10. Week 9 · Module 3 · Carrier Transport & Injection Phenomena — Continuity Equation and Surface Phenomena
11. Week 10 · Module 4 · Junctions, Interfaces & Characterisation — The p-n Junction Diode
12. Week 11 · Module 4 · Junctions, Interfaces & Characterisation — Metal–Semiconductor Interfaces
13. Week 12 · Module 4 · Junctions, Interfaces & Characterisation — Interface Characterisation and CAT 2
14. Week 13 · Module 5 · Advanced Device Technology — Bipolar Junction Transistors (BJTs)
15. Week 14 · Module 5 · Advanced Device Technology — Field-Effect Transistors (MOSFETs)
16. Week 15 · Module 5 · Advanced Device Technology — Revision and Final Exam Preparation
17. Revision & Practicals — Answer Key — All 50 Examination Questions
18. Revision & Practicals — Laboratory and Practical Component

ORIENTATION · COURSE

Start Here

Course: ELE 717 Solid State Electronics (3 credits, compulsory), Postgraduate Diploma, Nigerian Defence Academy.
Lecturer: Engr. Mrs Ruby Chinyere Beremeh Onyesoh.

This course follows the lecturer's own **15-week plan** (Lecture 2). There is one page per week, grouped into the five modules of the plan. Week 1 follows the issued lecture notes slide by slide. Weeks 2–15 teach each topic on the plan from scratch, because the notes for those weeks have not been issued yet.

Each week ends with **solved questions**:

- the lecturer's own **revision and calculation questions** (Week 1);
- every one of the **50 PGD examination questions** (Q1–Q50), filed under the week that teaches it;
- **CAT practice tests** with answers in Weeks 6 and 12.

The 15-week plan

Week	Topic (from the lecture plan)	Exam questions
Module 1	**Material Physics & Semiconductor Growth (Weeks 1–3)**	
1	Introduction to semiconductor materials: elemental vs compound (Si, Ge, GaAs); crystal structures, Miller indices and bonding	Q1–Q8
2	Material growth: purification of silicon (metallurgical to electronic grade); Czochralski bulk growth; epitaxy (VPE, LPE, MBE)	Q9–Q13
3	Impurity control and doping: diffusion and ion implantation; controlled doping profiles and evaluation of impurity distribution	Q14–Q21
Module 2	**Energy Bands & Carrier Statistics (Weeks 4–6)**	
4	Energy bands and carrier concentrations: quantum mechanics (Schrödinger, Kronig–Penney); band gaps; intrinsic and extrinsic concentrations	Q22–Q29, Q34
5	Carrier statistics: Fermi–Dirac function; density of states; Fermi level against temperature and doping	Q30–Q33
6	Continuous Assessment Test 1 and review: crystal structures and carrier statistics	CAT 1 practice
Module 3	**Carrier Transport & Injection Phenomena (Weeks 7–9)**	
7	Drift and diffusion currents: mobility, scattering, conductivity; diffusion, Fick's law, Einstein relation	Q35–Q40
8	High-field effects and excess carriers: velocity saturation, hot carriers; generation and recombination (SRH); lifetime and steady-state injection	Q41–Q44
9	Continuity equation and surface phenomena: minority-carrier continuity equation; surface states and surface recombination velocity	Q45–Q46
Module 4	**Junctions, Interfaces & Characterisation (Weeks 10–12)**	
10	The p-n junction: built-in potential, depletion width, field profile; Shockley equation under forward and reverse bias	Q47–Q48
11	Metal–semiconductor interfaces: Schottky barrier vs Ohmic contact; band diagrams in equilibrium and under bias	Q49
12	Interface characterisation: C-V profiling, I-V analysis; CAT 2	Q50, CAT 2 practice
Module 5	**Advanced Device Technology (Weeks 13–15)**	
13	Bipolar junction transistors: operation, minority-carrier profiles, emitter efficiency, base transport factor, α and β	—
14	MOSFETs: ideal MOS capacitor (accumulation, depletion, inversion), threshold voltage, linear and saturation regions	—
15	Revision and final exam preparation: formula sheet, problem-solving clinic, past-question review	All

After Week 15 there are two more pages: the **answer key** for all 50 exam questions, and the **laboratory practicals**.

Laboratory and practical component (3 hours/week)

The plan requires at least four practical or simulation experiments. Each one has its own section in the Laboratory page:

1. **Simulation of carrier transport:** modelling Fermi-level shifts in MATLAB or Python.
2. **I-V characterisation:** measuring and plotting the forward and reverse characteristics of a p-n junction and a Schottky diode.
3. **C-V profiling:** finding doping concentration and built-in potential from capacitance measurements.
4. **MOSFET parameter extraction:** measuring threshold voltage V_{th} and channel mobility.

Assessment

- **CAT 1** (Week 6): crystal structures and carrier statistics (Weeks 1–5).
- **CAT 2** (Week 12): transport, junctions and interfaces (Weeks 7–12).
- **Final examination:** the whole course. The 50-question PGD paper is the best guide to its style.

How to study each week

1. Read the week's notes once, straight through.
2. Cover the answers and attempt every question on paper.
3. For derivations, check you stated the assumptions; marks are given for them.
4. Tick the week as read (the button at the bottom of each page) to track your progress.

> Numbers to know without looking: E_g = 1.12 eV (Si), 0.66 eV (Ge), 1.42 eV (GaAs); n_i(Si) ≈ 1×10^{10} cm^{-3}; kT/q = 0.0259 V at 300 K; q = 1.6×10^{-19} C; Si lattice constant 5.43 Å; μ_n(Si) ≈ 1350 $cm^2/V\cdot s$.

Textbooks

Fundamentals of Solid State Electronics (C.-T. Sah, World Scientific, 1991); *Solid State Electronic Circuits* (Edward W. Maby); *Solid State Electronic Devices* (B. G. Streetman and S. K. Banerjee). Further reading: Sze and Ng, *Physics of Semiconductor Devices*; Neamen, *Semiconductor Physics and Devices*; Pierret, *Semiconductor Device Fundamentals*.

Introduction to Semiconductor Materials

Plan: elemental vs compound semiconductors (Si, Ge, GaAs); crystal structures, Miller indices and bonding mechanisms. This page follows the issued Lecture 2 notes in their own order, then goes to the depth the examination needs.

Learning objectives

By the end of this week you should be able to:

1. Define a semiconductor and distinguish it from a conductor and an insulator.
2. Describe the atomic structure of Si, Ge, Ga and As and identify their valence electrons.
3. Explain covalent bonding in silicon and electron–hole pair generation.
4. Distinguish intrinsic, n-type and p-type semiconductors, and donors from acceptors.
5. Compare elemental and compound semiconductors, and Si, Ge and GaAs.
6. Explain the energy-band picture, the band gap and direct versus indirect gaps.
7. Describe crystal lattices, unit cells and the diamond and zinc-blende structures.
8. Determine Miller indices of planes and the spacing between planes.

> **How to read this page.** Read it in order: each section builds on the one before. The tables and boxed equations are the parts to memorise; the paragraphs around them explain why they are true. The key points at the end summarise the week, and the solved revision and exam questions show how to write the answers.

Introduction

Semiconductor materials are the foundation of modern electronic and electrical systems. Almost every modern electronic device depends on them, including computers, mobile phones, solar cells, sensors, communication systems, power converters and control systems.

A **semiconductor** is a material whose electrical conductivity lies between that of a good conductor and an insulator. More importantly, its conductivity can be **controlled** by temperature, light, electric fields, magnetic fields, and the intentional addition of impurities known as **dopants**.

The most important semiconductor materials used in electronics are:

Material	Type	Where you meet it
Silicon (Si)	Elemental	ICs, microprocessors, solar cells, power devices
Germanium (Ge)	Elemental	Early transistors, infrared, SiGe high-speed transistors
Gallium arsenide (GaAs)	Compound (III–V)	Microwave, RF, LEDs, lasers
Gallium nitride (GaN)	Compound (III–V), wide gap	Blue/white LEDs, power and RF transistors
Silicon carbide (SiC)	Compound (IV–IV), wide gap	High-voltage, high-temperature power electronics
Indium phosphide (InP)	Compound (III–V)	Fibre-optic lasers and detectors, very high-speed circuits
Gallium phosphide (GaP)	Compound (III–V)	Red, yellow and green indicator LEDs

What is a semiconductor?

A semiconductor is a material whose electrical conductivity lies between that of a conductor and an insulator, and can be **significantly controlled** by temperature, impurities, light or applied electric fields. Examples: silicon (Si), germanium (Ge) and gallium arsenide (GaAs).

Its conductivity is **not fixed**, and that is exactly what makes it useful for electronic devices. Pure silicon at room temperature conducts poorly. When a small, controlled quantity of an impurity such as phosphorus or boron is introduced, its conductivity rises dramatically: one phosphorus atom per million silicon atoms raises the conductivity about 100,000 times. This process is called **doping**.

Classification of materials by electrical conductivity

Materials can broadly be classified into **conductors**, **semiconductors** and **insulators**.

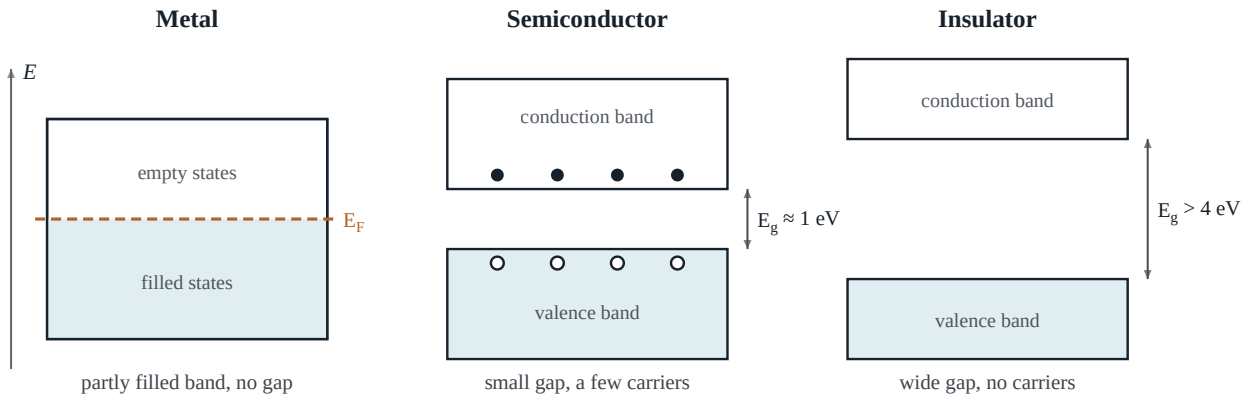
Conductors have a large number of free electrons available for electrical conduction. Examples: copper, aluminium, silver, gold, iron. Copper is widely used in electrical wiring because of its high conductivity. Characteristics:

- high electrical conductivity ($\sigma \approx 10^5$–10^6 S/cm);
- low electrical resistivity (Cu: 1.7×10^{-6} Ω·cm);
- large number of free charge carriers (about 10^{22} cm^{-3});
- small, or effectively overlapping, energy gap.

Insulators have very few free charge carriers under normal conditions. Examples: glass, rubber, plastic, ceramic, mica. They strongly resist the flow of electric current. Characteristics:

- very high resistivity (10^{10}–10^{18} Ω·cm);
- very low conductivity;
- large energy band gap (more than about 4 eV; $SiO_2 \approx 9$ eV);
- electrons strongly bound to their atoms.

Semiconductors have conductivity between conductors and insulators. Examples: silicon, germanium, gallium arsenide, silicon carbide, gallium nitride. Their conductivity can be modified through **temperature, doping, illumination, electric field, magnetic field and mechanical stress**. This controllability is one of the most important characteristics of semiconductor materials.



Band pictures at room temperature. In a metal the highest band is partly filled; in a semiconductor a small gap lets some electrons cross by thermal excitation; in an insulator the gap is too wide.

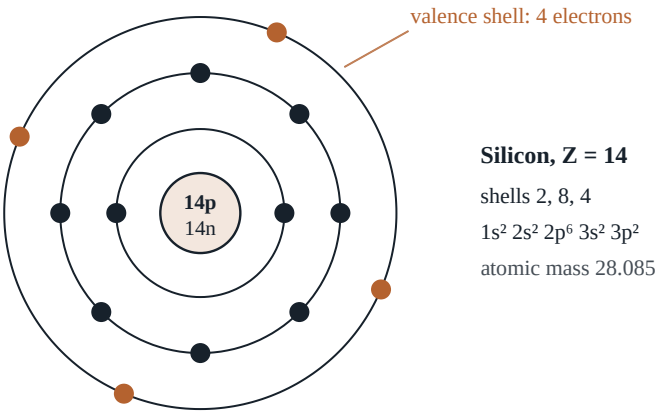
Comparison of conductors, semiconductors and insulators

Property	Conductor	Semiconductor	Insulator
Conductivity	Very high	Moderate, controllable	Very low
Resistivity	Very low	Intermediate	Very high
Band gap	Approximately zero	Small (≈ 0.5–3.5 eV)	Large (> 4 eV)
Free electrons	Many	Moderate, controllable	Very few
Temperature effect	Conductivity usually **decreases**	Conductivity usually **increases**	Small effect under normal conditions
Examples	Cu, Ag, Al	Si, Ge, GaAs	Glass, rubber, plastic
Main application	Wiring	Electronic devices	Electrical isolation

Why the temperature effect is opposite. In a metal the number of free electrons is fixed; heating only makes the lattice vibrate more, which scatters electrons, so resistance rises. In a semiconductor heating breaks covalent bonds and creates many more carriers (roughly as $e^{-E_g/2kT}$); the extra carriers far outweigh the extra scattering, so resistance falls.

Atomic structure of semiconductor materials

An atom consists of a **nucleus** containing **protons** (positive) and **neutrons** (neutral), with **electrons** (negative) occupying different energy levels (shells) around it, as in the Bohr model. The electrons in the outermost shell are the most important in determining the electrical properties of a material. These are the **valence electrons**.



Bohr model of silicon (Z = 14): shells hold 2, 8 and 4 electrons. The four outer electrons are the valence electrons.

Valence electrons

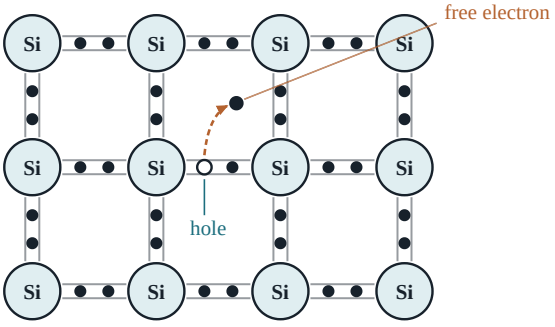
The electrons in the outermost energy level of an atom are its **valence electrons**. They take part in chemical bonding.

Atom	Z	Electron configuration	Valence electrons	Group
Silicon (Si)	14	$1s^2 2s^2 2p^6 3s^2 3p^2$ (shells 2, 8, 4)	4	IV (14)
Germanium (Ge)	32	$1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^{10} 4p^2$ (shells 2, 8, 18, 4)	4	IV (14)
Gallium (Ga)	31	$[Ar] 3d^{10} 4s^2 4p^1$	3	III (13)
Arsenic (As)	33	$[Ar] 3d^{10} 4s^2 4p^3$	5	V (15)

Silicon's atomic mass is 28.085. The difference in valence electrons between Ga (3) and As (5) is what allows them to combine into a compound semiconductor (average 4 per atom), and the difference between Si (4) and P (5) or B (3) is what makes doping work.

Covalent bonding in silicon

Silicon atoms form a **covalent bonding structure**. Each silicon atom has four valence electrons and **shares** them with four neighbouring silicon atoms, so each atom forms **four covalent bonds** and is surrounded by eight shared electrons (a full outer shell).



Two-dimensional picture of the silicon crystal. Each Si atom shares one electron pair with each of four neighbours. One bond has been broken by thermal energy, creating a free electron and a hole.

- The shared electrons hold the crystal structure together.

- At very low temperatures nearly all electrons stay in the covalent bonds, so electrical conductivity is very low (pure Si is an insulator at 0 K).
- When sufficient energy is supplied (heat, light), some electrons break away from their bonds and become mobile.

Electron–hole pair generation

When a valence electron receives sufficient energy (at least E_g = 1.12 eV in Si) from thermal excitation, it can leave its covalent bond:

- the electron becomes a **free electron**;
- the vacancy it leaves behind in the bond is called a **hole**.

  Energy supplied → electron–hole pair

- The electron carries negative charge, $q = -1.6\times10^{-19}$ C.
- The hole behaves as if it has positive charge, $q = +1.6\times10^{-19}$ C. A neighbouring bond electron can jump into the hole, so the hole moves in the opposite direction to that electron.
- **Both electrons and holes contribute to electrical conduction** in semiconductors. This two-carrier conduction is what separates semiconductors from metals.

The reverse process, a free electron falling back into a hole, is **recombination** (Week 8).

Intrinsic semiconductor

An **intrinsic semiconductor** is a pure semiconductor without intentionally introduced impurities. Examples: pure silicon, pure germanium. Every free electron came from a broken bond that also left a hole, so

$$n = p = n_i$$

where n is the electron concentration, p the hole concentration and n_i the **intrinsic carrier concentration** (Si: about 1×10^{10} cm^{-3} at 300 K; Ge: 2.4×10^{13} cm^{-3}). The number of electrons equals the number of holes. In thermal equilibrium the product is constant:

$$n \cdot p = n_i^2 \quad \text{(mass-action law, true for doped material too)}$$

Extrinsic semiconductor

An **extrinsic semiconductor** is one whose electrical properties have been deliberately modified by adding controlled impurities. This process is **doping**. There are two major types: **n-type** and **p-type**.

N-type semiconductor

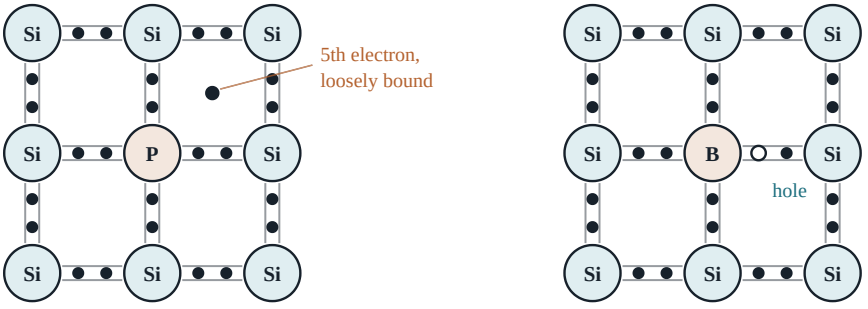
Produced by adding a **pentavalent impurity** (five valence electrons): phosphorus (P), arsenic (As) or antimony (Sb). Suppose phosphorus is added to silicon. Four of its five valence electrons form covalent bonds with neighbouring Si atoms; the fifth is only weakly bound (about 0.045 eV) and easily becomes a free electron. The phosphorus atom is called a **donor**, and it is left as a fixed positive ion.

- Majority carriers = **electrons**; minority carriers = **holes**.
- $n \approx N_D$ (donor concentration), $p = n_i^2/N_D$.

P-type semiconductor

Produced by adding a **trivalent impurity** (three valence electrons): boron (B), aluminium (Al), gallium (Ga) or indium (In). When boron is introduced into silicon, only three covalent bonds can be completed. An electron is missing from the fourth bond; this missing electron is a **hole**. A neighbouring electron easily moves in, so the hole wanders through the crystal. Boron is called an **acceptor**, and it is left as a fixed negative ion.

- Majority carriers = **holes**; minority carriers = **electrons**.
- $p \approx N_A$ (acceptor concentration), $n = n_i^2/N_A$.



n-type: phosphorus donor **p-type: boron acceptor**

Left: a phosphorus atom in silicon donates its fifth electron (n-type). Right: a boron atom leaves one bond incomplete, creating a hole (p-type).

Example 1.1: carriers in n-type silicon

Question. Silicon at 300 K is doped with 10^{16} cm^{-3} phosphorus atoms. Taking $n_i = 1\times10^{10}$ cm^{-3} and assuming every donor is ionised, calculate the electron and hole concentrations and state which carrier is the majority.

Given: $N_D = 1\times10^{16}$ cm^{-3}, $n_i = 1\times10^{10}$ cm^{-3}, T = 300 K, complete ionisation.

Find: n, p, and the majority carrier.

Solution.

Step 1. Each phosphorus atom gives one free electron, and $N_D \gg n_i$, so

$$n \approx N_D = 1\times10^{16}\ \text{cm}^{-3}$$

Step 2. The mass-action law $np = n_i^2$ gives the holes:

$$p = n_i^2/n = (1\times10^{10})^2 / (1\times10^{16}) = 10^{20}/10^{16} = 1\times10^{4}\ \text{cm}^{-3}$$

Answer: $n = 1\times10^{16}$ cm^{-3}, $p = 1\times10^{4}$ cm^{-3}. Electrons are the majority carriers; they outnumber the holes by 10^{12}.

Elemental semiconductors

An **elemental semiconductor** is made primarily from a single chemical element.

- **Silicon (Si):** atomic number 14, four valence electrons. Widely used in integrated circuits, power electronics and solar cells.
- **Germanium (Ge):** atomic number 32, four valence electrons. Higher carrier mobility than silicon, lower band gap. Used in specialised high-speed and infrared applications.

Compound semiconductors

Compound semiconductors consist of two or more different elements. Examples: GaAs, GaN, InP, GaP, CdTe. They can provide properties that are difficult to achieve with elemental semiconductors:

- **GaAs** has high electron mobility and is useful in high-frequency electronic systems.
- **GaN** has a wide band gap (3.4 eV) and is important in power electronics, LEDs, RF systems and high-temperature electronics.

They are named by the periodic-table groups of their elements: **III–V** (GaAs, GaN, InP, GaP), **II–VI** (CdTe, ZnSe), **IV–IV** (SiC, SiGe). By alloying (e.g. $Al_xGa_{1-x}As$) the band gap can be tuned, which is called band-gap engineering.

Silicon, germanium and gallium arsenide

Silicon

Silicon is the most widely used semiconductor material in modern electronics.

- Atomic number 14; four valence electrons; diamond-cubic crystal structure.
- Band gap $E_g \approx 1.12$ eV at room temperature.
- Used for microprocessors, memory chips, diodes, transistors, solar cells, power devices, sensors and integrated circuits.

Why it dominates: it is abundant (sand), cheap, grows an excellent native oxide (SiO_2), has a band gap large enough for low leakage up to about 150 °C, and can be grown as huge, nearly perfect crystals.

Germanium

Germanium was one of the earliest semiconductor materials, used extensively for early transistor technology (the first transistor, 1947, was germanium).

- Four valence electrons; diamond-cubic crystal structure.
- Band gap $E_g \approx 0.66$ eV at room temperature.
- **Advantages:** high carrier mobility; useful for high-speed applications; useful in infrared technology.
- **Disadvantages:** smaller band gap; higher leakage current; less suitable than silicon for high-temperature applications (works only to about 75 °C).

Gallium arsenide

Gallium arsenide is a compound semiconductor of gallium and arsenic (Ga + As), abbreviated **GaAs**.

- Band gap $E_g \approx 1.42$ eV at room temperature (direct).
- High electron mobility (about 8500 $cm^2/V{\cdot}s$, six times silicon), so it is particularly useful for high-frequency and optoelectronic applications.
- Applications: microwave circuits, satellite communication, radar, high-frequency amplifiers, LEDs, laser diodes, photodetectors, solar cells.

Comparison of Si, Ge and GaAs

Property	Silicon	Germanium	GaAs
Symbol	Si	Ge	GaAs
Type	Elemental	Elemental	Compound
Valence electrons	4	4	III–V compound (average 4)
Band gap	≈ 1.12 eV	≈ 0.66 eV	≈ 1.42 eV
Crystal structure	Diamond cubic	Diamond cubic	Zinc blende
Electron mobility	Moderate (1350)	High (3900)	Very high (8500)
Thermal stability	Good	Lower	Good
Cost	Relatively low	Higher	High
IC manufacturing	Excellent	Limited	Specialised
High-frequency use	Good	Good	Excellent
Optical applications	Limited	Good (infrared)	Excellent

(Mobilities in $cm^2/V{\cdot}s$ at 300 K, added for reference.)

Energy band concept

The electrical behaviour of a semiconductor is explained using **energy bands**. When many atoms come together to form a solid, the discrete energy levels of the individual atoms spread into closely spaced energy states called **energy bands**. The two most important are:

- **Valence band:** contains electrons involved in bonding.
- **Conduction band:** contains electrons that can move through the material and contribute significantly to conduction.

Between them is the **forbidden energy gap**, or **band gap E_g**.

Energy band gap

The band gap is the energy difference between the conduction band and the valence band:

$$E_g = E_C - E_V$$

where E_C is the conduction-band energy (bottom of the conduction band) and E_V the valence-band energy (top of the valence band). It is normally measured in **electron-volts** (1 eV = 1.6×10^{-19} J).

Why the band gap is important. It strongly affects semiconductor behaviour. A smaller band gap makes it easier for electrons to gain enough energy to enter the conduction band; a larger band gap makes it harder. The band gap influences:

- **electrical conductivity** ($n_i \propto e^{-E_g/2kT}$);
- **temperature dependence** of conductivity;
- **optical absorption** (only photons with $h\nu \geq E_g$ are absorbed);
- **light emission** (emitted wavelength $\lambda = 1.24/E_g$ μm);

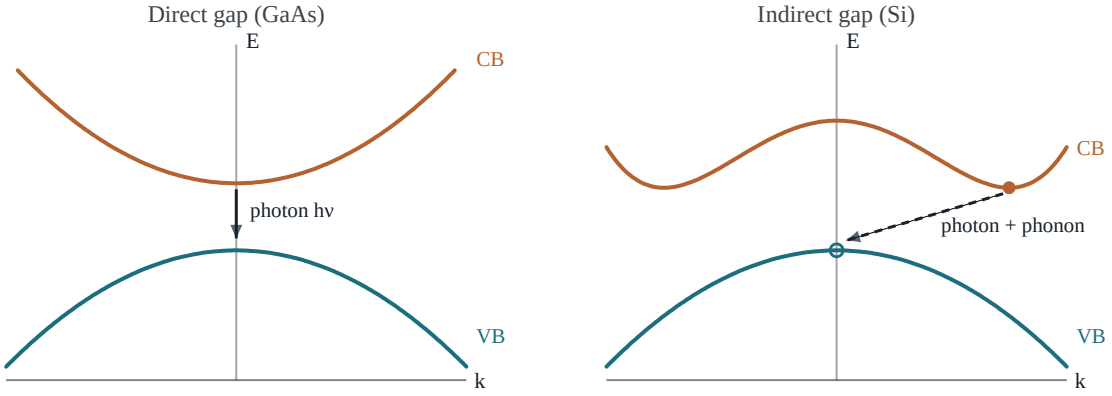
- **leakage current** (smaller gap, more leakage);
- **breakdown behaviour** (wider gap, higher breakdown field);
- **device operating temperature** (wider gap, hotter operation).

Direct and indirect band-gap semiconductors

Semiconductors are classified by the relationship between the valence-band maximum and the conduction-band minimum in momentum (k).

Direct band gap: electrons can transition between the conduction and valence bands without a significant change in momentum. Examples: GaAs, GaN, InP. These are suitable for **light-emitting devices**: LEDs, laser diodes, optical communication.

Indirect band gap: a momentum change is required during the transition, supplied by a lattice vibration (phonon). Examples: silicon, germanium. Silicon is therefore **not an efficient light emitter**.



E–k diagrams. In GaAs the band extrema line up at the same k, so an electron can drop straight down and emit a photon. In Si they are at different k, so a phonon must also take part.

Semiconductor crystal structure

A semiconductor is not simply a collection of independent atoms. The atoms are arranged in an ordered three-dimensional pattern called a **crystal lattice**, and this crystal structure determines many of its physical properties. The important concepts are: lattice, unit cell, crystal planes, crystal directions, Miller indices and atomic bonding.

Crystal lattice and unit cell

- A **crystal lattice** is a regular, repeating arrangement of atoms or points in space.
- A semiconductor crystal can be described by repeating a basic structural unit known as the **unit cell**.
- The unit cell contains enough information to reproduce the entire crystal by translation.

The cubic lattices

Structure	Atoms per cell	Coordination no.	Nearest-neighbour distance	APF
Simple cubic (SC)	1	6	a	0.52
Body-centred cubic (BCC)	2	8	$(\sqrt{3}/2)a$	0.68
Face-centred cubic (FCC)	4	12	$(\sqrt{2}/2)a$	0.74
Diamond cubic	8	4	$(\sqrt{3}/4)a$	0.34

Counting atoms in a cubic cell: a corner atom is shared by 8 cells (counts 1/8), a face atom by 2 (counts 1/2), a body atom by 1.

- SC: $8 \times 1/8 = 1$
- BCC: $8 \times 1/8 + 1 = 2$
- FCC: $8 \times 1/8 + 6 \times 1/2 = 4$
- Diamond: $8 \times 1/8 + 6 \times 1/2 + 4$ inside $= 8$

Silicon crystal structure

- Silicon has a **diamond-cubic** crystal structure.
- It can be regarded as **two interpenetrating face-centred cubic (FCC)** structures, displaced from each other by a quarter of the body diagonal.
- Each silicon atom has **four nearest-neighbour** silicon atoms, arranged at the corners of a tetrahedron (bond angle 109.5°).
- The bonding is predominantly **covalent**.
- This structure is responsible for many of silicon's useful semiconductor properties.

For Si ($a = 5.43$ Å): nearest-neighbour distance $= (\sqrt{3}/4)a = 2.35$ Å; atom density $= 8/a^3 = 5.0 \times 10^{22}$ atoms/cm^3.

Germanium crystal structure

- Germanium also has the diamond-cubic structure; each Ge atom forms four covalent bonds with neighbouring Ge atoms.
- Silicon and germanium therefore have similar bonding arrangements, but their atomic sizes ($a = 5.65$ Å for Ge) and electronic properties differ.

GaAs crystal structure

- GaAs forms the **zinc-blende** crystal structure under ordinary conditions.
- It consists of two interpenetrating FCC sublattices, one occupied by gallium and the other by arsenic. Each Ga has four As neighbours and vice versa.
- The bonding is mainly covalent with partial ionic character (As is more electronegative than Ga). This structure contributes to its important electronic and optical properties.

Lattice constant

The **lattice constant** is a characteristic dimension of a crystal unit cell. For a cubic crystal it is written **a** and measured in metres, nanometres or angstroms (1 Å $= 10^{-10}$ m $= 0.1$ nm). Si: 5.43 Å; Ge: 5.65 Å; GaAs: 5.65 Å.

The lattice constant determines: atomic spacing; crystal density; crystal matching between materials; epitaxial growth (a layer grows well only on a substrate with nearly the same a); and semiconductor heterostructures (e.g. AlGaAs on GaAs works because their lattice constants are almost identical).

Miller indices

Miller indices provide a standard notation for crystallographic planes and directions:

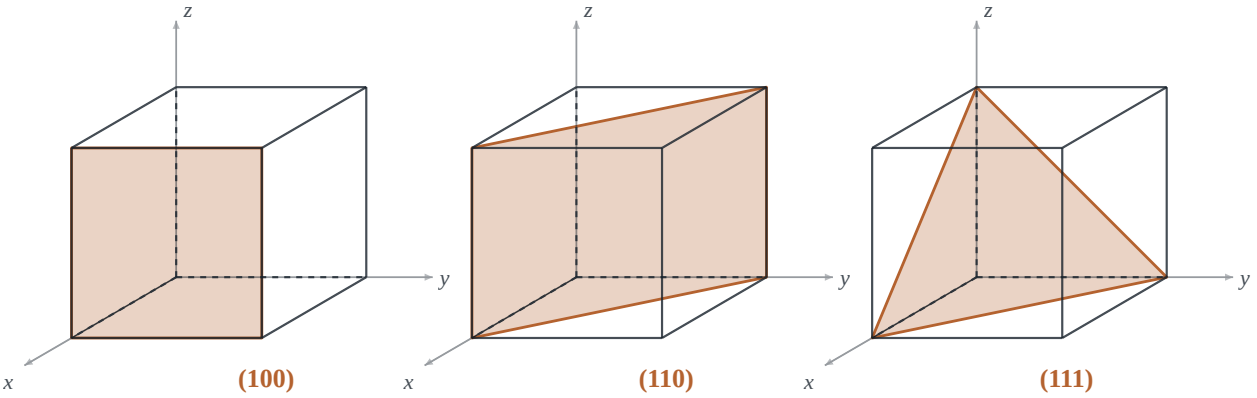
Notation	Meaning
(h k l)	A plane
{h k l}	A family of equivalent planes
[u v w]	A direction
⟨u v w⟩	A family of equivalent directions

They are important in wafer orientation, crystal growth, etching, diffraction and device fabrication.

Procedure

1. Determine the plane's intercepts on the x, y and z axes in lattice units.
2. Take the reciprocals of the intercepts.
3. Multiply by a common factor to obtain the smallest integers.
4. Write the result as (h k l).

- An intercept of ∞ gives an index of **0**, because the plane is parallel to that axis.
- **Negative** indices are shown with a bar, e.g. ($\bar{1}10$).



The three most important planes in a cubic crystal. MOS wafers are cut on (100); bipolar and some power wafers on (111).

Examples (from the lecture)

Example 1.2: Miller indices of a plane

Question. A crystal plane cuts the x, y and z axes at a, 2b and 3c respectively. Determine its Miller indices.

Given: intercepts a, 2b, 3c, i.e. 1, 2, 3 in lattice units.

Find: (h k l).

Solution.

Step	x	y	z
Intercepts (lattice units)	1	2	3
Reciprocals	1	1/2	1/3
Multiply by the LCM, 6	6	3	2

Answer: (6 3 2). Remember: Miller indices come from the **reciprocals** of the intercepts, not the intercepts themselves.

Example 1.3: a plane parallel to an axis

Question. A plane has intercepts x = a, y = 2b and is parallel to the z-axis (z = ∞). Determine its Miller indices.

Given: intercepts 1, 2, ∞ in lattice units.

Find: (h k l).

Solution.

Step	x	y	z
Intercepts	1	2	∞
Reciprocals	1	1/2	0
Multiply by 2	2	1	0

Answer: (2 1 0). The zero shows that the plane is parallel to the z-axis.

Interplanar spacing (cubic crystals)

$$d_{hkl} = a / \sqrt{(h^2 + k^2 + l^2)}$$

For Si, $d_{111} = 5.43/\sqrt{3} = 3.135$ Å and $d_{100} = 5.43$ Å. In cubic crystals the direction [h k l] is perpendicular to the plane (h k l).

Why orientation matters in processing

- **Surface atom density** differs by plane: in Si, (100) has 6.8×10^{14} atoms/cm^2, (111) has 7.8×10^{14} atoms/cm^2.
- **Oxidation rate:** (111) oxidises faster than (100).
- **Interface traps:** (100) gives the lowest Si/SiO_2 interface-state density, so **MOS technology uses (100) wafers**.
- **Anisotropic etching:** KOH etches (100) far faster than (111), giving V-grooves with 54.7° walls, used in MEMS.
- **Cleavage:** wafers break along {111} planes in Si and {110} in GaAs.
- **Epitaxy:** the substrate orientation and lattice constant set the orientation and quality of the grown layer.
- **Ion implantation channelling:** along low-index directions ions travel deep; wafers are tilted about 7° to avoid it.

Bonding mechanisms (exam depth)

The bonding mechanisms

Ionic bonding. One atom transfers electrons to another, producing positive and negative ions held together by electrostatic (Coulomb) attraction. Occurs between elements of very different electronegativity, e.g. Na^+Cl^-. The bond

is strong but **non-directional**. Electrons are tightly bound to the ions, so ionic solids are insulators with large gaps; they are hard, brittle and have high melting points.

Covalent bonding. Neighbouring atoms **share** pairs of valence electrons, one electron from each, so each atom completes its outer shell (8 electrons). The bond is strong and **directional**, which fixes the crystal geometry (tetrahedral, 109.5°). Electrons are localised in bonds at low temperature but can be freed by modest energy, which gives semiconducting behaviour. Examples: Si, Ge, diamond.

Metallic bonding. Valence electrons leave their atoms and form a delocalised "sea" or gas of electrons shared by the whole crystal, holding a lattice of positive ion cores together. Non-directional. The free electron sea explains the high electrical and thermal conductivity, lustre and ductility of metals. Examples: Cu, Al, Na.

Van der Waals bonding. Weak attraction between fluctuating dipoles in neutral atoms or molecules. Important in layered materials (graphite, MoS_2) and noble-gas solids, not in Si or GaAs.

Mixed ionic–covalent. Real compound semiconductors lie between the ideal limits.

	Ionic	Covalent	Metallic
Mechanism	Electron transfer, ion attraction	Electron-pair sharing	Shared electron sea
Directional?	No	Yes	No
Free electrons	None	Few (created thermally)	Very many
Electrical behaviour	Insulator	Semiconductor (or insulator)	Conductor
Example	NaCl	Si, Ge, C	Cu, Al

Defects in real crystals

Type	Dimension	Examples	Effect
Point	0-D	Vacancy, self-interstitial, substitutional or interstitial impurity, Frenkel pair (vacancy + interstitial)	Traps, recombination centres; vacancies assist diffusion
Line	1-D	Edge and screw dislocations	Leakage paths, recombination along the line, enhanced diffusion
Planar	2-D	Grain boundaries, stacking faults, twin boundaries, surfaces and interfaces	Scattering, interface traps, junction leakage
Volume	3-D	Voids, precipitates (e.g. SiO_2 clusters), inclusions	Gettering sites if controlled, device failure if not

Defects act as **scattering centres** (lower mobility), **recombination–generation centres** (shorter lifetime, more leakage), **traps** (charge storage and noise) and **diffusion paths**. Not all impurities are unwanted: deliberate dopants are substitutional point defects.

Key points of Week 1

- A semiconductor's conductivity lies between a conductor's and an insulator's and can be **controlled**, above all by doping.
- Conductors: bands overlap, conductivity falls when hot. Insulators: gap > 4 eV. Semiconductors: gap ≈ 1 eV, conductivity rises when hot.

- Si and Ge have 4 valence electrons and form 4 covalent bonds (diamond cubic). GaAs averages 4 per atom (zinc blende).
- Breaking a bond (energy ≥ E_g) makes an electron–hole pair; both carry current. Intrinsic: $n = p = n_i$.
- Pentavalent donors (P, As, Sb) → n-type, electrons majority. Trivalent acceptors (B, Al, Ga, In) → p-type, holes majority. $np = n_i^2$.
- $E_g = E_C - E_V$ sets conductivity, leakage, light emission ($\lambda = 1.24/E_g$ μm) and maximum temperature. Direct gap (GaAs) emits light; indirect (Si) does not.
- Miller indices: intercepts → reciprocals → smallest integers → (h k l); ∞ gives 0. $d = a/\sqrt{(h^2 + k^2 + l^2)}$.

Lecture revision questions solved

These are the lecturer's own Week 1 revision questions (Lecture 2, pages 41–43), in her order.

R1. Define a semiconductor.

A semiconductor is a material whose electrical conductivity lies between that of a conductor and an insulator ($\rho \approx 10^{-3}$ to 10^8 Ω·cm), and whose conductivity can be significantly controlled by temperature, light, electric and magnetic fields, and especially by adding small amounts of impurities (doping). Examples: Si, Ge, GaAs.

R2. Distinguish between a conductor, semiconductor and insulator.

A **conductor** has many free electrons and overlapping (or no) band gap, very high conductivity, and its conductivity falls as temperature rises (Cu, Ag, Al). An **insulator** has a large band gap (> 4 eV), very few free carriers, very high resistivity, and is used for isolation (glass, rubber, mica). A **semiconductor** has a small band gap (Si 1.12 eV), a moderate and controllable number of carriers (both electrons and holes), and its conductivity rises with temperature and with doping (Si, Ge, GaAs). See the comparison table above.

R3. What are valence electrons?

Valence electrons are the electrons in the outermost energy level (shell) of an atom. They take part in chemical bonding and determine the electrical properties of a material. Silicon has 4, gallium 3, arsenic and phosphorus 5, boron 3.

R4. Why does silicon have four valence electrons?

Silicon has atomic number Z = 14, so its 14 electrons fill the shells as 2, 8 and 4 (configuration $1s^2 2s^2 2p^6 3s^2 3p^2$). The first two shells (n = 1 and n = 2) are complete; the third shell (n = 3) holds only $3s^2 3p^2$, i.e. **4 electrons**. These four outer electrons are the valence electrons, which places silicon in Group IV (14) of the periodic table.

R5. Explain covalent bonding in silicon.

Each silicon atom has four valence electrons. In the crystal it shares one electron with each of four neighbouring Si atoms, and each neighbour shares one back, so each pair of atoms is joined by a shared **electron pair** (a covalent bond). Each atom therefore forms four covalent bonds, arranged tetrahedrally at 109.5° (diamond-cubic structure), and is surrounded by eight shared electrons, a stable full shell. The shared electrons hold the crystal together. At low temperature all electrons are locked in bonds, so pure Si hardly conducts; when energy of at least 1.12 eV is supplied, a bond breaks and creates a free electron and a hole.

R6. Define intrinsic semiconductor.

An intrinsic semiconductor is a pure semiconductor with no intentionally added impurities, in which carriers are produced only by thermal breaking of bonds. Electrons and holes are therefore equal in number: $n = p = n_i$ (Si: $n_i \approx 1\times10^{10}$ cm^{-3} at 300 K).

R7. Define extrinsic semiconductor.

An extrinsic semiconductor is one whose electrical properties have been deliberately modified by adding controlled amounts of impurity atoms (doping). It is n-type if doped with donors (electrons are majority carriers) or p-type if doped with acceptors (holes are majority carriers). $n \neq p$, but $np = n_i^2$ still holds.

R8. What is doping?

Doping is the deliberate, controlled introduction of small amounts of specific impurity atoms (typically 10^{14}–10^{20} cm^{-3}) into a pure semiconductor to increase its conductivity and set its carrier type. Pentavalent dopants (P, As, Sb) give n-type material; trivalent dopants (B, Al, Ga, In) give p-type. Methods are covered in Week 3 (diffusion and ion implantation).

R9. Differentiate between n-type and p-type semiconductors.

	n-type	**p-type**
Impurity	Pentavalent (P, As, Sb)	Trivalent (B, Al, Ga, In)
Impurity name	Donor	Acceptor
Majority carriers	Electrons	Holes
Minority carriers	Holes	Electrons
Fixed ion left	Positive	Negative
Concentrations	$n \approx N_D$, $p = n_i^2/N_D$	$p \approx N_A$, $n = n_i^2/N_A$
Fermi level	Near the conduction band	Near the valence band

R10. What is a donor impurity?

A donor is a pentavalent impurity atom (P, As or Sb in silicon) with five valence electrons. Four form covalent bonds with neighbouring Si atoms; the fifth is weakly bound (≈ 0.045 eV) and is "donated" to the conduction band as a free electron. The donor becomes a fixed positive ion and makes the material n-type.

R11. What is an acceptor impurity?

An acceptor is a trivalent impurity atom (B, Al, Ga or In in silicon) with three valence electrons. It completes only three bonds; the incomplete fourth bond "accepts" an electron from a neighbouring bond, creating a mobile hole in the valence band. The acceptor becomes a fixed negative ion and makes the material p-type.

R12. Define energy band gap.

The energy band gap E_g is the energy difference between the bottom of the conduction band and the top of the valence band, $E_g = E_C - E_V$. It is the minimum energy needed to free an electron from a covalent bond and create an electron–hole pair. It is measured in eV: Si 1.12 eV, Ge 0.66 eV, GaAs 1.42 eV.

R13. Differentiate between direct and indirect band-gap semiconductors.

In a **direct** band-gap semiconductor (GaAs, GaN, InP) the conduction-band minimum and valence-band maximum occur at the same momentum, so an electron can fall across the gap and emit a photon without any change in momentum. Light emission is efficient: LEDs, laser diodes, optical communication. In an **indirect** band-gap semiconductor (Si, Ge) they occur at different momenta, so a transition also needs a phonon (lattice vibration) to supply the momentum. This is unlikely, so light emission is very inefficient; the energy is mostly lost as heat. Silicon is therefore not used for LEDs or lasers.

R14. State the crystal structure of silicon.

Silicon has the **diamond-cubic** crystal structure: two interpenetrating face-centred cubic lattices displaced by one quarter of the body diagonal. Each atom has four nearest neighbours at the corners of a tetrahedron, bonded covalently. Lattice constant a = 5.43 Å, 8 atoms per unit cell, nearest-neighbour distance 2.35 Å.

R15. State the crystal structure of GaAs.

GaAs has the **zinc-blende** crystal structure: two interpenetrating FCC sublattices, one of gallium atoms and one of arsenic atoms, displaced by a quarter of the body diagonal. Each Ga atom has four As nearest neighbours and each As has four Ga neighbours, tetrahedrally arranged. Lattice constant a = 5.65 Å. The bonding is covalent with partial ionic character.

R16. What are Miller indices?

Miller indices are a set of three integers (h k l), obtained from the reciprocals of a plane's intercepts on the crystal axes (in lattice units) reduced to the smallest whole numbers, that identify the orientation of a crystal plane. The same notation gives directions [u v w], families of planes {h k l} and families of directions ⟨u v w⟩. They are used to specify wafer orientation, etching and cleavage planes, epitaxial growth and X-ray diffraction.

R17. Why is silicon widely used in integrated circuits?

1. It is abundant and cheap (sand, quartz), with a mature refining industry.
2. It grows a stable, high-quality native oxide, SiO_2, used for masking, passivation, isolation and the MOSFET gate insulator.
3. Its 1.12 eV band gap gives low leakage and operation up to about 150 °C.
4. Large, nearly defect-free single crystals (300 mm wafers) can be grown.
5. It is mechanically strong and conducts heat well.
6. Its doping by diffusion and ion implantation is well understood and controllable.

R18. State three applications of GaAs.

Any three of: microwave and RF amplifiers (mobile phones, radar); satellite communication circuits; high-frequency transistors (HEMTs, MESFETs); LEDs; laser diodes for optical-fibre communication; photodetectors; high-efficiency solar cells for satellites.

R19. Define carrier mobility.

Carrier mobility μ is the average drift velocity acquired by a charge carrier per unit applied electric field: $\mu = v_d/E$, in $cm^2/V{\cdot}s$. It measures how easily electrons or holes move through the crystal. In silicon $\mu_n \approx 1350$ and $\mu_p \approx 480$ $cm^2/V{\cdot}s$. (Full treatment in Week 7.)

R20. Explain why semiconductor conductivity increases with temperature.

Conductivity is $\sigma = q(n\mu_n + p\mu_p)$. As temperature rises, more valence electrons gain enough thermal energy to cross the band gap, breaking covalent bonds and creating electron–hole pairs. The intrinsic carrier concentration rises exponentially, $n_i \propto T^{3/2} e^{-E_g/2kT}$: in Si it roughly doubles every 8 °C. Mobility does fall slightly with temperature because of increased lattice scattering ($\mu \propto T^{-3/2}$), but this is a weak power law against an exponential increase in carriers, so the overall conductivity increases. This is the opposite of a metal, where the carrier number is fixed and only scattering increases.

Lecture calculation questions solved

C1. An n-type semiconductor has an electron concentration of 5×10^{16} cm^{-3}. If the electron mobility is 1200 cm^2/V·s, calculate its conductivity, assuming hole conduction is negligible.

With hole conduction neglected, $\sigma = q\, n\, \mu_n$:

$$\sigma = (1.6\times10^{-19}\ \text{C})(5\times10^{16}\ \text{cm}^{-3})(1200\ \text{cm}^2/\text{V·s})$$
$$= 1.6\times10^{-19} \times 6\times10^{19}$$
$$= 9.6\ \text{S/cm} \quad (\Omega^{-1}\text{cm}^{-1})$$

σ = 9.6 S/cm (= 960 S/m), so the resistivity is $\rho = 1/\sigma$ = **0.104 Ω·cm**. (With $q = 1.602\times10^{-19}$ C, σ = 9.61 S/cm.)

C2. A semiconductor has $n = 2 \times 10^{16}$ cm^{-3}, $p = 5 \times 10^{15}$ cm^{-3}, μ_n = 1300 cm^2/V·s, μ_p = 450 cm^2/V·s. Calculate its conductivity using $\sigma = q(n\mu_n + p\mu_p)$.

$$n\mu_n = 2\times10^{16} \times 1300 = 2.6\times10^{19}$$
$$p\mu_p = 5\times10^{15} \times 450 = 2.25\times10^{18}$$
$$n\mu_n + p\mu_p = 2.825\times10^{19} \quad (\text{cm·V·s})^{-1}$$

$$\sigma = 1.6\times10^{-19} \times 2.825\times10^{19} = 4.52\ \text{S/cm}$$

$\sigma \approx$ 4.52 S/cm, $\rho = 1/\sigma \approx$ **0.221 Ω·cm**. The electrons contribute 92 % of the conductivity (2.6 of 2.825), the holes 8 %.

C3. Explain why GaAs is preferred to silicon for some high-frequency applications.

1. **Higher electron mobility**: about 8500 cm^2/V·s against 1350 for Si, because of its small electron effective mass ($0.067m_0$). Electrons cross a transistor faster, giving higher transconductance and cut-off frequency (f_T), and lower noise.
2. **Higher peak electron velocity** (about 2×10^7 cm/s against 1×10^7 cm/s), so short transit times in small devices.
3. **Semi-insulating substrates** ($\rho \approx 10^8$ Ω·cm) reduce parasitic capacitance and substrate losses, ideal for microwave integrated circuits (MMICs) with on-chip transmission lines.
4. It supports **heterostructure devices** (AlGaAs/GaAs HEMTs and HBTs) with very high mobility two-dimensional electron gases.
5. **Larger band gap** (1.42 eV) gives better radiation hardness and higher-temperature operation than Si.

Silicon is still preferred where cost, integration density and hole mobility matter.

C4. A crystal plane has intercepts 2a, 3b and ∞c. Determine its Miller indices.

Step	x	y	z
Intercepts	2	3	∞
Reciprocals	1/2	1/3	0
Multiply by 6	3	2	0

Miller indices (3 2 0). The zero shows that the plane is parallel to the z-axis. For a cubic crystal, $d_{320} = a/\sqrt{(9 + 4 + 0)} = a/\sqrt{13}$.

C5. Explain the importance of semiconductor band gap in electronic and optoelectronic device design.

Electronic devices.

1. **Intrinsic carrier concentration and leakage:** $n_i \propto e^{-E_g/2kT}$. A wider gap gives far fewer thermally generated carriers, so lower reverse leakage and less noise (Si leaks less than Ge).
2. **Maximum operating temperature:** a device stops working when n_i approaches the doping level. Ge fails at about 75 °C, Si at about 150–200 °C, and SiC and GaN (3.3–3.4 eV) work above 300 °C.
3. **Breakdown field and power handling:** wider gap means a higher critical field (Si 0.3 MV/cm, GaN 3.3 MV/cm), so thinner, lower-resistance high-voltage devices.
4. **Turn-on voltage:** the diode forward voltage rises with E_g (Ge ≈ 0.3 V, Si ≈ 0.7 V, GaAs ≈ 1.1 V).

Optoelectronic devices.

1. **Emission wavelength:** an LED or laser emits at $\lambda = hc/E_g$, i.e. $\lambda(\mu m) = 1.24/E_g(eV)$. Designers choose or alloy materials for the colour: GaAs 0.87 μm (infrared), GaP green-yellow, GaN 365 nm (blue/UV with InGaN), InGaAsP 1.3–1.55 μm for optical fibres.
2. **Absorption and detection:** only photons with $h\nu \geq E_g$ are absorbed. Si detects up to 1.1 μm; Ge and InGaAs are used for longer wavelengths.
3. **Solar cells:** the band gap sets the trade-off between absorbing more of the spectrum and the voltage obtained; the optimum single-junction gap is about 1.1–1.4 eV (Si, GaAs).
4. **Direct or indirect:** only direct-gap materials emit light efficiently.

Examination questions solved (Q1–Q8)

Q1. Define a semiconductor and explain how its electrical conductivity differs from that of a conductor and an insulator.

Definition. A semiconductor is a material whose conductivity lies between that of conductors and insulators (resistivity roughly 10^{-3} to 10^{8} Ω·cm) and can be controlled by temperature, light, electric field and, above all, by adding small amounts of impurity (doping). Examples: Si, Ge, GaAs.

Difference, explained with energy bands.

1. **Conductor.** The valence and conduction bands overlap, or the highest band is only partly filled. There is no gap, so a huge number of free electrons (about 10^{22} cm^{-3}) is always present. Conductivity is very high ($\sigma \approx 10^{6}$ S/cm) and

almost independent of impurities. It **decreases** as temperature rises because lattice scattering increases.

2. **Insulator.** The valence band is full and the conduction band is empty, separated by a wide gap (E_g > about 4 eV, e.g. $SiO_2 \approx 9$ eV). Thermal energy at room temperature ($kT \approx 0.026$ eV) cannot lift electrons across, so there are practically no carriers. Conductivity is extremely small ($\sigma < 10^{-10}$ S/cm).
3. **Semiconductor.** Same full/empty arrangement but a **small gap** (Si 1.12 eV). At room temperature a small fraction of electrons is thermally excited into the conduction band, leaving holes behind, so both electrons and holes conduct. Conductivity is moderate and **increases** with temperature (negative temperature coefficient of resistance), and can be raised by many orders of magnitude by doping.

Summary. Conductors have no gap and fixed, abundant carriers; insulators have a wide gap and essentially no carriers; semiconductors have a small gap, two carrier types, and a carrier density that can be engineered.

Q2. Distinguish between elemental semiconductors and compound semiconductors, giving two examples of each.

	Elemental	**Compound**
Composition	One chemical element	Two or more elements
Periodic table	Group IV (14)	Combinations such as III–V, II–VI, IV–IV
Crystal structure	Diamond cubic	Zinc blende or wurtzite
Bonding	Purely covalent	Covalent with partial ionic character
Band gap	Fixed by the element; Si and Ge are indirect	Adjustable by choice of elements and alloying; many are direct
Processing	Mature, cheap (especially Si)	More difficult and costly; stoichiometry must be controlled
Typical use	ICs, power, detectors	Optoelectronics, RF, high power
Examples	**Silicon (Si), Germanium (Ge)**	**Gallium arsenide (GaAs), Indium phosphide (InP)**; also GaN, SiC, CdTe

Q3. Explain why silicon is widely used in semiconductor device fabrication.

1. **Abundance and cost.** Si makes up about 28 % of the Earth's crust as SiO_2 (sand, quartz). Raw material is cheap and the refining industry is enormous.
2. **Excellent native oxide.** Silicon oxidises to SiO_2, which is stable, insoluble in water, an excellent insulator ($E_g \approx 9$ eV, breakdown ≈ 10 MV/cm) and forms a very clean interface with Si (interface trap density about 10^{10} $cm^{-2}eV^{-1}$). SiO_2 is used as a diffusion mask, surface passivation, field isolation and the MOSFET gate dielectric. No other semiconductor has a comparable oxide.
3. **Suitable band gap.** At 1.12 eV it is large enough to give low intrinsic concentration and low leakage, so devices work up to about 150 °C (Ge fails above about 75 °C), yet small enough for good conduction when doped.
4. **Mature crystal growth.** Very large (300 mm), nearly defect-free, highly pure single crystals are routinely grown by the Czochralski method.
5. **Mechanical and thermal strength.** Si is hard, can be handled as thin wafers, and conducts heat well (about 3 times better than GaAs), which matters in power devices and dense ICs.
6. **Well-understood doping.** B, P, As and Sb dope it predictably by diffusion and ion implantation.
7. **Elemental.** No stoichiometry to control and no toxic constituents, unlike GaAs.

Its main limitation is the indirect gap, which makes it a poor light emitter; that is why lasers and LEDs use compound semiconductors.

Q4. Compare the properties and applications of silicon (Si), germanium (Ge), and gallium arsenide (GaAs).

Use the comparison table of Si, Ge and GaAs above as the core of the answer, then comment:

- **Structure and bonding.** Si and Ge are elemental, diamond cubic, purely covalent. GaAs is a III–V compound, zinc blende, mainly covalent with partial ionic character (As is more electronegative than Ga).
- **Band gap.** Ge (0.66 eV) < Si (1.12 eV) < GaAs (1.42 eV). The larger the gap, the lower n_i, the lower the leakage and the higher the maximum operating temperature. Si and Ge are **indirect**; GaAs is **direct**, so only GaAs emits light efficiently.
- **Mobility.** GaAs electrons are fastest (8500), then Ge (3900), then Si (1350) $cm^2/V{\cdot}s$. Hence GaAs and Ge suit high-speed work, though GaAs hole mobility is poor.
- **Oxide.** Only Si has an excellent native oxide, which is why MOS technology and VLSI are silicon.
- **Cost and processing.** Si cheapest and most mature; Ge more expensive; GaAs most expensive, brittle and toxic.

Applications.

- **Si:** microprocessors, memories, all digital and analogue ICs, MOSFETs and BJTs, power devices (IGBTs, thyristors), solar cells, sensors and MEMS.
- **Ge:** infrared detectors and lenses, high-speed SiGe heterojunction transistors, substrates for multi-junction space solar cells; historically the first transistors.
- **GaAs:** microwave and RF amplifiers in mobile phones and radar, satellite communication, HEMTs and MESFETs, LEDs, laser diodes for optical fibre links, high-efficiency solar cells for satellites.

Q5. What is a crystal lattice? Explain the significance of crystal structure in semiconductor materials.

Crystal lattice. A crystal lattice is an infinite, periodic three-dimensional array of points in space such that every point has identical surroundings. It is generated by repeating a unit cell (edges a, b, c and angles α, β, γ) by translation. Attaching a basis of one or more atoms to each lattice point gives the real crystal: crystal = lattice + basis. Si, for example, is an FCC lattice with a two-atom basis, which produces the diamond structure.

Significance of crystal structure.

1. **It produces the energy bands.** The periodic potential of the lattice splits the atomic levels into allowed bands and forbidden gaps (Bloch's theorem, Kronig–Penney model). The size and type (direct or indirect) of the band gap depend on the structure and the atomic spacing.
2. **Carrier mobility.** In a perfect periodic lattice electrons move without scattering; departures from periodicity (vibrations, impurities, defects, grain boundaries) cause scattering. Single crystals therefore have much higher mobility than polycrystalline or amorphous material.
3. **Anisotropy.** Properties such as surface atom density, oxidation rate, etch rate, interface trap density, piezoelectric response and mobility differ with crystal direction. Wafer orientation ((100) or (111)) is chosen accordingly.
4. **Doping and diffusion.** Dopant atoms sit on substitutional lattice sites to become active. Open structures (diamond, APF 0.34) allow interstitial diffusion.

5. **Epitaxy and heterostructures.** Layers can be grown defect-free only on substrates with matching structure and lattice constant (e.g. AlGaAs on GaAs).
6. **Mechanical behaviour.** Cleavage planes, hardness and how wafers are diced all follow from the structure and bond directions.
7. **Defects.** Departures from the ideal structure (vacancies, dislocations) create states in the gap that act as traps and recombination centres, shortening carrier lifetime.

Q6. Define Miller indices and explain how they are used to identify crystal planes.

Definition. Miller indices are a set of three smallest integers (h k l), proportional to the **reciprocals of the intercepts** a plane makes on the crystal axes (measured in units of the lattice constants), that specify the orientation of that plane and all planes parallel to it.

How they identify a plane (procedure):

1. Choose the origin at a lattice point not on the plane. Find the intercepts on x, y, z in units of a, b, c: say p, q, r.
2. Take reciprocals 1/p, 1/q, 1/r. A plane parallel to an axis has an infinite intercept and so a zero index.
3. Multiply by the lowest common multiple of the denominators to obtain integers.
4. Reduce to the smallest integers and enclose in round brackets (h k l). Negative indices carry a bar.

Use and notation. (h k l) names a particular plane; {h k l} the family of symmetry-equivalent planes; [u v w] a direction and ⟨u v w⟩ a family of directions. In cubic crystals [h k l] is normal to (h k l), and the spacing between planes is $d = a/\sqrt{(h^2 + k^2 + l^2)}$. Miller indices are used to specify wafer orientation ((100) for MOS, (111) for bipolar), etch planes, cleavage planes and X-ray diffraction peaks.

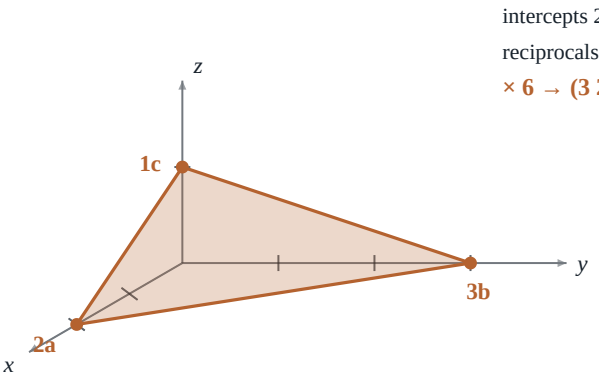
Example. Intercepts a, 2b, 3c → reciprocals 1, 1/2, 1/3 → × 6 → **(6 3 2)**.

Q7. Determine the Miller indices of a crystal plane having intercepts of 2a, 3b, and 1c.

Step	x	y	z
Intercepts (in units of a, b, c)	2	3	1
Reciprocals	1/2	1/3	1/1
Multiply by the LCM of the denominators, 6	3	2	6

Miller indices = (3 2 6).

Check: the integers 3, 2, 6 have no common factor, so no further reduction is possible. If the crystal is cubic with lattice constant a, the spacing of these planes is $d_{326} = a/\sqrt{(9 + 4 + 36)} = a/\sqrt{49} =$ **a/7**.



The (3 2 6) plane, cutting the axes at 2a, 3b and 1c.

Q8. Explain the difference between covalent, ionic, and metallic bonding and identify the dominant bonding mechanism in silicon.

Covalent bonding. Adjacent atoms share pairs of valence electrons so that each completes a stable outer shell. The shared electrons are localised between the two atoms, making the bond strong and directional. Covalent solids are hard, have high melting points, and are semiconductors or insulators depending on how tightly the bonds hold their electrons.

Ionic bonding. An electropositive atom gives up valence electrons to an electronegative atom; the resulting positive and negative ions attract electrostatically. The bond is strong but non-directional, there are no free electrons, and ionic solids (NaCl, MgO) are brittle insulators that conduct only when molten or dissolved.

Metallic bonding. Each atom releases its valence electrons into a common delocalised electron gas that binds the positive ion cores. The bond is non-directional, and the mobile electron gas gives high electrical and thermal conductivity, ductility and metallic lustre.

Key differences: how the electrons are held (shared between pairs / transferred to ions / shared by all atoms), directionality (yes / no / no), and the resulting electrical behaviour (semiconductor or insulator / insulator / conductor).

Bonding in silicon: covalent. Si has four valence electrons. In the diamond-cubic lattice each Si atom forms four sp^3 covalent bonds with four nearest neighbours at 109.5°, sharing one electron pair with each so that every atom has a full octet. Because Si–Si bonds are between identical atoms there is no ionic character. The modest bond strength (equivalent to E_g = 1.12 eV) is what lets thermal energy break a few bonds at room temperature, creating electron–hole pairs and giving silicon its semiconducting behaviour.

Practice questions

These extra questions practise this week's ideas. Work each one on paper before reading the solution; the calculation questions follow the same Question, Given, Find, Solution, Answer layout as the worked examples.

Practice 1.1: why germanium leaks more than silicon

Question. Explain why a germanium diode has a much larger reverse leakage current than a silicon diode of the same doping.

Answer. Leakage is carried by minority carriers, whose concentration is n_i^2/N. Germanium's band gap (0.66 eV) is much smaller than silicon's (1.12 eV), so far more bonds are broken thermally: n_i(Ge) ≈ 2.4×10^{13} cm^{-3} against 1×10^{10}

cm^{-3} for Si. Since leakage $\propto n_i^2$, the Ge diode has about $(2.4\times10^3)^2 \approx 6\times10^6$ times more minority carriers and a correspondingly larger leakage current.

Practice 1.2: Miller indices with a negative intercept

Question. A plane cuts the x-axis at −a, the y-axis at 2b and is parallel to the z-axis. Determine its Miller indices.

Given: intercepts −1, 2, ∞ (lattice units).

Find: (h k l).

Solution.

Reciprocals: −1, 1/2, 0

Multiply by 2: −2, 1, 0

Answer: ($\bar{2}$ 1 0), written with a bar over the 2 (in plain text (−2 1 0)).

Practice 1.3: interplanar spacing in germanium

Question. Germanium has a lattice constant of 5.65 Å. Calculate the spacing of the (110) and (111) planes.

Given: a = 5.65 Å.

Find: d_{110} and d_{111}.

Solution.

$d_{110} = a/\sqrt{(1 + 1 + 0)} = 5.65/\sqrt{2} = 3.995$ Å

$d_{111} = a/\sqrt{(1 + 1 + 1)} = 5.65/\sqrt{3} = 3.262$ Å

Answer: d_{110} = 4.00 Å, d_{111} = 3.26 Å.

Practice 1.4: atom density and mass density of silicon

Question. Silicon has the diamond structure with a = 5.43 Å and atomic mass 28.085 g/mol. Calculate the number of atoms per cm^3 and the density of silicon in g/cm^3.

Given: $a = 5.43\times10^{-8}$ cm, 8 atoms per unit cell, M = 28.085 g/mol, N_A(Avogadro) $= 6.022\times10^{23}$ mol^{-1}.

Find: atoms per cm^3 and density ρ.

Solution.

Cell volume $a^3 = (5.43\times10^{-8})^3 = 1.601\times10^{-22}$ cm^3

Atoms per $cm^3 = 8/a^3 = 8/1.601\times10^{-22} = 5.0\times10^{22}$ cm^{-3}

Mass of one cell $= 8 \times 28.085/6.022\times10^{23} = 3.731\times10^{-22}$ g

$\rho = 3.731\times10^{-22} / 1.601\times10^{-22} = 2.33$ g/cm^3

Answer: 5.0×10^{22} atoms/cm^3; ρ = 2.33 g/cm^3.

Practice 1.5: atom density of GaAs

Question. GaAs has the zinc-blende structure with a = 5.65 Å. Each unit cell contains 4 Ga and 4 As atoms; the molar mass of GaAs is 144.64 g/mol. Find the total atom density and the density of GaAs.

Given: $a = 5.65\times10^{-8}$ cm, 8 atoms (4 GaAs pairs) per cell, M = 144.64 g/mol.

Find: atoms per cm^3 and ρ.

Solution.

$a^3 = (5.65\times10^{-8})^3 = 1.804\times10^{-22}$ cm^3

Atoms per $cm^3 = 8/a^3 = 4.44\times10^{22}$ cm^{-3}

$\rho = 4 \times 144.64 / (6.022\times10^{23} \times 1.804\times10^{-22}) = 578.6/108.6 = 5.33$ g/cm^3

Answer: 4.44×10^{22} atoms/cm^3 (half Ga, half As); ρ = 5.33 g/cm^3.

Practice 1.6: the effect of doping on conductivity

Question. Compare the conductivity of intrinsic silicon ($n_i = 1\times10^{10}$ cm^{-3}) with that of silicon doped with 1×10^{15} cm^{-3} donors. Use $\mu_n = 1350$ and $\mu_p = 480$ $cm^2/V\cdot s$.

Given: $n_i = 10^{10}$ cm^{-3}, $N_D = 10^{15}$ cm^{-3}, $\mu_n = 1350$, $\mu_p = 480$ $cm^2/V\cdot s$.

Find: $\sigma_{intrinsic}$, σ_{doped} and their ratio.

Solution.

$\sigma_i = qn_i(\mu_n + \mu_p) = 1.6\times10^{-19} \times 10^{10} \times 1830 = 2.93\times10^{-6}$ S/cm

$\sigma = qN_D\mu_n = 1.6\times10^{-19} \times 10^{15} \times 1350 = 0.216$ S/cm

ratio $= 0.216/2.93\times10^{-6} = 7.4\times10^4$

Answer: doping with only one atom in 5×10^7 raises the conductivity about 74 000 times.

Practice 1.7: which light can silicon absorb?

Question. What is the longest wavelength of light that silicon ($E_g = 1.12$ eV) can absorb? Can a silicon photodiode detect the 1.55 µm light used in optical-fibre links?

Given: $E_g = 1.12$ eV.

Find: λ_{max}, and whether 1.55 µm is absorbed.

Solution.

$\lambda_{max} = 1.24/E_g = 1.24/1.12 = 1.11$ µm

A 1.55 µm photon has energy $1.24/1.55 = 0.80$ eV < 1.12 eV

Answer: λ_{max} = 1.11 µm; no, 1.55 µm photons do not have enough energy to create electron–hole pairs in silicon. Fibre receivers use Ge or InGaAs.

Objectives check

Use this table to confirm you have met this week's objectives. Cover the Answer column and answer each check question from memory. If you cannot, go back to the section in the last column before moving on.

#	Objective	Check yourself	Answer	Where it is taught
1	Define a semiconductor and distinguish it from a conductor and an insulator.	Why does a semiconductor conduct better when it is heated, while a metal conducts worse?	Heat breaks bonds and creates many more electron–hole pairs ($\propto e^{-E_g/2kT}$); in a metal the carrier number is fixed and only scattering increases.	Classification of materials
2	Describe the atomic structure of Si, Ge, Ga and As and identify their valence electrons.	How many valence electrons do Si, Ga and As have?	Si 4, Ga 3, As 5 (GaAs averages 4 per atom).	Atomic structure; valence electrons
3	Explain covalent bonding in silicon and electron–hole pair generation.	What two carriers are created when a covalent bond in Si breaks, and what minimum energy is needed?	A free electron and a hole; at least $E_g = 1.12$ eV.	Covalent bonding; electron–hole pairs
4	Distinguish intrinsic, n-type and p-type semiconductors, and donors from acceptors.	Si is doped with 10^{17} cm^{-3} boron. Is it n- or p-type, and what is the electron concentration ($n_i = 10^{10}$ cm^{-3})?	p-type; $n = n_i^2/N_A = 10^3$ cm^{-3}.	Intrinsic and extrinsic semiconductors
5	Compare elemental and compound semiconductors, and Si, Ge and GaAs.	Give one reason GaAs is preferred to Si for high-frequency work and one reason Si is preferred for ICs.	GaAs: electron mobility ≈ 8500 vs 1350 cm^2/V·s. Si: excellent native SiO_2 (also cheaper, larger wafers).	Elemental and compound; comparison table
6	Explain the energy-band picture, the band gap and direct versus indirect gaps.	Why is silicon not used for LEDs?	Its gap is indirect: a phonon is needed to conserve momentum, so radiative recombination is very inefficient.	Energy band concept; direct and indirect band gaps
7	Describe crystal lattices, unit cells and the diamond and zinc-blende structures.	How many atoms are in the diamond-cubic unit cell, and how many nearest neighbours does each atom have?	8 atoms per cell; 4 nearest neighbours.	Semiconductor crystal structure
8	Determine Miller indices of planes and the spacing between planes.	Find the Miller indices of a plane with intercepts 3a, ∞, c.	Reciprocals 1/3, 0, 1 → × 3 → (1 0 3).	Miller indices; Examples 1.2–1.3

Material Growth Techniques

Plan: purification of silicon (metallurgical to electronic grade); the Czochralski bulk crystal growth method; epitaxial growth techniques (VPE, LPE, MBE).

Learning objectives

By the end of this week you should be able to:

1. Describe the conversion of quartzite into metallurgical-grade silicon.
2. Explain how MG-Si is purified into electronic-grade silicon.
3. Explain the principle and steps of the Czochralski method.
4. State the advantages and limitations of CZ growth and compare it with float-zone growth.
5. Use the segregation coefficient to predict doping along an ingot.
6. Explain the principles of VPE, LPE and MBE and compare them.

The big picture

To make a chip you need three things, in this order:

1. **Extremely pure silicon.** A device stops working if more than about one atom in a billion is an unwanted impurity. Sand is nowhere near that pure.
2. **One single crystal.** The atoms must be lined up perfectly across the whole wafer (Week 1). Polycrystalline silicon, with grains pointing in different directions, scatters carriers and leaks current.
3. **Thin layers on top**, with different doping or even a different material, grown so that they continue the crystal perfectly.

This week covers each step: **purification** (sand → pure polysilicon), **crystal growth** (polysilicon → one big crystal, by the Czochralski method), and **epitaxy** (growing thin crystalline layers on top).

Part A: Purification of silicon

The whole route, from sand to wafer:

> quartzite (SiO_2) → metallurgical-grade Si (98 %) → trichlorosilane $SiHCl_3$ → purified $SiHCl_3$ → electronic-grade polysilicon → single crystal → wafers

Stage	Product	Purity
Arc furnace (reduction with carbon)	Metallurgical-grade Si (MG-Si)	About 98–99 %
Chlorination, distillation, Siemens CVD	Electronic-grade polysilicon (EG-Si)	99.9999999 % ("nine nines", 9N) or better
Czochralski or float-zone growth	Single-crystal ingot, doped as required	Dopant added deliberately

Step 1: making metallurgical-grade silicon (MG-Si)

Quartzite, a fairly pure form of SiO_2 (sand, quartz), is mixed with carbon (coal, coke, wood chips) and heated to about **2000 °C** in an electric arc furnace. Carbon pulls the oxygen off the silicon:

$$SiO_2 + 2C \rightarrow Si + 2CO$$

Molten silicon is drained from the furnace. This **MG-Si** is about 98 % pure. The 2 % of impurities (Fe, Al, Ca, C, B, P) is still about 10^{20} impurity atoms per cm^3, a hundred million times too many for devices. MG-Si is good enough for making steel and aluminium alloys, but not chips.

Step 2: turning silicon into a liquid that can be purified

Solids are hard to purify; liquids are easy to purify by distillation. So the next step is to convert silicon into a liquid compound. MG-Si powder is reacted with hydrogen chloride gas at about **300 °C**:

$$Si + 3HCl \rightarrow SiHCl_3 + H_2$$

The product, **trichlorosilane ($SiHCl_3$)**, is a liquid that boils at only 32 °C. The impurities also react with HCl and form their own chlorides ($FeCl_3$, BCl_3, PCl_3 and so on), which boil at different temperatures.

Step 3: fractional distillation

The trichlorosilane is purified by **fractional distillation**, the same process used to separate petrol from crude oil. Because $SiHCl_3$ and each impurity chloride boil at different temperatures, repeated distillation leaves $SiHCl_3$ with impurities at only a few **parts per billion**.

Step 4: Siemens process (back to solid silicon)

The pure $SiHCl_3$ is now turned back into silicon. It is mixed with pure hydrogen and passed over thin silicon rods heated electrically to about **1100 °C** inside a reactor. Silicon deposits on the hot rods (chemical vapour deposition, CVD):

$$SiHCl_3 + H_2 \rightarrow Si + 3HCl$$

Over many hours the rods grow into thick bars of **electronic-grade polycrystalline silicon (EG-Si)**, 9N to 11N pure. The HCl is recycled.

EG-Si is pure but still **polycrystalline** (many small grains) and undoped. The next step turns it into a single crystal and adds the doping.

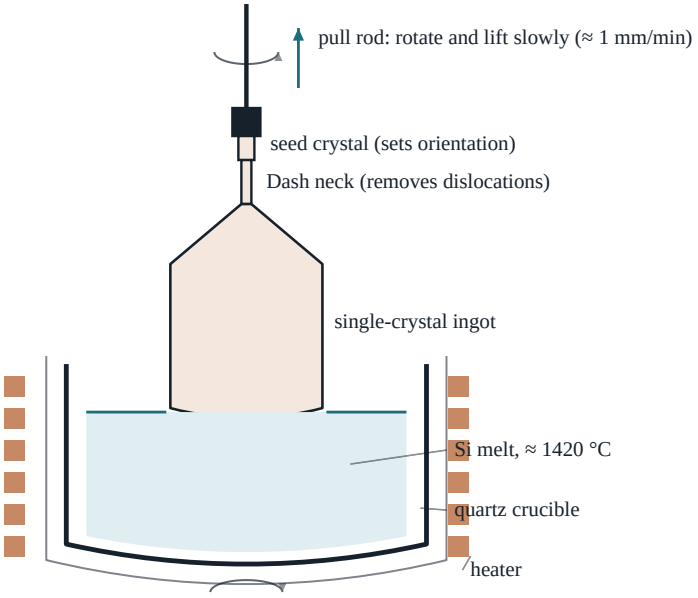
> **In plain words.** Steps 2–4 are a round trip: solid silicon → liquid compound → solid silicon. The trip is worth it because the liquid can be distilled, and distillation removes almost all the impurities.

Part B: Czochralski (CZ) crystal growth

The principle

Dip a small **seed crystal** into molten silicon and pull it out very slowly. Silicon from the melt freezes onto the seed, and as each atom freezes it lines up with the atoms already there. So the growing piece continues the seed's crystal, and you end up with one large single crystal (an **ingot**) with the same orientation as the seed.

It is like pulling a candle out of hot wax, except that each new layer copies the arrangement of the layer before it.



The Czochralski puller. The seed is dipped into the melt and slowly withdrawn while seed and crucible rotate in opposite directions; the melt freezes onto the seed with the seed's orientation.

The equipment

- A **quartz (SiO_2) crucible** holding the molten silicon, inside a graphite holder.
- **Heaters** (RF induction or resistance) keeping the melt just above silicon's melting point, **1414 °C**.
- A **pull rod** holding the seed, which rotates and rises slowly.
- A chamber filled with **argon** gas, so the hot silicon does not react with air.
- Sensors and control electronics to keep the diameter constant.

The steps

1. **Load.** Chunks of EG-Si and a carefully weighed amount of dopant (boron for p-type; phosphorus, arsenic or antimony for n-type) go into the crucible.
2. **Melt.** The chamber is filled with argon and the charge is melted at about 1420 °C.
3. **Dip.** The seed (cut to the wanted orientation, e.g. ⟨100⟩) is lowered until it just touches the melt.
4. **Neck (Dash technique).** The seed is pulled up quickly, growing a thin neck about 3 mm across. Dipping a cold seed into hot liquid creates crystal defects (dislocations); in a thin neck they run out of the sides and disappear, so the crystal below is dislocation-free.
5. **Shoulder and body.** The pull rate is reduced so the crystal widens to the full diameter (up to 300 mm). It is then pulled at about **1 mm per minute** while the seed and crucible rotate in **opposite directions** (10–30 rpm) to stir the melt and keep temperature and doping even.
6. **Tail.** At the end the pull rate is increased so the diameter narrows to a cone before the crystal leaves the melt. A sudden break would send defects back up the crystal.
7. **Cool** slowly. A typical ingot is 300 mm wide and up to 2 m long.

Controlling the diameter: pulling faster, or making the melt hotter, gives a thinner crystal; pulling slower or cooler gives a thicker one.

After growth the ingot is ground to an exact diameter, marked with a flat or notch to show its orientation, sliced into wafers with a wire saw, then lapped, etched and polished to a mirror finish.

Dopant segregation: why the doping changes along the ingot

When silicon freezes, impurity atoms "prefer" to stay in the liquid. The **segregation coefficient** measures this:

$$k_0 = C_s / C_l$$

C_s is the concentration in the solid and C_l in the liquid where they meet. For most dopants $k_0 < 1$: the crystal takes in less dopant than the melt contains.

Impurity	B	P	As	Sb	Al	Ga	O	Fe
k_0 in Si	0.8	0.35	0.3	0.023	0.002	0.008	0.25	8×10^{-6}

Consequence. The dopant left behind builds up in the shrinking pool of melt. So the end of the ingot that grows last is more heavily doped than the start. If C_0 is the starting melt concentration and f the fraction already frozen,

$$C_s = k_0\, C_0\, (1 - f)^{k_0 - 1}$$

Example 2.1: dopant segregation in a CZ crystal

Question. A boron-doped silicon crystal is to be grown by the Czochralski method with a boron concentration of 1×10^{16} cm^{-3} at the seed end. The segregation coefficient of boron in silicon is $k_0 = 0.8$. (a) What boron concentration must the melt contain at the start? (b) What is the boron concentration in the crystal when half of the melt has solidified?

Given: $C_s(\text{start}) = 1\times10^{16}$ cm^{-3}, $k_0 = 0.8$, f = 0.5 for part (b).

Find: (a) C_0, the initial melt concentration; (b) C_s at f = 0.5.

Solution.

(a) At the start, $C_s = k_0 C_0$, so

$$C_0 = C_s/k_0 = 1\times10^{16} / 0.8 = 1.25\times10^{16}\ \text{cm}^{-3}$$

(b) Use $C_s = k_0 C_0 (1 - f)^{k_0 - 1}$ with f = 0.5:

$$C_s = 0.8 \times 1.25\times10^{16} \times (0.5)^{-0.2} = 1\times10^{16} \times 1.149 = 1.15\times10^{16}\ \text{cm}^{-3}$$

Answer: (a) $C_0 = 1.25\times10^{16}$ cm^{-3}; (b) $C_s = 1.15\times10^{16}$ cm^{-3}, about 15 % higher than at the seed end, because rejected boron builds up in the melt.

The same effect also **purifies**: iron ($k_0 \approx 10^{-5}$) is almost entirely rejected into the melt.

Float-zone (FZ) growth, for comparison

A polysilicon rod is held vertically and a narrow molten zone, heated by an RF coil, is moved slowly along it, starting from a seed. **No crucible** touches the molten silicon, so there is almost no oxygen or carbon contamination, and very pure, high-resistivity silicon (up to 10^4 Ω·cm) can be made. FZ is used for power devices and radiation detectors, but wafers are smaller (≤ 200 mm) and more expensive.

Part C: Epitaxial growth

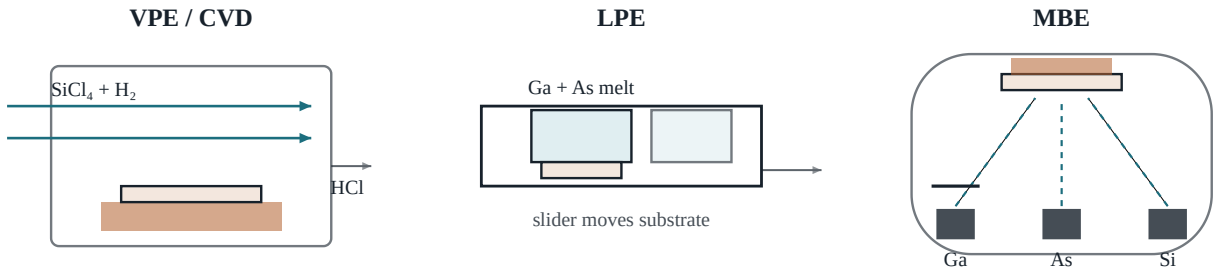
What epitaxy is and why we need it

Epitaxy (Greek *epi*, "upon", and *taxis*, "arrangement") is the growth of a thin single-crystal layer on a single-crystal wafer (the **substrate**), so that the new layer continues the substrate's crystal structure and orientation.

Why not just make everything in the bulk crystal? Because devices need **thin layers with different properties stacked on each other**, with sharp boundaries:

- a lightly doped layer on a heavily doped substrate (bipolar transistors, power diodes);
- a layer of a different material, such as AlGaAs on GaAs or InGaAs on InP (lasers, LEDs, high-speed transistors);
- a layer cleaner than the CZ substrate (less oxygen, fewer defects).

Homoepitaxy: same material (Si on Si). **Heteroepitaxy:** different materials (GaAs on Si, AlGaAs on GaAs). In heteroepitaxy the two **lattice constants** must be nearly equal (Week 1); otherwise the mismatch creates defects.



The three methods side by side: gas reacting on a hot surface (VPE), a cooling saturated melt (LPE), and atomic beams in ultra-high vacuum (MBE).

Vapour-phase epitaxy (VPE), also called CVD epitaxy

Principle. Gases containing silicon flow over a heated wafer. At the hot surface they react or break down, leaving silicon atoms behind. The atoms move about on the surface until they find the right place in the crystal, so the layer grows as a continuation of the wafer.

For silicon, the common reactions are:

$$SiCl_4 + 2H_2 \rightarrow Si + 4HCl \quad (\approx 1150\text{–}1250\ ^\circ C)$$

$$SiH_2Cl_2 \rightarrow Si + 2HCl \quad (\approx 1000\text{–}1100\ ^\circ C)$$

$$SiH_4 \rightarrow Si + 2H_2 \quad (\approx 600\text{–}1000\ ^\circ C)$$

Doping is done by adding a little dopant gas: $\mathbf{B_2H_6}$ (diborane) for p-type, $\mathbf{PH_3}$ (phosphine) or $\mathbf{AsH_3}$ (arsine) for n-type.

The growth rate is limited by one of two steps: how fast gas can reach the surface (**mass-transport limited**, at high temperature) or how fast the reaction happens on the surface (**reaction-rate limited**, at lower temperature).

For compound semiconductors the metal-organic version, **MOCVD**, is used: $(CH_3)_3Ga + AsH_3 \rightarrow GaAs + 3CH_4$ at about 600–750 °C. Nearly all commercial LEDs are made this way.

- Growth rate 0.1–5 µm/min; good uniformity; many wafers at once: the industrial workhorse.
- Problems: dopant from the substrate can evaporate and re-enter the layer (**autodoping**), and at high temperature dopant diffuses up from the substrate (**out-diffusion**), blurring the boundary.

Liquid-phase epitaxy (LPE)

Principle. Dissolve the semiconductor in a molten metal until no more will dissolve (a **saturated solution**), like sugar in hot water. Bring the wafer into contact with it and cool slowly. As the liquid cools it can hold less, so the extra semiconductor comes out of solution and crystallises onto the wafer, continuing its crystal.

Example: GaAs dissolved in molten gallium, at about **700–900 °C**, cooled at 0.1–1 °C per minute. A **sliding graphite boat** with several wells of different melts moves the wafer from one melt to the next to build up layers.

- Growth happens close to equilibrium, so crystal quality is excellent and defects few; the equipment is simple and cheap.
- But thickness control is poor (about 0.1 µm at best), boundaries are not sharp, surfaces are rough, and the area is small.
- Used for LEDs and the first heterostructure lasers.

Molecular-beam epitaxy (MBE)

Principle. In an **ultra-high vacuum** (about 10^{-10} Torr, roughly a ten-trillionth of atmospheric pressure), each element (Ga, As, Al, dopants) is heated in its own small oven (**effusion cell**). Atoms evaporate and travel in straight lines, as a beam, to the heated wafer, because there is no air to collide with. They land, move around, and join the crystal. Each oven has a **shutter** that can open or close in a fraction of a second, switching that element on or off.

Example: GaAs from Ga and As cells, wafer at about 550–650 °C.

- Growth is slow, about **1 µm per hour**, which is about **one atomic layer per second**. So layers can be controlled to **a single atomic layer**.
- Boundaries are atomically sharp. Ideal for quantum wells, superlattices, high-electron-mobility transistors (HEMTs) and advanced lasers.
- Growth can be watched while it happens by **RHEED** (electron diffraction), which counts the atomic layers.
- Very expensive equipment, slow, low throughput.

Comparing the three at a glance

	VPE (CVD)	**LPE**	**MBE**
Atoms arrive from	A gas, by chemical reaction	A liquid solution, by precipitation	A beam of atoms, in vacuum
Temperature	Highest (Si: 1000–1250 °C)	Medium (700–900 °C)	Lowest (500–650 °C)
Thickness control	Good (tens of nm)	Poor (~0.1 µm)	Excellent (one atomic layer)
Speed	Fast	Fast	Slow
Cost	Moderate	Low	Very high
Best for	Mass production (Si epi, LEDs by MOCVD)	Simple, thick, high-quality layers	Atomic-scale research and advanced devices

Key points of Week 2

- MG-Si (98 %) is made from quartz and carbon in an arc furnace: $SiO_2 + 2C \rightarrow Si + 2CO$.

- It is purified by converting it to liquid $SiHCl_3$, distilling, then depositing EG-Si by the Siemens process: $SiHCl_3 + H_2 \rightarrow Si + 3HCl$.
- CZ growth: a seed is pulled slowly from the melt, rotating; the crystal copies the seed. Dash necking removes dislocations.
- CZ gives large, cheap, dislocation-free crystals, but oxygen from the crucible and segregation ($k_0 < 1$) are its limitations.
- Epitaxy grows thin crystalline layers on a wafer: VPE from gas (production), LPE from a cooling melt (simple), MBE from atomic beams in vacuum (atomic precision).

Examination questions solved (Q9–Q13)

Q9. Describe the major stages involved in converting metallurgical-grade silicon (MG-Si) into electronic-grade silicon (EG-Si).

Starting point: production of MG-Si. Quartzite is reduced with carbon in an electric arc furnace at about 2000 °C: $SiO_2 + 2C \rightarrow Si + 2CO$. The product is about 98 % pure, with Fe, Al, Ca, B, P and C as the main impurities.

Stage 1: Hydrochlorination (conversion to a volatile compound). Pulverised MG-Si is reacted with anhydrous HCl at about 300 °C, usually with a catalyst in a fluidised bed: $Si + 3HCl \rightarrow SiHCl_3 + H_2$. Silicon becomes trichlorosilane, a liquid boiling at 32 °C. Impurities form their own chlorides.

Stage 2: Purification by fractional distillation. Because $SiHCl_3$ and the impurity chlorides have different boiling points, repeated fractional distillation separates them, leaving $SiHCl_3$ with electrically active impurities at parts-per-billion levels.

Stage 3: Reduction and deposition (Siemens process). The purified $SiHCl_3$ is mixed with high-purity hydrogen and passed over electrically heated thin silicon filaments (about 1100 °C) in a CVD reactor: $SiHCl_3 + H_2 \rightarrow Si + 3HCl$. Silicon deposits on the rods, which grow into thick polycrystalline bars of **electronic-grade silicon** (purity 9N–11N, i.e. 99.9999999 % or better). HCl and unreacted chlorosilanes are recovered and recycled.

Stage 4 (to make device material): single-crystal growth. The EG-Si polysilicon is broken into chunks, melted with the required dopant and grown into a single-crystal ingot by Czochralski or float-zone growth. Segregation during freezing purifies it further. The ingot is then shaped, sliced, lapped, etched and polished into wafers.

Q10. Explain the principle and operation of the Czochralski (CZ) crystal growth method.

Principle. Crystal pulling from the melt. A small single-crystal seed of known orientation is brought into contact with molten silicon held slightly above its melting point (1414 °C). Heat is conducted away up the seed, so the melt at the seed–melt interface solidifies. The freezing atoms take up positions that continue the seed's lattice, so the growing ingot is a single crystal with the same orientation as the seed. Pull rate and temperature control the diameter.

Equipment. Furnace (quartz crucible in a graphite susceptor, RF or resistance heater); crystal-pulling mechanism (seed holder on a rotating, slowly rising shaft; crucible rotation and lift); ambient control (argon flow at reduced pressure); diameter and temperature control system.

Operation.

1. Polysilicon EG-Si and a precise amount of dopant are loaded into the crucible. The chamber is purged with argon and the charge is melted.

2. The seed is lowered to touch the melt surface.
3. A thin neck is pulled quickly (Dash necking) to eliminate dislocations.
4. The pull rate and temperature are lowered so the crystal widens (shoulder) to the target diameter.
5. The body is grown at constant diameter with typical pull rate about 1 mm/min while seed and crucible counter-rotate to stir the melt and keep it thermally symmetric.
6. At the end the diameter is tapered to a tail cone and the crystal is lifted clear and cooled slowly.

The finished ingot is then characterised for orientation, resistivity and defects before being sliced into wafers.

Q11. State four advantages and two limitations of the Czochralski method.

Advantages:

1. **Large diameters**: 200 mm and 300 mm ingots are routine, which lowers cost per chip.
2. **Dislocation-free crystals** of controlled orientation ((100), (111)) from the seed, thanks to Dash necking.
3. **Controlled doping**: dopant added to the melt gives the required type and resistivity.
4. **High throughput and low cost**: mature, highly automated, relatively fast growth.
5. **Oxygen content strengthens the wafer** against warping and slip, and allows **intrinsic gettering** (oxygen precipitates in the bulk trap metallic impurities away from the device surface).

Limitations:

1. **Contamination from the crucible**: the quartz crucible dissolves slowly, introducing oxygen ($\approx 10^{18}$ cm^{-3}) and the graphite parts introduce carbon. Oxygen forms thermal donors and precipitates, and limits the attainable resistivity (CZ cannot easily exceed about 100 Ω·cm), so FZ is preferred for high-power and detector devices.
2. **Non-uniform axial doping** because of segregation ($k_0 < 1$): the dopant concentration rises from seed end to tail end, so resistivity varies along the ingot. Melt convection also causes radial non-uniformity and growth striations.

(Also acceptable: the crucible is consumed in each run; high energy consumption; batch process.)

Q12. Explain the principles of vapour-phase epitaxy (VPE), liquid-phase epitaxy (LPE), and molecular-beam epitaxy (MBE).

Epitaxy is the growth of a thin single-crystal film on a single-crystal substrate such that the film adopts the substrate's crystal structure and orientation. The three methods differ in the phase from which atoms arrive.

Vapour-phase epitaxy (VPE). The semiconductor is supplied as a gas-phase chemical compound carried by hydrogen over a heated substrate inside a reactor. The compound decomposes or is reduced at the hot surface, and the released atoms move to lattice sites. For silicon the standard reaction is hydrogen reduction of silicon tetrachloride at about 1200 °C, $SiCl_4 + 2H_2 \rightarrow Si + 4HCl$; silane $SiH_4 \rightarrow Si + 2H_2$ is used at lower temperature. Doping is achieved by adding gases such as PH_3, AsH_3 or B_2H_6. The rate is set either by mass transport of reactants through the gas boundary layer or by the surface reaction. MOCVD is the metal-organic version used for III–V compounds.

Liquid-phase epitaxy (LPE). The substrate is placed in contact with a saturated solution of the semiconductor in a molten metal solvent (e.g. GaAs in liquid Ga). On slow, controlled cooling the solubility decreases, the solution becomes supersaturated and the semiconductor precipitates epitaxially on the substrate. Growth occurs near thermodynamic equilibrium. A sliding graphite boat moves the substrate under successive melts to grow multilayers.

Molecular-beam epitaxy (MBE). In ultra-high vacuum ($\approx 10^{-10}$ Torr), elemental sources in heated effusion cells produce collision-free molecular beams aimed at a heated substrate. Atoms adsorb, migrate and incorporate in the

lattice. Each beam has a shutter, so composition and doping can be switched abruptly. Growth is slow (≈ 1 monolayer/s) and is monitored in real time by RHEED, so thickness is controlled to atomic precision.

Q13. Compare VPE, LPE, and MBE in terms of growth mechanism, temperature, precision, and applications.

Feature	VPE (CVD)	LPE	MBE
Source phase	Gas (chlorosilanes, silane, metal-organics)	Liquid solution (melt)	Atomic or molecular beams from solid sources
Growth mechanism	Chemical reaction / decomposition at the hot surface	Precipitation from a supersaturated solution on cooling; near equilibrium	Physical deposition: adsorption and surface migration; far from equilibrium
Environment	Atmospheric or reduced pressure, H_2 carrier	Inert or H_2, graphite boat	Ultra-high vacuum, 10^{-10}–10^{-11} Torr
Typical temperature	High: Si 1000–1250 °C; MOCVD 600–750 °C	Intermediate: 700–900 °C (GaAs)	Low: 400–650 °C (GaAs ≈ 600 °C)
Growth rate	0.1–5 μm/min	0.1–1 μm/min	≈ 1 μm/h (≈ 1 monolayer/s)
Thickness precision	≈ tens of nm; good uniformity	Poor, ≈ 0.1 μm; rough surface	Excellent, one atomic layer
Interface abruptness	Moderate (autodoping, out-diffusion)	Poor (graded)	Atomically abrupt
Crystal purity	Good	Very high, low deep-level defects	Very high (UHV)
In-situ monitoring	Limited	None	RHEED, mass spectrometry
Cost and throughput	Moderate cost, high throughput (batch)	Low cost, low throughput, small area	Very high cost, low throughput
Applications	Si epi layers for BJTs, CMOS, power devices; MOCVD for LEDs, lasers, solar cells	LEDs, early heterostructure lasers, thick layers, photodetectors	Quantum wells, superlattices, HEMTs, HBTs, quantum-cascade and VCSEL lasers, research

Conclusion. VPE is the industrial choice for volume production, LPE is a simple low-cost method giving high-quality thick layers with poor control, and MBE gives the ultimate atomic-scale control for advanced heterostructure devices at high cost.

Practice questions

These extra questions practise this week's ideas. Work each one on paper before reading the solution; the calculation questions follow the same Question, Given, Find, Solution, Answer layout as the worked examples.

Practice 2.1: the purification route

Question. List the stages from quartzite to electronic-grade silicon, giving the chemical reaction and approximate temperature of each.

Answer.

Stage	Reaction	Temperature
1. Arc furnace → MG-Si (98 %)	$SiO_2 + 2C \rightarrow Si + 2CO$	≈ 2000 °C
2. Hydrochlorination	$Si + 3HCl \rightarrow SiHCl_3 + H_2$	≈ 300 °C
3. Fractional distillation of $SiHCl_3$	(physical separation by boiling point)	≈ 32 °C boiling point
4. Siemens CVD → EG-Si polysilicon	$SiHCl_3 + H_2 \rightarrow Si + 3HCl$	≈ 1100 °C

Practice 2.2: phosphorus segregation along an ingot

Question. An n-type CZ crystal must have 5×10^{15} cm^{-3} phosphorus at the seed end. The segregation coefficient of P in Si is 0.35. Find the starting melt concentration and the phosphorus concentration in the crystal when 80 % of the melt has solidified.

Given: $C_s(0) = 5\times10^{15}$ cm^{-3}, $k_0 = 0.35$, $f = 0.8$.

Find: C_0 and C_s at $f = 0.8$.

Solution.

$C_0 = C_s/k_0 = 5\times10^{15}/0.35 = 1.43\times10^{16}$ cm^{-3}

$C_s = k_0 C_0 (1 - f)^{k_0 - 1} = 5\times10^{15} \times (0.2)^{-0.65} = 5\times10^{15} \times 2.85 = 1.42\times10^{16}$ cm^{-3}

Answer: $C_0 = 1.43\times10^{16}$ cm^{-3}; at $f = 0.8$, $C_s = 1.42\times10^{16}$ cm^{-3}, almost three times the seed-end value. This is why P-doped CZ ingots vary so much in resistivity along their length.

Practice 2.3: how much silicon in an ingot?

Question. A CZ ingot is 300 mm in diameter and 1.5 m long. Find its mass (density 2.33 g/cm^3) and how many wafers it gives if each wafer is 775 µm thick and the saw removes another 200 µm per cut.

Given: diameter 30 cm, length 150 cm, $\rho = 2.33$ g/cm^3, 775 + 200 = 975 µm per wafer.

Find: mass and number of wafers.

Solution.

Volume $= \pi r^2 L = \pi \times 15^2 \times 150 = 1.06\times10^5$ cm^3

Mass $= 1.06\times10^5 \times 2.33 = 2.47\times10^5$ g = 247 kg

Wafers = 150 cm / 0.0975 cm = 1538

Answer: about 247 kg and roughly 1500 wafers (fewer after the shoulder and tail are cut off).

Practice 2.4: MBE growth time

Question. A 0.5 µm GaAs layer is grown by MBE at 1 µm/h. How long does it take, and how many monolayers (each a/2 = 0.2825 nm thick) are deposited? What is the average monolayer rate?

Given: thickness 500 nm, rate 1 µm/h, monolayer 0.2825 nm.

Find: time, number of monolayers, rate.

Solution.

t = 0.5 µm / (1 µm/h) = 0.5 h = 1800 s

monolayers = 500 nm / 0.2825 nm = 1770

rate = 1770/1800 ≈ 1 monolayer per second

Answer: 30 minutes, about 1770 monolayers, at about one atomic layer per second, which is why MBE can control thickness to a single layer.

Practice 2.5: lattice mismatch

Question. Calculate the lattice mismatch for (a) Ge on Si (a = 5.65 and 5.43 Å) and (b) AlAs on GaAs (a = 5.661 and 5.653 Å). Which pair is suitable for defect-free heteroepitaxy?

Given: lattice constants above. $f = (a_{layer} - a_{sub})/a_{sub}$.

Find: f for each pair.

Solution.

(a) f = (5.65 − 5.43)/5.43 = 0.041 = 4.1 %

(b) f = (5.661 − 5.653)/5.653 = 0.0014 = 0.14 %

Answer: 4.1 % and 0.14 %. AlAs/GaAs is almost perfectly matched and grows defect-free, which is why AlGaAs/GaAs heterostructures are so widely used. Ge on Si quickly forms misfit dislocations unless the layer is very thin or graded.

Practice 2.6: CZ operating conditions

Question. Why is the CZ puller filled with argon, and why do the seed and crucible rotate in opposite directions?

Answer. Argon is inert: it stops the hot silicon reacting with oxygen and nitrogen, and the gas flow carries away SiO evaporating from the melt. Counter-rotation stirs the melt, so the temperature around the growing crystal is uniform (giving a round, constant-diameter crystal) and dopant is mixed evenly (giving uniform radial resistivity).

Objectives check

Use this table to confirm you have met this week's objectives. Cover the Answer column and answer each check question from memory. If you cannot, go back to the section in the last column before moving on.

#	Objective	Check yourself	Answer	Where it is taught
1	Describe the conversion of quartzite into metallurgical-grade silicon.	Write the arc-furnace reaction and state the purity of MG-Si.	$SiO_2 + 2C \rightarrow Si + 2CO$; about 98 % pure.	Part A, Step 1
2	Explain how MG-Si is purified into electronic-grade silicon.	Why is silicon converted into $SiHCl_3$ before purification?	$SiHCl_3$ is a liquid (boils at 32 °C) that can be purified by fractional distillation; solids cannot.	Part A, Steps 2–4
3	Explain the principle and steps of the Czochralski method.	What is the purpose of the Dash neck?	Growing a thin neck lets dislocations from the thermal shock run out of the crystal, so the ingot is dislocation-free.	Part B
4	State the advantages and limitations of CZ growth and compare it with float-zone growth.	Why does FZ silicon reach higher resistivity than CZ silicon?	No crucible touches the melt, so there is almost no oxygen or carbon contamination.	Part B; Float-zone
5	Use the segregation coefficient to predict doping along an ingot.	For phosphorus ($k_0 = 0.35$), is the tail end of a CZ ingot more or less heavily doped than the seed end?	More heavily doped: rejected phosphorus builds up in the melt.	Dopant segregation; Example 2.1
6	Explain the principles of VPE, LPE and MBE and compare them.	Which epitaxy method gives single-atomic-layer thickness control, and why?	MBE: slow growth (≈ one monolayer per second) and shuttered atomic beams in ultra-high vacuum.	Part C

Impurity Control and Doping

Plan: diffusion and ion implantation mechanisms; controlled doping profiles and evaluation of impurity distribution. This page follows your lecturer's handwritten notes on diffusion and ion implantation, including every calculation in them.

Learning objectives

By the end of this week you should be able to:

1. Explain why doping is needed and name the common dopants of silicon.
2. Explain diffusion and state Fick's first law and the temperature dependence of D.
3. Distinguish constant-source (erfc) and limited-source (Gaussian) diffusion and calculate their profiles.
4. Describe the step-by-step diffusion process and the atomic diffusion mechanisms.
5. Describe ion implantation, its equipment and the Gaussian implant profile.
6. Calculate implant peak concentration, junction depth, dose and beam current.
7. Compare diffusion with ion implantation, and explain channelling and annealing.
8. Describe how a doping profile is evaluated.

The big picture

A pure silicon wafer is almost useless. To make a diode, a transistor or a chip, we must put **dopant atoms** (boron, phosphorus, arsenic) into chosen parts of the wafer, in chosen amounts, down to a chosen depth. There are two ways to do this:

1. **Diffusion:** heat the wafer in an atmosphere rich in dopant. The dopant atoms creep in by themselves, the way a drop of ink spreads through water.
2. **Ion implantation:** turn the dopant atoms into ions, accelerate them with a high voltage, and fire them into the wafer like tiny bullets.

The purpose of both is the same: to change the electrical properties of the material and form the **n-type and p-type regions** needed for diodes, transistors, MOSFETs and integrated circuits.

Common dopants for silicon:

Dopant	Symbol	Makes silicon
Phosphorus	P	n-type
Arsenic	As	n-type
Antimony	Sb	n-type
Boron	B	p-type
Gallium	Ga	p-type

Part A: Diffusion

What diffusion is

Diffusion is the movement of dopant atoms from a region of **high concentration** to a region of **low concentration**, driven by thermal energy.

In processing, the silicon wafer is heated to a high temperature (about 900–1200 °C) and exposed to a dopant source. Dopant atoms enter the silicon surface and gradually move into the crystal.

Why atoms move "downhill"

Your notes illustrate this with gas molecules in a box. Suppose 21 of 32 oxygen molecules start on the left side of a box. Every molecule moves randomly, left or right with equal chance. Nothing pushes them in any particular direction. But because there are more molecules on the left, **more of them happen to cross from left to right than from right to left**. The result is a net flow towards the emptier side.

- **Concentration difference:** more particles on one side (21 of 32 on the left) creates a **concentration gradient**.
- **Net diffusion:** random motion carries particles both ways, but more move from the crowded side to the less crowded side.
- **Dynamic equilibrium:** eventually both sides have similar concentrations. Particles still move randomly in both directions, but the two flows now cancel, so there is no net change.

> **In plain words.** Diffusion is not a force. It is just statistics: if you have more atoms here than there, random jumping will, on average, carry atoms from here to there. The steeper the difference, the faster the flow.

The basic process is:

Dopant source → silicon surface → dopant penetration → desired dopant profile

Fick's first law: how fast atoms flow

$$J = -D \, dC/dx$$

Symbol	Meaning	Unit
J	**dopant flux**: number of dopant atoms crossing 1 cm^2 each second	atoms/cm^2·s
D	**diffusion coefficient**: how easily the dopant moves in silicon	cm^2/s
C	dopant concentration	atoms/cm^3
x	distance into the semiconductor	cm
dC/dx	the **concentration gradient**: how fast C changes with depth	atoms/cm^4

Reading the equation. The flow is proportional to how steeply the concentration changes. No gradient, no flow. The **minus sign** says the flow goes from high concentration towards low concentration: if C falls as x increases (dC/dx negative), J is positive, meaning atoms flow in the +x direction, deeper into the wafer.

(The J here is a flow of atoms, not an electric current. Later weeks use J for current density.)

The diffusion coefficient depends strongly on temperature

$$D = D_0 \, e^{-E_a/kT}$$

Symbol	Meaning
D_0	diffusion constant (cm^2/s), a property of the dopant–silicon pair
E_a	**activation energy** (eV): the energy barrier an atom must jump over to move one lattice site
k	Boltzmann constant, 8.617×10^{-5} eV/K
T	absolute temperature (K)

Because T sits inside an exponential, **increasing temperature greatly increases the diffusion rate**. Typical E_a is 3–4 eV, so D rises about tenfold for every 50–100 °C. This is why diffusion only happens in a furnace: at room temperature D is so small that dopants do not move at all in a lifetime.

How far do atoms travel? The diffusion length

The depth to which dopants spread grows with the **square root** of time:

$$\text{diffusion length} \approx \sqrt{(Dt)}$$

Your notes use the form **L = √(2Dt)**. Other books write $2\sqrt{(Dt)}$, which is what appears inside the erfc formula below. They are the same idea with a different constant in front. The key fact is that **depth ∝ √(Dt)**: diffusing four times as long only doubles the depth. In a calculation, use whichever form the question gives.

Two types of diffusion

A. Constant-source diffusion (predeposition)

The surface of the semiconductor is kept at an approximately **constant dopant concentration C_s,** because the source (dopant gas or glass) is present all the time. The profile is:

$$C(x, t) = C_s \, \text{erfc}\left(x / 2\sqrt{(Dt)} \right)$$

Symbol	Meaning
C_s	surface concentration ($atoms/cm^3$), fixed
x	depth
D	diffusion coefficient
t	diffusion time
erfc	the **complementary error function**, erfc(u) = 1 − erf(u). You read it from a table.

Characteristics:

1. The dopant source remains available throughout the process.
2. The surface concentration stays approximately constant.
3. It produces an erfc concentration profile.
4. It is often used for the **initial doping step**.

What erfc looks like. erfc(0) = 1, so at the surface C = C_s. As the argument u = x/2√(Dt) grows, erfc falls: erfc(0.5) = 0.48, erfc(1) = 0.16, erfc(2) = 0.005, erfc(3) ≈ 2×10^{-5}. So the concentration is highest at the surface and falls off steeply with depth. As time goes on, 2√(Dt) gets bigger, so the same depth corresponds to a smaller u, and the profile pushes deeper.

u	erfc(u)	u	erfc(u)
0	1.000	1.0	0.157
0.25	0.724	1.5	0.034
0.5	0.480	2.0	4.7×10^{-3}
0.527	0.456	2.5	4.1×10^{-4}
0.75	0.289	3.0	2.2×10^{-5}

B. Limited-source diffusion (drive-in)

After a dopant has been introduced into the surface, the wafer is heated **without** supplying any more dopant. The fixed amount already there spreads deeper. The profile is approximately **Gaussian**:

$$C(x, t) = (Q / \sqrt{(\pi Dt)})\, e^{-x^2/4Dt}$$

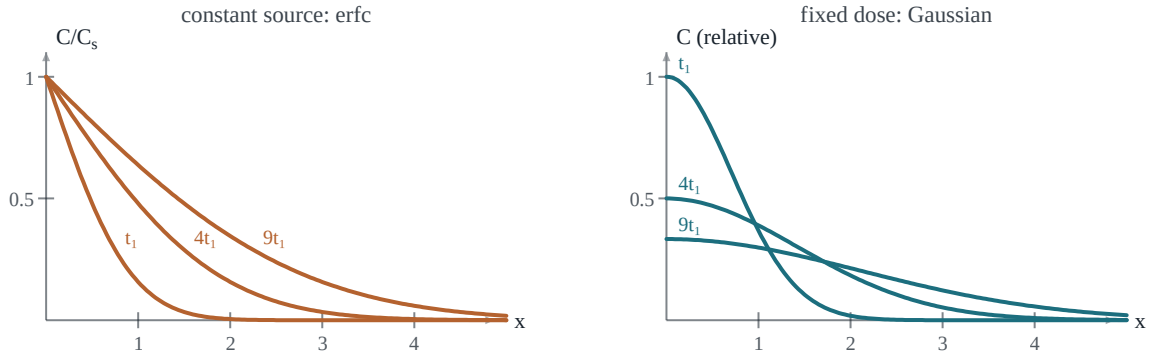
Symbol	Meaning
Q	total dopant dose per unit area (atoms/cm^2), fixed
D	diffusion coefficient
t	time

Characteristics:

1. The dopant penetrates deeper.
2. The **surface concentration decreases** (the same atoms are spread over a greater depth).
3. The total dopant quantity is approximately **conserved**.
4. It is used to obtain a desired **junction depth**.

The surface concentration is the value at x = 0, where $e^0 = 1$:

$$C(0, t) = Q / \sqrt{(\pi Dt)}$$



Left: constant-source (erfc) profiles at three times; the surface value stays fixed while the profile deepens. Right: limited-source (Gaussian) profiles; the area under each curve (the dose) is the same, so the surface value falls as the profile spreads.

> **In plain words.** Constant source: keep pouring ink on the surface, and the colour soaks deeper while the surface stays dark. Limited source: put one drop on, stop, and wait; the colour spreads deeper but gets paler everywhere.

Junction depth

If the wafer was already doped with the opposite type, at background concentration N_B (for example n-type 10^{16} cm^{-3} and we diffuse boron in), a **p-n junction** forms at the depth x_j where the diffused concentration has fallen to equal the background:

$$C(x_j) = N_B$$

erfc profile: $x_j = 2\sqrt{(Dt)} \cdot \text{erfc}^{-1}(N_B/C_s)$

Gaussian profile: $x_j = 2\sqrt{(Dt)} \cdot \sqrt{(\ln(C_s/N_B))}$

Step-by-step diffusion process

1. **Wafer preparation.** A high-purity silicon wafer is cleaned to remove (a) dust, (b) organic contamination, (c) metallic impurities and (d) native surface oxide and contamination.
2. **Dopant source preparation.** A suitable dopant source is selected: boron for p-type, phosphorus or arsenic for n-type.
3. **Heating.** The wafer is placed in a diffusion furnace and heated to a high temperature.
4. **Dopant introduction.** The dopant is supplied to the surface of the silicon.
5. **Dopant penetration.** Thermal energy allows dopant atoms to move from the surface into the silicon lattice.
6. **Drive-in.** Additional heating may be used to drive the dopant deeper into the wafer.
7. **Cooling.** The wafer is cooled under controlled conditions.
8. **Characterisation.** The resulting dopant profile is measured to determine (1) junction depth, (2) sheet resistance, (3) dopant concentration and (4) electrical properties.

Diffusion at the atomic level

Dopant atoms move through the silicon crystal mainly by two mechanisms:

A. Vacancy mechanism. A dopant atom sitting on a lattice site can only move when a neighbouring site is empty (a **vacancy**). It jumps into the vacancy:

atom → vacancy → new lattice position

This is how the main dopants (B, P, As, Sb) move. It is slow, because it needs a vacancy to be next to the atom at the right moment.

B. Interstitial mechanism. A small atom moves through the **spaces between** the normal lattice positions without needing a vacancy. This is much faster. Metals such as gold, copper and iron move this way.

Some dopants use both: they move through interstitial sites, or by interactions involving both interstitials and vacancies. Which mechanism a dopant uses determines how quickly it diffuses.

Advantages and disadvantages of diffusion

Advantages	Disadvantages
Relatively simple process	Requires high temperature
Can process many wafers simultaneously	Dopants spread laterally (sideways) as well as vertically
Suitable for high-temperature processing	Less precise depth control
Good for introducing large dopant concentrations	Difficult to form extremely shallow junctions
Historically important in semiconductor fabrication	High-temperature exposure can affect previously fabricated structures

---

Diffusion calculations (from the lecture notes)

Q1. Phosphorus is diffused into silicon with $D = 2\times10^{-13}$ cm^2/s for t = 500 s. Find the approximate diffusion distance using $L = \sqrt{(2Dt)}$.

Given: $D = 2\times10^{-13}$ cm^2/s, t = 500 s. **Find:** $L = \sqrt{(2Dt)}$.

Solution.

$$L = \sqrt{(2 \times 2\times10^{-13} \times 500)} = \sqrt{(2\times10^{-10})} = 1.414\times10^{-5} \text{ cm}$$

Convert to micrometres (1 cm = 10^4 μm):

$$L = 1.414\times10^{-5} \times 10^4 = 0.141 \text{ μm}$$

L ≈ 0.141 μm. Note how small this is: dopants move only a fraction of a micrometre in minutes.

Q2. A dopant concentration gradient in silicon is 4×10^{18} cm^{-4} and $D = 3\times10^{-14}$ cm^2/s. Find the diffusion flux using Fick's first law.

Given: $dC/dx = 4\times10^{18}$ cm^{-4}, $D = 3\times10^{-14}$ cm^2/s. **Find:** the flux $J = -D\, dC/dx$.

Solution.

$$J = -D\, dC/dx = -(3\times10^{-14})(4\times10^{18}) = -1.2\times10^5 \text{ atoms/cm}^2\cdot\text{s}$$

$|J| = 1.2\times10^5$ atoms/cm^2·s. The minus sign means the dopant moves from the region of higher concentration towards lower concentration (against the direction in which C increases).

Q4. Find D at T = 1273 K for a dopant with $D_0 = 0.5$ cm^2/s and $E_a = 3.5$ eV.

Given: $D_0 = 0.5$ cm^2/s, $E_a = 3.5$ eV, T = 1273 K, $k = 8.617\times10^{-5}$ eV/K. **Find:** $D = D_0 e^{-E_a/kT}$.

Solution.

$$kT = 8.617\times10^{-5} \times 1273 = 0.1097 \text{ eV}$$

$$E_a/kT = 3.5/0.1097 = 31.90$$

$$D = 0.5 \times e^{-31.90} = 0.5 \times 1.39\times10^{-14} = 6.9\times10^{-15} \text{ cm}^2/\text{s}$$

$D \approx 6.9\times10^{-15}$ cm^2/s. Try T = 1373 K (100 °C hotter): kT = 0.1183 eV, $E_a/kT = 29.58$, $D = 7.1\times10^{-14}$ cm^2/s, ten times larger.

Q5. Constant-source diffusion: $C_s = 1\times10^{20}$ cm^{-3}, $D = 1\times10^{-13}$ cm^2/s, t = 3600 s. Find C at x = 0.2 μm.

Given: $C_s = 1\times10^{20}$ cm^{-3}, $D = 1\times10^{-13}$ cm^2/s, t = 3600 s, x = 0.2 μm. **Find:** $C(x, t) = C_s\,\mathrm{erfc}(x/2\sqrt{(Dt)})$.

Solution.

Step 1, convert the depth: $x = 0.2\ \mu m = 2\times10^{-5}$ cm.

Step 2, the diffusion length: $\sqrt{(Dt)} = \sqrt{(1\times10^{-13}\times 3600)} = \sqrt{(3.6\times10^{-10})} = 1.897\times10^{-5}$ cm.

Step 3, the erfc argument:

$$u = x / 2\sqrt{(Dt)} = 2\times10^{-5} / (2 \times 1.897\times10^{-5}) = 0.527$$

Step 4, from the table: erf(0.527) = 0.544, so erfc(0.527) = 1 − 0.544 = 0.456.

Step 5:

$$C = C_s \times \mathrm{erfc}(u) = 1\times10^{20} \times 0.456 = 4.56\times10^{19}\ \text{atoms/cm}^3$$

$C(0.2\ \mu m) = 4.56\times10^{19}$ cm^{-3}, a little under half the surface value.

Q6. Drive-in: dose $Q = 2\times10^{15}$ atoms/cm^2, $D = 5\times10^{-14}$ cm^2/s, t = 1000 s. Find the concentration at x = 0.

Given: $Q = 2\times10^{15}$ cm^{-2}, $D = 5\times10^{-14}$ cm^2/s, t = 1000 s, x = 0. **Find:** $C(0, t) = Q/\sqrt{(\pi Dt)}$.

Solution.

At x = 0, $e^{-0} = 1$, so

$$C(0, t) = Q / \sqrt{(\pi Dt)}$$

$$Dt = 5\times10^{-14} \times 1000 = 5\times10^{-11}\ \text{cm}^2$$

$$\pi Dt = 1.571\times10^{-10},\quad \sqrt{(\pi Dt)} = 1.253\times10^{-5}\ \text{cm}$$

$$C(0, t) = 2\times10^{15} / 1.253\times10^{-5} = 1.60\times10^{20}\ \text{atoms/cm}^3$$

Surface concentration = 1.60×10^{20} cm^{-3}.

Why x = 0? x is the distance from the silicon surface: x = 0 is exactly at the surface; x = 0.1 μm is 0.1 μm below it; any x > 0 means below the surface. The question asks for the surface concentration, so we set x = 0 and the exponential becomes 1. (The dose is atoms per cm^2 because it is the total number of atoms under each square centimetre of surface.)

Part B: Ion implantation

What ion implantation is

Ion implantation is a doping technique in which dopant atoms are converted into **ions**, **accelerated** to high energy, and physically **injected** into the semiconductor wafer.

Unlike thermal diffusion, ion implantation allows the **dopant dose** and **implantation depth** to be controlled very precisely.

Basic flowchart:

Dopant source → ioniser → mass analyser → accelerator → beam → wafer

Main components: ion source, mass analyser, acceleration system, beam focusing system, scanning system, the semiconductor wafer, and a vacuum chamber (the ions must not hit air molecules on the way).

Step-by-step ion implantation

1. **Dopant ion generation.** A dopant-containing material (boron, phosphorus, arsenic) is introduced into an ion source, where its atoms are converted into positively charged ions.
2. **Ion extraction.** An electric field pulls the ions out of the source.
3. **Mass analysis.** The ion beam passes through a magnetic mass analyser. A magnet bends the path of each ion by an amount that depends on its **mass-to-charge ratio**, so only the wanted ion goes through the exit slit. This prevents unwanted species being implanted.
4. **Acceleration.** The selected ions are accelerated through a high electric potential. Their kinetic energy is **E = qV**, where E is the ion energy, q the ion charge and V the accelerating voltage. For a singly charged ion E = eV, so an ion accelerated through 100 kV has 100 keV.
5. **Beam focusing and scanning.** The ion beam is focused and scanned across the wafer so that the dopant is distributed uniformly.
6. **Implantation.** The energetic ions enter the silicon. As they travel through it, they collide with silicon nuclei and electrons and lose energy. Eventually they stop at a particular average depth.
7. **Annealing.** Implantation damages the crystal, because energetic ions knock silicon atoms out of place. The wafer is therefore heated (**thermal annealing**), which repairs crystal damage and moves the dopant atoms into proper lattice positions where they become electrically active.
8. **The implantation profile.** The implanted dopant normally produces an approximately **Gaussian** distribution.

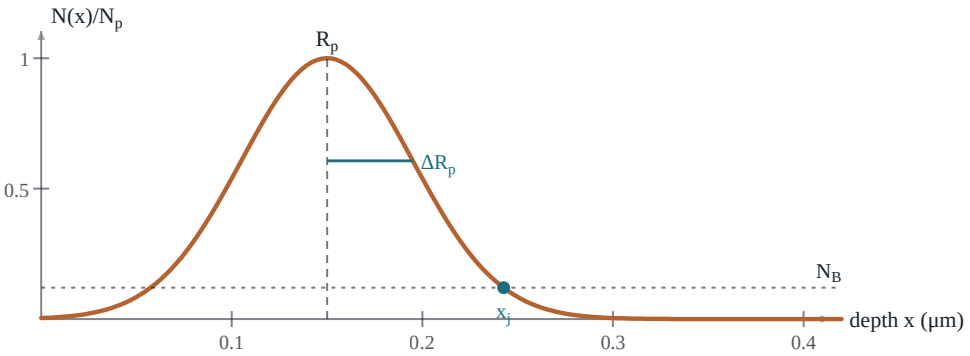
The implanted profile

$$C(x) = (Q / (\sqrt{(2\pi)}\ \Delta R_p))\ \exp[\ -(x - R_p)^2 / (2\Delta R_p^{\,2})\ ]$$

Symbol	Meaning
Q	implanted dose (ions/cm^2)
R_p	**projected range**: the average depth at which ions stop
ΔR_p	**straggle**: the spread of stopping depths about R_p (standard deviation)
x	depth

The peak is at $x = R_p$, below the surface, and its height is

$$N_p = Q / (\sqrt{(2\pi)}\ \Delta R_p) \approx 0.4\ Q/\Delta R_p$$



An implanted profile is a bell curve centred at R_p below the surface, with width set by ΔR_p. The junction forms where it falls to the background doping N_B.

> **In plain words.** Fire a million identical bullets into sand at the same speed. They do not all stop at exactly the same depth, but most stop near an average depth (R_p), with a spread (ΔR_p). That is the bell curve.

Implantation parameters

- **Implantation dose** Q = number of implanted ions per unit area, usually expressed in **ions/cm^2**.
- **Implantation energy**, usually expressed in **keV or MeV**. A higher implant energy generally produces a **deeper** implant.

The dose comes from the beam current I, the time t and the area A:

$$Q = (I\,t)/(q\,A) \qquad \text{(singly charged ions)}$$

How ion implantation works inside silicon

ion enters → collisions with atoms → loses energy → changes direction → eventually stops

There are two major energy-loss mechanisms:

Nuclear stopping. The implanted ion collides with silicon **nuclei**. This causes (i) deflection of the ion and (ii) displacement of silicon atoms, which is **lattice damage**. The nuclear stopping power is

$$S_n = -(1/N_{Si})\,(dE/dx)_{nuclear}$$

Electronic stopping. The ion interacts with the **electrons** in the silicon, losing energy through excitation and ionisation. At high energies electronic stopping is particularly important. It causes little damage.

$$S_e = -(1/N_{Si})\,(dE/dx)_{electronic}$$

Here N_{Si} is the number of silicon atoms per cm^3 (5×10^{22} cm^{-3}) and dE/dx the energy lost per unit distance. Heavy ions (As) lose energy mainly to nuclei and stop shallow; light ions (B) lose it mainly to electrons and go deeper.

Channelling

Channelling occurs when ions enter the crystal along certain crystallographic directions and travel relatively deeply through open channels between rows of atoms. This produces an unexpectedly deep dopant distribution.

To reduce channelling, manufacturers may:

1. **tilt** the wafer relative to the ion beam (typically 7°);
2. **rotate** the wafer;
3. use suitable implantation conditions;
4. sometimes introduce a thin **amorphous layer** (a screen oxide, or a pre-amorphising implant) so the crystal channels are hidden.

Example: forming an n-type region both ways

Suppose we want to convert part of a silicon wafer into an n-type region.

Using diffusion: silicon wafer → phosphorus source → high temperature → P atoms diffuse into Si → n-type region

Using ion implantation: phosphorus source → P^+ ions → mass analysis → acceleration → implantation into Si → annealing → n-type region

Diffusion versus ion implantation

Feature	Diffusion	Ion implantation
Dopant introduction	Thermal movement	Accelerated ions
Main driving force	Concentration gradient	Ion kinetic energy
Temperature	High	Relatively low (implant near room temperature)
Depth control	Less precise	Highly precise
Dose control	Moderate	Excellent
Crystal damage	Low	Significant
Annealing	Sometimes required	Normally required
Profile	erfc or Gaussian, maximum at the surface	Approximately Gaussian, peak below the surface
Modern IC use	Less dominant for shallow junctions	Widely used
Equipment	Diffusion furnace	Ion implanter

Ion implantation calculations (from the lecture notes)

P1. A silicon wafer is implanted with boron, $Q = 1.0\times10^{13}$ cm^{-2}, $R_p = 0.5$ μm, $\Delta R_p = 0.05$ μm. (a) Find the peak concentration N_p. (b) Find the boron concentration at the surface ($x = 0$).

Given: $Q = 1.0\times10^{13}$ cm^{-2}, $R_p = 0.5$ μm, $\Delta R_p = 0.05$ μm. **Find:** (a) N_p; (b) $N(0)$.

Solution.

(a) Convert $\Delta R_p = 0.05$ μm $= 5.0\times10^{-6}$ cm.

$$N_p = Q / (\sqrt{(2\pi)}\,\Delta R_p) = 1.0\times10^{13} / (2.5066 \times 5.0\times10^{-6}) = 1.0\times10^{13} / 1.253\times10^{-5}$$
$$= 7.98\times10^{17} \text{ cm}^{-3}$$

(b) At x = 0:

$$N(0) = N_p \exp(-R_p^2/2\Delta R_p^2), \quad R_p/\Delta R_p = 0.5/0.05 = 10$$

$$N(0) = N_p \exp(-10^2/2) = N_p\, e^{-50} = 7.98\times10^{17} \times 1.93\times10^{-22} = 1.54\times10^{-4}\ \text{cm}^{-3}$$

Effectively **zero boron at the surface**: the whole dose is buried around 0.5 µm.

P2. An n-type substrate with $C_B = 1.0\times10^{15}$ cm^{-3} is implanted with boron to form a p-type region. Peak $N_p = 1.0\times10^{17}$ cm^{-3} at $R_p = 0.4$ µm, $\Delta R_p = 0.05$ µm. Find the junction depth x_j (where the boron concentration equals C_B).

Given: $C_B = 1.0\times10^{15}$ cm^{-3}, $N_p = 1.0\times10^{17}$ cm^{-3}, $R_p = 0.4$ µm, $\Delta R_p = 0.05$ µm. **Find:** x_j where $N(x_j) = C_B$.

Solution.

At the junction $N(x_j) = C_B$:

$$C_B = N_p \exp(-(x_j - R_p)^2/2\Delta R_p^2)$$

$$\ln(N_p/C_B) = (x_j - R_p)^2/2\Delta R_p^2$$

$$x_j = R_p + \Delta R_p \sqrt{(2\ln(N_p/C_B))}$$

Numbers: $N_p/C_B = 10^{17}/10^{15} = 100$; $\ln(100) = 4.605$; $\times 2 = 9.21$; $\sqrt{9.21} = 3.035$.

$$x_j = 0.4\ \mu\text{m} + 0.05\ \mu\text{m} \times 3.035 = 0.4 + 0.152 = 0.552\ \mu\text{m}$$

The p-n junction is formed **0.552 µm below the silicon surface**. (There is also a second junction on the shallow side, at $R_p - 0.152 = 0.248$ µm, because the buried p-region has n-type silicon above it as well as below it.)

P3. A 300 mm diameter wafer needs a boron dose $Q = 2.0\times10^{14}$ ions/cm^2. (a) Find the total number of ions. (b) If the implanter uses singly charged ions ($q = 1.6\times10^{-19}$ C) and processes the wafer in 60 s, what beam current is needed?

Given: diameter 300 mm, $Q = 2.0\times10^{14}$ cm^{-2}, $q = 1.6\times10^{-19}$ C, t = 60 s. **Find:** (a) total ions N; (b) beam current I.

Solution.

(a) Radius r = 300/2 = 150 mm = 15 cm. Area $A = \pi r^2 = 3.142 \times 225 = 706.9$ cm^2.

$$N_{total} = Q \times A = 2.0\times10^{14} \times 706.9 = 1.414\times10^{17}\ \text{ions}$$

(b) Total charge delivered = $N_{total} \times q = 1.414\times10^{17} \times 1.6\times10^{-19} = 0.0226$ C.

$$I = Q_{charge} / t = 0.0226 / 60 = 3.77\times10^{-4}\ \text{A} = 0.377\ \text{mA}$$

The implanter needs a beam current of **0.377 mA** to deliver the target dose in one minute.

P4. A photoresist mask must protect part of a wafer during a boron implant. In photoresist $R_p = 0.8$ µm and $\Delta R_p = 0.08$ µm. The substrate under the mask must receive no more than 0.1 % of the peak concentration. Find the minimum mask thickness x_{mask}.

Given: $R_p = 0.8$ µm, $\Delta R_p = 0.08$ µm in resist, $N(x_{mask})/N_p \le 10^{-3}$. **Find:** minimum x_{mask}.

Solution.

At the bottom of the mask the concentration must be 10^{-3} of the peak:

$$N(x_{mask})/N_p = \exp(-(x_{mask} - R_p)^2/2\Delta R_p^2) = 10^{-3}$$

Take logarithms. $\ln(10^{-3}) = -3 \ln 10 = -6.908$. The minus signs cancel:

$$(x_{mask} - R_p)^2/2\Delta R_p^2 = 6.908$$
$$x_{mask} - R_p = \Delta R_p \sqrt{(2 \times 6.908)} = \Delta R_p \sqrt{13.816} = 3.717\ \Delta R_p$$
$$x_{mask} = 0.8 + 0.08 \times 3.717 = 0.8 + 0.297 = 1.097\ \mu m$$

To keep transmission below 0.1 % the photoresist must be **at least 1.10 µm thick**. In real manufacturing engineers add a safety margin and use about 1.3–1.4 µm.

Part C: Evaluating the impurity distribution

After doping, the engineer must check that the dose, the profile and the junction depth are what was intended (step 8 of the diffusion process). The main methods:

Method	What it measures	How it works
Four-point probe	Sheet resistance R_s (Ω/□)	Pass current through the outer two of four probes in a line; measure voltage across the inner two
Groove-and-stain	Junction depth x_j	Grind a shallow groove, stain it (p and n regions take different colours), measure under a microscope
Spreading resistance profiling	Carrier concentration against depth	Step two probes down a shallow bevel, measuring resistance at each point
C-V profiling	Doping against depth	Capacitance of a junction against reverse bias (Week 12)
SIMS	All dopant atoms against depth	Sputter the surface with an ion beam and count ejected dopant ions in a mass spectrometer
Hall measurement	Carrier type, concentration, mobility	Hall voltage in a magnetic field (Week 7)

Sheet resistance, explained

A diffused layer is a thin conducting skin of thickness x_j. Its resistance is described by the **sheet resistance**:

$$R_s = \rho / x_j \quad \text{[ohms per square, Ω/□]}$$

"Per square" means: any square piece of the layer, whatever its size, has resistance R_s between two opposite edges. (A bigger square is longer, which raises R, but also wider by the same factor, which lowers R; the two cancel.) So a resistor L long and W wide made from the layer has

$$R = R_s \times (L/W) = R_s \times (\text{number of squares})$$

Measured with a four-point probe on a large thin sample:

$$R_s = 4.532 \times (V/I)$$

Example 3.1: sheet resistance from a four-point probe

Question. A four-point probe on a boron-diffused wafer reads V = 2.0 mV when I = 1.0 mA is passed. The junction depth is 1.0 µm. Calculate the sheet resistance of the layer and its average resistivity.

Given: V = 2.0 mV, I = 1.0 mA, x_j = 1.0 μm = 1.0×10^{-4} cm, R_s = 4.532 V/I.

Find: R_s and the average resistivity ρ.

Solution.

Step 1. Sheet resistance:

$$R_s = 4.532 \times V/I = 4.532 \times (2.0\times10^{-3} / 1.0\times10^{-3}) = 4.532 \times 2.0 = 9.06\ \Omega/\square$$

Step 2. Average resistivity, from $R_s = \rho/x_j$:

$$\rho = R_s \times x_j = 9.06 \times 1.0\times10^{-4} = 9.1\times10^{-4}\ \Omega\cdot\text{cm}$$

Answer: R_s = 9.06 Ω/□, ρ ≈ 9×10^{-4} Ω·cm. A resistivity this low corresponds to an average boron concentration of a few times 10^{19} cm^{-3}.

Junction depth by groove and stain

A cylindrical groove of radius R is ground into the surface and stained. If the stained chord is a wide at the surface and b wide at the junction:

$$x_j \approx (a^2 - b^2) / (8R)$$

Key points of Week 3

- Doping puts controlled amounts of B (p-type) or P, As, Sb (n-type) into chosen regions.
- Diffusion: random thermal motion carries dopant from high to low concentration. Fick: $J = -D\,dC/dx$.
- $D = D_0 e^{-E_a/kT}$: hotter means much faster. Depth grows as √(Dt).
- Constant source → erfc profile, fixed surface concentration. Limited source → Gaussian, fixed dose, surface concentration falls.
- Ion implantation: ions accelerated to E = qV, Gaussian profile peaked at R_p with spread ΔR_p, dose Q = It/qA, annealing required, channelling avoided by tilting.
- Junction depth is where the dopant concentration equals the background doping.
- Profiles are checked by sheet resistance, junction depth, SIMS, spreading resistance and C-V.

Examination questions solved (Q14–Q21)

Q14. What is semiconductor doping? Explain why doping is necessary in semiconductor device fabrication.

Doping is the intentional introduction of controlled, small quantities of specific impurity atoms into a pure (intrinsic) semiconductor to change its electrical properties. Group V donors (P, As, Sb) make silicon **n-type**; Group III acceptors (B, Al, Ga) make it **p-type**. Typical levels are 10^{14} to 10^{20} cm^{-3}.

Why it is necessary:

1. **Intrinsic conductivity is too low and uncontrolled.** Pure Si has only $n_i \approx 10^{10}$ cm^{-3} carriers, giving $\rho = 1/(qn_i(\mu_n + \mu_p)) \approx 3\times10^5$ Ω·cm. Doping with 10^{16} cm^{-3} P raises n a million times, to ρ ≈ 0.5 Ω·cm.

2. **It sets the carrier type.** Devices depend on regions where electrons or holes dominate. Without n- and p-type regions there can be no p-n junction, no diode, no BJT and no MOSFET.
3. **Temperature stability.** In an extrinsic semiconductor the carrier density is fixed by N_D or N_A over a wide temperature range (the extrinsic region) instead of varying exponentially with T as n_i does.
4. **Device design.** Doping profiles set the built-in potential, depletion width, breakdown voltage, threshold voltage of MOSFETs, current gain of BJTs, sheet resistance of resistors and the quality of Ohmic contacts (heavily doped n^+, p^+ regions).
5. **Selective, local control.** Using masks, different areas of one chip can be given different doping, which is what makes integrated circuits possible.

Q15. Differentiate between n-type and p-type semiconductors.

Feature	n-type	p-type
Dopant added	Pentavalent (Group V): P, As, Sb	Trivalent (Group III): B, Al, Ga, In
Name of dopant	Donor	Acceptor
Effect of dopant	Donates a free electron to the conduction band	Accepts an electron, creating a hole in the valence band
Fixed ion left behind	Positive donor ion N_D^+	Negative acceptor ion N_A^-
Majority carriers	Electrons	Holes
Minority carriers	Holes	Electrons
Carrier densities	$n \approx N_D$, $p = n_i^2/N_D$	$p \approx N_A$, $n = n_i^2/N_A$
Impurity level	E_D just below E_c	E_A just above E_v
Fermi level	Above E_i, towards E_c	Below E_i, towards E_v
Conductivity	$\sigma \approx qN_D\mu_n$	$\sigma \approx qN_A\mu_p$
Hall coefficient	Negative	Positive

Both remain electrically neutral overall: every free electron (or hole) is balanced by a fixed ion.

Q16. Explain the difference between donor impurities and acceptor impurities, giving suitable examples.

Donor impurities have **one more** valence electron than the host atom they replace. In Si, Group V atoms such as **phosphorus, arsenic and antimony** have five valence electrons; four form covalent bonds with neighbouring Si atoms, and the fifth is bound only by about 0.04–0.05 eV. It is easily thermally released into the conduction band, leaving a positively charged, immobile donor ion. Donors introduce an energy level E_D just below the conduction band and make the material n-type.

Acceptor impurities have **one fewer** valence electron than the host. In Si, Group III atoms such as **boron, aluminium, gallium and indium** have three valence electrons and leave one bond incomplete. A nearby valence electron moves in with very little energy (≈ 0.045 eV for B), so the acceptor becomes a negative ion and a mobile hole is created in the valence band. Acceptors introduce a level E_A just above the valence band and make the material p-type.

	Donor	Acceptor
Valence electrons (in Si)	5	3
Gives / takes	Gives an electron	Takes an electron (gives a hole)
Ionised charge	+	−
Level	Near E_c	Near E_v
Examples in Si	P, As, Sb	B, Al, Ga, In
Examples in GaAs	S, Se, Te (on As site); Si on Ga site	Zn, Be, Mg (on Ga site); C on As site

Q17. Describe the mechanism of dopant diffusion in a semiconductor.

Diffusion is the net movement of dopant atoms from a region of high concentration to one of low concentration, driven by random thermal jumps of the atoms in the lattice. At room temperature it is negligible; at 900–1200 °C (silicon) it is significant.

Atomic mechanisms:

1. **Vacancy (substitutional) diffusion.** Substitutional dopants (B, P, As, Sb) occupy lattice sites. A dopant can jump only when an adjacent lattice site is vacant. The rate depends on the vacancy concentration (which rises steeply with temperature) and the jump probability. High activation energy (≈ 3–4 eV), so D is small (10^{-15} to 10^{-12} cm^2/s).
2. **Interstitial diffusion.** Small atoms or those that do not bond into the lattice (Au, Cu, Fe, Na, Li, H) move between lattice sites through the open spaces of the diamond structure. Low activation energy (0.5–1.5 eV), so they diffuse many orders of magnitude faster. These are usually unwanted contaminants.
3. **Interstitialcy (kick-out) and dissociative mechanisms.** A dopant is pushed off its site by a silicon self-interstitial, travels interstitially, then returns to a substitutional site. This explains enhanced diffusion of B and P during oxidation (oxidation-enhanced diffusion).

Temperature dependence. $D = D_0 \exp(-E_a/kT)$, so D roughly doubles every 20–30 °C.

Macroscopic description. The flux obeys Fick's first law $F = -D\,\partial C/\partial x$, and the profile evolves by Fick's second law $\partial C/\partial t = D\,\partial^2 C/\partial x^2$. In practice a two-step process is used: predeposition from a constant surface source (erfc profile) followed by drive-in of the fixed dose (Gaussian profile). Diffusion is masked by SiO_2 windows, and the dopant spreads sideways under the mask edge by about 0.8 of its depth (lateral diffusion).

Q18. State and explain Fick's first and second laws of diffusion.

Fick's first law. The diffusion flux is proportional to the negative of the concentration gradient:

$$F = -D\,\partial C/\partial x$$

- F: number of atoms crossing unit area per unit time ($cm^{-2}s^{-1}$).
- C(x, t): impurity concentration (cm^{-3}).
- D: diffusion coefficient (cm^2/s), $D = D_0 \exp(-E_a/kT)$.
- The minus sign: atoms move from high to low concentration. If there is no gradient, there is no net flux.

It describes the flux at one instant but not how the profile changes.

Fick's second law. Consider a thin slice of thickness dx. Atoms entering minus atoms leaving equals the increase stored (continuity, no generation or loss):

$$\partial C/\partial t = -\partial F/\partial x$$

Substituting the first law (with D constant):

$$\partial C/\partial t = D\,\partial^2 C/\partial x^2$$

This is the diffusion equation. It says the concentration rises where the profile is curved upward (concave) and falls where it is curved downward, so diffusion always smooths the profile. Its solutions, set by the boundary conditions, are the complementary-error-function profile (constant surface source) and the Gaussian profile (fixed dose) described in Part A above.

Q19. A semiconductor is doped by diffusion with a constant surface concentration of 1 × 10^{20} cm^{-3}. Explain how the impurity concentration varies with depth.

A constant surface concentration is the **predeposition** (unlimited source) case. Solve Fick's second law with:

- $C(0, t) = C_s = 1\times10^{20}$ cm^{-3} (held fixed for all t)
- $C(\infty, t) = 0$ (deep in the wafer)
- $C(x, 0) = 0$ (no dopant at the start)

The solution is the **complementary error function**:

$$C(x, t) = C_s\,\mathrm{erfc}\left(x / 2\sqrt{(Dt)} \right) = 1\times10^{20} \cdot \mathrm{erfc}\left(x / 2\sqrt{(Dt)} \right) \quad cm^{-3}$$

How it varies with depth:

1. At the surface (x = 0), erfc(0) = 1, so $C = 1\times10^{20}$ cm^{-3}, independent of time.
2. The concentration **decreases monotonically** with depth. Near the surface it falls almost linearly; deeper it falls faster than exponentially, because $\mathrm{erfc}(u) \approx e^{-u^2}/(u\sqrt{\pi})$ for large u.
3. The depth scale is the diffusion length $2\sqrt{(Dt)}$. At $x = 2\sqrt{(Dt)}$, erfc(1) = 0.157, so $C = 1.57\times10^{19}$ cm^{-3}. At $x = 4\sqrt{(Dt)}$, erfc(2) = 4.7×10^{-3}, so $C = 4.7\times10^{17}$ cm^{-3}.
4. As time increases, the profile penetrates deeper (as √t) while the surface stays at 10^{20} cm^{-3}, so the total dose grows: $Q = (2/\sqrt{\pi})C_s\sqrt{(Dt)}$.
5. If the wafer has background doping N_B of the opposite type, a junction forms where $C(x_j) = N_B$.

Worked illustration (assume boron into n-Si with $N_B = 1\times10^{16}$ cm^{-3}, $D = 1\times10^{-13}$ cm^2/s at about 1100 °C, t = 1 h = 3600 s):

$$\sqrt{(Dt)} = \sqrt{(1\times10^{-13} \times 3600)} = 1.90\times10^{-5}\ \mathrm{cm} = 0.190\ \mu\mathrm{m}$$

$$2\sqrt{(Dt)} = 0.379\ \mu\mathrm{m}$$

Junction: $\mathrm{erfc}(u) = N_B/C_s = 10^{16}/10^{20} = 10^{-4} \Rightarrow u \approx 2.75$

$x_j = 2.75 \times 0.379\ \mu\mathrm{m} \approx 1.04\ \mu\mathrm{m}$

Dose: $Q = (2/\sqrt{\pi})(1\times10^{20})(1.90\times10^{-5}) = 2.14\times10^{15}$ $atoms/cm^2$

Depth x (μm)	u = x/2√(Dt)	erfc(u)	C (cm^{-3})
0	0	1	1.0×10^{20}
0.19	0.5	0.480	4.8×10^{19}
0.38	1.0	0.157	1.6×10^{19}
0.57	1.5	0.034	3.4×10^{18}
0.76	2.0	4.7×10^{-3}	4.7×10^{17}
1.04	2.75	1.0×10^{-4}	1.0×10^{16} (junction)

So the impurity concentration is highest at the surface and falls off as an erfc curve, reaching the background doping about 1 μm below the surface for these conditions.

Q20. Explain the principle of ion implantation and identify its major advantages over thermal diffusion.

Principle. Ion implantation introduces dopant atoms into a semiconductor by bombarding it with a beam of energetic dopant ions. The dopant is ionised in an ion source, the ions are extracted, selected by mass in a magnetic analyser, accelerated to an energy of a few keV to MeV and scanned uniformly across the wafer. Each ion penetrates the crystal, losing energy by nuclear and electronic collisions, and stops at a depth that depends on its energy and mass. The resulting distribution is approximately Gaussian, peaking at the projected range R_p with straggle ΔR_p. The dose is measured exactly by integrating the beam current in a Faraday cup ($\Phi = It/qA$). Because the impact damages the lattice and leaves dopants on non-lattice sites, the wafer is then annealed (furnace or RTA) to repair the damage and electrically activate the dopants.

Advantages over thermal diffusion:

1. **Precise, reproducible dose control** (±1 % or better) over a very wide range, 10^{11} to 10^{16} cm^{-2}, by measuring charge. Diffusion dose depends on solid solubility, time and temperature and is much less accurate.
2. **Independent control of depth and dose.** Depth is set by energy, dose by charge. In diffusion both depend on the same Dt product.
3. **Low-temperature process.** Implantation is done near room temperature; only a short anneal is needed. Existing profiles are not disturbed and the thermal budget is small, which is vital for shallow junctions.
4. **Wide choice of masks**, including photoresist, which diffusion cannot use.
5. **Buried and retrograde profiles.** The peak can be placed below the surface (e.g. retrograde wells, buried layers); diffusion always peaks at the surface.
6. **Very shallow junctions** with low-energy implants.
7. **Less lateral spread** under the mask edge, allowing smaller devices and self-aligned structures.
8. **Purity.** Mass separation guarantees only the chosen species is implanted.
9. **Not limited by solid solubility**: concentrations above solubility can be obtained (though not all are activated).
10. **Uniformity** across large wafers, and the process is clean (vacuum, dry).

Q21. Compare diffusion and ion implantation as semiconductor doping techniques.

Feature	Thermal diffusion	Ion implantation
Principle	Dopant moves in from a surface source by random thermal jumps (Fick's laws)	Energetic ions are driven into the crystal and stop by collisions
Process temperature	High, 900–1200 °C	Room temperature, plus anneal (RTA ≈ 1000 °C for seconds)
Profile shape	erfc or Gaussian, always maximum at the surface	Gaussian peaking at R_p below the surface; buried and retrograde profiles possible
Dose control	Limited (± 5–10 %), set by solubility, T and t	Excellent (± 1 %), set by measured charge
Depth control	Coupled to dose through Dt	Independent: set by energy
Junction depth	Relatively deep; hard to make very shallow junctions	Very shallow junctions possible
Lateral spread	Large (≈ 0.8 × depth)	Small (≈ straggle)
Masking	SiO_2, Si_3N_4 (must survive high T)	Photoresist, oxide, nitride, polysilicon
Lattice damage	None	Significant; needs annealing; amorphisation at high dose
Channelling	Not applicable	Must be avoided (7° tilt, screen oxide)
Throughput and cost	Batch of 100+ wafers, cheap equipment	Single wafer or small batch, expensive equipment
Purity	Depends on source and furnace cleanliness	Mass-analysed, very pure
Concentration limit	Limited by solid solubility	Can exceed solubility
Typical use today	Drive-in, deep wells, solar cells, emitter diffusion in older processes	Source/drain, threshold-adjust, wells, LDD, channel stops in all modern CMOS

Summary: diffusion is simple, cheap and damage-free but gives coarse, coupled control of dose and depth and needs high temperature. Ion implantation gives precise, independent, low-temperature control of dose and depth and is the standard technique for modern VLSI, at the cost of expensive equipment and lattice damage that must be annealed. In practice the two are often combined: implant to set the dose, then a thermal drive-in to set the depth.

Practice questions

These extra questions practise this week's ideas. Work each one on paper before reading the solution; the calculation questions follow the same Question, Given, Find, Solution, Answer layout as the worked examples.

Practice 3.1: how far does boron move?

Question. Boron diffuses in silicon with $D = 1\times10^{-14}$ cm^2/s. Find the characteristic diffusion length $\sqrt{(Dt)}$ after 2 hours, and the scale length $2\sqrt{(Dt)}$ used in the erfc profile.

Given: $D = 1\times10^{-14}$ cm^2/s, $t = 2$ h $= 7200$ s.

Find: $\sqrt{(Dt)}$ and $2\sqrt{(Dt)}$.

Solution.

$\sqrt{(Dt)} = \sqrt{(1\times10^{-14} \times 7200)} = \sqrt{(7.2\times10^{-11})} = 8.49\times10^{-6}$ cm = 0.085 µm

$2\sqrt{(Dt)}$ = 0.17 µm

Answer: $\sqrt{(Dt)}$ = 0.085 µm, $2\sqrt{(Dt)}$ = 0.17 µm.

Practice 3.2: predeposition junction depth and dose

Question. Boron is predeposited into n-type silicon ($N_B = 1\times10^{17}$ cm^{-3}) from a constant source with $C_s = 1\times10^{20}$ cm^{-3}, $D = 5\times10^{-14}$ cm^2/s, for 30 minutes. Find the junction depth and the dose introduced.

Given: $C_s = 10^{20}$ cm^{-3}, $N_B = 10^{17}$ cm^{-3}, $D = 5\times10^{-14}$ cm^2/s, t = 1800 s.

Find: x_j and Q.

Solution.

$\sqrt{(Dt)} = \sqrt{(5\times10^{-14} \times 1800)} = 9.49\times10^{-6}$ cm, $\quad 2\sqrt{(Dt)} = 1.90\times10^{-5}$ cm

$\text{erfc}(u) = N_B/C_s = 10^{-3} \quad \Rightarrow \quad u = 2.33$ (from erfc tables)

$x_j = u \times 2\sqrt{(Dt)} = 2.33 \times 1.90\times10^{-5} = 4.4\times10^{-5}$ cm = 0.44 µm

$Q = (2/\sqrt{\pi})\, C_s \sqrt{(Dt)} = 1.128 \times 10^{20} \times 9.49\times10^{-6} = 1.07\times10^{15}$ cm^{-2}

Answer: x_j = 0.44 µm, $Q = 1.07\times10^{15}$ cm^{-2}.

Practice 3.3: driving in the dose

Question. The dose of Practice 3.2 is driven in, with no further source, at $D = 1\times10^{-13}$ cm^2/s for 1 hour. Find the new surface concentration and junction depth ($N_B = 1\times10^{17}$ cm^{-3}).

Given: $Q = 1.07\times10^{15}$ cm^{-2}, $D = 10^{-13}$ cm^2/s, t = 3600 s, $N_B = 10^{17}$ cm^{-3}.

Find: C(0) and x_j.

Solution.

$\sqrt{(\pi Dt)} = \sqrt{(\pi \times 3.6\times10^{-10})} = 3.36\times10^{-5}$ cm

$C(0) = Q/\sqrt{(\pi Dt)} = 1.07\times10^{15} / 3.36\times10^{-5} = 3.2\times10^{19}$ cm^{-3}

$2\sqrt{(Dt)} = 2\sqrt{(3.6\times10^{-10})} = 3.79\times10^{-5}$ cm

$x_j = 2\sqrt{(Dt)}\ \sqrt{(\ln(C(0)/N_B))} = 3.79\times10^{-5} \times \sqrt{(\ln 318)} = 3.79\times10^{-5} \times 2.40 = 9.1\times10^{-5}$ cm

Answer: $C(0) = 3.2\times10^{19}$ cm^{-3} (down from 10^{20}), x_j = 0.91 µm (up from 0.44 µm). Drive-in trades surface concentration for depth.

Practice 3.4: dose from beam current

Question. An implanter delivers a singly charged boron beam of 50 µA for 200 s, uniformly over a 150 mm wafer. What dose is implanted?

Given: $I = 50\times10^{-6}$ A, t = 200 s, r = 7.5 cm, $q = 1.6\times10^{-19}$ C.

Find: Q (ions/cm^2).

Solution.

$A = \pi \times 7.5^2 = 176.7\ \text{cm}^2$

$Q = It/(qA) = (50\times10^{-6} \times 200)/(1.6\times10^{-19} \times 176.7) = 10^{-2}/2.83\times10^{-17} = 3.5\times10^{14}\ \text{cm}^{-2}$

Answer: $Q = 3.5\times10^{14}$ ions/cm^2.

Practice 3.5: shallow implanted junction

Question. Arsenic is implanted into p-type silicon ($N_B = 1\times10^{16}\ \text{cm}^{-3}$) with $Q = 5\times10^{14}\ \text{cm}^{-2}$, $R_p = 0.10\ \mu\text{m}$ and $\Delta R_p = 0.03\ \mu\text{m}$. Find the peak concentration and the depth of the junction below the peak.

Given: $Q = 5\times10^{14}\ \text{cm}^{-2}$, $R_p = 0.10\ \mu\text{m}$, $\Delta R_p = 0.03\ \mu\text{m} = 3\times10^{-6}\ \text{cm}$, $N_B = 10^{16}\ \text{cm}^{-3}$.

Find: N_p and x_j.

Solution.

$N_p = Q/(\sqrt{(2\pi)}\Delta R_p) = 5\times10^{14} / (2.507 \times 3\times10^{-6}) = 6.6\times10^{19}\ \text{cm}^{-3}$

$x_j = R_p + \Delta R_p \sqrt{(2 \ln(N_p/N_B))} = 0.10 + 0.03 \times \sqrt{(2 \ln 6650)} = 0.10 + 0.03 \times 4.20 = 0.226\ \mu\text{m}$

Answer: $N_p = 6.6\times10^{19}$ cm^{-3}, $x_j = 0.23\ \mu$m. (Here the surface concentration is high enough that the n^+ region reaches the surface, so there is only the one junction.)

Practice 3.6: diffusion coefficient of phosphorus

Question. For phosphorus in silicon $D_0 = 3.85\ \text{cm}^2/\text{s}$ and $E_a = 3.66$ eV. Find D at 1100 °C.

Given: $D_0 = 3.85\ \text{cm}^2/\text{s}$, $E_a = 3.66$ eV, $T = 1373$ K, $k = 8.617\times10^{-5}$ eV/K.

Find: D.

Solution.

$kT = 8.617\times10^{-5} \times 1373 = 0.1183$ eV

$E_a/kT = 3.66/0.1183 = 30.93$

$D = 3.85 \times e^{-30.93} = 3.85 \times 3.7\times10^{-14} = 1.4\times10^{-13}\ \text{cm}^2/\text{s}$

Answer: $D \approx 1.4\times10^{-13}$ cm^2/s.

Practice 3.7: annealing and tilt

Question. Give two reasons an implanted wafer must be annealed, and explain why implants are usually done with the wafer tilted by about 7°.

Answer. Annealing (1) repairs the lattice damage caused by the ions knocking silicon atoms out of place (heavy implants even make the surface amorphous), and (2) moves the dopant atoms onto substitutional lattice sites where they become electrically active donors or acceptors. Tilting by about 7° stops the beam lining up with the open channels between rows of atoms along low-index directions; ions entering a channel travel much deeper than R_p predicts (channelling), giving an unwanted deep tail.

Objectives check

Use this table to confirm you have met this week's objectives. Cover the Answer column and answer each check question from memory. If you cannot, go back to the section in the last column before moving on.

#	Objective	Check yourself	Answer	Where it is taught
1	Explain why doping is needed and name the common dopants of silicon.	Which of P, B, As, Ga make silicon n-type?	P and As (Group V); B and Ga make it p-type.	The big picture
2	Explain diffusion and state Fick's first law and the temperature dependence of D.	By roughly what factor does D change when the furnace is 100 °C hotter ($E_a \approx 3.5$ eV, ~1000 °C)?	About tenfold (Q4: $6.9\times10^{-15} \rightarrow 7.1\times10^{-14}$ cm^2/s).	Part A; Q4
3	Distinguish constant-source (erfc) and limited-source (Gaussian) diffusion and calculate their profiles.	In which type of diffusion does the surface concentration fall with time?	Limited-source (drive-in): the fixed dose spreads deeper, so $C(0) = Q/\sqrt{(\pi Dt)}$ falls.	Two types of diffusion; Q5–Q6
4	Describe the step-by-step diffusion process and the atomic diffusion mechanisms.	Which mechanism, vacancy or interstitial, is faster, and which do B and P use?	Interstitial is faster; B and P move mainly by the vacancy mechanism.	Step-by-step process; atomic level
5	Describe ion implantation, its equipment and the Gaussian implant profile.	What does the mass analyser do?	Bends the ion beam with a magnet so only ions of the chosen mass-to-charge ratio reach the wafer.	Part B
6	Calculate implant peak concentration, junction depth, dose and beam current.	$Q = 10^{14}$ cm^{-2} with $\Delta R_p = 0.04$ µm: find N_p.	$N_p = Q/(\sqrt{(2\pi)}\Delta R_p) = 10^{14}/(2.507 \times 4\times10^{-6}) = 1.0\times10^{19}$ cm^{-3}.	Implantation calculations P1–P4
7	Compare diffusion with ion implantation, and explain channelling and annealing.	Why is the wafer tilted about 7° during implantation?	To stop ions travelling deep along open crystal channels (channelling).	Channelling; comparison table
8	Describe how a doping profile is evaluated.	What does a four-point probe measure, and how is it calculated?	Sheet resistance, $R_s = 4.532$ V/I (Ω/□).	Part C; Example 3.1

Energy Bands and Carrier Concentrations

Plan: quantum-mechanics foundations (Schrödinger equation overview, Kronig–Penney model); energy band gaps, allowed and forbidden states; intrinsic and extrinsic carrier concentrations.

Learning objectives

By the end of this week you should be able to:

1. State the basic ideas of quantum mechanics needed for solids.
2. State the time-independent Schrödinger equation and explain its terms.
3. Explain how discrete atomic levels become energy bands.
4. Explain the Kronig–Penney model and how it produces allowed and forbidden bands.
5. Explain E–k diagrams, effective mass and holes.
6. Distinguish direct and indirect band gaps and calculate emission wavelengths.
7. Derive n_i and calculate intrinsic and extrinsic carrier concentrations.

The big picture

In Week 1 we simply stated that a solid has a **valence band**, a **conduction band** and a **gap** between them. This week explains **why**. The answer needs one idea from quantum mechanics: **an electron behaves like a wave, and a wave trapped in a region can only vibrate in certain ways**. In a single atom that gives sharp energy levels. In a crystal, where the electron wave runs through a regular row of atoms, it gives broad bands of allowed energies separated by forbidden gaps.

You do not need to solve hard equations for this course. You need to understand three steps:

1. Electrons are waves, described by the Schrödinger equation.
2. Trapped waves have only certain allowed energies.
3. A periodic crystal turns those allowed energies into bands with gaps (the Kronig–Penney model shows this).

Then Part C uses the bands to count the carriers.

Part A: Quantum-mechanics foundations

Why classical physics is not enough

Classical physics treats an electron as a tiny ball. That picture cannot explain why copper conducts and glass does not, or why an atom emits light only at certain colours. Three ideas from early quantum theory fix this:

- **Energy comes in packets (quanta).** Light of frequency ν comes in photons of energy $E = h\nu$. Electrons in atoms can only have certain energies.
- **Particles behave as waves.** An electron with momentum p behaves as a wave of wavelength $\lambda = h/p$ (de Broglie). We often write $p = \hbar k$, where $k = 2\pi/\lambda$ is the **wave vector**: k is just "how many waves per unit length", and it measures the electron's momentum.

- **Uncertainty.** We cannot know an electron's position and momentum exactly at the same time. So instead of a definite position we describe it by a **wave function** ψ, and $|\psi|^2$ tells us where it is likely to be found.

Constants: $h = 6.626\times10^{-34}$ J·s, $\hbar = h/2\pi = 1.055\times10^{-34}$ J·s, electron mass $m_0 = 9.11\times10^{-31}$ kg.

The Schrödinger equation

The time-independent Schrödinger equation (in one dimension) is

$$-(\hbar^2/2m)\, d^2\psi/dx^2 + V(x)\,\psi = E\,\psi$$

Read it as an energy balance: **kinetic energy + potential energy = total energy**.

Term	Meaning
$\psi(x)$	the **wave function**. ψ itself is not measurable, but $\|\psi(x)\|^2\,dx$ is the probability of finding the electron between x and x + dx
$-(\hbar^2/2m)\, d^2\psi/dx^2$	the **kinetic energy** term. $d^2\psi/dx^2$ measures how sharply the wave curves: a more tightly wiggling wave means more kinetic energy
$V(x)\,\psi$	the **potential energy** term. V(x) is the potential energy at each point (for example the pull of the atomic nuclei)
E	the **total energy**, a constant. The equation only has sensible solutions for certain values of E: that is how quantisation appears

The wave function must be finite, single-valued and smooth (continuous, with a continuous slope), and the total probability must be 1.

Two simple cases that explain everything

A free electron (no forces, V = 0). The solution is a travelling wave, $\psi = Ae^{ikx}$, with energy

$$E = \hbar^2k^2/2m$$

Any energy is allowed. The graph of E against k is a smooth parabola. This is the starting point for electrons in a solid.

An electron in a box of width L with walls it cannot pass. The wave must be zero at both walls, just as a guitar string is fixed at both ends. So only whole numbers of half-waves fit, and only certain energies are allowed:

$$E_n = n^2\pi^2\hbar^2/2mL^2, \quad n = 1, 2, 3, \ldots$$

> **In plain words.** A guitar string can only play certain notes, because only certain wavelengths fit between its fixed ends. An electron trapped in a region is the same: only certain wavelengths fit, so only certain energies are allowed. That is what "quantised energy levels" means.

From atoms to bands

One isolated silicon atom has sharp energy levels (1s, 2s, 2p, 3s, 3p). Now bring two atoms close together. Their outer wave functions overlap, and each level **splits into two** slightly different levels. (Two identical pendulums joined by a spring do the same: instead of one frequency they now have two, swinging together and swinging opposite.)

Bring **N atoms** together and each level splits into **N levels**. A crystal has about 5×10^{22} atoms per cm^3, so each level splits into so many closely spaced levels that they form a practically continuous **band**. Between bands are ranges of

energy where there are **no** levels at all: the **forbidden gaps**.

In silicon, the outer 3s and 3p levels (4 electrons per atom) mix and split into two bands:

- a lower band with room for 4 electrons per atom: the **valence band**. Silicon has exactly 4 outer electrons per atom, so this band is **completely full** at 0 K;
- an upper band, also with room for 4 per atom: the **conduction band**, **empty** at 0 K;
- separated by the band gap $\mathbf{E_g = 1.12\ eV}$.

Bloch's theorem

In a perfect crystal the potential energy repeats with the spacing of the atoms a: V(x + a) = V(x). Bloch showed that the electron wave in such a crystal is

$$\psi_k(x) = u_k(x)\, e^{ikx}, \quad \text{with } u_k(x + a) = u_k(x)$$

A travelling wave (e^{ikx}) multiplied by a pattern that repeats from atom to atom (u_k). The important consequence: **an electron in a perfect crystal travels without being scattered by the atoms**. The atoms are not obstacles; they just shape the wave. Only imperfections (vibrations, impurities, defects) scatter electrons. This is why carriers in a crystal can move so freely (Week 7).

The Kronig–Penney model: how a periodic crystal makes bands and gaps

The real potential inside a crystal is complicated. Kronig and Penney (1931) replaced it with the simplest repeating potential that can be solved exactly: a row of **rectangular wells and barriers**.

- Each **well** (width a, V = 0) represents the region near an atom, where the electron is attracted.
- Each **barrier** (height V_0, width b) represents the region between atoms.
- The pattern repeats forever with period a + b.

What they did:

1. Solved the Schrödinger equation inside a well (where the solution oscillates) and inside a barrier (where it decays).
2. Joined the pieces smoothly at every edge (ψ and dψ/dx continuous).
3. Applied Bloch's theorem (the solution must repeat, up to a phase factor, from one period to the next).
4. Simplified by making the barriers very thin and very high.

The result is one condition that the energy must satisfy:

$$P \cdot \sin(\alpha a)/(\alpha a) + \cos(\alpha a) = \cos(ka)$$

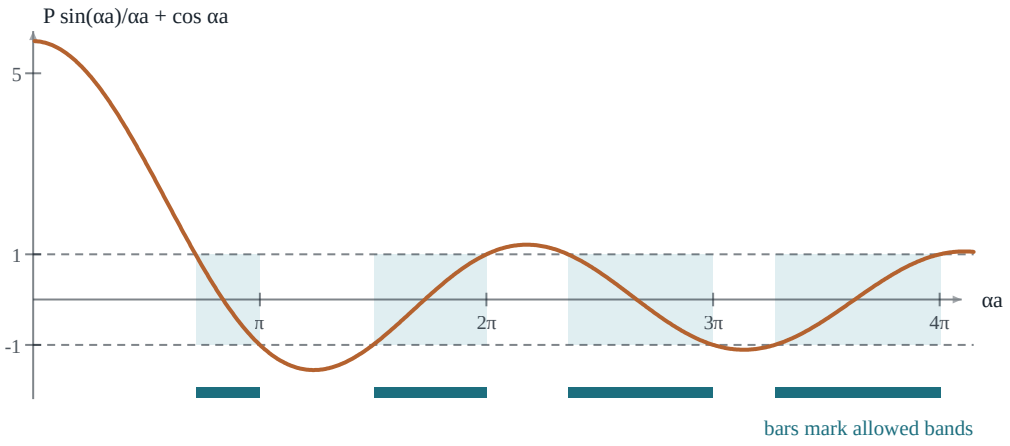
$$\alpha = \sqrt{(2mE)}/\hbar, \qquad P = m\, V_0\, b\, a\, /\, \hbar^2$$

Symbol	Meaning
α	depends on the electron energy E: larger E, larger α
a	well width (the atomic spacing)
k	the electron's wave vector (momentum)
P	barrier strength: how strongly the electron is held by each atom

How to read it: the key step. The right-hand side, cos(ka), is a cosine. **A cosine can only be between −1 and +1.** The left-hand side depends on the energy through α. So:

- for energies where the left-hand side is **between −1 and +1**, there is a k that satisfies the equation: these energies are **allowed**;
- for energies where the left-hand side is **greater than +1 or less than −1**, no real k works: these energies are **forbidden**.

Plot the left-hand side against αa (i.e. against energy). It wiggles like a fading cosine, starting above +1 and repeatedly overshooting the ±1 lines. Every time it is inside the ±1 strip you have a **band**; every time it pokes outside you have a **gap**.



The Kronig–Penney condition plotted against αa for P = 3π/2. Shaded strips are the energies where the curve stays between −1 and +1: the allowed bands. Where the curve goes outside ±1 there are no solutions: the forbidden gaps.

What the model teaches:

1. The energy spectrum is a series of **allowed bands separated by forbidden gaps**.
2. The band edges occur at ka = nπ, i.e. k = ±nπ/a, the **Brillouin-zone boundaries**. Physically, at these wavelengths the electron wave reflects off the rows of atoms (Bragg reflection), forming standing waves, which split the energies and open a gap.
3. Bands get **wider** at higher energy (the curve's wiggles fade, so it spends longer inside ±1).
4. If the atoms hold their electrons very tightly (P → ∞), the bands shrink to the sharp levels of single atoms.
5. If the barriers vanish (P → 0), the gaps close and the electron is free: $E = \hbar^2k^2/2m$.

> **In plain words.** A wave running through a regular row of obstacles is fine at most wavelengths, but at certain wavelengths the reflections from every row add up and the wave cannot travel. Those wavelengths correspond to the forbidden energies: the band gaps.

Part B: Band gaps, allowed and forbidden states

E–k diagrams and effective mass

The **band structure** of a material is a graph of the allowed energies E against the wave vector k. Near the bottom of the conduction band (and the top of the valence band) the curve looks like a parabola, just like a free electron's $E = \hbar^2k^2/2m$, but with a different curvature. We describe that curvature by an **effective mass** $m*$:

$$1/m^* = (1/\hbar^2)\, d^2E/dk^2$$

The electron then responds to forces **as if** it were a free particle of mass m^*. The effect of all the crystal's atoms is packed into this one number.

- A **sharply curved** band means small m^*: a light, fast, high-mobility carrier (GaAs electrons, $m^* = 0.067m_0$).
- A **flatter** band means large m^*: a heavy, slow carrier (Si holes, about $0.56m_0$).

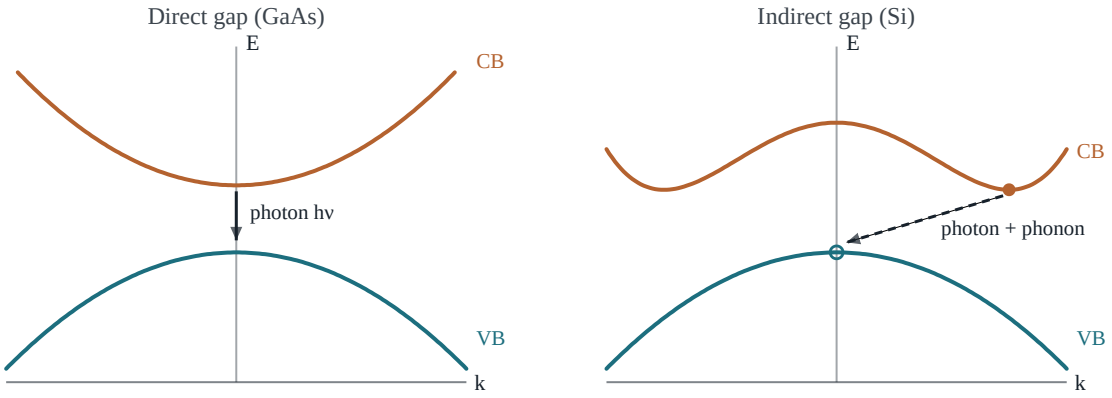
Near the **top** of the valence band the curve bends downward. An electron there would have negative mass. It is simpler to describe the **missing** electron instead: it behaves as a positive particle with positive mass, the **hole**.

Valence band, conduction band and forbidden gap

- **Valence band:** the highest band that is completely full at 0 K. Its top is E_v. Its electrons are the bonding electrons. A full band cannot carry current (for every electron moving one way there is one moving the other), but a band with a few **holes** can.
- **Conduction band:** the next band up, empty at 0 K. Its bottom is E_c. Electrons here move freely and carry current.
- **Forbidden gap:** $E_g = E_c - E_v$. In a perfect crystal there are no states here at all. Impurities and defects can add a few states inside it (donor, acceptor and trap levels).

The gap shrinks slightly with temperature: for Si, 1.17 eV at 0 K and 1.12 eV at 300 K.

Direct and indirect band gaps



In a direct-gap material (GaAs) the bottom of the conduction band sits directly above the top of the valence band in k. In an indirect-gap material (Si) they are at different k, so an electron dropping across the gap must also change its momentum.

When an electron falls from the conduction band to the valence band it must lose energy **and** keep momentum balanced. A photon carries plenty of energy but almost no momentum. So on an E–k diagram, a photon can only make **vertical** jumps.

- **Direct gap** (GaAs, InP, GaN): the conduction-band minimum is directly above the valence-band maximum. The electron can drop straight down and give out a photon. Light emission is fast and efficient: **LEDs and lasers**.
- **Indirect gap** (Si, Ge): the minimum is at a different k. To drop across, the electron must also get rid of momentum, which needs a **phonon** (a lattice vibration) at the same instant. That is unlikely, so the electron usually loses its energy as heat instead. **Silicon is a poor light emitter.**

The wavelength of light from a band-to-band transition is

$$\lambda\ (\mu m) = 1.24 / E_g\ (eV)$$

GaAs: 1.24/1.42 = 0.87 μm (near infrared). Si: 1.24/1.12 = 1.11 μm, which is also the longest wavelength a silicon photodetector can absorb.

Part C: Intrinsic and extrinsic carrier concentrations

Now that we have bands, we can count carriers. The full method (Fermi–Dirac statistics, density of states) is Week 5. Here are the results you need this week.

The two key equations

The number of electrons in the conduction band and holes in the valence band depend on how far the **Fermi level** E_F is from each band edge:

$$n = N_c\, e^{-(E_c - E_F)/kT}, \qquad p = N_v\, e^{-(E_F - E_v)/kT}$$

N_c and N_v are the **effective densities of states** (Si at 300 K: 2.8×10^{19} and 1.04×10^{19} cm^{-3}). The closer E_F is to E_c, the more electrons; the closer to E_v, the more holes.

Intrinsic semiconductor

In pure silicon every free electron came from a broken bond, which also left a hole, so $n = p = n_i$. Multiply the two equations above; E_F cancels:

$$n\,p = N_c\,N_v\, e^{-(E_c - E_v)/kT} = N_c\,N_v\, e^{-E_g/kT}$$

With $n = p = n_i$:

$$n_i = \sqrt{(N_c\,N_v)}\; e^{-E_g/2kT}$$

For Si at 300 K, $n_i \approx 1\times10^{10}$ cm^{-3}. The Fermi level of an intrinsic semiconductor, E_i, is almost exactly at **mid-gap**.

What controls n_i: the band gap (exponentially: wider gap, far fewer carriers) and temperature (exponentially: hotter, far more carriers). Ge (0.66 eV) has $n_i = 2.4\times10^{13}$; GaAs (1.42 eV) has only about 2×10^{6} cm^{-3}.

The mass-action law

Because E_F cancelled, the product is the same whether or not the material is doped:

$$n\,p = n_i^2$$

Extrinsic (doped) semiconductor

With full ionisation and doping well above n_i:

- **n-type:** $n \approx N_D$, $p = n_i^2/N_D$.
- **p-type:** $p \approx N_A$, $n = n_i^2/N_A$.
- **Both present (compensation):** use the net doping. The exact result from charge neutrality is

$$n = (N_D - N_A)/2 + \sqrt{\left((N_D - N_A)/2\right)^2 + n_i^2}$$

Example 4.1: how doping shifts both carrier concentrations

Question. Silicon at 300 K ($n_i = 1\times10^{10}$ cm^{-3}) is doped with $N_D = 1\times10^{16}$ cm^{-3}. Calculate n and p, and compare each with its intrinsic value.

Given: $N_D = 1\times10^{16}$ cm^{-3}, $n_i = 1\times10^{10}$ cm^{-3}.

Find: n, p, and the factors n/n_i and n_i/p.

Solution.

$$n \approx N_D = 1\times10^{16} \text{ cm}^{-3}$$

$$p = n_i^2/n = 10^{20}/10^{16} = 1\times10^{4} \text{ cm}^{-3}$$

$$n/n_i = 10^{16}/10^{10} = 10^{6}, \quad n_i/p = 10^{10}/10^{4} = 10^{6}$$

Answer: $n = 10^{16}$ cm^{-3}, $p = 10^{4}$ cm^{-3}. Doping raised the electrons a million times above n_i and lowered the holes by the same factor, so $np = n_i^2$ still holds.

Temperature regions (n-type): at very low temperature the donors are not all ionised (**freeze-out**, $n < N_D$); over the normal range $n \approx N_D$ (**extrinsic**); at high temperature n_i overtakes N_D and the material behaves as intrinsic (**intrinsic region**). Week 5 works through these in detail.

Key points of Week 4

- Electrons are waves; the Schrödinger equation is kinetic + potential = total energy, and only certain energies give acceptable waves.
- Bringing N atoms together splits each level into N levels, forming bands; between them are forbidden gaps.
- Kronig–Penney: $P\,\sin(\alpha a)/(\alpha a) + \cos(\alpha a) = \cos(ka)$. Energies where the left side is between ±1 are allowed (bands); elsewhere forbidden (gaps).
- Effective mass $m*$ describes the band curvature. A missing electron near the top of the valence band acts as a positive hole.
- Direct gap: photon emission is efficient (GaAs). Indirect gap: a phonon is needed, so emission is poor (Si). $\lambda = 1.24/E_g$ μm.
- $n_i = \sqrt{(N_c N_v)}e^{-E_g/2kT}$; $np = n_i^2$; $n \approx N_D$ in n-type, $p \approx N_A$ in p-type.

Examination questions solved (Q22–Q29, Q34)

Q22. Explain the basic assumptions of the quantum mechanical model of electrons in a semiconductor.

1. **Wave nature of electrons.** Each electron is described by a wave function ψ obeying the Schrödinger equation; $|\psi|^2$ gives the probability of finding it at a point. Energy and momentum ($p = \hbar k$) are related through the band structure $E(k)$.
2. **Adiabatic (Born–Oppenheimer) approximation.** The nuclei are much heavier than electrons and are treated as fixed at their lattice sites. Their vibrations (phonons) are added later as a perturbation that causes scattering.
3. **Perfect periodic crystal.** The ion cores form an infinite, perfectly periodic lattice, so the potential energy is periodic, $V(r + R) = V(r)$ for every lattice vector R. Surfaces, impurities and defects are treated separately.
4. **One-electron (independent-electron) approximation.** Each electron moves independently in an average periodic potential produced by the ion cores and all the other electrons (mean-field or Hartree approximation), so a many-

body problem becomes a single-particle one.

5. **Bloch's theorem** follows: wave functions are modulated plane waves $\psi = u_k(r)e^{ik\cdot r}$.
6. **Pauli exclusion principle.** Each quantum state (k and spin) holds at most one electron, so the states fill from the bottom up. This, with the Fermi–Dirac distribution, gives the occupation of the bands.
7. **Effective-mass approximation.** Near the band edges, electrons and holes respond to external forces as free particles with an effective mass $m*$ that contains the effect of the crystal potential.
8. **Thermal equilibrium statistics.** In equilibrium, electrons are distributed among the available states according to the Fermi–Dirac function.

Q23. State the time-independent Schrödinger equation and explain the physical meaning of its terms.

In one dimension:

$$-(\hbar^2/2m)\, d^2\psi(x)/dx^2 + V(x)\,\psi(x) = E\,\psi(x)$$

In three dimensions: $-(\hbar^2/2m)\nabla^2\psi(r) + V(r)\psi(r) = E\psi(r)$, compactly $\hat{H}\psi = E\psi$.

Meaning of each term:

- **$\psi(x)$, the wave function**: the amplitude of the electron's probability wave. It has no direct physical meaning, but $|\psi(x)|^2\,dx$ is the probability of finding the electron between x and x + dx. ψ must be finite, single-valued, continuous and normalised.
- **$\hbar = h/2\pi$**, the reduced Planck constant (1.055×10^{-34} J·s); **m**, the electron mass (or the effective mass $m*$ in a crystal).
- **$-(\hbar^2/2m)\, d^2\psi/dx^2$**: the **kinetic-energy** term. It comes from $p^2/2m$ with the momentum operator $p = -i\hbar\, d/dx$. The more sharply ψ curves, the higher the kinetic energy.
- **$V(x)\psi$**: the **potential-energy** term. V(x) is the potential energy of the electron at x, e.g. the periodic potential of the crystal ions, a barrier, or a well.
- **E**: the **total energy** of the electron, a constant (the equation is valid for stationary states whose energy does not change with time). Acceptable solutions exist only for particular values of E (eigenvalues), which is how quantisation arises.
- The left-hand side as a whole is the **Hamiltonian operator** $\hat{H}$ acting on ψ.

The full time-dependent wave function is $\Psi(x, t) = \psi(x)e^{-iEt/\hbar}$; the time-independent equation is obtained when V does not depend on time.

Q24. Explain the basic principle of the Kronig-Penney model and how it leads to the formation of energy bands.

Principle. The real periodic potential of a crystal is complicated. Kronig and Penney (1931) replaced it with the simplest periodic potential that can be solved exactly: a one-dimensional infinite array of **rectangular potential wells and barriers**. Each well (V = 0, width a) represents the region near an ion; each barrier (height V_0, width b) represents the region between ions. The period is a + b.

Method.

1. Solve Schrödinger's equation separately in a well (oscillating solutions, $\alpha^2 = 2mE/\hbar^2$) and in a barrier (exponential solutions, $\beta^2 = 2m(V_0 - E)/\hbar^2$).

2. Apply continuity of ψ and dψ/dx at each boundary.
3. Apply **Bloch's theorem**, $\psi(x + (a + b)) = e^{ik(a+b)}\psi(x)$, because the potential is periodic.
4. A non-trivial solution exists only if a determinant vanishes. Taking the limit of narrow, high barriers ($b \to 0$, $V_0 \to \infty$, with $V_0 b$ finite) gives:

   $$P \cdot \sin(\alpha a)/(\alpha a) + \cos(\alpha a) = \cos(ka), \qquad P = mV_0 ba/\hbar^2$$

How bands form. The right-hand side, cos(ka), must lie between −1 and +1 for real k. Plot the left-hand side f(αa) against αa (i.e. against energy). It starts at 1 + P, oscillates like a damped cosine, and repeatedly rises above +1 and falls below −1. Only the energy ranges where $-1 \le f(\alpha a) \le +1$ are solutions: these are the **allowed energy bands**. The ranges where $|f| > 1$ have no real k and therefore no electron states: these are the **forbidden energy gaps**.

Results:

- The spectrum consists of alternating allowed bands and forbidden gaps.
- Discontinuities in E(k) occur at $k = n\pi/a$ (Brillouin-zone boundaries), where Bragg reflection of the electron wave occurs.
- The width of allowed bands increases with energy (because the 1/αa factor damps the oscillation).
- As $P \to \infty$ (electrons tightly bound), bands shrink to discrete atomic levels. As $P \to 0$ (free electrons), the gaps vanish and $E = \hbar^2 k^2/2m$.

The model explains in a simple way why solids have bands and gaps, which is the basis of the classification into metals, semiconductors and insulators.

Q25. Distinguish between valence band, conduction band, and forbidden energy gap.

	Valence band	**Conduction band**	**Forbidden energy gap**
What it is	The highest band that is completely filled with electrons at 0 K	The lowest band above the valence band, empty at 0 K	The energy range between E_v and E_c in which no electron states exist (in a perfect crystal)
Edge	Top at E_v	Bottom at E_c	Width $E_g = E_c - E_v$
Electrons	Bound in covalent bonds; cannot conduct while the band is full	Free; move through the crystal and conduct	None (only impurity or defect levels)
Carriers	Holes (empty states) conduct here	Electrons conduct here	—
Origin	Bonding combinations of the atomic sp^3 orbitals	Antibonding combinations	Splitting of bonding and antibonding states
Values	—	—	Si 1.12 eV, Ge 0.66 eV, GaAs 1.42 eV, SiO_2 ≈ 9 eV

An electron needs at least E_g to jump from the valence to the conduction band; it then leaves a hole behind. The width of the gap decides whether a material is a metal (no gap), a semiconductor (small gap) or an insulator (large gap).

Q26. Explain the difference between a direct band-gap and an indirect band-gap semiconductor.

On the E–k diagram, look at where the **conduction-band minimum** and the **valence-band maximum** occur.

	Direct band gap	Indirect band gap
Band-edge positions	CB minimum and VB maximum at the **same k** (usually k = 0, the Γ point)	CB minimum at a **different k** from the VB maximum
Transition across the gap	Vertical: an electron can drop and emit a photon directly, conserving momentum	Needs a change of momentum, which a photon cannot supply; a **phonon** must take part
Process	Two-particle (electron + photon)	Three-particle (electron + photon + phonon), much less probable
Radiative recombination	Fast, efficient; lifetime ≈ ns	Slow, inefficient; recombination is mostly non-radiative via traps; lifetime μs–ms
Absorption near E_g	Sharp, strong ($\alpha \approx 10^4$ cm^{-1} just above E_g)	Weak, gradual; thicker material needed to absorb light
Examples	GaAs (1.42 eV), InP, GaN, InGaAs, CdTe	Si (1.12 eV), Ge (0.66 eV), GaP, AlAs, SiC
Applications	LEDs, laser diodes, efficient thin-film solar cells, fast photodetectors	ICs, transistors, power devices, Si solar cells and photodiodes (absorb but do not emit well)

The emitted wavelength is $\lambda = hc/E_g$, i.e. $\lambda(\mu m) = 1.24/E_g(eV)$. GaAs emits at 0.87 μm. Silicon's indirect gap is why there is no practical silicon laser.

Q27. What is meant by an intrinsic semiconductor? Derive the expression for intrinsic carrier concentration.

Intrinsic semiconductor. A pure semiconductor crystal with no significant impurities or defects, in which free carriers are produced **only** by thermal excitation of electrons from the valence band to the conduction band. Each excited electron leaves a hole, so

$$n = p = n_i$$

and the Fermi level lies at the intrinsic level E_i, very near the middle of the gap.

Derivation.

Step 1: electrons. The electron concentration is the density of states times the occupation probability, integrated over the conduction band:

$$n = \int_{E_c}^{\infty} g_c(E)\, f(E)\, dE, \qquad g_c(E) = (4\pi(2m_n^*)^{3/2}/h^3)\,\sqrt{(E - E_c)}$$

Step 2: Boltzmann approximation. For $E_c - E_F \gg kT$, $f(E) \approx \exp(-(E - E_F)/kT)$. Then

$$n = (4\pi(2m_n^*)^{3/2}/h^3)\,\exp(-(E_c - E_F)/kT) \int_0^{\infty} \sqrt{x}\cdot e^{-x/kT}\, dx, \quad x = E - E_c$$

Step 3: the integral. $\int_0^{\infty} x^{1/2}\, e^{-x/kT}\, dx = (kT)^{3/2}\,\Gamma(3/2) = (kT)^{3/2}\,(\sqrt{\pi}/2)$. Hence

$$n = N_c \exp(-(E_c - E_F)/kT), \qquad N_c = 2(2\pi m_n^* kT/h^2)^{3/2}$$

Step 4: holes. Similarly, with $1 - f(E) \approx \exp(-(E_F - E)/kT)$ in the valence band,

$$p = N_v \exp(-(E_F - E_v)/kT), \qquad N_v = 2(2\pi m_p^* kT/h^2)^{3/2}$$

Step 5: the intrinsic condition. Multiply n and p. E_F cancels:

$$n\,p = N_c\,N_v \exp(-(E_c - E_v)/kT) = N_c\,N_v \exp(-E_g/kT)$$

In an intrinsic semiconductor $n = p = n_i$, so $n \cdot p = n_i^2$, and

$$n_i = \sqrt{(N_c\, N_v)} \cdot \exp(-E_g / 2kT)$$

Substituting N_c and N_v gives the explicit temperature dependence:

$$n_i = 2\,(2\pi kT/h^2)^{3/2}\,(m_n{*}\, m_p{*})^{3/4} \exp(-E_g / 2kT) \quad \propto T^{3/2}\, e^{-E_g/2kT}$$

Step 6: the intrinsic Fermi level. Setting $n = p$ and solving:

$$E_i = (E_c + E_v)/2 + (kT/2)\ln(N_v/N_c) = (E_c + E_v)/2 + (3kT/4)\ln(m_p{*}/m_n{*})$$

Numerical check for Si at 300 K: $\sqrt{(2.8\times10^{19} \times 1.04\times10^{19})} = 1.71\times10^{19}$; $\exp(-1.12/(2 \times 0.0259)) = \exp(-21.6) = 4.1\times10^{-10}$; $n_i \approx 7\times10^{9}$ cm^{-3}, the same order as the accepted 1×10^{10} cm^{-3} (the difference comes from the exact values of E_g and the effective masses).

Q28. Explain the difference between intrinsic and extrinsic carrier concentrations.

	Intrinsic carrier concentration	**Extrinsic carrier concentration**
Material	Pure (undoped) semiconductor	Doped semiconductor
Source of carriers	Thermal breaking of covalent bonds only (band-to-band excitation)	Mainly ionised dopants (donors or acceptors), plus a small thermal contribution
Electron vs hole	$n = p = n_i$	Unequal: $n \gg p$ (n-type) or $p \gg n$ (p-type); majority and minority carriers
Formula	$n_i = \sqrt{(N_c N_v)}\exp(-E_g/2kT)$	$n \approx N_D$, $p = n_i^2/N_D$ (n-type); $p \approx N_A$, $n = n_i^2/N_A$ (p-type)
Typical value (Si, 300 K)	1×10^{10} cm^{-3}	10^{14}–10^{20} cm^{-3} majority
Temperature dependence	Very strong, exponential	Nearly constant over the extrinsic range; becomes intrinsic at high T
Controlled by	Material (E_g) and temperature	Doping level, chosen by the designer
Fermi level	At E_i, near mid-gap	Shifted toward E_c (n-type) or E_v (p-type)
Conductivity	Low ($\sigma \approx 3\times10^{-6}$ S/cm for Si)	Much higher, set by doping

The two are linked by the mass-action law $\mathbf{np = n_i^2}$: doping increases one carrier type and decreases the other so that their product equals the intrinsic value squared.

Q29. Silicon has an intrinsic carrier concentration of approximately 1.0 × 10^{10} cm^{-3} at a specified temperature. If the electron and hole concentrations are equal, determine the electron and hole concentrations.

If $n = p$, the semiconductor is intrinsic (undoped, or exactly compensated). From the mass-action law:

$$n \cdot p = n_i^2 \quad \text{and} \quad n = p \;\Rightarrow\; n^2 = n_i^2 \;\Rightarrow\; n = n_i$$

$n = p = n_i = 1.0\times10^{10}$ cm^{-3}

Check: $n\cdot p = (1.0\times10^{10})^2 = 1.0\times10^{20}$ cm^{-6} $= n_i^2$. ✓

At this concentration, using $\mu_n = 1350$ and $\mu_p = 480$ cm^2/V·s, the intrinsic conductivity would be $\sigma = qn_i(\mu_n + \mu_p) = 1.602\times10^{-19} \times 10^{10} \times 1830 = 2.9\times10^{-6}$ S/cm, i.e. $\rho \approx 3.4\times10^{5}$ Ω·cm.

Q34. An n-type semiconductor has a donor concentration of 5×10^{16} cm^{-3}. Assuming complete ionization, determine the approximate majority-carrier concentration.

With complete ionisation and $N_D \gg n_i$ ($5\times10^{16} \gg 10^{10}$), every donor contributes one electron:

$$n_0 \approx N_D = 5 \times 10^{16}\ \text{cm}^{-3}$$

Majority-carrier (electron) concentration ≈ 5×10^{16} cm^{-3}.

Additional results (assuming silicon at 300 K, $n_i = 1\times10^{10}$ cm^{-3}):

Minority holes: $p_0 = n_i^2/n_0 = (10^{10})^2 / (5\times10^{16}) = 2\times10^{3}$ cm^{-3}

Fermi level: $E_F - E_i = kT \ln(n_0/n_i) = 0.0259 \times \ln(5\times10^{6}) = 0.0259 \times 15.42 = 0.40$ eV above E_i

Conductivity: $\sigma \approx q\, n_0\, \mu_n = 1.602\times10^{-19} \times 5\times10^{16} \times 1350 \approx 10.8$ S/cm ($\rho \approx 0.093$ Ω·cm)

The thermal contribution n_i is negligible, which is why the majority-carrier density is simply the donor density. (Mobility falls at this doping to about 1000 cm^2/V·s, so the real σ is nearer 8 S/cm.)

Practice questions

These extra questions practise this week's ideas. Work each one on paper before reading the solution; the calculation questions follow the same Question, Given, Find, Solution, Answer layout as the worked examples.

Practice 4.1: energy levels in a quantum well

Question. An electron is confined in a 1 nm wide well with infinitely high walls. Calculate the energies of the first two levels in eV.

Given: $L = 1\times10^{-9}$ m, $m = 9.11\times10^{-31}$ kg, $h = 6.626\times10^{-34}$ J·s.

Find: E_1 and E_2.

Solution. $E_n = n^2h^2/(8mL^2)$ (the same as $n^2\pi^2\hbar^2/2mL^2$).

$E_1 = (6.626\times10^{-34})^2 / (8 \times 9.11\times10^{-31} \times 10^{-18}) = 6.02\times10^{-20}$ J = 0.376 eV

$E_2 = 4E_1 = 1.50$ eV

Answer: $E_1 = 0.38$ eV, $E_2 = 1.50$ eV. Confinement on the nanometre scale gives level spacings comparable to band gaps, the basis of quantum-well lasers.

Practice 4.2: wavelength of a thermal electron

Question. An electron in a semiconductor at 300 K has a kinetic energy of about (3/2)kT = 0.0388 eV. Find its de Broglie wavelength and compare it with the silicon lattice constant.

Given: $E = 0.0388$ eV $= 6.21\times10^{-21}$ J, $m = 9.11\times10^{-31}$ kg.

Find: $\lambda = h/p$.

Solution.

$p = \sqrt{(2mE)} = \sqrt{(2 \times 9.11\times10^{-31} \times 6.21\times10^{-21})} = 1.06\times10^{-25}$ kg·m/s

$\lambda = h/p = 6.626\times10^{-34} / 1.06\times10^{-25} = 6.2\times10^{-9}$ m = 6.2 nm

Answer: λ ≈ 6 nm, about 11 times the lattice constant (0.543 nm). The electron is a wave spread over many atoms, which is why a wave description is needed.

Practice 4.3: emission wavelengths

Question. Find the band-to-band emission wavelength of GaN (3.4 eV), GaP (2.26 eV) and InP (1.35 eV), and say where each lies in the spectrum.

Given: E_g values above; $\lambda(\mu m) = 1.24/E_g(eV)$.

Find: λ for each.

Solution.

GaN: 1.24/3.4 = 0.365 μm

GaP: 1.24/2.26 = 0.549 μm

InP: 1.24/1.35 = 0.919 μm

Answer: GaN 365 nm (near ultraviolet), GaP 549 nm (green), InP 919 nm (near infrared). (GaP is indirect, so its LEDs need special doping; GaN and InP are direct.)

Practice 4.4: intrinsic concentration of GaAs

Question. For GaAs at 300 K, $N_c = 4.7\times10^{17}$ cm^{-3}, $N_v = 7.0\times10^{18}$ cm^{-3} and $E_g = 1.42$ eV. Calculate n_i.

Given: N_c, N_v, E_g above; 2kT = 0.0518 eV.

Find: n_i.

Solution.

$\sqrt{(N_c N_v)} = \sqrt{(4.7\times10^{17} \times 7.0\times10^{18})} = 1.81\times10^{18}$ cm^{-3}

$e^{-E_g/2kT} = e^{-1.42/0.0518} = e^{-27.4} = 1.25\times10^{-12}$

$n_i = 1.81\times10^{18} \times 1.25\times10^{-12} = 2.3\times10^{6}$ cm^{-3}

Answer: $n_i \approx 2\times10^6$ cm^{-3}, about 4000 times smaller than in silicon, which is why GaAs can be made semi-insulating.

Practice 4.5: how fast n_i rises with temperature

Question. Ignoring the small change of band gap with temperature, by what factor does n_i of silicon ($E_g = 1.12$ eV) increase between 300 K and 350 K?

Given: $n_i \propto T^{3/2} e^{-E_g/2kT}$, $E_g = 1.12$ eV, $k = 8.617\times10^{-5}$ eV/K.

Find: $n_i(350)/n_i(300)$.

Solution.

$(350/300)^{3/2} = 1.26$

$\exp[\ (E_g/2k)(1/300 - 1/350)\ ] = \exp[\ 6499 \times 4.76\times10^{-4}\ ] = e^{3.09} = 22.1$

ratio = 1.26 × 22.1 = 27.8

Answer: about 28 times. A 50 K rise gives a 28-fold increase, almost all from the exponential.

Practice 4.6: compensated silicon

Question. Silicon contains $N_D = 1\times10^{16}$ cm^{-3} and $N_A = 4\times10^{15}$ cm^{-3}. Find n and p at 300 K ($n_i = 1\times10^{10}$ cm^{-3}).

Given: $N_D = 10^{16}$, $N_A = 4\times10^{15}$, $n_i = 10^{10}$ cm^{-3}.

Find: n and p.

Solution.

$N_D > N_A$, so the material is n-type with net doping $N_D - N_A = 6\times10^{15}$ cm^{-3} $\gg n_i$

$n \approx 6\times10^{15}$ cm^{-3}

$p = n_i^2/n = 10^{20}/6\times10^{15} = 1.7\times10^{4}$ cm^{-3}

Answer: $n = 6\times10^{15}$ cm^{-3}, $p = 1.7\times10^{4}$ cm^{-3}.

Practice 4.7: why bands widen

Question. Using the Kronig–Penney model, explain why allowed bands become wider at higher energies.

Answer. The left-hand side of P sin(αa)/(αa) + cos(αa) = cos(ka) is a cosine plus a term whose amplitude falls as 1/(αa). At low energy (small αa) the P sin(αa)/(αa) term is large, so the curve swings far beyond ±1 and spends only short stretches inside the allowed strip: narrow bands, wide gaps. As energy increases, αa increases, that term shrinks, and the curve is mostly the plain cosine, which stays between ±1 almost everywhere: wide bands, narrow gaps. Physically, high-energy electrons are less affected by the atomic barriers and behave more like free electrons.

Objectives check

Use this table to confirm you have met this week's objectives. Cover the Answer column and answer each check question from memory. If you cannot, go back to the section in the last column before moving on.

#	Objective	Check yourself	Answer	Where it is taught		
1	State the basic ideas of quantum mechanics needed for solids.	What is the meaning of	ψ	^{2}?	The probability density of finding the electron at a point.	Why classical physics is not enough
2	State the time-independent Schrödinger equation and explain its terms.	Which term of $-(\hbar^2/2m)d^2\psi/dx^2 + V\psi = E\psi$ is the kinetic energy?	$-(\hbar^2/2m)\,d^2\psi/dx^2$.	The Schrödinger equation		
3	Explain how discrete atomic levels become energy bands.	Into how many levels does one atomic level split when N atoms form a crystal?	N levels, so close together that they form a band.	From atoms to bands		
4	Explain the Kronig–Penney model and how it produces allowed and forbidden bands.	In $P\,\sin(\alpha a)/(\alpha a) + \cos(\alpha a) = \cos(ka)$, which energies are forbidden?	Those for which the left-hand side is greater than +1 or less than −1.	The Kronig–Penney model		
5	Explain E–k diagrams, effective mass and holes.	Does a strongly curved band give a heavy or a light effective mass?	Light: $1/m^* = (1/\hbar^2)\,d^2E/dk^2$ is large.	E–k diagrams and effective mass		
6	Distinguish direct and indirect band gaps and calculate emission wavelengths.	What wavelength does GaAs (E_g = 1.42 eV) emit?	$\lambda = 1.24/1.42 = 0.87$ μm.	Direct and indirect band gaps		
7	Derive n_i and calculate intrinsic and extrinsic carrier concentrations.	Write n_i in terms of N_c, N_v and E_g.	$n_i = \sqrt{(N_c N_v)}\,e^{-E_g/2kT}$.	Part C; Example 4.1; Q27		

Carrier Statistics

Plan: the Fermi–Dirac distribution function; density of states; Fermi energy level calculation as a function of temperature and doping. This page follows the order of your lecturer's carrier-statistics note (sections 7–26) and adds the background those sections assume.

Learning objectives

By the end of this week you should be able to:

1. Define and use the Fermi–Dirac distribution function.
2. Explain the physical meaning of the Fermi level and the shape of f(E) at 0 K.
3. Explain the density of states and the effective density of states.
4. Calculate electron and hole concentrations and apply the mass action law.
5. Determine the position of the Fermi level for given doping.
6. Apply the charge-neutrality condition, including compensated material.
7. Explain the effect of temperature on carrier concentration and the Fermi level.
8. Distinguish degenerate and non-degenerate semiconductors.

The big picture

Carrier statistics is the method for answering one question: **how many free electrons and holes does a piece of semiconductor contain?** Everything else in the course (conductivity, diodes, transistors) depends on that number.

Think of the conduction band as a large theatre:

- The **seats** are the allowed energy states. How many seats there are at each energy is the **density of states**.
- Whether a seat is **occupied** is a matter of probability. The probability is given by the **Fermi–Dirac function**.
- The number of electrons = (number of seats) × (probability each is filled), added up over all energies.

  number of electrons = Σ (states available) × (probability occupied)

The **Fermi level E_F** is the single number that sets those probabilities. Once you know where E_F sits in the band gap, you know n and p. That is why most of this week is about finding E_F.

1. The Fermi–Dirac distribution function

Electrons obey the **Pauli exclusion principle**: no two can occupy the same state. In thermal equilibrium, the probability that a state at energy E is occupied by an electron is:

$$f(E) = 1 / (1 + e^{(E - E_F)/kT})$$

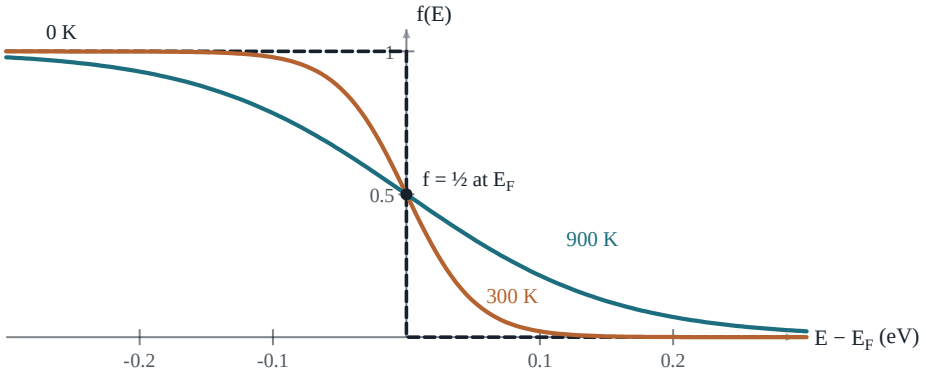
Symbol	Meaning
f(E)	probability (between 0 and 1) that a state at energy E holds an electron
E	energy of the state (eV)
E_F	the **Fermi energy** or Fermi level (eV)
k	Boltzmann constant, 8.617×10^{-5} eV/K
T	absolute temperature (K); kT = 0.02585 eV at 300 K

Three things to remember about f(E)

1. **At E = E_F**, the exponent is zero, $e^0 = 1$, and f = 1/(1 + 1) = ½. The Fermi level is the energy where a state has a 50 % chance of being occupied, at any temperature.
2. **Below E_F** (E < E_F), the exponent is negative, so f is close to 1: states are almost certainly full.
3. **Above E_F** (E > E_F), the exponent is positive, so f is close to 0: states are almost certainly empty.

The change from "full" to "empty" happens over a few kT (about ±0.1 eV at room temperature).

At absolute zero (T = 0 K), kT → 0 and the change becomes instantaneous: f = 1 for every state below E_F and f = 0 for every state above. f(E) is a **sharp step**. A pure semiconductor at 0 K therefore has a full valence band, an empty conduction band, and no free carriers.



The Fermi–Dirac function. At 0 K it is a step at E_F. At 300 K and 900 K the step is smoothed over a few kT. It always passes through ½ at E_F.

Probability that a state is empty. A state is either full or empty, so

probability empty = 1 − f(E)

An empty state in the valence band is a **hole**, so 1 − f(E) is the probability of finding a hole.

Calculating an occupation probability

At 300 K (kT = 0.02585 eV), a state 0.10 eV **above** E_F:

$(E - E_F)/kT = 0.10/0.02585 = 3.868$

$f = 1/(1 + e^{3.868}) = 1/(1 + 47.85) = 0.0205$

About a **2 % chance** of being occupied. A state 0.10 eV **below** E_F has $f = 1/(1 + e^{-3.868}) = 0.980$, a 98 % chance; its probability of being empty is 2 %. The function is symmetric about E_F.

The Boltzmann approximation

If $E - E_F$ is more than about 3kT (0.08 eV at 300 K), $e^{(E-E_F)/kT}$ is much larger than 1, and we can drop the "1 +":

$$f(E) \approx e^{-(E - E_F)/kT} \qquad \text{(Maxwell–Boltzmann approximation)}$$

This is accurate to better than 5 %. For the 0.10 eV example: $e^{-3.868} = 0.0209$, against the exact 0.0205. Almost every calculation in this course uses this approximation, because the conduction band is usually well above E_F.

2. Density of states

The **density of states** g(E) is the number of allowed electron states per unit energy, per unit volume, at energy E (units: states per eV per cm^3).

$$\text{Conduction band: } g_c(E) \propto (m_n{*})^{3/2} \sqrt{(E - E_c)} \quad \text{for E above } E_c$$

$$\text{Valence band: } g_v(E) \propto (m_p{*})^{3/2} \sqrt{(E_v - E)} \quad \text{for E below } E_v$$

$$\text{In the gap: } g(E) = 0$$

In words: there are **no states in the gap**; just above E_c the number of states starts at zero and grows as the square root of how far you go into the band. The effective masses $m_n{*}$ and $m_p{*}$ describe the shape of each band; a heavier mass means more states.

Effective density of states

Adding up "states × probability" over the whole conduction band (with the Boltzmann approximation) gives a very simple result. It is **as if** all the conduction-band states were squeezed into one level at E_c, with a total number N_c of them. N_c is the **effective density of states**:

$$N_c = 2\,(2\pi\, m_n{*}\, kT / h^2)^{3/2}, \qquad N_v = 2\,(2\pi\, m_p{*}\, kT / h^2)^{3/2}$$

For silicon at 300 K: $\mathbf{N_c = 2.8\times10^{19}\ cm^{-3}}$, $\mathbf{N_v = 1.04\times10^{19}\ cm^{-3}}$. Both grow with temperature as $T^{3/2}$.

7. Electron concentration

The electron concentration in the conduction band is obtained by adding up available states × occupation probability:

$$n = \int g_c(E)\, f(E)\, dE \qquad \text{(from } E_c \text{ to the top of the band)}$$

For a **non-degenerate** semiconductor, f(E) is replaced by the Maxwell–Boltzmann form, and the integral gives:

$$n = N_c\, e^{-(E_c - E_F)/kT}$$

or, measuring from the intrinsic level E_i,

$$n = n_i\, e^{(E_F - E_i)/kT}$$

Reading it. $E_c - E_F$ is the distance from the Fermi level up to the conduction-band edge. The smaller that distance, the larger n, and the dependence is exponential: every 60 mV that E_F moves towards E_c multiplies n by 10.

8. Hole concentration

The probability that a valence-band state is **empty** is 1 − f(E). The hole concentration is

$$p = \int g_v(E)\,[1 - f(E)]\,dE \qquad \text{(over the valence band)}$$

For a non-degenerate semiconductor this gives:

$$p = N_v\, e^{-(E_F - E_v)/kT}$$

and

$$p = n_i\, e^{(E_i - E_F)/kT}$$

These four equations (two for n, two for p) are the fundamental equations of semiconductor carrier statistics.

9. Position of the Fermi level

Turn the electron equation round to find E_F when n is known. From $n = N_c\, e^{-(E_c - E_F)/kT}$, divide by N_c and take the natural logarithm:

$$\ln(n/N_c) = -(E_c - E_F)/kT$$

Therefore:

$$E_c - E_F = kT \ln(N_c/n)$$

Similarly, from the hole concentration:

$$E_F - E_v = kT \ln(N_v/p)$$

And, using the intrinsic level:

$$E_F - E_i = kT \ln(n/n_i), \qquad E_i - E_F = kT \ln(p/n_i)$$

These equations let the Fermi level be calculated when the carrier concentration is known.

Example 5.1: Fermi level from the electron concentration

Question. An n-type silicon sample at 300 K has an electron concentration of 1×10^{16} cm^{-3}. Taking $N_c = 2.8\times10^{19}$ cm^{-3} and kT = 0.02585 eV, find how far the Fermi level lies below the conduction-band edge.

Given: $n = 1\times10^{16}$ cm^{-3}, $N_c = 2.8\times10^{19}$ cm^{-3}, kT = 0.02585 eV.

Find: $E_c - E_F$.

Solution.

Use $E_c - E_F = kT \ln(N_c/n)$:

$$N_c/n = 2.8\times10^{19} / 1\times10^{16} = 2800, \quad \ln(2800) = 7.94$$

$$E_c - E_F = 0.02585 \times 7.94 = 0.205 \text{ eV}$$

Answer: $E_c - E_F = 0.205$ eV: the Fermi level is 0.205 eV below the conduction band.

10. Intrinsic semiconductor

A semiconductor with no intentional donor or acceptor doping is called **intrinsic**. Every free electron came from a broken bond that also left a hole, so

$$n = p = n_i$$

where n_i is the **intrinsic carrier concentration**. Multiplying the n and p equations, E_F cancels out:

$$n \cdot p = N_c\, N_v\, e^{-(E_c - E_v)/kT} = N_c\, N_v\, e^{-E_g/kT}$$

and since $n = p = n_i$,

$$n_i = \sqrt{(N_c\, N_v)} \cdot e^{-E_g/2kT}$$

where $E_g = E_c - E_v$ is the band-gap energy. This equation shows that the intrinsic carrier concentration depends strongly on temperature and on the band gap. For Si at 300 K, $n_i \approx 1\times10^{10}$ cm^{-3} (many textbooks use 1.5×10^{10}; the accurate modern value is 9.65×10^{9}).

11. Intrinsic Fermi level

For an intrinsic semiconductor, n = p. Set the two expressions equal:

$$N_c\, e^{-(E_c - E_i)/kT} = N_v\, e^{-(E_i - E_v)/kT}$$

Solving for E_i gives

$$E_i = (E_c + E_v)/2 + (kT/2)\ln(N_v/N_c)$$

If N_c and N_v were equal, the logarithm would be zero, and then

$$E_i = (E_c + E_v)/2$$

Thus the intrinsic Fermi level is approximately at the **middle of the forbidden gap**. For Si the correction is $(0.02585/2)\ln(1.04/2.8) = -0.013$ eV, only 13 meV below mid-gap.

12. Mass action law

For a semiconductor in thermal equilibrium:

$$n \cdot p = n_i^2$$

This is the **mass action law**. It holds for intrinsic and doped (non-degenerate) material alike, because when we multiplied n by p above, E_F cancelled.

- For an **intrinsic** semiconductor n = p, therefore $n = p = n_i$.
- For an **n-type** semiconductor where $n \gg p$: $n \approx N_D$, and therefore $\mathbf{p = n_i^2/N_D}$.
- For a **p-type** semiconductor where $p \gg n$: $p \approx N_A$, and therefore $\mathbf{n = n_i^2/N_A}$.

> **In plain words.** n × p is fixed. Add donors, and the extra electrons fill in holes, so the hole count drops by exactly the same factor the electron count rises. Doping one carrier up pushes the other down.

13. Effect of doping on the Fermi level

Doping changes the concentration of carriers, and therefore shifts the Fermi level.

N-type. Donor atoms introduce additional electrons, so the Fermi level moves **towards the conduction band**:

$$E_c - E_F = kT \ln(N_c/n), \quad E_F - E_i = kT \ln(n/n_i)$$

If donors are fully ionised and $N_D \gg n_i$, then $n \approx N_D$, and

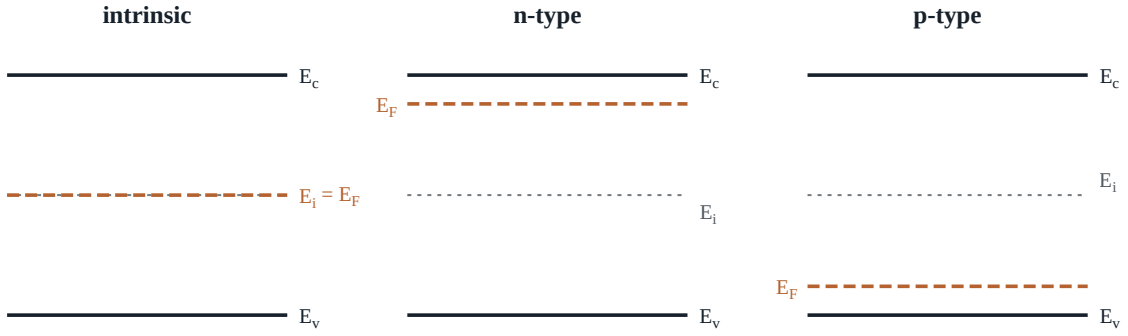
$$E_c - E_F = kT \ln(N_c/N_D), \quad E_F - E_i = kT \ln(N_D/n_i)$$

P-type. Acceptor atoms create holes, so the Fermi level moves **towards the valence band**:

$$E_F - E_v = kT \ln(N_v/p), \quad E_i - E_F = kT \ln(p/n_i)$$

If acceptors are fully ionised, $p \approx N_A$, so

$$E_F - E_v = kT \ln(N_v/N_A), \quad E_i - E_F = kT \ln(N_A/n_i)$$



$E_F - E_i = kT \ln(N_D/n_i)$ for n-type; $E_i - E_F = kT \ln(N_A/n_i)$ for p-type

Where the Fermi level sits: at mid-gap (E_i) when intrinsic, near E_c for n-type, near E_v for p-type.

How big are the shifts? Each factor of 10 in doping moves E_F by $kT \ln 10 = 0.0595$ eV ≈ 60 mV at 300 K.

Doping (Si, 300 K)	**$E_F - E_i$**	**$E_c - E_F$**
Intrinsic	0	0.56 eV
$N_D = 10^{14}$ cm^{-3}	0.238 eV	0.324 eV
$N_D = 10^{16}$ cm^{-3}	0.357 eV	0.205 eV
$N_D = 10^{18}$ cm^{-3}	0.476 eV	0.086 eV

14. Charge neutrality in a doped semiconductor

A piece of semiconductor is electrically neutral overall. The positive charges (holes and ionised donors) must equal the negative charges (electrons and ionised acceptors):

$$p + N_D^+ = n + N_A^-$$

where p is the hole concentration, n the electron concentration, N_D^+ the concentration of ionised donors and N_A^- the concentration of ionised acceptors. If donors and acceptors are completely ionised ($N_D^+ = N_D$, $N_A^- = N_A$):

$$p + N_D = n + N_A$$

This is the **charge-neutrality condition**. Combined with $np = n_i^2$ it gives n exactly:

$$n = (N_D - N_A)/2 + \sqrt{\left((N_D - N_A)/2\right)^2 + n_i^2}$$

When $N_D - N_A \gg n_i$, this reduces to $n \approx N_D - N_A$. When both dopants are present they partly cancel, which is called **compensation**.

15. N-type semiconductor

When silicon is doped with a donor impurity (phosphorus P, arsenic As, antimony Sb), it becomes n-type. A donor contributes an extra electron. If $N_D \gg n_i$ and all donors are ionised:

$$n \approx N_D$$

Using the mass action law $np = n_i^2$, we obtain:

$$p \approx n_i^2 / N_D$$

16. P-type semiconductor

When silicon is doped with an acceptor impurity (boron B, aluminium Al, gallium Ga), it becomes p-type. An acceptor creates a hole. If $N_A \gg n_i$, then:

$$p \approx N_A$$

and:

$$n \approx n_i^2 / N_A$$

17. Fermi level as a function of temperature

Temperature has a major effect on carrier concentration and Fermi level. There are generally three regions:

Region 1: Freeze-out region. At very low temperature (below about 100 K in Si), donor and acceptor atoms are not fully ionised: there is not enough thermal energy to free the extra electron from each donor. Consequently there are relatively few free carriers, and $n < N_D$.

Region 2: Extrinsic region. At moderate temperature (roughly 150–450 K for Si), the dopants are fully ionised:

$$\text{n-type: } n \approx N_D \qquad \text{p-type: } p \approx N_A$$

The carrier concentration is set by the doping and hardly changes with temperature. This is the region used in most basic semiconductor calculations, and the region where devices operate.

Region 3: Intrinsic region. At sufficiently high temperature, thermally generated electron–hole pairs become dominant. Eventually

$$n_i \gg N_D \quad \text{so} \quad n \approx p \approx n_i$$

The semiconductor behaves more and more like an intrinsic semiconductor, and E_F returns to E_i. The doping no longer controls the material, which is why every device has a maximum operating temperature.

Example 5.2: Fermi level against temperature

Question. Silicon is doped with $N_D = 1\times10^{15}$ cm^{-3}. Using the intrinsic concentrations in the table ($n_i = 1\times10^{10}$ cm^{-3} at 300 K, rising with temperature), calculate the electron concentration and the position of the Fermi level relative to E_i at 300, 400, 500 and 600 K. At what temperature does the material become effectively intrinsic?

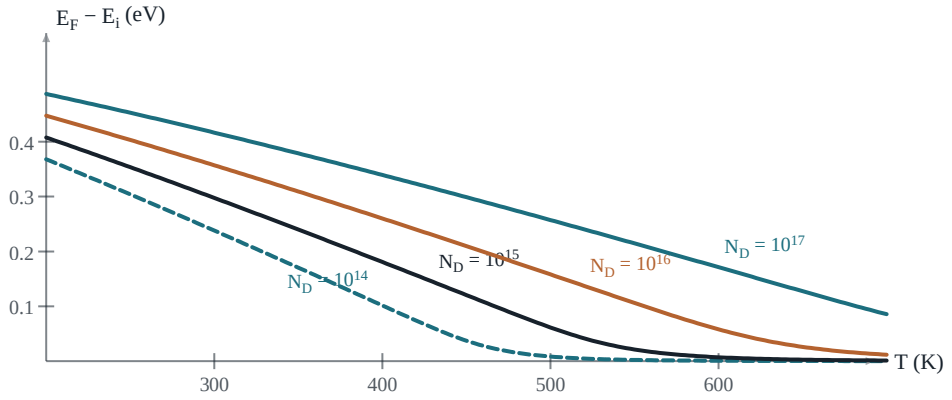
Given: $N_D = 1\times10^{15}$ cm^{-3}, complete ionisation; $n_i(T)$ from $n_i \propto T^{3/2}e^{-E_g(T)/2kT}$, scaled to 1×10^{10} cm^{-3} at 300 K.

Find: n and $E_F - E_i$ at each temperature.

Solution. At each temperature use the exact charge-neutrality result, $n = N_D/2 + \sqrt{((N_D/2)^2 + n_i^2)}$, then $E_F - E_i = kT\ln(n/n_i)$.

T (K)	kT (eV)	n_i (cm^{-3})	n (cm^{-3})	$E_F - E_i = kT\ln(n/n_i)$ (eV)
300	0.0259	1.0×10^{10}	1.00×10^{15}	$0.0259 \times \ln(1.0\times10^{5}) = 0.298$
400	0.0345	5.3×10^{12}	1.00×10^{15}	$0.0345 \times \ln(189) = 0.181$
500	0.0431	2.6×10^{14}	1.06×10^{15}	$0.0431 \times \ln(4.16) = 0.061$
600	0.0517	3.7×10^{15}	4.18×10^{15}	$0.0517 \times \ln(1.15) = 0.007$

Answer: $E_F - E_i$ falls from **0.298 eV at 300 K to 0.007 eV at 600 K**. n_i passes N_D at about **550 K**; above that the material is effectively intrinsic and the doping no longer controls it.



Fermi level relative to E_i against temperature for silicon at four donor concentrations (cm^{-3}). Heavier doping keeps the Fermi level away from mid-gap up to a higher temperature.

18. Worked example: n-type silicon

Example 5.3: majority and minority carriers

Question. Silicon at 300 K is doped with $N_D = 1\times10^{16}$ cm^{-3}. Assuming $n_i = 1\times10^{10}$ cm^{-3} and complete donor ionisation, calculate the electron and hole concentrations and identify the majority and minority carriers.

Given: $N_D = 1\times10^{16}$ cm^{-3}, $n_i = 1\times10^{10}$ cm^{-3}, T = 300 K.

Find: n, p; majority and minority carriers.

Solution.

Step 1: electron concentration. Since $N_D \gg n_i$ ($10^{16} \gg 10^{10}$), every donor contributes one electron:

$$n \approx N_D = 1\times10^{16}\ \text{cm}^{-3}$$

Step 2: hole concentration, from the mass-action law:

$$p = n_i^2/n = (1\times10^{10})^2 / (1\times10^{16}) = 10^{20}/10^{16} = 1\times10^{4}\ \text{cm}^{-3}$$

Answer: $n = 1\times10^{16}$ cm^{-3}, $p = 1\times10^{4}$ cm^{-3}. Electrons are the majority carriers, holes the minority carriers. (With $n_i = 1.5\times10^{10}$ cm^{-3}, $p = 2.25\times10^{4}$ cm^{-3}; the method is identical.)

19. Worked example: Fermi level of n-type silicon

Example 5.4: Fermi level of n-type silicon

Question. For the silicon of Example 5.3 ($n = N_D = 1\times10^{16}$ cm^{-3}, $n_i = 1\times10^{10}$ cm^{-3}) at 300 K, calculate the position of the Fermi level relative to the intrinsic level, and estimate how far it lies below the conduction band.

Given: $n = 1\times10^{16}$ cm^{-3}, $n_i = 1\times10^{10}$ cm^{-3}, $kT = 0.02585$ eV, $E_g = 1.12$ eV.

Find: $E_F - E_i$, and $E_c - E_F$.

Solution.

$$E_F - E_i = kT \ln(n/n_i) = 0.02585 \times \ln(10^{16}/10^{10}) = 0.02585 \times \ln(10^{6})$$

Since $\ln(10^{6}) = 6 \times 2.303 = 13.82$,

$$E_F - E_i = 0.02585 \times 13.82 = 0.357 \text{ eV}$$

E_i is close to mid-gap, 0.56 eV below E_c, so

$$E_c - E_F \approx 0.56 - 0.357 = 0.20 \text{ eV}$$

Answer: $E_F = E_i + 0.357$ eV, about 0.20 eV below E_c: the Fermi level has moved up towards the conduction band. (With $n_i = 1.5\times10^{10}$ cm^{-3}: $E_F - E_i = 0.02585 \ln(6.67\times10^{5}) = 0.347$ eV.)

20. Worked example: p-type silicon

Example 5.5: carriers and Fermi level in p-type silicon

Question. Silicon at 300 K is doped with $N_A = 1\times10^{17}$ cm^{-3} boron atoms. Taking $n_i = 1\times10^{10}$ cm^{-3}, calculate the hole and electron concentrations and the position of the Fermi level.

Given: $N_A = 1\times10^{17}$ cm^{-3}, $n_i = 1\times10^{10}$ cm^{-3}, $kT = 0.02585$ eV.

Find: p, n and $E_i - E_F$.

Solution.

$$p \approx N_A = 1\times10^{17} \text{ cm}^{-3}$$

$$n = n_i^2/p = 10^{20}/10^{17} = 1\times10^{3} \text{ cm}^{-3}$$

$$E_i - E_F = kT \ln(p/n_i) = 0.02585 \times \ln(10^{7}) = 0.02585 \times 16.12 = 0.417 \text{ eV}$$

Answer: $p = 1\times10^{17}$ cm^{-3}, $n = 1\times10^{3}$ cm^{-3}, $E_F = E_i - 0.417$ eV: the Fermi level moves down towards the valence band, about $0.56 - 0.417 = 0.14$ eV above E_v.

21. Degenerate and non-degenerate semiconductors

The Maxwell–Boltzmann approximation is valid only when the Fermi level is far enough from the band edges. For a **non-degenerate** semiconductor:

$$E_c - E_F \gg kT \ \text{(n-type)}, \quad E_F - E_v \gg kT \ \text{(p-type)}$$

In practice "≫" means more than about 3kT.

In a **degenerate** semiconductor the doping is very high (in Si, above about 10^{18}–10^{19} cm^{-3}) and the Fermi level approaches, or even enters, the conduction or valence band. Then the full Fermi–Dirac statistics must be used, and the material behaves almost like a metal. Degenerate n^+ and p^+ layers are used for Ohmic contacts (Week 11) and tunnel diodes.

Feature	Non-degenerate	Degenerate
Doping	Relatively low or moderate	Very high
Statistics	Maxwell–Boltzmann approximation	Fermi–Dirac
Fermi level	Inside the band gap, more than 3kT from the edges	Near or inside a band
Carrier concentration	Simple exponential equations	More complex

Check. For Si with $N_D = 10^{18}$ cm^{-3}, $E_c - E_F = 0.02585 \ln(2.8\times10^{19}/10^{18}) = 0.086$ eV = 3.3kT, just non-degenerate. At 10^{19} cm^{-3}, $E_c - E_F = 0.027$ eV ≈ 1kT: degenerate.

22. Relationship between carrier concentration and Fermi level

The electron concentration increases exponentially as the Fermi level approaches the conduction band, $n = N_c\, e^{-(E_c - E_F)/kT}$. Similarly, the hole concentration increases as the Fermi level approaches the valence band, $p = N_v\, e^{-(E_F - E_v)/kT}$. Therefore:

- Fermi level closer to E_c → more electrons (n-type).
- Fermi level closer to E_v → more holes (p-type).
- Fermi level at E_i → $n = p = n_i$ (intrinsic).

23. Temperature dependence of intrinsic carrier concentration

Recall $n_i = \sqrt{(N_c N_v)}\, e^{-E_g/2kT}$. Since N_c and N_v are each proportional to $T^{3/2}$, their product's square root is proportional to $T^{3/2}$, so approximately

$$n_i \propto T^{3/2}\, e^{-E_g/2kT}$$

The **exponential term** is especially important. As temperature increases, kT increases, the negative exponent $-E_g/2kT$ gets smaller in magnitude, and $e^{-E_g/2kT}$ increases rapidly. The $T^{3/2}$ factor adds a smaller increase. For silicon, n_i roughly doubles every 8 °C near room temperature, and rises about 25,000 times between 300 K and 500 K.

Therefore semiconductor conductivity generally **increases with temperature**, because more electron–hole pairs are thermally generated.

24. Formula sheet for carrier statistics

Quantity	Formula
Fermi–Dirac function	$f(E) = 1/(1 + e^{(E - E_F)/kT})$
Electron concentration	$n = N_c\, e^{-(E_c - E_F)/kT} = n_i\, e^{(E_F - E_i)/kT}$
Hole concentration	$p = N_v\, e^{-(E_F - E_v)/kT} = n_i\, e^{(E_i - E_F)/kT}$
Effective density of states	$N_c = 2(2\pi m_n * kT/h^2)^{3/2}$, $N_v = 2(2\pi m_p * kT/h^2)^{3/2}$
Intrinsic carrier concentration	$n_i = \sqrt{(N_c N_v)}\, e^{-E_g/2kT}$
Mass action law	$np = n_i^2$
Intrinsic Fermi level	$E_i = (E_c + E_v)/2 + (kT/2)\ln(N_v/N_c) \approx$ mid-gap
N-type Fermi level	$E_F - E_i = kT \ln(N_D/n_i)$
P-type Fermi level	$E_i - E_F = kT \ln(N_A/n_i)$
Charge neutrality	$p + N_D^+ = n + N_A^-$

25. Summary: what you should be able to do

1. Explain the concept of carrier statistics in semiconductors.
2. Define and use the Fermi–Dirac distribution function.
3. Explain the physical meaning of the Fermi energy level.
4. Calculate the probability of electron occupation of an energy state.
5. Explain the density of states in the conduction and valence bands.
6. Calculate electron and hole concentrations.
7. Calculate the intrinsic carrier concentration.
8. Apply the mass action law.
9. Determine the position of the Fermi level.
10. Explain the effect of donor and acceptor doping on the Fermi level.
11. Distinguish between intrinsic, n-type and p-type semiconductors.
12. Explain the effect of temperature on carrier concentration.
13. Distinguish between degenerate and non-degenerate semiconductors.
14. Apply the charge-neutrality equation to doped semiconductors.

---

26. Mid-module evaluation, solved

These are the lecturer's mid-module evaluation questions. The numerical values in the printed note did not come through in the copy provided, so the calculations below use the standard silicon values stated in each question. If your copy gives different numbers, follow the same steps.

Part A: short-answer questions

A1. Define carrier statistics. Carrier statistics is the study of how electrons and holes are distributed among the allowed energy states of a semiconductor in thermal equilibrium. It combines the density of states with the Fermi–Dirac probability to calculate the electron and hole concentrations and the position of the Fermi level.

A2. What is the Fermi energy level? The Fermi level E_F is the energy at which the probability of a state being occupied by an electron is exactly one half. At 0 K it is the boundary between filled and empty states. Its position relative to E_c and E_v determines the electron and hole concentrations, and in equilibrium it is constant throughout a material or device.

A3. State the Fermi–Dirac distribution function. $f(E) = 1/(1 + e^{(E - E_F)/kT})$, where f(E) is the probability that a state of energy E is occupied, E_F the Fermi level, k Boltzmann's constant and T the absolute temperature.

A4. What is the probability that a state at $E = E_F$ is occupied? Exactly **½ (50 %)**, because the exponent is zero: $f = 1/(1 + e^0) = 1/2$. This is true at any temperature.

A5. Define density of states. The density of states g(E) is the number of allowed electron energy states per unit energy per unit volume at energy E (states/eV·cm^3). In a parabolic band, $g_c(E) \propto \sqrt{(E - E_c)}$ in the conduction band and $g_v(E) \propto \sqrt{(E_v - E)}$ in the valence band; it is zero in the band gap.

A6. What is meant by effective density of states? The effective density of states (N_c for the conduction band, N_v for the valence band) is the number of states that, if all placed at the band edge, would give the same carrier concentration as the real distributed band. It lets the carrier concentration be written simply as $n = N_{ce}^{-(E_c - E_F)/kT}$. For Si at 300 K, $N_c = 2.8\times10^{19}$ and $N_v = 1.04\times10^{19}$ cm^{-3}.

A7. Distinguish between intrinsic and extrinsic semiconductors. An intrinsic semiconductor is pure: its carriers come only from thermal breaking of bonds, so $n = p = n_i$ and $E_F \approx E_i$ (mid-gap). An extrinsic semiconductor is doped with donors or acceptors: one carrier type dominates ($n \approx N_D$ or $p \approx N_A$), E_F moves towards E_c or E_v, and the carrier concentration is set by the doping rather than by temperature.

A8. Define majority and minority carriers. Majority carriers are the more numerous carrier type in a doped semiconductor (electrons in n-type, holes in p-type). Minority carriers are the less numerous type (holes in n-type, electrons in p-type). In n-type Si with $N_D = 10^{16}$ cm^{-3}, there are 10^{16} majority electrons and only 10^4 minority holes per cm^3.

A9. State the mass action law. In thermal equilibrium the product of the electron and hole concentrations is constant at a given temperature: $np = n_i^2$, regardless of doping (for a non-degenerate semiconductor).

A10. Explain why doping changes the Fermi level. The Fermi level is tied to the carrier concentrations by $n = n_{ie}^{(E_F - E_i)/kT}$. Donors add electrons, so n rises above n_i; for the equation to hold, E_F must rise above E_i (towards E_c), by $kT\ln(N_D/n_i)$. Acceptors add holes, so p rises and E_F must fall towards E_v by $kT\ln(N_A/n_i)$. The heavier the doping, the larger the shift.

Part B: calculation questions

B1. At 300 K, find the probability that a state 0.10 eV above the Fermi level is occupied (kT = 0.02585 eV).

$$f = 1/(1 + e^{0.10/0.02585}) = 1/(1 + e^{3.868}) = 1/(1 + 47.85) = 0.0205$$

The probability is **0.0205, about 2 %**.

B2. Silicon has $N_D = 1\times10^{16}$ cm^{-3} and $n_i = 1\times10^{10}$ cm^{-3}. Assuming complete ionisation, find (a) n, (b) p, (c) the majority carrier.

(a) $n \approx N_D = 1\times10^{16}\ cm^{-3}$

(b) $p = n_i^2/n = 10^{20}/10^{16} = 1\times10^{4}\ cm^{-3}$

(c) Electrons are the majority carriers (n-type).

B3. P-type silicon has $N_A = 5\times10^{16}\ cm^{-3}$ and $n_i = 1\times10^{10}\ cm^{-3}$. Find (a) p, (b) n, (c) the minority carrier.

(a) $p \approx N_A = 5\times10^{16}\ cm^{-3}$

(b) $n = n_i^2/p = 10^{20}/(5\times10^{16}) = 2\times10^{3}\ cm^{-3}$

(c) Electrons are the minority carriers.

B4. At 300 K an n-type semiconductor has $N_D = 1\times10^{17}\ cm^{-3}$ and $n_i = 1\times10^{10}\ cm^{-3}$. Find the Fermi level relative to the intrinsic level.

$$E_F - E_i = kT \ln(N_D/n_i) = 0.02585 \times \ln(10^7) = 0.02585 \times 16.12 = 0.417\ eV$$

The Fermi level is **0.417 eV above E_i**.

B5. Explain, using equations, why increasing temperature increases the intrinsic carrier concentration.

$$n_i = \sqrt{(N_c N_v)}\, e^{-E_g/2kT}, \quad \text{with } N_c, N_v \propto T^{3/2} \quad \Rightarrow \quad n_i \propto T^{3/2}\, e^{-E_g/2kT}$$

Two factors increase with T. The $T^{3/2}$ factor grows slowly: more states become effectively available as the electrons' thermal energy spreads. The exponential grows very fast: $E_g/2kT$ becomes smaller as T increases, so $e^{-E_g/2kT}$ increases steeply. Physically, a higher temperature gives more electrons enough thermal energy to cross the band gap, breaking more covalent bonds and creating more electron–hole pairs. For Si, n_i is about $10^{10}\ cm^{-3}$ at 300 K, 5×10^{12} at 400 K and 2.6×10^{14} at 500 K.

Examination questions solved (Q30–Q33)

Q30. Explain the Fermi-Dirac distribution function and its significance in semiconductor physics.

The function. Electrons are indistinguishable particles of half-integer spin that obey the Pauli exclusion principle (fermions). In thermal equilibrium at temperature T, the probability that an available state of energy E is occupied by an electron is given by Fermi–Dirac statistics:

$$f(E) = 1 / (1 + \exp((E - E_F)/kT))$$

where E_F is the Fermi level and k is Boltzmann's constant.

Properties:

- $0 \le f(E) \le 1$ (at most one electron per state).
- $f(E_F) = ½$ at any temperature: E_F is the energy with a 50 % chance of occupation.
- f is symmetric about E_F: the probability that a state ΔE above E_F is occupied equals the probability that a state ΔE below E_F is empty, so $1 - f(E_F + \Delta E) = f(E_F - \Delta E)$.
- The transition from 1 to 0 happens over a few kT (about 0.1 eV at 300 K).
- For $E - E_F > 3kT$ it reduces to the Boltzmann factor $\exp(-(E - E_F)/kT)$.

Significance:

1. **Gives the carrier concentrations.** $n = \int g_c\, f\, dE$ and $p = \int g_v(1 - f)dE$. Without f(E) we could not calculate n, p or n_i.
2. **Defines the Fermi level**, the single most useful quantity in device physics. Its position relative to E_c and E_v tells immediately whether the material is n-type, p-type or intrinsic and how heavily doped.
3. **Constant Fermi level in equilibrium.** E_F is flat across any system in equilibrium; this determines band bending and built-in potential in p-n junctions, Schottky contacts and MOS structures.
4. **Explains temperature dependence** of carrier concentration, intrinsic behaviour, freeze-out and degeneracy.
5. **Out of equilibrium**, the concept is extended to quasi-Fermi levels E_{Fn} and E_{Fp}, which are used to describe biased devices.

Q31. Write the Fermi-Dirac distribution equation and explain what happens to the distribution at absolute zero.

$$f(E) = 1 / (1 + e^{(E - E_F)/kT})$$

At T = 0 K, $kT \to 0$, so the exponent $(E - E_F)/kT \to \pm\infty$:

- For $\mathbf{E < E_F}$: $(E - E_F)/kT \to -\infty$, $e^{-\infty} = 0$, so **f(E) = 1/(1 + 0) = 1**. Every state below E_F is certainly filled.
- For $\mathbf{E > E_F}$: $(E - E_F)/kT \to +\infty$, $e^{+\infty} \to \infty$, so **f(E) = 0**. Every state above E_F is certainly empty.

The distribution becomes a **sharp rectangular step** at $E = E_F$. The Fermi level is then the boundary between completely filled and completely empty states: the highest occupied energy at 0 K.

For a semiconductor this means the valence band is completely full and the conduction band completely empty at 0 K. There are no free carriers, and **a pure semiconductor is a perfect insulator at absolute zero**.

As T increases, the step softens over a range of about ±3kT: some states just below E_F become empty and some just above become filled. This tail reaching into the conduction band is what produces free electrons at room temperature. At $E = E_F$, f = ½ always. (See the Fermi–Dirac figure above.)

$E - E_F$	f at 0 K	f at 300 K (kT = 0.0259 eV)
−0.1 eV	1	0.979
0	½ (undefined step)	0.500
+0.1 eV	0	0.021
+0.3 eV	0	9.3×10^{-6}

Q32. Define the density of states in a semiconductor and explain its importance in determining carrier concentration.

Definition. The density of states g(E) is the number of allowed electron energy states (including both spins) per unit energy interval per unit volume of the crystal, at energy E. Unit: $cm^{-3}eV^{-1}$ (or $m^{-3}J^{-1}$).

For the conduction and valence bands (parabolic band approximation):

$$g_c(E) = (4\pi\,(2m_n{*})^{3/2} / h^3)\,\sqrt{(E - E_c)}, \quad E \geq E_c$$

$$g_v(E) = (4\pi\,(2m_p{*})^{3/2} / h^3)\,\sqrt{(E_v - E)}, \quad E \leq E_v$$

$$g(E) = 0 \text{ inside the gap}$$

Physical origin. In a crystal of volume V, the allowed wave vectors k are quantised, each occupying $(2\pi)^3/V$ of k-space. The number of states with energy up to E is the volume of the sphere of radius k divided by that, times 2 for spin.

Differentiating with respect to E, using $E - E_c = \hbar^2k^2/2m*$, gives the square-root law. The effective mass enters because it describes the band's curvature: heavier mass means more states per unit energy.

Importance in determining carrier concentration:

1. The carrier concentration is the product of **available states** and **probability of occupation**: $n = \int g_c(E)f(E)dE$, $p = \int g_v(E)[1 - f(E)]dE$. The Fermi function alone says nothing about how many electrons there are; g(E) supplies the number of places they can go.
2. It produces the **effective densities of states** N_c and N_v, which appear in every carrier-concentration formula ($n = N_c\, e^{-(E_c - E_F)/kT}$).
3. Because g(E) is zero in the gap, no carriers exist there even though f(E) is non-zero, which explains why the gap is "forbidden".
4. It explains differences between materials. GaAs has a small m_n* ($0.067m_0$), so its N_c (4.7×10^{17} cm^{-3}) is 60 times smaller than Si's; for the same doping its Fermi level sits much closer to E_c and it becomes degenerate at lower doping.
5. It sets the $T^{3/2}$ factor in N_c, N_v and n_i, and determines where the Fermi level lies relative to mid-gap (E_i shifts by $(3/4)kT \ln(m_p*/m_n*)$).
6. It also governs optical absorption and the density of electrons available for tunnelling and contact behaviour.

Q33. Explain how the Fermi energy level changes with doping in n-type and p-type semiconductors.

In an intrinsic semiconductor $E_F = E_i$, near mid-gap. Doping shifts it so that the carrier concentrations satisfy $n = n_i\, e^{(E_F - E_i)/kT}$ and $p = n_i\, e^{(E_i - E_F)/kT}$.

n-type (donor doping N_D). $n \approx N_D > n_i$, so E_F moves **up, towards E_c**:

$$E_F - E_i = kT \ln(N_D/n_i), \qquad E_c - E_F = kT \ln(N_c/N_D)$$

p-type (acceptor doping N_A). $p \approx N_A > n_i$, so E_F moves **down, towards E_v**:

$$E_i - E_F = kT \ln(N_A/n_i), \qquad E_F - E_v = kT \ln(N_v/N_A)$$

Behaviour:

1. The shift is **logarithmic**: each tenfold increase in doping moves E_F by $kT \ln 10 \approx 0.06$ eV at 300 K.
2. **Heavier doping → closer to the band edge.** At very high doping ($\approx N_c$ or N_v) E_F reaches or enters the band and the semiconductor becomes **degenerate** (n^+, p^+).
3. **Compensation** (both N_D and N_A) shifts E_F according to the net doping $N_D - N_A$.
4. **Temperature:** as T rises, n_i increases and E_F moves back toward E_i; at high enough T every semiconductor becomes intrinsic. At very low T (freeze-out) E_F rises above E_D in n-type material.
5. In equilibrium the Fermi level must be the same everywhere; when n and p regions meet, bands bend to keep E_F flat.

Example (Si, 300 K, $n_i = 10^{10}$ cm^{-3}):

Doping	E_F position
Intrinsic	$E_F = E_i$
$N_D = 10^{15}$ cm^{-3}	$E_F - E_i = 0.0259 \ln(10^5) = 0.298$ eV
$N_D = 10^{17}$ cm^{-3}	$E_F - E_i = 0.0259 \ln(10^7) = 0.417$ eV
$N_A = 10^{16}$ cm^{-3}	$E_i - E_F = 0.0259 \ln(10^6) = 0.358$ eV

(See the Fermi-level figure above.)

Practice questions

These extra questions practise this week's ideas. Work each one on paper before reading the solution; the calculation questions follow the same Question, Given, Find, Solution, Answer layout as the worked examples.

Practice 5.1: accuracy of the Boltzmann approximation

Question. Calculate the occupation probability of a state 3kT above the Fermi level using (a) the Fermi–Dirac function and (b) the Boltzmann approximation. What is the error?

Given: $E - E_F = 3kT$.

Find: f(FD), f(Boltzmann) and the percentage error.

Solution.

(a) $f = 1/(1 + e^3) = 1/21.09 = 0.0474$

(b) $f \approx e^{-3} = 0.0498$

error $= (0.0498 - 0.0474)/0.0474 = 5$ %

Answer: 0.0474 against 0.0498, a 5 % error: this is why "more than 3kT from E_F" is the rule for using the Boltzmann approximation.

Practice 5.2: carriers from the Fermi-level position

Question. In silicon at 300 K the Fermi level is 0.30 eV below the conduction band. Find the probability that a state at E_c is occupied, and the electron concentration.

Given: $E_c - E_F = 0.30$ eV, $kT = 0.0259$ eV, $N_c = 2.8\times10^{19}$ cm^{-3}.

Find: $f(E_c)$ and n.

Solution.

$f(E_c) \approx e^{-0.30/0.0259} = e^{-11.58} = 9.3\times10^{-6}$

$n = N_c\, e^{-(E_c - E_F)/kT} = 2.8\times10^{19} \times 9.3\times10^{-6} = 2.6\times10^{14}$ cm^{-3}

Answer: $f = 9.3\times10^{-6}$, $n = 2.6\times10^{14}$ cm^{-3}.

Practice 5.3: doping for a chosen Fermi level

Question. What donor concentration places the Fermi level of silicon 0.20 eV above the intrinsic level at 300 K?

Given: $E_F - E_i = 0.20$ eV, $n_i = 10^{10}$ cm^{-3}, $kT = 0.02585$ eV.

Find: N_D.

Solution.

$$N_D = n_i\, e^{(E_F - E_i)/kT} = 10^{10} \times e^{0.20/0.02585} = 10^{10} \times e^{7.74} = 2.3\times10^{13}\ \text{cm}^{-3}$$

Answer: $N_D \approx 2.3\times10^{13}\ \text{cm}^{-3}$, a very light doping.

Practice 5.4: compensated p-type silicon

Question. Silicon is doped with $N_A = 3\times10^{16}\ \text{cm}^{-3}$ and $N_D = 1\times10^{16}\ \text{cm}^{-3}$. Find p, n and the Fermi-level position at 300 K.

Given: $N_A = 3\times10^{16}$, $N_D = 10^{16}$, $n_i = 10^{10}\ \text{cm}^{-3}$, $kT = 0.02585$ eV.

Find: p, n and $E_i - E_F$.

Solution.

$$p \approx N_A - N_D = 2\times10^{16}\ \text{cm}^{-3}$$

$$n = n_i^2/p = 10^{20}/2\times10^{16} = 5\times10^{3}\ \text{cm}^{-3}$$

$$E_i - E_F = kT \ln(p/n_i) = 0.02585 \times \ln(2\times10^{6}) = 0.375\ \text{eV}$$

Answer: $p = 2\times10^{16}\ \text{cm}^{-3}$, $n = 5\times10^{3}\ \text{cm}^{-3}$, $E_F = E_i - 0.375$ eV.

Practice 5.5: lightly doped germanium

Question. Germanium ($n_i = 2.4\times10^{13}\ \text{cm}^{-3}$) is doped with $N_D = 1\times10^{14}\ \text{cm}^{-3}$. Find n and p using the full charge-neutrality equation.

Given: $N_D = 10^{14}\ \text{cm}^{-3}$, $N_A = 0$, $n_i = 2.4\times10^{13}\ \text{cm}^{-3}$.

Find: n and p.

Solution.

$$n = N_D/2 + \sqrt{(N_D/2)^2 + n_i^2} = 5\times10^{13} + \sqrt{2.5\times10^{27} + 5.76\times10^{26}}$$

$$= 5\times10^{13} + 5.55\times10^{13} = 1.05\times10^{14}\ \text{cm}^{-3}$$

$$p = n_i^2/n = 5.76\times10^{26} / 1.05\times10^{14} = 5.5\times10^{12}\ \text{cm}^{-3}$$

Answer: $n = 1.05\times10^{14}\ \text{cm}^{-3}$, $p = 5.5\times10^{12}\ \text{cm}^{-3}$. Thermal generation adds 5 % to the electron count, so $n \approx N_D$ is only approximate here.

Practice 5.6: is it degenerate?

Question. Silicon is doped with $N_D = 5\times10^{18}\ \text{cm}^{-3}$. Find $E_c - E_F$ and decide whether the Boltzmann approximation is valid.

Given: $N_D = 5\times10^{18}\ \text{cm}^{-3}$, $N_c = 2.8\times10^{19}\ \text{cm}^{-3}$, $kT = 0.02585$ eV.

Find: $E_c - E_F$ in eV and in units of kT.

Solution.

$$E_c - E_F = kT \ln(N_c/N_D) = 0.02585 \times \ln(5.6) = 0.02585 \times 1.72 = 0.045\ \text{eV} = 1.7kT$$

Answer: 0.045 eV, only 1.7kT: less than 3kT, so the material is (nearly) degenerate and full Fermi–Dirac statistics should be used.

Practice 5.7: where exactly is E_i?

Question. Show that the intrinsic level of silicon lies slightly below mid-gap, and by how much.

Given: $E_i = (E_c + E_v)/2 + (kT/2)\ln(N_v/N_c)$, $N_c = 2.8\times10^{19}$, $N_v = 1.04\times10^{19}$ cm^{-3}.

Find: E_i − mid-gap.

Solution.

$$(kT/2)\ln(N_v/N_c) = (0.02585/2) \times \ln(1.04/2.8) = 0.0129 \times (-0.990) = -0.0128 \text{ eV}$$

Answer: E_i is 13 meV below mid-gap, because the valence band has fewer effective states ($N_v < N_c$), so E_F must sit a little closer to it to make p equal n.

Objectives check

Use this table to confirm you have met this week's objectives. Cover the Answer column and answer each check question from memory. If you cannot, go back to the section in the last column before moving on.

#	Objective	Check yourself	Answer	Where it is taught
1	Define and use the Fermi–Dirac distribution function.	Find the probability that a state 0.2 eV above E_F is occupied at 300 K.	$f = 1/(1 + e^{0.2/0.02585}) = 1/(1 + 2290) = 4.4\times10^{-4}$.	Section 1
2	Explain the physical meaning of the Fermi level and the shape of f(E) at 0 K.	What is f(E) below and above E_F at 0 K?	1 below E_F, 0 above: a sharp step.	Section 1
3	Explain the density of states and the effective density of states.	Why are there no carriers in the band gap even though f(E) is not zero there?	The density of states is zero in the gap: there are no states to occupy.	Section 2
4	Calculate electron and hole concentrations and apply the mass action law.	Si with $N_A = 10^{15}$ cm^{-3}: find p and n ($n_i = 10^{10}$ cm^{-3}).	$p = 10^{15}$, $n = 10^5$ cm^{-3}.	Sections 7–8, 12, 15–16; Examples 5.3, 5.5
5	Determine the position of the Fermi level for given doping.	Find $E_F - E_i$ for $N_D = 10^{14}$ cm^{-3}.	$0.02585 \ln(10^4) = 0.238$ eV.	Sections 9, 13; Examples 5.1, 5.4
6	Apply the charge-neutrality condition, including compensated material.	Si has $N_D = 5\times10^{16}$ and $N_A = 2\times10^{16}$ cm^{-3}. Find n.	$n \approx N_D - N_A = 3\times10^{16}$ cm^{-3}.	Section 14
7	Explain the effect of temperature on carrier concentration and the Fermi level.	Name the three temperature regions and what n equals in the middle one.	Freeze-out, extrinsic ($n \approx N_D$), intrinsic.	Sections 17, 23; Example 5.2
8	Distinguish degenerate and non-degenerate semiconductors.	When must full Fermi–Dirac statistics be used instead of the Boltzmann approximation?	When E_F is within about 3kT of a band edge (very heavy doping).	Section 21

Continuous Assessment Test 1 and Review

Plan: mid-module evaluation of crystal structures and carrier statistics (Weeks 1–5).

Learning objectives

By the end of this week you should be able to:

1. Revise crystal structures, Miller indices and bonding (Week 1).
2. Revise energy bands and direct/indirect gaps (Week 4).
3. Calculate occupation probabilities with the Fermi–Dirac function (Week 5).
4. Calculate n, p and E_F for doped and compensated silicon and germanium (Weeks 4–5).

This page has a one-screen review of what CAT 1 covers, followed by a practice test with full solutions. The questions are written in the style of the lecturer's revision questions and the PGD exam paper. They are practice questions, not the actual CAT.

Review: what you must be able to do

Crystals (Week 1)

- Define lattice, basis, unit cell, lattice constant, coordination number, APF.
- State the structures: Si and Ge diamond cubic, GaAs zinc blende; 8 atoms per cell; 4 nearest neighbours.
- Find Miller indices from intercepts (intercepts → reciprocals → clear fractions); handle ∞ (index 0) and negative intercepts (bar).
- Use $d_{hkl} = a/\sqrt{(h^2 + k^2 + l^2)}$.

Materials and bonding (Week 1)

- Compare conductors, semiconductors and insulators; Si, Ge and GaAs.
- Explain covalent bonding, electron–hole pairs, n-type and p-type doping.

Bands and statistics (Weeks 4–5)

- Schrödinger equation terms; Kronig–Penney: allowed bands where $|P \sin(\alpha a)/\alpha a + \cos \alpha a| \leq 1$.
- Direct and indirect gaps; $\lambda(\mu m) = 1.24/E_g(eV)$.
- $f(E) = 1/(1 + e^{(E-E_F)/kT})$; at 0 K a step; Boltzmann approximation for $E - E_F > 3kT$.
- $n = N_c\, e^{-(E_c-E_F)/kT}$, $p = N_v\, e^{-(E_F-E_v)/kT}$, $n_i = \sqrt{(N_c N_v)} e^{-E_g/2kT}$, $np = n_i^2$.
- Charge neutrality and compensation; $E_F - E_i = kT \ln(n/n_i)$.

> **Also revise:** the lecturer's own mid-module evaluation (ten short-answer and five calculation questions on carrier statistics) is solved at the end of Week 5, section 26.

Practice test (60 minutes)

Use $q = 1.6\times10^{-19}$ C, $kT = 0.0259$ eV at 300 K, Si: $n_i = 1\times10^{10}$ cm^{-3}, $N_c = 2.8\times10^{19}$ cm^{-3}, $\mu_n = 1350$, $\mu_p = 480$ cm^2/V·s; Ge: $n_i = 2.4\times10^{13}$ cm^{-3}, $\mu_n = 3900$, $\mu_p = 1900$ cm^2/V·s.

1. Silicon has lattice constant 5.43 Å. Calculate (a) the number of atoms per cm^3 and (b) the nearest-neighbour distance.
2. Find the Miller indices of planes with intercepts (a) 3a, 2b, c and (b) a, −b, ∞. For (a), find the interplanar spacing in Si.
3. Show that the atomic packing factor of the diamond structure is 0.34.
4. State three differences between direct and indirect band-gap semiconductors and give the emission wavelength of GaAs.
5. Find the probability that a state (a) 0.1 eV above E_F and (b) kT above E_F is occupied at 300 K.
6. At what temperature is there a 10^{-4} probability that a state 0.30 eV above E_F is occupied?
7. In a Si sample at 300 K the Fermi level is 0.25 eV below E_c. Find n and p. Is it n- or p-type?
8. Si is doped with $N_A = 1\times10^{16}$ cm^{-3} and $N_D = 3\times10^{15}$ cm^{-3}. Find n, p and the Fermi level position at 300 K.
9. Germanium is doped with $N_D = 1\times10^{15}$ cm^{-3}. Find n and p at 300 K, without assuming n_i is negligible.
10. Calculate the conductivity and resistivity of intrinsic germanium at 300 K.

Solutions

1. Atom density and nearest neighbours in Si

Question. Silicon has lattice constant 5.43 Å. Calculate (a) the number of atoms per cm^3 and (b) the nearest-neighbour distance.

Solution.

The diamond unit cell holds 8 atoms (8 corners × ⅛ + 6 faces × ½ + 4 inside).

(a) $N = 8/a^3 = 8/(5.43\times10^{-8}\text{ cm})^3 = 8/(1.601\times10^{-22}\text{ cm}^3) = 5.0\times10^{22}$ atoms/cm^3

(b) $r_{nn} = (\sqrt{3}/4)a = 0.433 \times 5.43$ Å $= 2.35$ Å

2. Miller indices

Question. Find the Miller indices of planes with intercepts (a) 3a, 2b, c and (b) a, −b, ∞. For (a), find the interplanar spacing in Si.

Solution.

	x	y	z
(a) Intercepts	3	2	1
Reciprocals	1/3	1/2	1
× 6	2	3	6

(a) **(2 3 6)**. $d_{236} = 5.43/\sqrt{(4 + 9 + 36)} = 5.43/7 =$ **0.776 Å**.

(b) Intercepts 1, −1, ∞ → reciprocals 1, −1, 0 → **(1 1̄ 0)**, a plane parallel to z.

3. Packing factor of diamond

Question. Show that the atomic packing factor of the diamond structure is 0.34.

Solution.

Nearest neighbours touch along the body diagonal at distance $(\sqrt{3}/4)a$, so the atomic radius is $r = (\sqrt{3}/8)a$. With 8 atoms per cell:

$$\text{APF} = 8 \times (4/3)\pi r^3 / a^3 = 8 \times (4/3)\pi\,(\sqrt{3}/8)^3 = (32\pi/3)(3\sqrt{3}/512) = \pi\sqrt{3}/16 = 0.34$$

Only 34 % of the volume is filled, an open structure that allows interstitial impurities.

4. Direct and indirect gaps

Question. State three differences between direct and indirect band-gap semiconductors and give the emission wavelength of GaAs.

Solution.

(1) In a direct gap the conduction-band minimum and valence-band maximum are at the same k; in an indirect gap they are at different k. (2) Direct transitions need only a photon; indirect transitions also need a phonon. (3) Direct-gap materials emit light efficiently (GaAs, GaN, InP: LEDs, lasers); indirect ones do not (Si, Ge). Also: absorption near E_g is strong in direct-gap and weak in indirect-gap material; radiative lifetime is ns in direct and much longer in indirect.

$$\text{GaAs: } \lambda = 1.24/1.42 = 0.873\ \mu\text{m (near infrared)}$$

5. Occupation probabilities

Question. Find the probability that a state (a) 0.1 eV above E_F and (b) kT above E_F is occupied at 300 K.

Solution.

$$\text{(a) } f = 1/(1 + e^{0.1/0.0259}) = 1/(1 + e^{3.861}) = 1/(1 + 47.5) = 0.0206$$

$$\text{(b) } f = 1/(1 + e^{1}) = 1/3.718 = 0.269$$

6. Temperature for a given probability

Question. At what temperature is there a 10^{-4} probability that a state 0.30 eV above E_F is occupied?

Solution.

Since f is small, use the Boltzmann form: $f \approx e^{-(E - E_F)/kT} = 10^{-4}$.

$$0.30/kT = \ln(10^4) = 9.21 \quad \Rightarrow \quad kT = 0.0326\ \text{eV}$$

$$T = 0.0326/8.617\times10^{-5} = 378\ \text{K}$$

(The exact form $1/(1 + e^x) = 10^{-4}$ gives x = 9.21 as well, to three figures.)

7. Carrier densities from E_F

Question. In a Si sample at 300 K the Fermi level is 0.25 eV below E_c. Find n and p. Is it n- or p-type?

Solution.

$$n = N_c\, e^{-(E_c - E_F)/kT} = 2.8\times10^{19} \times e^{-0.25/0.0259} = 2.8\times10^{19} \times e^{-9.65}$$

$$= 2.8\times10^{19} \times 6.4\times10^{-5} = 1.8\times10^{15}\ \text{cm}^{-3}$$

$$p = n_i^2/n = 10^{20}/1.8\times10^{15} = 5.6\times10^{4}\ \text{cm}^{-3}$$

$n \gg p$, so the sample is **n-type** (doped with about 1.8×10^{15} cm^{-3} donors).

8. Compensated silicon

Question. Si is doped with $N_A = 1\times10^{16}$ cm^{-3} and $N_D = 3\times10^{15}$ cm^{-3}. Find n, p and the Fermi level position at 300 K.

Solution.

$N_A > N_D$, so the material is p-type with net doping $N_A - N_D = 7\times10^{15}$ cm$^{-3} \gg n_i$.

$$p \approx 7\times10^{15}\ \text{cm}^{-3}$$

$$n = n_i^2/p = 10^{20}/7\times10^{15} = 1.4\times10^{4}\ \text{cm}^{-3}$$

$$E_i - E_F = kT \ln(p/n_i) = 0.0259 \times \ln(7\times10^{5}) = 0.0259 \times 13.46 = 0.349\ \text{eV}$$

The Fermi level is 0.349 eV below E_i. Both dopants still scatter carriers, so mobility is that for 1.3×10^{16} cm^{-3} total impurities.

9. Lightly doped germanium

Question. Germanium is doped with $N_D = 1\times10^{15}$ cm^{-3}. Find n and p at 300 K, without assuming n_i is negligible.

Solution.

$n_i = 2.4\times10^{13}$ is not negligible compared with $N_D = 10^{15}$, so use the exact neutrality solution:

$$n = N_D/2 + \sqrt{(N_D/2)^2 + n_i^2} = 5\times10^{14} + \sqrt{2.5\times10^{29} + 5.76\times10^{26}}$$

$$= 5\times10^{14} + 5.006\times10^{14} = 1.0006\times10^{15}\ \text{cm}^{-3}$$

$$p = n_i^2/n = 5.76\times10^{26}/1.0006\times10^{15} = 5.76\times10^{11}\ \text{cm}^{-3}$$

n is still essentially N_D, but the minority concentration (5.8×10^{11}) is 10^7 times larger than in silicon with the same doping (10^5): this is why Ge diodes leak so much.

10. Intrinsic germanium

Question. Calculate the conductivity and resistivity of intrinsic germanium at 300 K.

Solution.

$$\sigma_i = q\, n_i\, (\mu_n + \mu_p) = 1.6\times10^{-19} \times 2.4\times10^{13} \times (3900 + 1900)$$

$$= 1.6\times10^{-19} \times 1.392\times10^{17} = 0.0223\ \text{S/cm}$$

$$\rho_i = 1/\sigma_i = 44.9\ \Omega\cdot\text{cm}$$

For comparison, intrinsic Si is about 3×10^{5} Ω·cm, nearly 10^4 times higher, because its band gap is larger.

Practice questions

These extra questions practise this week's ideas. Work each one on paper before reading the solution; the calculation questions follow the same Question, Given, Find, Solution, Answer layout as the worked examples.

Practice 6.1: intercepts from Miller indices

Question. A plane has Miller indices (1 2 3). Find a set of intercepts it makes on the axes.

Given: (h k l) = (1 2 3).

Find: intercepts.

Solution. Intercepts are the reciprocals of the indices: 1, 1/2, 1/3. Multiplying by 6 to clear fractions gives an equivalent parallel plane.

Answer: a, b/2, c/3 (or, equivalently, 6a, 3b, 2c).

Practice 6.2: band gap from an absorption edge

Question. A semiconductor stops absorbing light at wavelengths longer than 1.1 μm. Estimate its band gap.

Given: λ_{max} = 1.1 μm.

Find: E_g.

Solution.

$$E_g = 1.24/\lambda = 1.24/1.1 = 1.13 \text{ eV}$$

Answer: $E_g \approx 1.13$ eV (silicon).

Practice 6.3: hole concentration from the Fermi level

Question. In p-type silicon at 300 K the Fermi level is 0.10 eV above the valence band. Find the hole concentration.

Given: $E_F - E_v = 0.10$ eV, $N_v = 1.04\times10^{19}$ cm^{-3}, kT = 0.0259 eV.

Find: p.

Solution.

$$p = N_v\, e^{-(E_F - E_v)/kT} = 1.04\times10^{19} \times e^{-3.86} = 1.04\times10^{19} \times 0.0211 = 2.2\times10^{17} \text{ cm}^{-3}$$

Answer: $p = 2.2\times10^{17}$ cm^{-3}.

Practice 6.4: probability of a hole

Question. What is the probability that a state 0.05 eV below the Fermi level is empty (holds a hole) at 300 K?

Given: $E = E_F - 0.05$ eV, kT = 0.0259 eV.

Find: 1 − f(E).

Solution.

$$1 - f = 1 - 1/(1 + e^{-0.05/0.0259}) = 1/(1 + e^{0.05/0.0259}) = 1/(1 + 6.89) = 0.127$$

Answer: 12.7 %.

Practice 6.5: short-answer revision

Question. (a) State the mass-action law. (b) What is the coordination number of the diamond structure? (c) Name the structure of GaAs. (d) Which carrier is the minority carrier in boron-doped silicon?

Answer. (a) $np = n_i^2$ in thermal equilibrium. **(b)** 4. **(c)** Zinc blende. **(d)** Electrons.

Objectives check

Use this table to confirm you have met this week's objectives. Cover the Answer column and answer each check question from memory. If you cannot, go back to the section in the last column before moving on.

#	Objective	Check yourself	Answer	Where it is taught
1	Revise crystal structures, Miller indices and bonding (Week 1).	Find the Miller indices for intercepts 2a, 2b, 4c.	1/2, 1/2, 1/4 → × 4 → (2 2 1).	Review; practice Q1–Q3
2	Revise energy bands and direct/indirect gaps (Week 4).	State one property that distinguishes a direct-gap semiconductor.	Efficient light emission (also: CB minimum and VB maximum at the same k).	Practice Q4
3	Calculate occupation probabilities with the Fermi–Dirac function (Week 5).	What is f at $E = E_F + kT$?	$1/(1 + e) = 0.269$.	Practice Q5–Q6
4	Calculate n, p and E_F for doped and compensated silicon and germanium (Weeks 4–5).	Si with E_F 0.25 eV below E_c: n-type or p-type?	n-type ($n = 1.8\times10^{15}$ cm^{-3}).	Practice Q7–Q10

Drift and Diffusion Currents

Plan: carrier mobility, scattering mechanisms and conductivity; diffusion of carriers, Fick's law and the Einstein relation.

Learning objectives

By the end of this week you should be able to:

1. Explain thermal motion, drift velocity and mobility.
2. Explain the scattering mechanisms and the factors that affect mobility.
3. Derive and use the drift current density and conductivity.
4. Explain carrier diffusion and calculate diffusion current with Fick's law.
5. State, explain and use the Einstein relation.

The big picture

Weeks 4–5 told us **how many** carriers there are. This week asks **how they move**, because moving charge is current. There are exactly two reasons a carrier moves in a particular direction:

1. **Drift:** an electric field pushes it. Like water flowing downhill because of gravity.
2. **Diffusion:** there are more carriers in one place than another, and random motion spreads them out. Like a drop of ink spreading in still water (the same idea as dopant diffusion in Week 3, but for electrons and holes, and much faster).

Every current in every device in this course is drift, diffusion, or a mixture. At the end of the week we will see that the two are tied together by a surprisingly simple rule, the **Einstein relation**.

Part A: Drift, mobility and conductivity

Random thermal motion

Even with no voltage applied, electrons in a semiconductor are not still. At room temperature they rush about at the **thermal velocity**, about 10^7 cm/s (100 km/s), bouncing off vibrating atoms and impurities roughly every 0.1–1 picosecond (the **mean free time** τ_m). Their directions are random, so on average they go nowhere: no current.

Drift in an electric field

Now apply an electric field E. Between collisions each electron is pushed by the field and gains a little extra speed in one direction. At each collision that extra speed is lost (the direction is randomised again). So on top of the random motion there is a small, steady average **drift velocity**:

$$v_d = \mu E$$

For electrons the drift is against the field (they are negative); for holes it is along the field.

> **In plain words.** Picture a crowd milling about randomly in a square. Tilt the square slightly. People still wander randomly, but on average the crowd now slides slowly downhill. The slow average slide is the drift velocity; the random wandering is much faster but cancels out.

Mobility

The constant μ is the **mobility**: how much drift velocity you get per unit field.

$$\mu = v_d / E = q\tau_m / m^* \qquad [\text{cm}^2/\text{V}\cdot\text{s}]$$

From the formula, mobility is high when:

- **collisions are rare** (long τ_m): fewer things to bump into;
- **the effective mass is small** (m^* small): the carrier is easy to accelerate.

Material	μ_n (cm^2/V·s)	μ_p (cm^2/V·s)
Si	1350	480
Ge	3900	1900
GaAs	8500	400

Electrons are faster than holes in all three, mainly because of their smaller effective mass.

Example 7.1: drift velocity

Question. An electric field of 1000 V/cm is applied to a silicon sample. Calculate the drift velocity of the electrons and of the holes.

Given: E = 1000 V/cm, μ_n = 1350 cm^2/V·s, μ_p = 480 cm^2/V·s.

Find: v_{dn} and v_{dp}.

Solution.

$$v_{dn} = \mu_n E = 1350 \times 1000 = 1.35\times10^6 \text{ cm/s}$$
$$v_{dp} = \mu_p E = 480 \times 1000 = 4.8\times10^5 \text{ cm/s}$$

Answer: electrons 1.35×10^6 cm/s, holes 4.8×10^5 cm/s, both far below the random thermal velocity of about 10^7 cm/s.

What scatters carriers

A perfectly regular, perfectly still crystal would not scatter electrons at all (Bloch's theorem, Week 4). Anything that breaks the perfect regularity causes scattering:

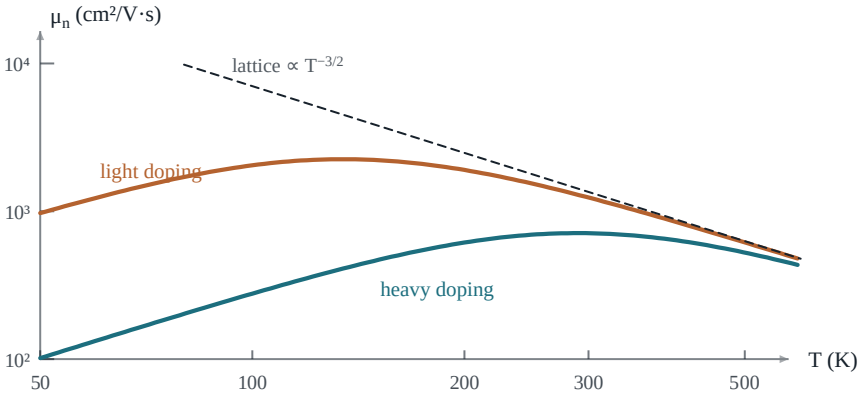
1. **Lattice (phonon) scattering.** The atoms vibrate with heat. The hotter, the more they vibrate, and the more they scatter. Mobility falls as $\mu_L \propto T^{-3/2}$.
2. **Ionised-impurity scattering.** The charged donor and acceptor ions deflect passing carriers, like a magnet deflecting a passing steel ball. Slow carriers are deflected more, so this is worst at **low** temperature and **high** doping: $\mu_I \propto T^{3/2}/N_I$.
3. **Neutral-impurity scattering**, important only at very low temperature.
4. **Carrier–carrier scattering**, at very high carrier densities.

5. **Defect scattering** from dislocations, grain boundaries and vacancies.
6. **Surface scattering.** In a MOSFET channel, carriers are pressed against the rough Si/SiO_2 surface; channel mobility is only about half the bulk value.
7. **Polar and alloy scattering** in compound semiconductors and alloys.

When several mechanisms act, their scattering rates add, so the reciprocals of the mobilities add (**Matthiessen's rule**):

$$1/\mu = 1/\mu_L + 1/\mu_I + \dots$$

The **lowest** mobility dominates, just as the narrowest section of a pipe limits the flow.



Mobility against temperature. At low temperature impurity scattering limits mobility; at high temperature lattice scattering does. Heavier doping lowers the curve and moves its peak to higher temperature.

What affects mobility (summary)

- **Temperature:** near room temperature, higher T → more lattice vibration → lower μ.
- **Doping:** more impurities → more scattering → lower μ (Si electrons: 1350 at light doping, about 1000 at 10^{17}, about 100 at 10^{20} cm^{-3}).
- **Effective mass:** smaller $m*$ → higher μ.
- **Electric field:** at very high fields μ falls (velocity saturation, Week 8).
- **Crystal quality, surfaces, orientation and strain.**

Drift current density

Current density J is the charge crossing 1 cm^2 per second. It equals (charge per carrier) × (number of carriers per cm^3) × (their average speed). For electrons, the charge is $-q$ and the velocity is $-\mu_n E$; the two minus signs cancel:

$$J_n = (-q)(n)(-\mu_n E) = qn\mu_n E$$

For holes:

$$J_p = (+q)(p)(+\mu_p E) = qp\mu_p E$$

Electrons move against the field but carry negative charge, so their **current** is in the same direction as the hole current. The total drift current is

$$J = q(n\mu_n + p\mu_p)\,E = \sigma E$$

This is just Ohm's law, with **conductivity** and **resistivity**

$$\sigma = q(n\mu_n + p\mu_p), \quad \rho = 1/\sigma$$

For n-type material $n \gg p$, so $\sigma \approx qN_D\mu_n$. For p-type, $\sigma \approx qN_A\mu_p$.

Example 7.2: conductivity and resistance of a silicon bar

Question. A bar of n-type silicon doped with $N_D = 1\times10^{16}$ cm^{-3} is 1 cm long and has a cross-section of 1 mm^2. Calculate its conductivity, resistivity and resistance.

Given: $N_D = 1\times10^{16}$ cm^{-3}, $\mu_n = 1350$ cm^2/V·s, L = 1 cm, A = 1 mm^2 = 0.01 cm^2, $q = 1.6\times10^{-19}$ C.

Find: σ, ρ and R.

Solution.

Step 1. n-type, so $\sigma \approx qN_D\mu_n$:

$$\sigma = 1.6\times10^{-19} \times 10^{16} \times 1350 = 2.16 \text{ S/cm}$$

Step 2. $\rho = 1/\sigma = 1/2.16 = 0.46$ Ω·cm

Step 3. $R = \rho L/A$:

$$R = 0.46 \times 1 / 0.01 = 46\ \Omega$$

Answer: σ = 2.16 S/cm, ρ = 0.46 Ω·cm, R = 46 Ω.

Sheet resistance. For a thin layer of thickness t, $R_s = \rho/t$ in ohms per square (Ω/□), measured with a four-point probe (Week 3).

Hall effect

Put a sample carrying current I_x in a magnetic field B_z. The magnetic force pushes the moving carriers sideways, so charge piles up on one side and a small **Hall voltage** V_H appears across the sample. Its **sign** shows whether the carriers are electrons or holes, and its **size** gives their concentration:

$$R_H = 1/qp \text{ (p-type, positive)}, \quad R_H = -1/qn \text{ (n-type, negative)}$$

Combined with a resistivity measurement it gives the mobility, $\mu = |R_H|\sigma$. This is the standard way to measure carrier type, concentration and mobility.

Part B: Diffusion, Fick's law and the Einstein relation

Diffusion of carriers

Suppose there are more electrons on the left of a sample than on the right, for example because light shines on the left. Each electron moves randomly. With more of them on the left, more cross any plane from left to right than from right to left. There is a net flow towards the right, with no field at all. This is **carrier diffusion**.

The flow obeys **Fick's law** (the same law as in Week 3, now for carriers):

$$\text{flux of electrons} = -D_n\, dn/dx, \quad \text{flux of holes} = -D_p\, dp/dx$$

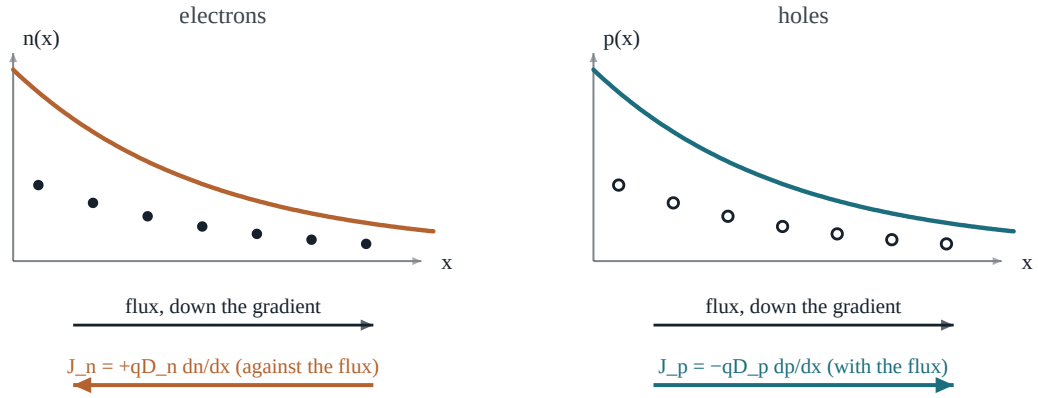
D_n and D_p are the carrier **diffusion coefficients** (Si at 300 K: $D_n \approx 35$ cm^2/s, $D_p \approx 12$ cm^2/s). These are about 10^{14} times larger than dopant diffusion coefficients: carriers diffuse across a device in nanoseconds, while dopant atoms need hours in a furnace.

Diffusion current density

Multiply each particle flux by the charge of the carrier:

$$J_n(\text{diff}) = (-q)(-D_n\, dn/dx) = +qD_n\, dn/dx$$
$$J_p(\text{diff}) = (+q)(-D_p\, dp/dx) = -qD_p\, dp/dx$$

Watch the signs. Both electrons and holes flow from high concentration to low. But because the electron charge is negative, the **electron current** points **opposite** to the electron flow, while the hole current points the same way as the hole flow.



Both carriers diffuse down their concentration gradient. Because electrons are negative, the electron diffusion current points against their flow, while the hole current points with it.

Example 7.3: electron diffusion current

Question. In a silicon sample the electron concentration falls linearly from 1×10^{17} cm^{-3} to zero over a distance of 1 μm. Taking $D_n = 35$ cm^2/s, calculate the electron diffusion current density.

Given: $\Delta n = 1\times10^{17}$ cm^{-3} over $\Delta x = 1$ μm $= 1\times10^{-4}$ cm, $D_n = 35$ cm^2/s, $q = 1.6\times10^{-19}$ C.

Find: $J_n(\text{diff})$.

Solution.

Step 1. The concentration gradient:

$$dn/dx = -10^{17} / 10^{-4} = -10^{21}\ \text{cm}^{-4}$$

Step 2. $J_n = qD_n\, dn/dx$:

$$J_n = 1.6\times10^{-19} \times 35 \times (-10^{21}) = -5.6\times10^{3}\ \text{A/cm}^2$$

Answer: $|J_n| = 5.6\times10^3$ A/cm^2, flowing in the −x direction (opposite to the electron flow, because electrons are negative).

A large current, from a modest gradient. Diffusion is not a small effect: it carries the whole current in a diode's neutral regions and in a transistor's base.

The total current equations

When a field and a gradient are both present, simply add drift and diffusion:

$$J_n = qn\mu_n E + qD_n \, dn/dx$$

$$J_p = qp\mu_p E - qD_p \, dp/dx$$

$$J = J_n + J_p$$

These two equations, together with the continuity equation (Week 9) and Poisson's equation, describe every semiconductor device.

The Einstein relation

Drift and diffusion are two responses of the same carriers to two different pushes, and both are limited by the **same collisions**. So it is no surprise that μ and D are related:

$$D_n / \mu_n = D_p / \mu_p = kT/q$$

kT/q is the **thermal voltage**, 0.0259 V (about 26 mV) at 300 K.

Why it is true (the equilibrium argument). Take an n-type sample whose doping is higher on the left than on the right, and leave it alone (no battery).

1. Electrons diffuse from the heavily doped left side towards the right.
2. As they leave, they uncover positive donor ions on the left. Positive charge on the left and extra electrons on the right create an internal electric field.
3. The field pulls electrons back to the left: a drift current opposite to the diffusion current.
4. With no battery, no net current can flow, so in equilibrium **drift exactly cancels diffusion**: $0 = qn\mu_n E + qD_n \, dn/dx$.
5. In equilibrium the Fermi level is flat (Week 5). Using $n = N_c \, e^{-(E_c - E_F)/kT}$, the gradient dn/dx can be written in terms of the field: $dn/dx = -(qn/kT)E$.
6. Substituting: $0 = qn\mu_n E - qD_n(qn/kT)E$, which gives $D_n/\mu_n = kT/q$.

Using it. Mobility is easy to measure (Hall effect, conductivity), so D follows immediately:

	μ (cm^2/V·s)	**D = μ × 0.0259 (cm^2/s)**
Si electrons	1350	35.0
Si holes	480	12.4
GaAs electrons	8500	220
Ge electrons	3900	101

> **In plain words.** A carrier that rarely collides both drifts easily in a field (high μ) and spreads quickly by itself (high D). The ratio between the two depends only on temperature.

Key points of Week 7

- Drift: $v_d = \mu E$, with $\mu = q\tau/m*$. Electrons are faster than holes; GaAs electrons are fastest.
- Mobility is limited by lattice scattering (worse when hot) and impurity scattering (worse when heavily doped); $1/\mu = \Sigma 1/\mu_i$.
- Drift current $J = q(n\mu_n + p\mu_p)E = \sigma E$; $\sigma \approx qN_D\mu_n$ for n-type.
- Diffusion: carriers flow down their concentration gradient. $J_n = qD_n \, dn/dx$, $J_p = -qD_p \, dp/dx$.

- Total current = drift + diffusion.
- Einstein relation: $D/\mu = kT/q = 0.0259$ V at 300 K.

Examination questions solved (Q35–Q40)

Q35. Define carrier mobility and explain the factors that affect electron and hole mobility in semiconductors.

Definition. Carrier mobility μ is the drift velocity acquired per unit applied electric field:

$$\mu = |v_d| / E = q\tau_m / m^* \qquad [\text{cm}^2/\text{V}\cdot\text{s}]$$

where τ_m is the mean free time between collisions and m^* the effective mass. It measures how easily carriers move through the crystal under a field. Electron mobility μ_n and hole mobility μ_p are different (Si: 1350 and 480 $\text{cm}^2/\text{V}\cdot\text{s}$).

Factors affecting mobility:

1. **Effective mass.** $\mu \propto 1/m^*$. Electrons generally have lighter effective mass than holes, so $\mu_n > \mu_p$. GaAs electrons ($m^* = 0.067m_0$) have $\mu_n \approx 8500$ $\text{cm}^2/\text{V}\cdot\text{s}$.
2. **Temperature.** Two opposing effects: (a) Lattice scattering increases with T, $\mu_L \propto T^{-3/2}$. (b) Ionised-impurity scattering decreases with T, $\mu_I \propto T^{3/2}/N_I$, since faster carriers are deflected less. Mobility therefore rises at low T, peaks, and falls at high T. Near room temperature lattice scattering usually dominates and μ falls as T rises.
3. **Doping (impurity) concentration.** More ionised impurities mean more Coulomb scattering: μ falls as $N_D + N_A$ increases (Si $\mu_n \approx 1350$ at 10^{14} cm^{-3} but about 100 at 10^{20} cm^{-3}). Compensated material has low mobility even though net doping is small.
4. **Electric field.** At low field μ is constant. At high field carriers heat up, emit optical phonons, and drift velocity saturates, so the effective mobility v_d/E falls.
5. **Crystal defects** (dislocations, grain boundaries, vacancies) add scattering; polycrystalline and amorphous material have much lower mobility.
6. **Surface and interface scattering.** In MOSFET channels, roughness and interface charge reduce mobility to roughly half the bulk value, and it falls further with higher gate field.
7. **Crystal orientation and strain.** Mobility depends on direction in anisotropic bands; strain alters effective masses (strained-Si technology).
8. **Carrier concentration** at high injection (carrier–carrier scattering).

These are combined using Matthiessen's rule, $1/\mu = 1/\mu_L + 1/\mu_I + \dots$

Q36. Explain the mechanisms responsible for carrier scattering in semiconductor materials.

In a perfectly periodic, motionless lattice an electron would move without scattering (Bloch's theorem). Scattering happens whenever something breaks that perfect periodicity. The main mechanisms:

1. **Lattice (phonon) scattering.** Above 0 K the atoms vibrate about their positions, distorting the periodic potential. Carriers exchange energy and momentum with these lattice vibrations (phonons). *Acoustic-phonon* scattering is low-energy and nearly elastic, dominant at room temperature in Si; its rate increases with T, giving $\mu_L \propto T^{-3/2}$. *Optical-phonon* scattering involves high-energy phonons (63 meV in Si); it matters at high temperatures and high

fields and causes velocity saturation. *Polar-optical and piezoelectric* scattering occur in compound semiconductors (GaAs), where the vibrating ionic bonds create electric fields that scatter carriers strongly.

2. **Ionised-impurity scattering.** Ionised donors (+) and acceptors (−) have Coulomb fields that deflect passing carriers (Rutherford-like scattering). Slower carriers spend more time near the ion and are deflected more, so it is strongest at low temperature and high doping: $\mu_I \propto T^{3/2}/N_I$.
3. **Neutral-impurity scattering.** Non-ionised impurity atoms scatter weakly by their distorted potential. Matters only at low temperatures when dopants are frozen out.
4. **Carrier–carrier scattering.** Coulomb interactions between carriers. Important at very high carrier densities (heavily doped or highly injected regions, e.g. in power devices and solar cells).
5. **Defect scattering.** Dislocations (charged lines in n-type material), grain boundaries, stacking faults, vacancies and clusters disturb the lattice and scatter carriers.
6. **Surface and interface scattering.** At surfaces and at the Si/SiO_2 interface, surface roughness, interface traps and oxide charges scatter carriers confined near the interface (MOSFET channels).
7. **Alloy scattering.** In alloys such as $Al_xGa_{1-x}As$ or SiGe, the random distribution of different atoms makes the potential non-periodic.
8. **Intervalley scattering.** Carriers move between equivalent or non-equivalent conduction-band minima by absorbing or emitting phonons (basis of the Gunn effect).

The total rate is the sum of the individual rates, $1/\tau = \Sigma\, 1/\tau_i$, giving Matthiessen's rule $1/\mu = \Sigma\, 1/\mu_i$. Near room temperature in lightly doped Si, lattice scattering dominates; in heavily doped Si or at low temperature, ionised-impurity scattering dominates.

Q37. Derive the expression for drift current density due to electrons and holes.

Assumptions: uniform semiconductor, steady state, uniform field E in the +x direction, low enough field that $v_d \propto E$.

Step 1: drift velocity. An electron experiences force $F = -qE$. Between collisions (mean free time τ_m), it accelerates by $a = -qE/m_n{*}$. Averaging over all electrons, momentum gained from the field equals momentum lost in collisions:

$$m_n{*}\, v_{dn} / \tau_m = -qE \quad \Rightarrow \quad v_{dn} = -(q\tau_m/m_n{*})\, E = -\mu_n E$$

Similarly, holes (charge +q) drift with the field:

$$v_{dp} = +(q\tau_{mp}/m_p{*})\, E = +\mu_p E$$

Step 2: current density from charge flow. Current density is charge density × velocity. Consider a bar of cross-section A. In time dt, all carriers within distance $v_d \cdot dt$ of a cross-section pass through it; that is (v_d dt A) × concentration carriers.

For electrons, charge density −qn and velocity v_{dn}:

$$J_n = (-qn)(v_{dn}) = (-qn)(-\mu_n E) = qn\mu_n E$$

For holes, charge density +qp and velocity v_{dp}:

$$J_p = (+qp)(+\mu_p E) = qp\mu_p E$$

Electrons move against the field but carry negative charge, so their current is in the same direction as the hole current: both are along E.

Step 3: total drift current density.

$$J_{drift} = J_n + J_p = q(n\mu_n + p\mu_p) E$$

Comparing with Ohm's law $J = \sigma E$:

$$\sigma = q(n\mu_n + p\mu_p), \quad \rho = 1/\sigma = 1 / [q(n\mu_n + p\mu_p)]$$

For an n-type sample ($n \gg p$), $\sigma \approx qn\mu_n \approx qN_D\mu_n$; for p-type, $\sigma \approx qN_A\mu_p$. For an intrinsic sample, $\sigma_i = qn_i(\mu_n + \mu_p)$.

Q38. A semiconductor has an electron concentration of 2×10^{16} cm^{-3}, electron mobility of 1350 $cm^2/V{\cdot}s$, and an electric field of 10 V/cm. Calculate the electron drift current density.

Given: $n = 2\times10^{16}$ cm^{-3}, $\mu_n = 1350$ $cm^2/V{\cdot}s$, $E = 10$ V/cm, $q = 1.602\times10^{-19}$ C.

$$\begin{aligned} J_n &= q\, n\, \mu_n\, E \\ &= (1.602\times10^{-19}\ \text{C})(2\times10^{16}\ \text{cm}^{-3})(1350\ \text{cm}^2/\text{V}{\cdot}\text{s})(10\ \text{V/cm}) \\ &= (1.602\times10^{-19})(2.7\times10^{20})\ \ \text{A/cm}^2 \\ &= 43.3\ \text{A/cm}^2 \end{aligned}$$

$J_n \approx 43.3$ A/cm^2 ($= 4.33\times10^5$ A/m^2).

Units check: $C \cdot cm^{-3} \cdot cm^2/(V{\cdot}s) \cdot V/cm = C/(s{\cdot}cm^2) = A/cm^2$. ✓

Supporting quantities:

- Drift velocity $v_d = \mu_n E = 1350 \times 10 = 1.35\times10^4$ cm/s (well below $v_{sat} = 10^7$ cm/s, so the linear model is valid).
- Conductivity $\sigma = qn\mu_n = 4.33$ S/cm, resistivity $\rho = 0.231\ \Omega{\cdot}cm$, and $J = \sigma E = 4.33 \times 10 = 43.3$ A/cm^2. ✓

Q39. Explain diffusion current in a semiconductor and state Fick's law.

Diffusion current is the current produced by the net movement of carriers from a region of higher concentration to one of lower concentration, caused by their random thermal motion. No electric field is needed. If the concentration varies with position, more carriers cross any plane from the high-concentration side than from the low side simply because there are more of them, so a net flux results. Because the carriers are charged, this flux is a current.

Diffusion current is important wherever carrier gradients exist: in the neutral regions of a forward-biased p-n junction (injected minority carriers diffusing away from the junction), in the base of a bipolar transistor, under non-uniform illumination, and in non-uniformly doped material.

Fick's (first) law. The particle flux is proportional to the negative of the concentration gradient:

$$F = -D\, dC/dx$$

where D is the diffusion coefficient (cm^2/s). Applied to electrons and holes:

$$\text{Electron flux: } F_n = -D_n\, dn/dx \qquad \text{Hole flux: } F_p = -D_p\, dp/dx$$

Diffusion current densities (multiply by the charge, −q for electrons, +q for holes):

$$J_{n,diff} = +q\, D_n\, dn/dx$$
$$J_{p,diff} = -q\, D_p\, dp/dx$$

In three dimensions, $J_n = qD_n\nabla n$ and $J_p = -qD_p\nabla p$.

Example. If the electron concentration falls linearly from 10^{17} cm^{-3} to 0 over 1 μm in Si (D_n = 35 cm^2/s):

$$dn/dx = -10^{17}/10^{-4} = -10^{21}\ \text{cm}^{-4}$$

$$J_n = q\,D_n\,dn/dx = (1.602\times10^{-19})(35)(-10^{21}) = -5.6\times10^{3}\ \text{A/cm}^2$$

The negative sign means the current flows in the −x direction, while the electrons themselves flow in +x (down their gradient).

Total current when a field is present: $J_n = qn\mu_n E + qD_n\,dn/dx$ and $J_p = qp\mu_p E - qD_p\,dp/dx$.

Q40. State and explain the Einstein relation between carrier mobility and diffusion coefficient.

Statement.

$$D_n/\mu_n = kT/q, \qquad D_p/\mu_p = kT/q$$

The ratio of diffusion coefficient to mobility equals the thermal voltage kT/q (0.0259 V at 300 K), for both electrons and holes, in a non-degenerate semiconductor.

Explanation. Drift and diffusion are two responses of the same carriers, both limited by the same random scattering events. Mobility describes the response to an electric field ($\mu = q\tau_m/m^*$); diffusion describes the response to a concentration gradient ($D = v_{th}^2\tau_m/3 \approx kT\tau_m/m^*$). Dividing gives $D/\mu = kT/q$: the scattering time and mass cancel, leaving only temperature.

Derivation (equilibrium argument). In a non-uniformly doped sample in thermal equilibrium, an internal field E develops such that the drift and diffusion currents cancel exactly:

$$J_n = qn\mu_n E + qD_n\,dn/dx = 0$$

In equilibrium E_F is constant and $n = n_i \exp((E_F - E_i)/kT)$. Since E_i varies with the electrostatic potential, $dE_i/dx = qE$, so

$$dn/dx = -(n/kT)\,dE_i/dx = -(qn/kT)\,E$$

Substituting:

$$qn\mu_n E - qD_n(qn/kT)E = 0 \quad \Rightarrow \quad D_n = (kT/q)\mu_n$$

Uses.

1. One measurement gives the other: mobility is easy to measure (Hall effect, conductivity), so D follows. Si electrons: $D_n = 0.0259 \times 1350 =$ **35 cm^2/s**; holes: $D_p = 0.0259 \times 480 =$ **12.4 cm^2/s**.
2. It links the drift and diffusion terms in the current equations and is needed to calculate diffusion lengths $L = \sqrt{(D\tau)}$ and diode saturation currents.
3. It shows why the built-in voltage of a junction is set by kT/q times a logarithm of a concentration ratio.

For degenerate semiconductors the relation is modified ($D/\mu > kT/q$), because carriers then obey Fermi–Dirac rather than Boltzmann statistics.

Practice questions

These extra questions practise this week's ideas. Work each one on paper before reading the solution; the calculation questions follow the same Question, Given, Find, Solution, Answer layout as the worked examples.

Practice 7.1: resistivity of p-type silicon

Question. Find the resistivity of silicon doped with 5×10^{15} cm^{-3} boron, taking $\mu_p = 470$ cm^2/V·s.

Given: $N_A = 5\times10^{15}$ cm^{-3}, $\mu_p = 470$ cm^2/V·s.

Find: ρ.

Solution.

$$\sigma = qN_A\mu_p = 1.6\times10^{-19} \times 5\times10^{15} \times 470 = 0.376 \text{ S/cm}$$

$$\rho = 1/\sigma = 2.66 \ \Omega\cdot\text{cm}$$

Answer: $\rho = 2.7\ \Omega\cdot$cm.

Practice 7.2: doping for a target resistivity

Question. What phosphorus concentration gives n-type silicon a resistivity of 1 Ω·cm? Take $\mu_n = 1300$ cm^2/V·s.

Given: $\rho = 1\ \Omega\cdot$cm, $\mu_n = 1300$ cm^2/V·s.

Find: N_D.

Solution.

$$N_D = 1/(q\mu_n\rho) = 1/(1.6\times10^{-19} \times 1300 \times 1) = 4.8\times10^{15} \text{ cm}^{-3}$$

Answer: $N_D \approx 4.8\times10^{15}$ cm^{-3}.

Practice 7.3: Hall measurement

Question. A semiconductor sample 0.01 cm thick carries I = 1 mA in a magnetic field of 0.1 T (10^{-5} Wb/cm^2). A Hall voltage of +2 mV is measured, and the resistivity is 4 Ω·cm. Find the carrier type, concentration and mobility.

Given: t = 0.01 cm, I = 10^{-3} A, B = 10^{-5} Wb/cm^2, $V_H = +2\times10^{-3}$ V, $\rho = 4\ \Omega\cdot$cm.

Find: type, p, μ.

Solution.

$$R_H = V_H\, t/(IB) = (2\times10^{-3} \times 0.01)/(10^{-3} \times 10^{-5}) = 2000 \text{ cm}^3/\text{C} \quad \text{(positive} \rightarrow \text{holes)}$$

$$p = 1/(qR_H) = 1/(1.6\times10^{-19} \times 2000) = 3.1\times10^{15} \text{ cm}^{-3}$$

$$\mu_p = R_H/\rho = 2000/4 = 500 \text{ cm}^2/\text{V}\cdot\text{s}$$

Answer: p-type, $p = 3.1\times10^{15}$ cm^{-3}, $\mu_p = 500$ cm^2/V·s.

Practice 7.4: combining scattering mechanisms

Question. In a silicon sample the lattice-limited mobility is 1400 cm^2/V·s and the impurity-limited mobility is 2500 cm^2/V·s. Find the total mobility.

Given: $\mu_L = 1400$, $\mu_I = 2500$ cm^2/V·s.

Find: μ.

Solution.

$1/\mu = 1/1400 + 1/2500 = 7.14\times10^{-4} + 4.0\times10^{-4} = 1.114\times10^{-3}$

$\mu = 897\ \text{cm}^2/\text{V}\cdot\text{s}$

Answer: $\mu \approx 900\ \text{cm}^2/\text{V}\cdot\text{s}$, lower than either mechanism alone.

Practice 7.5: drift plus diffusion

Question. In a region of silicon the hole concentration is $1\times10^{15}\ \text{cm}^{-3}$ and falls with a gradient of $-1\times10^{18}\ \text{cm}^{-4}$, while a field of 50 V/cm acts in the +x direction. Find the hole drift, diffusion and total current densities ($\mu_p = 480\ \text{cm}^2/\text{V}\cdot\text{s}$, $D_p = 12.4\ \text{cm}^2/\text{s}$).

Given: $p = 10^{15}\ \text{cm}^{-3}$, $dp/dx = -10^{18}\ \text{cm}^{-4}$, $E = 50$ V/cm, $\mu_p = 480$, $D_p = 12.4\ \text{cm}^2/\text{s}$.

Find: $J_p(\text{drift})$, $J_p(\text{diff})$, J_p.

Solution.

$J_p(\text{drift}) = qp\mu_pE = 1.6\times10^{-19} \times 10^{15} \times 480 \times 50 = 3.84\ \text{A/cm}^2$

$J_p(\text{diff}) = -qD_p\ dp/dx = -1.6\times10^{-19} \times 12.4 \times (-10^{18}) = +1.98\ \text{A/cm}^2$

$J_p = 3.84 + 1.98 = 5.82\ \text{A/cm}^2$

Answer: $3.84 + 1.98 = 5.8\ \text{A/cm}^2$, both in the +x direction.

Practice 7.6: Einstein relation at a higher temperature

Question. Holes in a sample have $\mu_p = 400\ \text{cm}^2/\text{V}\cdot\text{s}$ at 400 K. Find D_p.

Given: $\mu_p = 400\ \text{cm}^2/\text{V}\cdot\text{s}$, $T = 400$ K.

Find: D_p.

Solution.

$kT/q = 8.617\times10^{-5} \times 400 = 0.0345$ V

$D_p = (kT/q)\mu_p = 0.0345 \times 400 = 13.8\ \text{cm}^2/\text{s}$

Answer: $D_p = 13.8\ \text{cm}^2/\text{s}$.

Practice 7.7: why doping lowers mobility

Question. Explain why electron mobility in silicon falls from about 1350 to about 100 $\text{cm}^2/\text{V}\cdot\text{s}$ as the doping rises from 10^{14} to $10^{20}\ \text{cm}^{-3}$.

Answer. Each dopant is an ionised impurity with a Coulomb field that deflects passing carriers. Mobility is $\mu = q\tau_m/m*$; more ionised impurities mean more frequent scattering, a shorter mean free time τ_m, and so a lower μ. At $10^{14}\ \text{cm}^{-3}$ lattice scattering dominates and impurities hardly matter; at $10^{20}\ \text{cm}^{-3}$ impurity scattering dominates completely (Matthiessen: $1/\mu = 1/\mu_L + 1/\mu_I$, with $\mu_I \propto 1/N_I$).

Objectives check

Use this table to confirm you have met this week's objectives. Cover the Answer column and answer each check question from memory. If you cannot, go back to the section in the last column before moving on.

#	Objective	Check yourself	Answer	Where it is taught
1	Explain thermal motion, drift velocity and mobility.	Si electrons in a 500 V/cm field: find v_d.	$v_d = 1350 \times 500 = 6.75\times10^5$ cm/s.	Part A; Example 7.1
2	Explain the scattering mechanisms and the factors that affect mobility.	Which scattering mechanism dominates at low temperature in heavily doped Si?	Ionised-impurity scattering.	Scattering; factors affecting mobility
3	Derive and use the drift current density and conductivity.	Find σ for p-type Si with $N_A = 10^{16}$ cm^{-3} and $\mu_p = 480$ $cm^2/V{\cdot}s$.	$\sigma = 1.6\times10^{-19} \times 10^{16} \times 480 = 0.77$ S/cm.	Drift current density; Example 7.2
4	Explain carrier diffusion and calculate diffusion current with Fick's law.	Why does electron diffusion current point opposite to the electron flow?	Because the electron charge is negative.	Part B; Example 7.3
5	State, explain and use the Einstein relation.	Find D_n for GaAs ($\mu_n = 8500$ $cm^2/V{\cdot}s$) at 300 K.	$D = 0.0259 \times 8500 = 220$ cm^2/s.	The Einstein relation

High-Field Effects and Excess Carriers

Plan: velocity saturation and hot-carrier effects; generation and recombination mechanisms (Shockley–Read–Hall theory); carrier lifetime and steady-state excess carrier injection.

Learning objectives

By the end of this week you should be able to:

1. Explain velocity saturation and its effect on devices.
2. Explain hot-carrier effects and how they are reduced.
3. Explain excess carriers and low-level injection.
4. Describe the generation and recombination mechanisms.
5. Explain SRH recombination and calculate lifetime from trap data.
6. Calculate decay of excess carriers and steady-state injection levels.

The big picture

This week covers two separate stories.

Story 1: what happens at very high fields. Week 7 said $v_d = \mu E$: double the field, double the speed. That stops being true at high fields. Carriers hit a **speed limit**, and the fastest ones become **"hot"** and start damaging things. This matters in modern transistors, which are so small that even a volt across them gives enormous fields.

Story 2: what happens when we disturb the balance of carriers. Light, or a forward-biased junction, can add **extra** electrons and holes beyond the equilibrium numbers. These extras do not last: they **recombine** and disappear. How long they last, the **lifetime**, controls how diodes switch, how transistors amplify and how well solar cells work.

Part A: High-field effects

Velocity saturation

At low fields carriers lose the energy they pick up from the field to small lattice vibrations, so they stay "cool" and $v_d = \mu E$.

At high fields (above about 10^4 V/cm in Si) they pick up energy faster than they can lose it that way. Once a carrier has about 0.06 eV of extra energy (in Si), it can shake the lattice much harder by emitting an **optical phonon**, which takes that energy away in one go. So every time a carrier speeds up, it is quickly slowed down again. The drift velocity stops increasing and levels off at the **saturation velocity**:

$v_{sat} \approx 1\times10^7$ cm/s (Si electrons and holes, at 300 K)

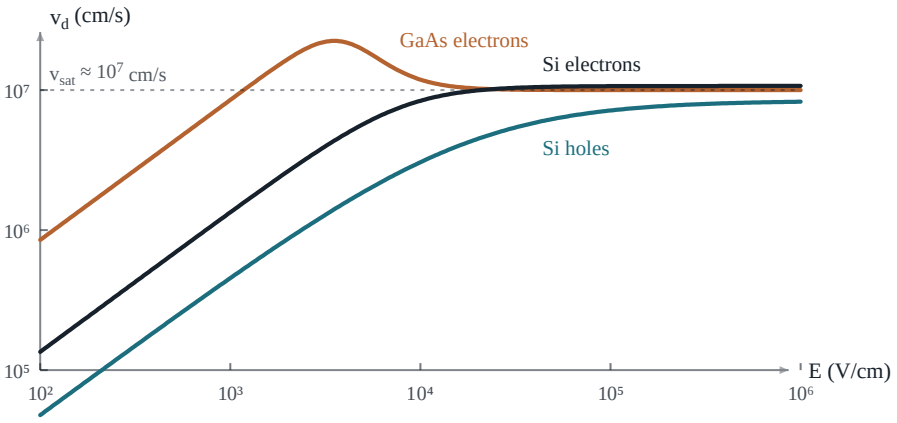
> **In plain words.** Pushing harder on a cyclist on a flat road makes them go faster, until air resistance grows so quickly that extra effort gives almost no extra speed. Optical phonons are the "air resistance" for carriers.

An empirical formula that joins the two regimes smoothly (Caughey–Thomas):

$$v_d = \mu E / [\, 1 + (\mu E / v_{sat})^{\beta} \,]^{1/\beta} \qquad (\beta \approx 2 \text{ for Si electrons, about 1 for holes})$$

At low field it reduces to μE; at high field to v_{sat}.

GaAs is different. Its electrons speed up to about 2×10^7 cm/s near 3.5 kV/cm, and then actually **slow down** at higher fields, because they jump into a different part of the conduction band where they are heavier. A region where more field gives less current is called **negative differential mobility**; it is used in Gunn diodes to make microwave oscillators.



Drift velocity against field. Silicon carriers level off near 10^7 cm/s. GaAs electrons overshoot to about 2×10^7 cm/s, then fall as they transfer to a heavier valley.

Why it matters: in a short-channel MOSFET the carriers are at v_{sat} across most of the channel, so the current depends on $(V_{GS} - V_T)$ rather than $(V_{GS} - V_T)^2$ (Week 14), and the transistor speed is limited to about $v_{sat}/(2\pi L)$.

Hot-carrier effects

When carriers gain energy much faster than they lose it, their average energy rises far above the normal thermal energy. They are called **hot carriers** (their energy is described by an "electron temperature" T_e much higher than the crystal's temperature). Hot carriers cause:

1. **Velocity saturation** and reduced mobility (above).
2. **Impact ionisation.** A carrier with energy above about $1.5E_g$ can knock an electron out of a covalent bond when it collides, creating a new electron–hole pair. The new carriers are accelerated too and do the same: an **avalanche**. This causes **avalanche breakdown** of junctions (Week 10) and is used deliberately in avalanche photodiodes.
3. **Hot-carrier injection** into the gate oxide of MOSFETs. Near the drain, some electrons gain enough energy (over 3.1 eV) to jump over the Si/SiO_2 barrier into the oxide, where they get trapped or create defects. The threshold voltage slowly shifts and the transistor degrades over months or years: a key **reliability** problem.
4. **Substrate current** from holes created by impact ionisation, which can trigger unwanted bipolar action or latch-up in CMOS.
5. **Velocity overshoot** in very short devices, where carriers cross before they have time to reach v_{sat}, briefly going faster.

Cures: the **lightly doped drain (LDD)** structure, which spreads the voltage drop and lowers the peak field; lower supply voltages; better oxides.

Part B: Generation and recombination

Equilibrium and excess carriers

In thermal equilibrium, electron–hole pairs are being created (**generated**) by heat and destroyed (**recombined**) all the time, at exactly equal rates. That balance is what keeps $n_0 p_0 = n_i^2$.

Now shine light on the sample, or inject carriers through a junction. We get **excess carriers**:

$$n = n_0 + \Delta n, \quad p = p_0 + \Delta p$$

Light creates electrons and holes in pairs, so usually $\Delta n = \Delta p$. Now $np > n_i^2$, recombination is faster than generation, and the extras start disappearing.

Low-level injection means the excess is much smaller than the majority concentration ($\Delta p \ll n_0$ in n-type). Then:

- the **majority** concentration hardly changes;
- the **minority** concentration may change enormously.

Example 8.1: what low-level injection does to each carrier

Question. n-type silicon with $N_D = 1\times10^{16}$ cm^{-3} (so $p_0 = 1\times10^{4}$ cm^{-3}) is illuminated, creating $\Delta n = \Delta p = 1\times10^{14}$ cm^{-3} excess carriers. Calculate the new electron and hole concentrations and the factor by which each has changed.

Given: $n_0 = 1\times10^{16}$ cm^{-3}, $p_0 = 1\times10^{4}$ cm^{-3}, $\Delta n = \Delta p = 1\times10^{14}$ cm^{-3}.

Find: n, p and their change factors.

Solution.

$$n = n_0 + \Delta n = 1\times10^{16} + 1\times10^{14} = 1.01\times10^{16}\ \text{cm}^{-3} \quad (n/n_0 = 1.01)$$

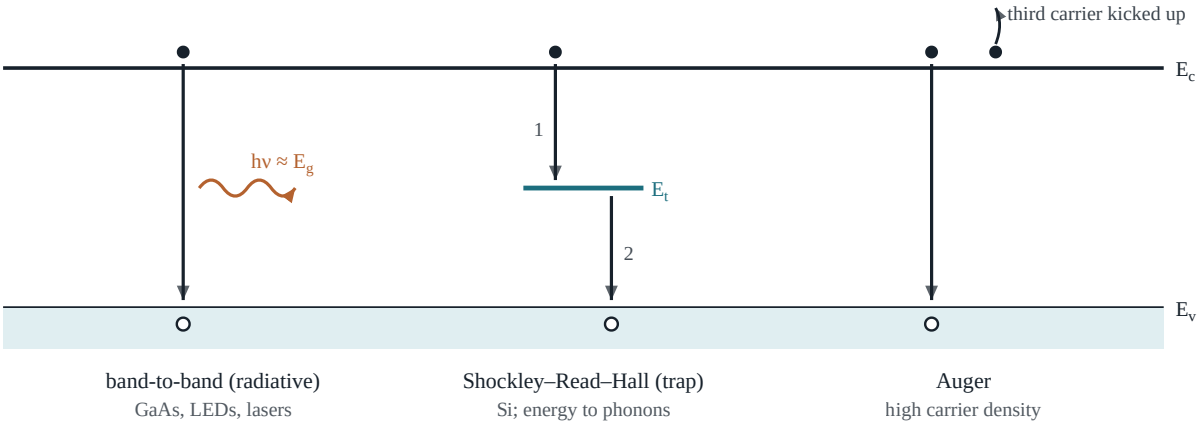
$$p = p_0 + \Delta p = 1\times10^{4} + 1\times10^{14} \approx 1\times10^{14}\ \text{cm}^{-3} \quad (p/p_0 = 10^{10})$$

Answer: The electrons change by only 1 %; the holes increase ten-billion-fold. This is why minority carriers control device behaviour: they are the ones that change.

How carriers are generated

1. **Thermal generation:** heat breaks a bond. Usually it happens in two small steps through a defect level in the gap, which is easier than one big jump.
2. **Optical generation:** a photon with energy $h\nu \geq E_g$ is absorbed and creates a pair. This is how photodiodes and solar cells work.
3. **Impact ionisation:** a hot carrier creates a pair (above).
4. **Tunnelling** across the gap at extremely high fields (Zener effect).

How carriers recombine



The three recombination paths. Band-to-band: the electron drops straight across and emits a photon. Shockley–Read–Hall: it drops in two steps via a trap in the gap, giving heat. Auger: the energy is handed to a third carrier.

1. **Direct (band-to-band, radiative).** An electron drops straight from the conduction band into a hole and emits a photon of energy ≈ E_g. Rate = B·np. Strong in direct-gap GaAs (LEDs, lasers); very weak in indirect-gap Si (Week 4).
2. **Indirect (trap-assisted, Shockley–Read–Hall, SRH).** The electron drops in two steps through a **trap**, a localised energy level in the gap created by a defect or an impurity atom such as gold, iron or copper. The energy goes into lattice vibrations (heat). **This is the dominant mechanism in silicon.**
3. **Auger.** The energy released is given to a third carrier, which is kicked high into its band and then cools down. Important at very high carrier densities.
4. **Surface recombination:** SRH through the many traps at a surface (Week 9).

Shockley–Read–Hall recombination: how it works

Picture a trap level E_t in the middle of the gap. It can be empty or hold one electron. Four things can happen to it:

Step	What happens	Result
1. Electron capture	A conduction-band electron falls into the empty trap	Trap now full
2. Electron emission	The trapped electron is shaken back up into the conduction band	Trap empty again
3. Hole capture	The trapped electron drops down into the valence band, filling a hole	A hole disappears; trap empty
4. Hole emission	A valence-band electron jumps up into the empty trap	A hole appears; trap full

Recombination = step 1 then step 3: an electron and a hole have both disappeared, via the trap. **Generation** = step 4 then step 2: an electron and a hole have both been created.

> **In plain words.** It is much easier to go down a staircase two steps at a time than to jump the whole flight. A trap in the middle of the gap is a landing half-way down. That is why a few impurity atoms can make recombination enormously faster.

Working out the balance of the four steps gives the **SRH recombination rate**:

$$U = (np - n_i^2) / [\, \tau_p(n + n_1) + \tau_n(p + p_1) \,]$$

$$\tau_n = 1/(\sigma_n \, v_{th} \, N_t), \quad \tau_p = 1/(\sigma_p \, v_{th} \, N_t)$$
$$n_1 = n_i \, e^{(E_t - E_i)/kT}, \quad p_1 = n_i \, e^{(E_i - E_t)/kT}$$

Symbol	Meaning
U	net recombination rate (pairs per cm^3 per second)
N_t	trap density (cm^{-3})
σ_n, σ_p	capture cross-sections: how "big a target" the trap is for electrons, holes (cm^2)
v_{th}	thermal velocity, about 10^7 cm/s
E_t	trap energy level

What the formula tells you:

1. **The top line, $np - n_i^2$, is the driving force.** If $np > n_i^2$ (extra carriers), U is positive: net recombination. If $np < n_i^2$ (carriers removed, as in a reverse-biased depletion region), U is negative: net **generation**, which is the source of diode leakage current. In equilibrium, $np = n_i^2$ and $U = 0$.
2. **U is largest when E_t is near mid-gap.** Traps near a band edge just catch and re-release carriers. **Mid-gap traps are the effective recombination centres.** Gold sits close to mid-gap, which is why it is used deliberately to make fast-switching diodes.
3. **More traps, faster recombination:** $\tau \propto 1/N_t$.
4. **Low-level injection in n-type** ($n \gg p$): the formula simplifies to

   $$U \approx \Delta p / \tau_p$$

The recombination rate is set by how fast the **minority** carriers (holes) get captured.

Example 8.2: lifetime from the trap density

Question. An n-type silicon wafer contains 1×10^{13} cm^{-3} mid-gap recombination centres with a hole capture cross-section of 1×10^{-15} cm^2. The thermal velocity is 1×10^7 cm/s. Calculate the minority-carrier (hole) lifetime.

Given: $N_t = 1\times10^{13}$ cm^{-3}, $\sigma_p = 1\times10^{-15}$ cm^2, $v_{th} = 1\times10^7$ cm/s.

Find: τ_p.

Solution.

$$\tau_p = 1/(\sigma_p \, v_{th} \, N_t) = 1 / (10^{-15} \times 10^7 \times 10^{13}) = 1/10^5 = 1\times10^{-5} \text{ s}$$

Answer: $\tau_p = 10$ µs. Doubling the trap density would halve the lifetime.

Part C: Lifetime and steady-state injection

Minority-carrier lifetime

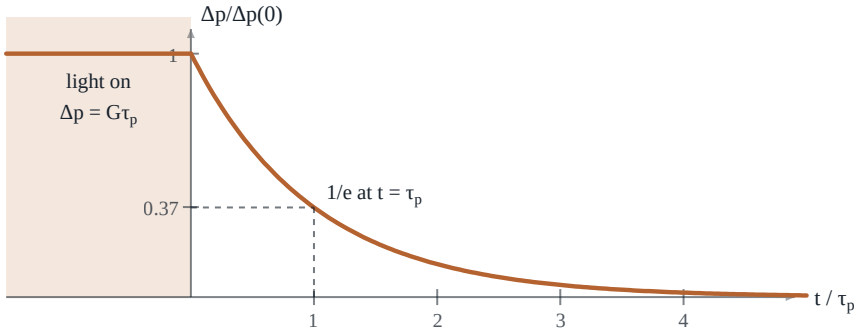
Switch off the light at $t = 0$. With $U = \Delta p/\tau_p$, the excess holes decay as

$$d\Delta p/dt = -\Delta p/\tau_p \quad \Rightarrow \quad \Delta p(t) = \Delta p(0) \, e^{-t/\tau_p}$$

τ_p is the **minority-carrier lifetime**: the average time an extra minority carrier survives before it recombines. After one lifetime, 37 % (1/e) of the excess remains; after five lifetimes, under 1 %.

Typical values: very pure Si, 1 µs to 1 ms; gold-doped Si for fast switching, 1–10 ns; GaAs, 1–10 ns (fast radiative recombination).

If several mechanisms act together, their rates add: $1/\tau = 1/\tau_{SRH} + 1/\tau_{radiative} + 1/\tau_{Auger}$.



Under steady illumination the excess holds at $\Delta p = G_L\tau_p$. When the light is switched off it decays exponentially, falling to 1/e in one lifetime.

Steady-state excess carrier injection

Now keep the light on, generating G_L pairs per cm^3 per second uniformly. The excess builds up until recombination exactly balances generation:

$$d\Delta p/dt = G_L - \Delta p/\tau_p = 0 \quad \Rightarrow \quad \Delta p = \Delta n = G_L\tau_p$$

The excess equals the generation rate times the lifetime. A long lifetime lets a large excess build up.

The extra carriers raise the conductivity, called **photoconductivity**:

$$\Delta\sigma = q(\mu_n + \mu_p)\,\Delta n = q(\mu_n + \mu_p)\,G_L\tau$$

Example 8.3: photoconductivity

Question. n-type silicon with $N_D = 1\times10^{15}$ cm^{-3} and $\tau_p = 10$ µs is uniformly illuminated, generating $G_L = 1\times10^{20}$ electron–hole pairs per cm^3 per second. Calculate the steady-state excess carrier concentration, the increase in conductivity, and compare it with the dark conductivity.

Given: $N_D = 1\times10^{15}$ cm^{-3}, $\tau_p = 10$ µs $= 1\times10^{-5}$ s, $G_L = 1\times10^{20}$ $cm^{-3}s^{-1}$, $\mu_n = 1350$, $\mu_p = 480$ $cm^2/V\cdot s$.

Find: Δn, $\Delta\sigma$ and σ_0.

Solution.

Step 1. Steady state, generation = recombination:

$$\Delta n = \Delta p = G_L\tau_p = 10^{20} \times 10^{-5} = 1\times10^{15}\ cm^{-3}$$

Step 2. Extra conductivity:

$$\Delta\sigma = q(\mu_n + \mu_p)\Delta n = 1.6\times10^{-19} \times (1350 + 480) \times 10^{15} = 0.293\ S/cm$$

Step 3. Dark conductivity:

$$\sigma_0 = qN_D\mu_n = 1.6\times10^{-19} \times 10^{15} \times 1350 = 0.216\ S/cm$$

Answer: $\Delta n = 10^{15}$ cm^{-3}, $\Delta\sigma = 0.293$ S/cm, against $\sigma_0 = 0.216$ S/cm: the conductivity more than doubles in the light.

This is how a light-dependent resistor works. (Here Δp equals n_0, so this is not low-level injection, and the single-lifetime formula is only approximate.)

Quasi-Fermi levels

Out of equilibrium, $np \neq n_i^2$, so a single Fermi level cannot describe both carriers. Instead we use **two**: one for electrons, E_{Fn}, and one for holes, E_{Fp}:

$$n = n_i\, e^{(E_{Fn} - E_i)/kT}, \quad p = n_i\, e^{(E_i - E_{Fp})/kT}, \quad np = n_i^2\, e^{(E_{Fn} - E_{Fp})/kT}$$

The **gap between them**, $E_{Fn} - E_{Fp}$, measures how far from equilibrium the material is. In a forward-biased diode it equals qV; in a solar cell it sets the open-circuit voltage. In equilibrium they merge back into one Fermi level.

Key points of Week 8

- At high fields drift velocity saturates at $v_{sat} \approx 10^7$ cm/s (optical-phonon emission).
- Hot carriers cause impact ionisation (avalanche) and oxide damage in MOSFETs; LDD reduces them.
- Excess carriers: $n = n_0 + \Delta n$. Under low-level injection only the minority concentration changes much.
- Recombination: band-to-band (light, GaAs), SRH via traps (heat, dominant in Si), Auger (high density).
- SRH rate $U = (np - n_i^2)/[\tau_p(n + n_1) + \tau_n(p + p_1)]$: mid-gap traps are most effective; $\tau = 1/(\sigma v_{th} N_t)$.
- Low-level n-type: $U = \Delta p/\tau_p$; after switch-off $\Delta p(t) = \Delta p(0)e^{-t/\tau_p}$; under steady light $\Delta p = G_L\tau_p$.

Examination questions solved (Q41–Q44)

Q41. Explain velocity saturation and hot-carrier effects in semiconductors.

Velocity saturation. At low fields, drift velocity is proportional to the field, $v_d = \mu E$, because carriers lose the energy they gain from the field to the lattice (acoustic phonons) and stay near thermal equilibrium. As the field increases (above about 10^3–10^4 V/cm in Si), carriers gain energy faster than acoustic-phonon scattering can remove it. Once their energy reaches the optical-phonon energy (≈ 0.063 eV in Si), they emit optical phonons very rapidly, which transfers energy to the lattice efficiently and limits further gain. The drift velocity therefore stops increasing and approaches a **saturation velocity**:

$$v_{sat} \approx 10^7 \text{ cm/s} \quad \text{(Si, 300 K)}$$

and is described empirically by $v_d = \mu E/[1 + (\mu E/v_{sat})^{\beta}]^{1/\beta}$. The effective mobility v_d/E falls at high field. In GaAs the electron velocity peaks at about 2×10^7 cm/s then decreases (negative differential mobility) because electrons transfer to the heavier L-valley.

Consequences: in short-channel MOSFETs the drain current becomes proportional to $(V_{GS} - V_T)$ rather than $(V_{GS} - V_T)^2$, and the maximum transconductance and transit-time-limited speed are set by v_{sat}: $f_T \approx v_{sat}/(2\pi L)$.

Hot-carrier effects. Under high fields, carriers acquire an average kinetic energy far greater than the thermal value $(3/2)kT$; they are described by an effective electron temperature $T_e \gg T$. These "hot" carriers cause:

1. **Impact ionisation**: carriers with energy above about $1.5E_g$ break bonds and create electron–hole pairs, leading to avalanche multiplication and **avalanche breakdown** of junctions.

2. **Hot-carrier injection into gate oxides**: hot electrons near the drain of a MOSFET surmount the 3.1 eV Si/SiO_2 barrier, become trapped in the oxide or create interface traps. The threshold voltage shifts and transconductance degrades over the device's life (hot-carrier degradation).
3. **Substrate and gate currents**: impact-generated holes flow to the substrate, which can forward-bias source junctions and trigger parasitic bipolar action or CMOS latch-up.
4. **Velocity overshoot**: in very short channels carriers transit before scattering brings them to steady state, briefly exceeding v_{sat}.
5. **Increased noise and light emission.**

Mitigation: lightly doped drain (LDD) structures to reduce the peak field, reduced supply voltage, graded junctions, and nitrided or hardened gate oxides.

Q42. What is meant by generation and recombination of charge carriers?

Generation is any process that creates free electron–hole pairs (or a free carrier from a dopant or trap) by giving an electron enough energy to move from the valence band (or a localised state) into the conduction band. The rate G is the number of pairs created per unit volume per unit time ($cm^{-3}s^{-1}$). Mechanisms:

- **Thermal generation** by lattice vibrations, directly or through trap levels in the gap.
- **Optical generation** by absorption of photons with $h\nu \geq E_g$ (photodetectors, solar cells).
- **Impact ionisation** by energetic carriers in high fields (avalanche).
- **Tunnelling** generation at very high fields (Zener).

Recombination is the reverse: a conduction-band electron falls back to the valence band and fills a hole, so both carriers are annihilated. The released energy E_g appears as a photon, as heat (phonons), or as kinetic energy of a third carrier. The rate R has the same units. Mechanisms:

- **Direct (band-to-band, radiative)** recombination, emitting a photon; dominant in direct-gap materials.
- **Indirect (trap-assisted, SRH)** recombination through defect or impurity levels in the gap, releasing heat; dominant in Si.
- **Auger** recombination, energy given to a third carrier; dominant at high carrier density.
- **Surface** recombination through surface states.

Equilibrium and non-equilibrium. In thermal equilibrium the two processes balance ($G_{th} = R_{th}$), giving $n_0 p_0 = n_i^2$. If excess carriers are introduced ($np > n_i^2$), net recombination $U = R - G > 0$ restores equilibrium; if carriers are removed ($np < n_i^2$, e.g. a reverse-biased depletion region), net generation occurs. For low-level injection the net recombination rate is $U = \Delta p/\tau_p$ (n-type), where τ_p is the minority-carrier lifetime.

Generation and recombination govern the leakage current of diodes, the operation of solar cells, LEDs and photodetectors, the gain of BJTs and the switching speed of p-n junctions.

Q43. Explain the Shockley-Read-Hall (SRH) recombination mechanism.

Principle. SRH (trap-assisted or indirect) recombination occurs through localised energy levels E_t in the forbidden gap introduced by crystal defects or impurities (e.g. Au, Fe, Cu, vacancies, dislocations). Instead of crossing the whole gap in one step, which in an indirect semiconductor requires an unlikely phonon-assisted transition, the electron makes two smaller steps via the trap, and the energy is released as lattice vibrations (phonons).

The four processes at a trap (density N_t, occupation probability f_t):

Process	Description	Rate
(a) Electron capture	CB electron captured by an empty trap	$r_a = v_{th}\sigma_n\, n\, N_t(1 - f_t)$
(b) Electron emission	Trapped electron emitted to CB	$r_b = e_n\, N_t\, f_t$
(c) Hole capture	Trapped electron drops to VB, annihilating a hole	$r_c = v_{th}\sigma_p\, p\, N_t\, f_t$
(d) Hole emission	VB electron captured by empty trap, leaving a hole	$r_d = e_p\, N_t(1 - f_t)$

Recombination = (a) then (c). **Generation** = (d) then (b).

Derivation outline. In steady state the trap occupation is constant, so the net electron capture rate equals the net hole capture rate: $r_a - r_b = r_c - r_d = U$. The emission coefficients follow from detailed balance in equilibrium: $e_n = v_{th}\sigma_n\, n_1$ and $e_p = v_{th}\sigma_p\, p_1$. Solving for f_t and substituting gives:

$$U = (np - n_i^2) / [\ \tau_p(n + n_1) + \tau_n(p + p_1)\ ]$$

with $\tau_n = 1/(\sigma_n\, v_{th}\, N_t)$, $\tau_p = 1/(\sigma_p\, v_{th}\, N_t)$, $n_1 = n_i\, e^{(E_t - E_i)/kT}$, $p_1 = n_i\, e^{(E_i - E_t)/kT}$.

Interpretation.

1. The driving force is $(np - n_i^2)$: positive for excess carriers (recombination), negative for depletion (generation), zero in equilibrium.
2. U is maximum when E_t is near mid-gap ($E_t \approx E_i$). Shallow levels near a band edge just re-emit carriers and are ineffective. **Deep, mid-gap levels are the effective recombination centres.**
3. $U \propto N_t$: more traps give faster recombination and shorter lifetime.
4. For low-level injection in n-type material ($n \gg p, n_1, p_1$): $U = (p - p_0)/\tau_p = \Delta p/\tau_p$. The rate is limited by capture of the minority carrier.
5. In a reverse-biased depletion region ($n, p \approx 0$): $U = -n_i/(\tau_p + \tau_n) \approx -n_i/2\tau_0$, a net generation that produces the generation leakage current $J_{gen} = qn_iW/2\tau_0$.

Applications: gold or platinum doping, or electron irradiation, is deliberately used to reduce lifetime in fast-switching diodes and thyristors. Conversely, gettering and clean processing reduce trap densities for long lifetimes in solar cells, CCDs and low-leakage devices.

Q44. Define minority-carrier lifetime and explain its significance in semiconductor devices.

Definition. The minority-carrier lifetime is the average time an excess minority carrier exists in a semiconductor before it recombines with a majority carrier. For low-level injection, τ_p (holes in n-type) or τ_n (electrons in p-type) is defined by

$$U = \Delta p / \tau_p$$

so that, when generation stops, the excess density decays exponentially:

$$\Delta p(t) = \Delta p(0)\, e^{-t/\tau_p}$$

τ_p is the time for the excess to fall to 1/e (37 %) of its initial value. With SRH recombination, $\tau_p = 1/(\sigma_p\, v_{th}\, N_t)$: lifetime is inversely proportional to the trap density. Typical values: 1 ns to 1 ms in Si depending on purity.

Significance:

1. **Diffusion length.** Injected minority carriers diffuse an average distance $L = \sqrt{(D\tau)}$ before recombining. In Si with $D_p = 12$ cm^2/s and $\tau_p = 1$ µs, $L_p = \sqrt{(12 \times 10^{-6})} = 35$ µm. L determines how far carriers travel and therefore device dimensions.
2. **Diode current.** The saturation current $J_s = qn_i^2(D_p/(L_pN_D) + D_n/(L_nN_A))$ depends on lifetime through L. Short lifetime raises leakage; generation current in the depletion region is $\propto 1/\tau$.
3. **Switching speed.** When a forward-biased diode is turned off, the stored minority charge $Q = I\tau$ must be removed; the reverse-recovery time is $\approx \tau$. Fast-recovery and switching diodes are gold- or platinum-doped to cut τ to nanoseconds.
4. **BJT current gain.** Electrons injected into a p-type base must cross it before recombining. The base transport factor $\approx 1 - W_B^2/2L_n^2$, so long lifetime ($L_n \gg W_B$) gives high β.
5. **Solar cells and photodetectors.** Photogenerated carriers must reach the junction before recombining; long lifetime (long L) means higher collection efficiency, current and efficiency.
6. **LEDs and lasers.** The radiative fraction of recombination, and the modulation speed (bandwidth $\propto 1/\tau$), depend on lifetime.
7. **Power devices.** In p-i-n diodes and IGBTs, long lifetime gives conductivity modulation and low on-state voltage but slow switching; there is a design trade-off.
8. **Material quality indicator.** Lifetime is very sensitive to metallic contamination, so it is measured (photoconductive decay, µ-PCD) to monitor crystal growth and process cleanliness.

Practice questions

These extra questions practise this week's ideas. Work each one on paper before reading the solution; the calculation questions follow the same Question, Given, Find, Solution, Answer layout as the worked examples.

Practice 8.1: when does velocity saturate?

Question. Estimate the field at which electron drift in silicon starts to saturate ($\mu_n = 1350$ cm^2/V·s, $v_{sat} = 1\times10^7$ cm/s). Is a 100 nm MOSFET channel with 1 V across it saturated?

Given: $\mu_n = 1350$ cm^2/V·s, $v_{sat} = 10^7$ cm/s, V = 1 V over L = 100 nm = 10^{-5} cm.

Find: E where $\mu E = v_{sat}$; the channel field.

Solution.

$E_{sat} \approx v_{sat}/\mu_n = 10^7/1350 = 7.4\times10^3$ V/cm

Channel field $\approx V/L = 1/10^{-5} = 10^5$ V/cm

Answer: $E_{sat} \approx 7.4$ kV/cm; the channel field (100 kV/cm) is over ten times larger, so the carriers are fully velocity-saturated.

Practice 8.2: gold as a lifetime killer

Question. A switching diode is doped with 1×10^{15} cm^{-3} gold atoms ($\sigma = 1\times10^{-15}$ cm^2, $v_{th} = 1\times10^7$ cm/s). Find the minority-carrier lifetime.

Given: $N_t = 10^{15}$ cm^{-3}, $\sigma = 10^{-15}$ cm^2, $v_{th} = 10^7$ cm/s.

Find: τ.

Solution.

$\tau = 1/(\sigma v_{th} N_t) = 1/(10^{-15} \times 10^{7} \times 10^{15}) = 10^{-7}$ s

Answer: τ = 100 ns, short enough for a fast-recovery diode.

Practice 8.3: how long until the excess is gone?

Question. After the light is switched off, excess holes in n-type silicon decay with τ_p = 5 µs. How long does it take for the excess to fall to 1 % of its initial value?

Given: τ_p = 5 µs, $\Delta p(t)/\Delta p(0) = 0.01$.

Find: t.

Solution.

$e^{-t/\tau} = 0.01 \quad \Rightarrow \quad t = \tau \ln(100) = 5\ \mu s \times 4.61 = 23\ \mu s$

Answer: t ≈ 23 µs (about 4.6 lifetimes).

Practice 8.4: steady-state injection level

Question. n-type silicon ($N_D = 1\times10^{16}$ cm^{-3}, τ_p = 20 µs) is uniformly illuminated with $G_L = 5\times10^{19}$ cm^{-3}s^{-1}. Find Δp and decide whether this is low-level injection.

Given: $G_L = 5\times10^{19}$ cm^{-3}s^{-1}, $\tau_p = 2\times10^{-5}$ s, $N_D = 10^{16}$ cm^{-3}.

Find: Δp and $\Delta p/n_0$.

Solution.

$\Delta p = G_L \tau_p = 5\times10^{19} \times 2\times10^{-5} = 1\times10^{15}$ cm^{-3}

$\Delta p/n_0 = 10^{15}/10^{16} = 0.1$

Answer: $\Delta p = 10^{15}$ cm^{-3}, 10 % of n_0: borderline but usually treated as low-level injection.

Practice 8.5: quasi-Fermi level splitting

Question. n-type silicon with $N_D = 1\times10^{16}$ cm^{-3} is illuminated so that $\Delta n = \Delta p = 1\times10^{14}$ cm^{-3}. Find $E_Fn - E_i$, $E_i - E_Fp$ and the splitting $E_Fn - E_Fp$ at 300 K.

Given: $n = 1.01\times10^{16}$, $p \approx 10^{14}$ cm^{-3}, $n_i = 10^{10}$ cm^{-3}, kT = 0.0259 eV.

Find: the quasi-Fermi levels and their splitting.

Solution.

$E_{Fn} - E_i = kT \ln(n/n_i) = 0.0259 \ln(1.01\times10^{6}) = 0.358$ eV

$E_i - E_{Fp} = kT \ln(p/n_i) = 0.0259 \ln(10^{4}) = 0.239$ eV

$E_{Fn} - E_{Fp} = 0.358 + 0.239 = 0.597$ eV

Answer: $E_{Fn} = E_i + 0.358$ eV, $E_{Fp} = E_i - 0.239$ eV, splitting ≈ 0.60 eV. The electron level hardly moves; the hole level moves a long way.

Practice 8.6: generation current in a depletion region

Question. A reverse-biased silicon junction has a depletion width of 1 μm, area 10^{-3} cm^2 and effective lifetime $\tau_0 = 1$ μs. Estimate the generation leakage current, $J_{gen} = qn_iW/2\tau_0$.

Given: $W = 10^{-4}$ cm, $A = 10^{-3}$ cm^2, $\tau_0 = 10^{-6}$ s, $n_i = 10^{10}$ cm^{-3}.

Find: J_{gen} and I_{gen}.

Solution.

$$J_{gen} = qn_iW/(2\tau_0) = 1.6\times10^{-19} \times 10^{10} \times 10^{-4} / (2\times10^{-6}) = 8\times10^{-8} \text{ A/cm}^2$$

$$I_{gen} = J_{gen}A = 8\times10^{-11} \text{ A}$$

Answer: $J_{gen} = 8\times10^{-8}$ A/cm^2, $I_{gen} = 80$ pA. In silicon this generation current usually exceeds the ideal diffusion leakage.

Practice 8.7: Auger versus SRH

Question. Explain why Auger recombination is important in heavily doped emitters but not in lightly doped bulk silicon.

Answer. Auger recombination needs three carriers to meet (the recombining electron and hole plus a third carrier that takes the energy), so its rate grows as n^2p or np^2, much faster than SRH, which grows only linearly with the excess. In lightly doped silicon carrier densities are low and three-carrier encounters are rare, so SRH dominates. In emitters doped to 10^{19}–10^{20} cm^{-3} the majority density is enormous, Auger becomes the fastest path and it limits the minority-carrier lifetime.

Objectives check

Use this table to confirm you have met this week's objectives. Cover the Answer column and answer each check question from memory. If you cannot, go back to the section in the last column before moving on.

#	Objective	Check yourself	Answer	Where it is taught
1	Explain velocity saturation and its effect on devices.	What is the saturation velocity in Si and what causes it?	About 10^7 cm/s; optical-phonon emission by energetic carriers.	Velocity saturation
2	Explain hot-carrier effects and how they are reduced.	How does a hot carrier cause avalanche breakdown?	It impact-ionises, creating new pairs that are accelerated and do the same.	Hot-carrier effects
3	Explain excess carriers and low-level injection.	Under low-level injection in n-type material, which carrier concentration changes significantly?	The minority (hole) concentration.	Equilibrium and excess carriers; Example 8.1
4	Describe the generation and recombination mechanisms.	Which recombination mechanism dominates in silicon?	Shockley–Read–Hall (trap-assisted).	Generation; recombination
5	Explain SRH recombination and calculate lifetime from trap data.	Why are mid-gap traps the most effective recombination centres?	They capture electrons and holes with similar ease; shallow traps just re-emit carriers.	SRH; Example 8.2
6	Calculate decay of excess carriers and steady-state injection levels.	With $G_L = 10^{19}$ cm^{-3}s^{-1} and $\tau = 2$ μs, find Δp.	$\Delta p = G_L\tau = 2\times10^{13}$ cm^{-3}.	Part C; Example 8.3

Continuity Equation and Surface Phenomena

Plan: derivation and application of the minority-carrier continuity equation; semiconductor surface states and surface recombination velocity.

Learning objectives

By the end of this week you should be able to:

1. Derive the continuity equation for holes and electrons.
2. Write and interpret the minority-carrier diffusion equation.
3. Solve the diffusion equation for the standard cases.
4. Calculate diffusion lengths and explain their importance.
5. Explain surface states and surface recombination velocity.
6. Calculate the effective lifetime of a wafer with recombining surfaces.

The big picture

Weeks 7 and 8 gave us all the separate pieces: carriers **flow** (drift and diffusion), and carriers are **created and destroyed** (generation and recombination). The **continuity equation** puts the pieces together into one equation.

It is nothing more than **book-keeping**, like a bank account:

> change in balance = money in − money out + interest earned − charges paid

For carriers in a small slice of semiconductor:

> change in carriers = carriers flowing in − carriers flowing out + carriers generated − carriers recombined

Solve this equation and you know where the carriers are and how they change with time. That is how we will get the diode equation in Week 10 and the transistor currents in Week 13.

The second half of the week is about **surfaces**, where the crystal ends and carriers get lost much faster than inside.

Deriving the continuity equation

Take a thin slice of semiconductor between x and x + dx, with cross-section area A. Count the holes in it. The number changes for three reasons:

1. **Flow.** Holes enter at x carried by the current $J_p(x)$, and leave at x + dx carried by $J_p(x + dx)$. Dividing a current by q turns it into a number of carriers per second, so the net inflow is $[J_p(x) - J_p(x + dx)]A/q$.
2. **Generation** inside the slice: G_p per cm^3 per second, times the volume A dx.
3. **Recombination** inside the slice: R_p per cm^3 per second, times A dx.

So:

> (rate of change of p) × A dx = $[J_p(x) - J_p(x + dx)]$ A/q + $(G_p - R_p)$ A dx

Divide by A dx. As dx $\to$ 0, $[J_p(x) - J_p(x + dx)]/dx$ becomes $-\partial J_p/\partial x$:

$$\partial p/\partial t = -(1/q)\, \partial J_p/\partial x + G_p - R_p$$

For electrons the sign of the flow term is opposite, because the electron charge is negative:

$$\partial n/\partial t = +(1/q)\, \partial J_n/\partial x + G_n - R_n$$

Reading it. If more current flows into the slice than out of it (current decreasing with x), carriers pile up. If generation beats recombination, carriers build up. Otherwise they fall.

The minority-carrier diffusion equation

Now apply it to the case that matters most: **holes as minority carriers in an n-type region**, with low-level injection.

- From Week 7, the hole current is $J_p = qp\mu_p E - qD_p\, \partial p/\partial x$.
- From Week 8, the net recombination is $\Delta p/\tau_p$.
- In a neutral region of a device the field is small, so the drift term can usually be dropped.

Substituting gives the **minority-carrier diffusion equation**:

$$\partial \Delta p/\partial t = D_p\, \partial^2 \Delta p/\partial x^2 - \Delta p/\tau_p + G_L$$

Term	Meaning
$\partial \Delta p/\partial t$	how fast the excess changes at this point
$D_p\, \partial^2 \Delta p/\partial x^2$	diffusion: carriers spread from where there are more to where there are fewer
$-\Delta p/\tau_p$	recombination: excess carriers die off with lifetime τ_p
G_L	external generation, e.g. by light

(If there is a field, add $-\mu_p E\, \partial \Delta p/\partial x$ for drift.) For electrons in p-type: $\partial \Delta n/\partial t = D_n\, \partial^2 \Delta n/\partial x^2 - \Delta n/\tau_n + G_L$.

The four standard solutions

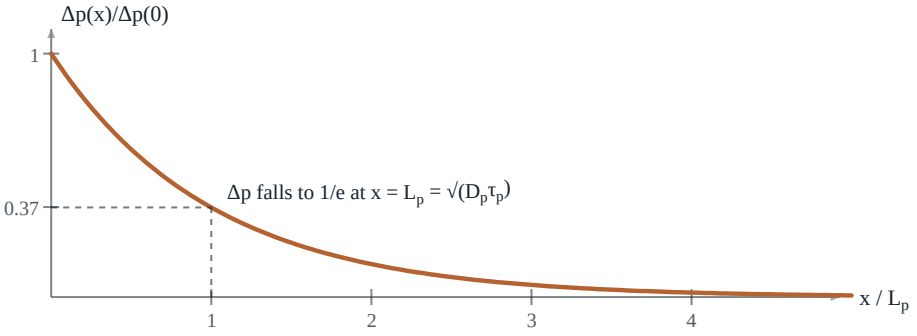
Situation	Equation becomes	Solution
Uniform sample, light switched off at t = 0	$d\Delta p/dt = -\Delta p/\tau_p$	$\Delta p(t) = \Delta p(0)\, e^{-t/\tau_p}$ (exponential decay in time)
Uniform sample, steady light	$0 = G_L - \Delta p/\tau_p$	$\Delta p = G_L \tau_p$
Steady injection at x = 0 into a long sample	$0 = D_p\, d^2\Delta p/dx^2 - \Delta p/\tau_p$	$\Delta p(x) = \Delta p(0)\, e^{-x/L_p}$ (exponential decay in space)
Steady injection into a short sample, $W \ll L_p$	$d^2\Delta p/dx^2 \approx 0$	$\Delta p(x) = \Delta p(0)(1 - x/W)$, a straight line

The diffusion length

The third case introduces the **diffusion length**:

$$L_p = \sqrt{(D_p\, \tau_p)}, \quad L_n = \sqrt{(D_n\, \tau_n)}$$

It is the average distance an injected minority carrier diffuses before it recombines. The excess falls to 1/e (37 %) at one diffusion length, and to under 1 % at five.



Steady-state profile when minority carriers are injected at x = 0 into a long n-type bar: an exponential decay with characteristic length L_p.

> **In plain words.** Inject carriers at one end of a bar and they spread out by diffusion, but they also keep dying by recombination. They spread a typical distance L_p before they die. The longer they live (τ) and the faster they spread (D), the further they get.

Example 9.1: diffusion length

Question. Holes are injected into n-type silicon in which $D_p = 12$ cm^2/s and the hole lifetime is 1 μs. Calculate the hole diffusion length, and the fraction of the injected excess remaining 100 μm from the injection point.

Given: $D_p = 12$ cm^2/s, $\tau_p = 1$ μs $= 1\times10^{-6}$ s, $x = 100$ μm.

Find: L_p, and $\Delta p(100\ \mu m)/\Delta p(0)$.

Solution.

Step 1. $L_p = \sqrt{(D_p\tau_p)} = \sqrt{(12 \times 10^{-6})} = 3.46\times10^{-3}$ cm $= 34.6$ μm

Step 2. $\Delta p(x)/\Delta p(0) = e^{-x/L_p}$:

$$e^{-100/34.6} = e^{-2.89} = 0.056$$

Answer: $L_p = 34.6$ μm; only **5.6 %** of the excess survives 100 μm from the injection point.

The hole current at the injection point comes from the slope of the profile there:

$$J_p(0) = -qD_p\, d\Delta p/dx\,|_0 = qD_p\Delta p(0)/L_p$$

This is exactly the formula that, applied at the edge of a junction, gives the Shockley diode equation (Week 10). The fourth case (straight-line profile in a short region) is what happens in a transistor's thin base (Week 13).

Where the continuity equation is used

- **Diodes:** the exponential profile in each neutral region gives the diode current (Week 10).
- **Bipolar transistors:** the straight-line profile in the thin base gives the collector current and the gain (Week 13).
- **Measuring lifetime:** the exponential decay after a light pulse (photoconductive decay) gives τ directly.
- **Haynes–Shockley experiment:** a pulse of carriers drifts and spreads; timing it gives μ, D and τ.
- **Solar cells and photodiodes:** with generation $G_L(x) = G_0e^{-\alpha x}$, the solution gives how many carriers are collected.
- **Device simulators** solve these equations numerically for every chip design.

Surfaces and surface states

Why surfaces are special

Inside the crystal, every silicon atom is bonded to four neighbours. At a surface the crystal simply stops: the surface atoms have bonds pointing out into nothing, called **dangling bonds**. Adsorbed impurities and defects in any oxide on top add more problems.

These create a high density of **surface states**: energy levels inside the band gap, located at the surface (typically 10^{10}–10^{15} per cm^2). They:

- act as **SRH traps** (Week 8), so carriers recombine much faster at the surface than in the bulk;
- **trap charge**, which bends the bands near the surface even with no voltage applied;
- cause noise and instability, and shift the threshold voltage in MOSFETs.

Surface recombination velocity

Recombination at a surface happens in an extremely thin layer, so it is counted **per unit area**, not per unit volume:

$$U_s = S \times \Delta p(0) \qquad [\text{carriers per cm}^2 \text{ per second}]$$

where $\Delta p(0)$ is the excess minority concentration at the surface and **S is the surface recombination velocity** (cm/s).

> **In plain words.** Imagine the excess carriers near the surface rushing into it at speed S and vanishing. A surface with S = 100 cm/s swallows carriers as fast as if they flowed into a drain at 100 cm/s. A perfect surface has S = 0; a surface that destroys every carrier that reaches it has $S \to \infty$.

For surface traps of density N_{st} (per cm^2):

$$S = \sigma_p \, v_{th} \, N_{st}$$

It enters the diffusion equation as a **boundary condition**: the carriers arriving at the surface by diffusion must equal the carriers recombining there,

$$D_p \, d\Delta p/dx \,|_{surface} = S \, \Delta p(0)$$

Surface	Typical S (cm/s)
Bare or freshly etched Si	$10^4 – 10^6$
Thermally oxidised Si	10 – 100
Best passivated Si (Al_2O_3, SiN_x, annealed SiO_2)	below 10
Ohmic metal contact	$\to \infty$ (Δp forced to 0)

How surfaces shorten the lifetime

In a wafer of thickness W with both surfaces recombining, the measured (effective) lifetime combines bulk and surface losses:

$$1/\tau_{eff} = 1/\tau_{bulk} + 2S/W \qquad \text{(for small S)}$$

Example 9.2: how surfaces reduce the lifetime

Question. A silicon wafer 300 μm thick has a bulk lifetime of 100 μs. Both surfaces are oxidised, giving a surface recombination velocity of 100 cm/s. Calculate the effective lifetime. Repeat for unprotected surfaces with S = 10^5 cm/s.

Given: τ_{bulk} = 100 μs = 1×10^{-4} s, W = 300 μm = 0.03 cm, S = 100 cm/s (then 10^5 cm/s).

Find: τ_{eff} in each case.

Solution.

Use $1/\tau_{eff} = 1/\tau_{bulk} + 2S/W$.

S = 100 cm/s: $2S/W = 2 \times 100/0.03 = 6.7\times10^3\ s^{-1}$, $1/\tau_{bulk} = 1\times10^4\ s^{-1}$

$1/\tau_{eff} = 1.67\times10^4\ s^{-1} \Rightarrow \tau_{eff} = 60\ \mu s$

S = 10^5 cm/s: $2S/W = 6.7\times10^6\ s^{-1} \gg 1/\tau_{bulk} \Rightarrow \tau_{eff} \approx 1/6.7\times10^6 = 0.15\ \mu s$

Answer: τ_{eff} = 60 μs with oxidised surfaces, but only about 0.15 μs with bare surfaces: an unprotected surface, not the bulk, then controls the lifetime.

With an unprotected surface (S = 10^5 cm/s), $2S/W = 6.7\times10^6\ s^{-1}$ and τ_{eff} falls to about **0.15 μs**: the surface, not the bulk, then controls everything.

Reducing surface recombination (passivation): grow thermal SiO_2 and anneal in hydrogen to tie up dangling bonds; deposit silicon nitride or aluminium oxide; add a heavily doped surface layer that repels minority carriers (a "back-surface field" in solar cells); and keep surfaces clean and damage-free.

Key points of Week 9

- Continuity equation = carrier book-keeping: $\partial p/\partial t = -(1/q)\partial J_p/\partial x + G - R$.
- Minority-carrier diffusion equation: $\partial \Delta p/\partial t = D_p\partial^2\Delta p/\partial x^2 - \Delta p/\tau_p + G_L$.
- Standard solutions: decay in time $e^{-t/\tau}$; steady light $\Delta p = G_L\tau$; long bar $e^{-x/L}$; short region a straight line.
- Diffusion length $L = \sqrt{(D\tau)}$: how far a minority carrier travels before recombining.
- Surfaces have dangling bonds → surface states → fast recombination, measured by S (cm/s).
- $1/\tau_{eff} = 1/\tau_{bulk} + 2S/W$; passivation keeps S small.

Examination questions solved (Q45–Q46)

Q45. Explain the continuity equation for minority carriers and discuss its application to semiconductor devices.

Statement. The continuity equation expresses conservation of carriers: in any small volume, the rate of increase of carrier concentration equals the net inflow due to current, plus generation, minus recombination.

$$\partial p/\partial t = -(1/q)\,\partial J_p/\partial x + G_p - R_p \qquad \text{(holes)}$$
$$\partial n/\partial t = +(1/q)\,\partial J_n/\partial x + G_n - R_n \qquad \text{(electrons)}$$

Derivation. For a slab of area A and thickness dx, the number of holes increases at the rate $(1/q)[J_p(x) - J_p(x + dx)]A$ from current flow and at $(G_p - R_p)A\,dx$ from generation and recombination. Dividing by A dx and taking the limit gives

the equations above.

Minority-carrier form. For holes in an n-type region, substitute the hole current $J_p = q\mu_{pp}E - qD_p\,\partial p/\partial x$ and, for low-level injection, the recombination rate $R_p - G_{th} = \Delta p/\tau_p$. With uniform doping:

$$\partial\Delta p/\partial t = D_p\,\partial^2\Delta p/\partial x^2 - \mu_p E\,\partial\Delta p/\partial x - \mu_p\,p\,\partial E/\partial x + G_L - \Delta p/\tau_p$$

In a quasi-neutral region $E \approx 0$, giving the **ambipolar / minority-carrier diffusion equation**:

$$\partial\Delta p/\partial t = D_p\,\partial^2\Delta p/\partial x^2 - \Delta p/\tau_p + G_L$$

For electrons in p-type material: $\partial\Delta n/\partial t = D_n\,\partial^2\Delta n/\partial x^2 - \Delta n/\tau_n + G_L$.

Applications:

1. **p-n junction diode.** In the neutral regions under steady forward bias, $0 = D_p\,d^2\Delta p/dx^2 - \Delta p/\tau_p$ gives $\Delta p(x) = \Delta p(x_n)e^{-(x - x_n)/L_p}$. The diffusion current at the depletion edge leads directly to the Shockley diode equation (Week 10).
2. **Bipolar transistor.** In a narrow base ($W_B \ll L_n$), solving the equation gives a nearly linear minority-carrier profile; the slope gives collector current, and the small recombination loss gives the base transport factor and current gain.
3. **Photoconductive decay and lifetime measurement.** For uniform illumination switched off at $t = 0$, $\Delta p(t) = \Delta p(0)e^{-t/\tau_p}$; measuring the decay of conductivity gives τ.
4. **Haynes–Shockley experiment.** A pulse of minority carriers drifts and spreads; the time-dependent solution, a moving and broadening Gaussian, gives μ_p, D_p and τ_p and was used to verify the Einstein relation.
5. **Solar cells and photodiodes.** With $G_L(x) = G_0e^{-\alpha x}$, the solution gives the collection efficiency and photocurrent and shows why the diffusion length must exceed the absorption depth.
6. **Switching transients.** The time-dependent equation gives stored charge and the reverse-recovery time of diodes.
7. **Device simulation.** Together with Poisson's equation and the current equations, the continuity equations are solved numerically in every TCAD simulator.

Q46. What is surface recombination velocity? Explain how semiconductor surface states affect carrier lifetime.

Surface recombination velocity. At a semiconductor surface, recombination happens in a very thin layer through surface states. It is described per unit area as

$$U_s = S \cdot \Delta p_s \qquad (\text{cm}^{-2}\,\text{s}^{-1})$$

where Δp_s is the excess minority-carrier density at the surface and **S is the surface recombination velocity** (cm/s). It is the effective velocity at which excess carriers move into the surface and are annihilated. For surface traps of density N_{st} (cm^{-2}) with capture cross-section σ_p:

$$S = \sigma_p\,v_{th}\,N_{st}$$

S enters the boundary condition for the diffusion equation: $D_p\,d\Delta p/dx = S\Delta p$ at the surface. $S = 0$ means an ideal surface; $S \to \infty$ means every carrier arriving recombines ($\Delta p_s = 0$, like an Ohmic contact). Typical values: unpassivated Si 10^4–10^6 cm/s; thermally oxidised Si 10–10^3 cm/s; well-passivated Si < 10 cm/s.

Surface states. The crystal periodicity ends at the surface, leaving unsaturated (dangling) bonds, as well as adsorbed atoms, oxide defects and contamination. These produce a high density of localised energy levels distributed throughout the band gap, many near mid-gap.

How they affect lifetime:

1. **Additional recombination path.** Mid-gap surface states are efficient SRH centres. Carriers that diffuse to the surface recombine there quickly, so the measured (effective) lifetime is lower than the bulk lifetime: $1/\tau_{eff} = 1/\tau_{bulk} + 1/\tau_{surface}$, with $1/\tau_{surface} \approx 2S/W$ for a wafer of thickness W recombining at both surfaces. In thin wafers and small devices surface recombination can dominate: a 300 µm wafer with $\tau_{bulk} = 100$ µs falls to $\tau_{eff} \approx 60$ µs at S = 100 cm/s and below 1 µs at $S = 10^5$ cm/s.
2. **Carrier depletion near the surface.** The excess carrier density falls towards the surface, creating a gradient that pulls more carriers there, so even carriers generated deep in the material are lost.
3. **Surface band bending.** Charge in surface states bends the bands, creating depletion or inversion layers. Under depletion, both electrons and holes are present in similar numbers near the surface, which maximises SRH recombination and generation.
4. **Leakage and noise.** Surface generation adds to reverse leakage current of junctions where they meet the surface; trapping and detrapping produces 1/f noise; charged states cause threshold-voltage drift and instability in MOS devices.
5. **Device performance.** High S reduces solar-cell efficiency (especially for blue light absorbed near the surface), reduces BJT current gain (surface recombination at the emitter–base perimeter), and degrades photodetector response.

Reduction (passivation): thermal oxidation (SiO_2) with forming-gas (H_2) anneal to tie up dangling bonds; silicon nitride and Al_2O_3 films with fixed charge for field-effect passivation; heavily doped surface layers (back-surface field) that repel minority carriers; and clean, damage-free surface preparation.

Practice questions

These extra questions practise this week's ideas. Work each one on paper before reading the solution; the calculation questions follow the same Question, Given, Find, Solution, Answer layout as the worked examples.

Practice 9.1: electron diffusion length

Question. Electrons in p-type silicon have $D_n = 25$ cm^2/s and $\tau_n = 10$ µs. Find the diffusion length.

Given: $D_n = 25$ cm^2/s, $\tau_n = 10^{-5}$ s.

Find: L_n.

Solution.

$$L_n = \sqrt{(D_n \tau_n)} = \sqrt{(25 \times 10^{-5})} = 1.58\times10^{-2} \text{ cm} = 158 \text{ µm}$$

Answer: $L_n \approx 160$ µm.

Practice 9.2: injected profile and current

Question. Holes are injected at x = 0 into a long n-type bar with $\Delta p(0) = 1\times10^{14}$ cm^{-3}, $D_p = 12$ cm^2/s and $L_p = 30$ µm. Find the hole current density at x = 0 and the excess hole concentration at x = 60 µm.

Given: $\Delta p(0) = 10^{14}$ cm^{-3}, $D_p = 12$ cm^2/s, $L_p = 3\times10^{-3}$ cm.

Find: $J_p(0)$ and $\Delta p(60$ µm$)$.

Solution.

$J_p(0) = qD_p\Delta p(0)/L_p = 1.6\times10^{-19} \times 12 \times 10^{14} / 3\times10^{-3} = 0.064$ A/cm^2

$\Delta p(60\ \mu m) = \Delta p(0)\, e^{-60/30} = 10^{14} \times e^{-2} = 1.35\times10^{13}$ cm^{-3}

Answer: $J_p(0) = 64$ mA/cm^2, $\Delta p(60\ \mu m) = 1.35\times10^{13}$ cm^{-3}.

Practice 9.3: diffusion across a short region

Question. Electrons are injected into a p-type region only 2 µm wide (much shorter than L_n), with $\Delta n = 1\times10^{15}$ cm^{-3} at one side and $\Delta n = 0$ at the other. Find the electron current density ($D_n = 25$ cm^2/s).

Given: $W = 2\times10^{-4}$ cm, $\Delta n(0) = 10^{15}$ cm^{-3}, $\Delta n(W) = 0$, $D_n = 25$ cm^2/s.

Find: J_n.

Solution. In a short region the profile is a straight line, so the gradient is $\Delta n(0)/W$:

$J_n = qD_n\Delta n(0)/W = 1.6\times10^{-19} \times 25 \times 10^{15} / 2\times10^{-4} = 20$ A/cm^2

Answer: $J_n = 20$ A/cm^2. This is exactly the situation in a BJT base (Week 13).

Practice 9.4: how good must the surface be?

Question. A 200 µm thick wafer has a bulk lifetime of 200 µs. What surface recombination velocity can be tolerated if the effective lifetime must be at least 50 µs?

Given: $\tau_{bulk} = 2\times10^{-4}$ s, $\tau_{eff} = 5\times10^{-5}$ s, $W = 0.02$ cm.

Find: S.

Solution.

$2S/W = 1/\tau_{eff} - 1/\tau_{bulk} = 2\times10^4 - 5\times10^3 = 1.5\times10^4$ s^{-1}

$S = 1.5\times10^4 \times 0.02 / 2 = 150$ cm/s

Answer: $S \le 150$ cm/s, achievable with a good thermal oxide.

Practice 9.5: surface recombination current

Question. At a surface with S = 1000 cm/s the excess hole concentration is 1×10^{13} cm^{-3}. Find the recombination rate per unit area and the equivalent current density lost to the surface.

Given: $S = 10^3$ cm/s, $\Delta p_s = 10^{13}$ cm^{-3}.

Find: U_s and $J = qU_s$.

Solution.

$U_s = S\Delta p_s = 10^3 \times 10^{13} = 10^{16}$ cm^{-2}s^{-1}

$J = qU_s = 1.6\times10^{-19} \times 10^{16} = 1.6\times10^{-3}$ A/cm^2

Answer: $U_s = 10^{16}$ cm^{-2}s^{-1}, $J = 1.6$ mA/cm^2.

Practice 9.6: no recombination

Question. What does the steady-state minority-carrier profile look like if the lifetime is infinite (no recombination) and there is no field or generation? Why?

Answer. The diffusion equation reduces to $D_p\, d^2\Delta p/dx^2 = 0$, so the second derivative is zero and **Δp is a straight line**. With no recombination, every carrier that enters a slice must leave it, so the current, and hence the slope, is the same everywhere.

Objectives check

Use this table to confirm you have met this week's objectives. Cover the Answer column and answer each check question from memory. If you cannot, go back to the section in the last column before moving on.

#	Objective	Check yourself	Answer	Where it is taught
1	Derive the continuity equation for holes and electrons.	Write the continuity equation for holes.	$\partial p/\partial t = -(1/q)\partial J_p/\partial x + G_p - R_p$.	Deriving the continuity equation
2	Write and interpret the minority-carrier diffusion equation.	What does the term $-\Delta p/\tau_p$ represent?	Loss of excess holes by recombination.	The minority-carrier diffusion equation
3	Solve the diffusion equation for the standard cases.	What is the steady-state profile for injection into a long bar?	$\Delta p(x) = \Delta p(0)e^{-x/L_p}$.	The four standard solutions
4	Calculate diffusion lengths and explain their importance.	Find L_n for $D_n = 30$ cm^2/s, $\tau_n = 10$ µs.	$L_n = \sqrt{(30 \times 10^{-5})} = 1.7\times10^{-2}$ cm = 173 µm.	Diffusion length; Example 9.1
5	Explain surface states and surface recombination velocity.	What value of S corresponds to an Ohmic metal contact?	$S \to \infty$ (Δp forced to zero).	Surfaces and surface states
6	Calculate the effective lifetime of a wafer with recombining surfaces.	Write $1/\tau_{eff}$ for a wafer of thickness W.	$1/\tau_{eff} = 1/\tau_{bulk} + 2S/W$.	How surfaces shorten the lifetime; Example 9.2

The p-n Junction Diode

Plan: built-in potential, depletion width and electric-field profile; ideal diode (Shockley) equation derivation under forward and reverse bias.

Learning objectives

By the end of this week you should be able to:

1. Explain the formation of the depletion region in a p-n junction.
2. Derive and calculate the built-in potential.
3. Calculate the depletion width and electric-field profile.
4. Explain the junction under forward and reverse bias and the law of the junction.
5. Derive and use the Shockley diode equation.
6. Explain breakdown and junction capacitance.

The big picture

A **p-n junction** is what you get when one part of a single crystal is p-type and the part next to it is n-type. It is the most important structure in electronics: it is a diode on its own, two of them make a bipolar transistor, and it appears in every MOSFET, solar cell, LED and laser.

Its key property is that it conducts easily one way and hardly at all the other way (**rectification**). This week explains why, using only ideas you already have:

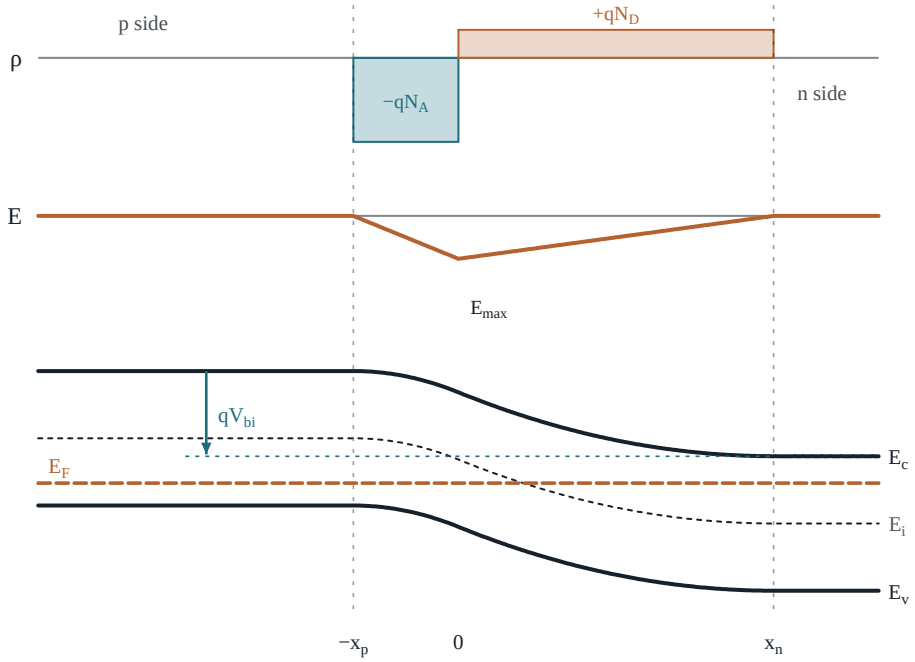
1. **Diffusion** (Week 7) makes carriers cross the junction, leaving fixed charges behind.
2. Those charges create an **electric field and a built-in voltage**, which stops further diffusion.
3. An applied voltage raises or lowers that barrier. Lowering it lets minority carriers be **injected**, and they **diffuse and recombine** (Weeks 8–9). Adding up that diffusion gives the diode equation.

Forming the junction: where the depletion region comes from

Imagine joining p-type silicon (lots of holes, doping N_A) and n-type silicon (lots of electrons, doping N_D) in one crystal.

1. **Diffusion starts.** There are many holes on the p side and very few on the n side, so holes diffuse from p into n. Electrons diffuse the other way, from n into p.
2. **Fixed charges are uncovered.** Each hole that leaves the p side leaves behind a negative acceptor ion (N_A^-), which is locked in the crystal and cannot move. Each electron that leaves the n side leaves behind a positive donor ion (N_D^+). The carriers that cross also recombine with the majority carriers on the other side and disappear.
3. **A charged layer forms.** Near the junction there is now a layer of **negative** ions on the p side and **positive** ions on the n side, with almost no mobile carriers in it. This is the **depletion region** (also called the space-charge region).
4. **The charges create a field.** Positive charge on the n side and negative on the p side create an electric field pointing from n to p.

5. **The field pushes back.** That field drives holes back to the p side and electrons back to the n side: a **drift** current opposite to the diffusion current.
6. **Balance.** The depletion region grows until, for each type of carrier, drift exactly cancels diffusion. Now no net current flows. The **Fermi level is flat** across the whole junction, as it must be in equilibrium (Week 5).



The abrupt p-n junction in equilibrium. Top: the fixed charge in the depletion region (negative on the p side, positive on the n side). Middle: the electric field, strongest at the junction. Bottom: the band diagram; the Fermi level is flat and the bands bend by qV_{bi}.

> **In plain words.** Two crowds on either side of a line start mixing. But everyone who crosses leaves a tied-up dog behind, and the dogs bark at anyone trying to follow. Soon the barking (the field) is strong enough to stop the crossing. The strip full of dogs and empty of people is the depletion region.

The built-in potential

The field in the depletion region corresponds to a voltage across it, the **built-in potential V_{bi}**. The n side is at a higher potential than the p side. In the band diagram, the bands on the p side sit higher than on the n side by qV_{bi}.

We can find V_{bi} from the Fermi-level positions of Week 5. On the p side, E_F is $kT \ln(N_A/n_i)$ below E_i. On the n side it is $kT \ln(N_D/n_i)$ above E_i. In the joined junction E_F is one flat line, so the bands must bend by the sum:

$$qV_{bi} = kT \ln(N_A/n_i) + kT \ln(N_D/n_i)$$

$$V_{bi} = (kT/q) \ln(N_A N_D / n_i^2)$$

Example 10.1: built-in potential

Question. A silicon p-n junction has $N_A = 1\times10^{17}$ cm^{-3} on the p side and $N_D = 1\times10^{16}$ cm^{-3} on the n side. Calculate the built-in potential at 300 K.

Given: $N_A = 1\times10^{17}$ cm^{-3}, $N_D = 1\times10^{16}$ cm^{-3}, $n_i = 1\times10^{10}$ cm^{-3}, $kT/q = 0.0259$ V.

Find: V_{bi}.

Solution.

$$V_{bi} = (kT/q)\ln(N_A N_D/n_i^2) = 0.0259 \times \ln(10^{17} \times 10^{16} / 10^{20})$$

$$= 0.0259 \times \ln(10^{13}) = 0.0259 \times 29.93 = 0.775 \text{ V}$$

Answer: $V_{bi} = 0.775$ V.

You **cannot** measure V_{bi} with a voltmeter across a diode: where the wires meet the semiconductor, contact potentials appear that exactly cancel it. It is an internal barrier, not a battery.

Depletion width and field: the depletion approximation

To calculate the width of the depletion region we make the **depletion approximation**: inside the region (from $-x_p$ on the p side to x_n on the n side) there are no mobile carriers at all, only the fixed ions; outside it the material is neutral. Then Poisson's equation (the link between charge and field) gives:

Charge balance. The total negative charge equals the total positive charge:

$$N_A\, x_p = N_D\, x_n$$

So the depletion region extends **further into the more lightly doped side**. If $N_A = 10 \times N_D$, then $x_n = 10 \times x_p$.

Width.

$$W = x_n + x_p = \sqrt{\left(\frac{2\varepsilon_s}{q}(V_{bi} - V_a)\left(\frac{1}{N_A} + \frac{1}{N_D}\right)\right)}$$

with $x_n = W\, N_A/(N_A + N_D)$ and $x_p = W\, N_D/(N_A + N_D)$. Here ε_s is the permittivity of silicon ($11.7 \times 8.854\times10^{-14} = 1.04\times10^{-12}$ F/cm) and V_a the applied voltage (positive for forward bias).

Peak field (at the junction):

$$E_{max} = qN_D\, x_n/\varepsilon_s = qN_A\, x_p/\varepsilon_s, \quad \text{and} \quad V_{bi} - V_a = \tfrac{1}{2}\, E_{max}\, W$$

For a **one-sided p^+n junction** ($N_A \gg N_D$), almost all of W is on the n side: $W \approx \sqrt{(2\varepsilon_s(V_{bi} - V_a)/qN_D)}$.

Example 10.2: depletion width and peak field

Question. For the junction of Example 10.1 at zero bias, calculate the total depletion width, how much of it lies on each side, and the peak electric field.

Given: $V_{bi} = 0.775$ V, $N_A = 10^{17}$ cm^{-3}, $N_D = 10^{16}$ cm^{-3}, $\varepsilon_s = 1.04\times10^{-12}$ F/cm, $q = 1.6\times10^{-19}$ C, $V_a = 0$.

Find: W, x_n, x_p and E_{max}.

Solution.

Step 1. Total width:

$$W = \sqrt{\left(\frac{2\varepsilon_s}{q}\, V_{bi}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)\right)}$$

$$= \sqrt{\left(2 \times 1.04\times10^{-12} \times 0.775 \times (10^{-17} + 10^{-16}) / 1.6\times10^{-19}\right)} = \sqrt{(1.10\times10^{-9})} = 3.32\times10^{-5} \text{ cm}$$

Step 2. Split, using $x_n = W\, N_A/(N_A + N_D)$:

$$x_n = 0.332 \times 10/11 = 0.30\ \mu\text{m}, \quad x_p = 0.332 - 0.30 = 0.03\ \mu\text{m}$$

Step 3. Peak field at the junction:

$$E_{max} = qN_D x_n/\varepsilon_s = 1.6\times10^{-19} \times 10^{16} \times 3.0\times10^{-5} / 1.04\times10^{-12} = 4.7\times10^{4}\ \text{V/cm}$$

Answer: W = 0.33 μm (0.30 μm on the n side, 0.03 μm on the p side), $E_{max} = 4.7\times10^4$ V/cm. The region is narrow but the field in it is large.

The junction under bias

Forward bias (p side made positive, $V_a > 0$). The applied voltage opposes the built-in voltage, so the barrier **falls** to $V_{bi} - V_a$ and the depletion region **narrows**. Diffusion now wins over drift. Holes pour from p into n, and electrons from n into p. On arrival they are minority carriers: they are **injected**. They diffuse away from the junction and recombine (Week 9). The current rises steeply (exponentially) with voltage.

Reverse bias (p side negative, $V_a < 0$). The applied voltage adds to the built-in voltage. The barrier **rises** to $V_{bi} + |V_a|$ and the depletion region **widens**. Diffusion is shut off. The only current left is the tiny drift of the few minority carriers that happen to wander to the edge of the depletion region and get swept across. This is the **reverse saturation current**, almost independent of voltage, until breakdown.

The law of the junction. The barrier controls how many carriers can be injected. At the edges of the depletion region, the minority-carrier concentrations become:

$$p_n(x_n) = p_{n0}\, e^{qV_a/kT}, \qquad n_p(-x_p) = n_{p0}\, e^{qV_a/kT}$$

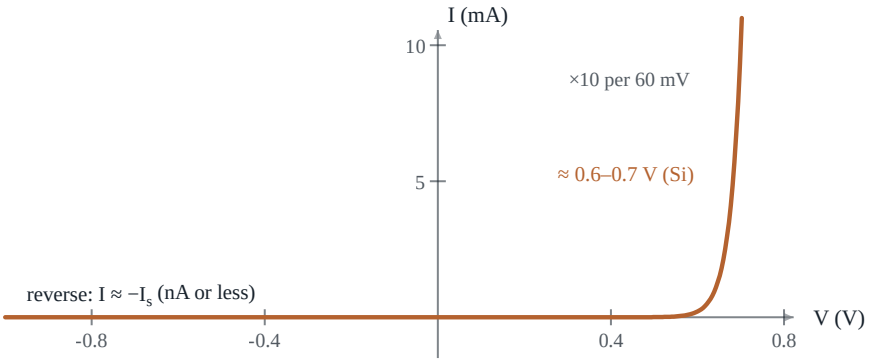
$p_{n0} = n_i^2/N_D$ and $n_{p0} = n_i^2/N_A$ are the equilibrium minority concentrations. Forward bias multiplies them by $e^{qV_a/kT}$; reverse bias makes them almost zero. Every 60 mV of forward bias multiplies the injected minority carriers by 10.

The ideal diode (Shockley) equation, in words

The full derivation is worked in exam question Q48 below. The idea in four steps:

1. **Injection.** The law of the junction gives the excess minority carriers at each edge of the depletion region: $\Delta p = p_{n0}(e^{qV/kT} - 1)$ on the n side.
2. **Diffusion and recombination.** In each neutral region the excess decays exponentially over a diffusion length (Week 9): $\Delta p(x) = \Delta p(x_n) e^{-(x - x_n)/L_p}$.
3. **Current at the edge.** The diffusion current at the edge is $qD_p \times \text{slope} = (qD_p/L_p)\Delta p(x_n)$. Same for electrons on the p side.
4. **Add them.** The total current is the sum of the hole and electron diffusion currents:

$$I = I_s\,(\,e^{qV/kT} - 1\,), \qquad I_s = qA\, n_i^2\,(\,D_p/(L_pN_D) + D_n/(L_nN_A)\,)$$



The ideal diode characteristic. Forward current grows ten times for every 60 mV at 300 K; reverse current is a tiny $-I_s$.

What the equation says:

- **Forward** ($V >$ about 0.1 V): the -1 is negligible and $I \approx I_s\, e^{qV/kT}$. Because I_s is tiny (10^{-15}–10^{-12} A for small Si diodes), a noticeable current needs about 0.6–0.7 V in Si (0.2–0.3 V in Ge).
- **Reverse** (V negative, more than a few kT/q): $e^{qV/kT} \to 0$ and $I \approx -I_s$, tiny and constant.
- $I_s \propto n_i^2$, so it rises steeply with temperature and is much larger in small-gap materials (Ge diodes leak far more than Si).

Example 10.3: saturation current and forward current of a diode

Question. A silicon p^+n diode has $N_D = 1\times10^{16}$ cm^{-3} on the n side, hole diffusion coefficient $D_p = 10$ cm^2/s, hole lifetime $\tau_p = 1$ µs and area 10^{-4} cm^2. Calculate the saturation current, and the forward current at 0.6 V and at 0.7 V (300 K).

Given: $N_D = 10^{16}$ cm^{-3}, $D_p = 10$ cm^2/s, $\tau_p = 10^{-6}$ s, $A = 10^{-4}$ cm^2, $n_i = 10^{10}$ cm^{-3}, $kT/q = 0.0259$ V.

Find: I_s, $I(0.6\text{ V})$, $I(0.7\text{ V})$.

Solution.

Step 1. Diffusion length: $L_p = \sqrt{(D_p\tau_p)} = \sqrt{(10 \times 10^{-6})} = 3.16\times10^{-3}$ cm

Step 2. For a p^+n diode only the hole term matters:

$$I_s \approx qAn_i^2D_p/(L_pN_D) = 1.6\times10^{-19} \times 10^{-4} \times 10^{20} \times 10 \,/\, (3.16\times10^{-3} \times 10^{16}) = 5.1\times10^{-16}\text{ A}$$

Step 3. Forward current, $I \approx I_s\, e^{qV/kT}$:

$$I(0.6\text{ V}) = 5.1\times10^{-16} \times e^{0.6/0.0259} = 5.1\times10^{-16} \times 1.15\times10^{10} = 5.9\times10^{-6}\text{ A}$$

$$I(0.7\text{ V}) = 5.1\times10^{-16} \times e^{0.7/0.0259} = 5.1\times10^{-16} \times 5.45\times10^{11} = 2.8\times10^{-4}\text{ A}$$

Answer: $I_s = 5.1\times10^{-16}$ A; I = 5.9 µA at 0.6 V and 280 µA at 0.7 V. An extra 0.1 V multiplies the current by about 47.

Breakdown

At large reverse voltage the current suddenly rises. Two mechanisms:

- **Avalanche breakdown.** Carriers crossing the strong depletion field gain enough energy to knock new electron–hole pairs out of the lattice (impact ionisation, Week 8), which multiply in a chain reaction. Typical of lightly doped

junctions; breakdown voltage above about 6 V; rises with temperature.

- **Zener breakdown.** In very heavily doped junctions the depletion region is extremely thin, and electrons **tunnel** straight through from the p-side valence band to the n-side conduction band. Breakdown voltage below about 4 V; falls with temperature. Used in Zener voltage regulators.

Junction capacitance

The depletion region is a layer of stored charge between two conducting regions, so it behaves as a **capacitor** with plate separation W:

$$C_j = \varepsilon_s A/W$$

Since W grows with reverse bias, C_j **falls** as the reverse voltage increases. Diodes used deliberately as voltage-controlled capacitors (in radio tuning) are called **varactors**. Measuring C_j against voltage is also how doping is measured (C-V profiling, Week 12). Under forward bias the stored minority carriers add a larger **diffusion capacitance**.

Key points of Week 10

- Diffusion across the junction uncovers fixed ions: the depletion region, with a field from n to p.
- Equilibrium: drift balances diffusion, E_F is flat, and the bands bend by $qV_{bi} = kT \ln(N_A N_D/n_i^2)$.
- $N_A x_p = N_D x_n$: the depletion region lies mostly on the lightly doped side. $W = \sqrt{((2\varepsilon_s/q)(V_{bi} - V)(1/N_A + 1/N_D))}$.
- Forward bias lowers the barrier and injects minority carriers (law of the junction); reverse bias raises it.
- Shockley equation: $I = I_s(e^{qV/kT} - 1)$, from minority-carrier diffusion in the neutral regions.
- Breakdown by avalanche (lightly doped) or Zener tunnelling (heavily doped); $C_j = \varepsilon_s A/W$.

Examination questions solved (Q47–Q48)

Q47. Explain the formation of the depletion region and built-in potential in a p-n junction.

Formation of the depletion region.

1. When p-type and n-type regions are formed in one crystal, the p side has a high hole concentration ($p_{p0} \approx N_A$) and the n side a high electron concentration ($n_{n0} \approx N_D$). The steep concentration gradients cause holes to diffuse from p to n and electrons from n to p.
2. Holes leaving the p side near the junction leave behind **immobile negative acceptor ions**; electrons leaving the n side leave behind **immobile positive donor ions**. Diffusing carriers also recombine with the majority carriers on the other side.
3. A region on both sides of the metallurgical junction is therefore stripped of mobile carriers and contains only fixed ionic charge: **negative on the p side, positive on the n side**. This is the **depletion region** or space-charge region, typically 0.1–10 μm wide.
4. The two charge layers are equal in magnitude, $N_A\, x_p = N_D\, x_n$, so the depletion region extends further into the more lightly doped side.
5. The space charge produces an **electric field** directed from the positive (n) side to the negative (p) side, maximum at the junction: $E_{max} = qN_D\, x_n/\varepsilon_s$.

6. This field drives drift currents of holes back to p and electrons back to n, opposing the diffusion. The depletion region grows until **drift exactly balances diffusion** for both carriers and the net current is zero. In equilibrium the Fermi level is constant throughout.

Built-in potential. The field corresponds to an electrostatic potential difference across the depletion region, the **built-in potential V_{bi}**, with the n side positive with respect to the p side. In the band diagram, the bands bend by qV_{bi} so that the flat Fermi level is near E_v on the p side and near E_c on the n side. It forms a potential barrier that majority carriers must surmount to cross.

Its value follows from the Fermi-level positions: $q\,V_{bi} = (E_i - E_F)_p + (E_F - E_i)_n = kT\ln(N_A/n_i) + kT\ln(N_D/n_i)$, so

$$V_{bi} = (kT/q)\ln(N_A N_D / n_i^2)$$

Width of the depletion region:

$$W = \sqrt{(2\varepsilon_s V_{bi}/q)(N_A + N_D)/(N_A N_D)}$$

Example (Si, $N_A = 10^{17}$, $N_D = 10^{16}$ cm^{-3}): $V_{bi} = 0.0259\ln(10^{13}) =$ **0.775 V**; $W =$ **0.33 µm**, of which 0.30 µm is on the n side; $E_{max} \approx 4.7\times10^4$ V/cm.

The built-in potential cannot drive current in an external circuit, because in equilibrium it is exactly balanced by the contact potentials at the metal–semiconductor contacts. Applied bias changes the barrier to $V_{bi} - V_a$, which is the basis of rectification.

Q48. Derive the ideal Shockley diode equation and explain the effect of forward and reverse bias.

Assumptions (Shockley):

1. Abrupt junction; depletion approximation (no carriers in the depletion region, neutral elsewhere).
2. Boltzmann statistics apply (non-degenerate).
3. **Low-level injection**: injected minority densities are much smaller than majority densities.
4. No generation or recombination inside the depletion region, so electron and hole currents are each constant across it.
5. All applied voltage V appears across the depletion region; neutral regions have negligible resistance (zero field).
6. Long neutral regions (longer than the diffusion lengths) terminated by Ohmic contacts.

Step 1: law of the junction. In equilibrium $n_{n0}/n_{p0} = \exp(qV_{bi}/kT)$. With bias V the barrier becomes $V_{bi} - V$, and (quasi-equilibrium across the depletion region) the minority densities at its edges become:

$$p_n(x_n) = p_{n0}\exp(qV/kT), \qquad n_p(-x_p) = n_{p0}\exp(qV/kT)$$

So the excess minority densities at the edges are:

$$\Delta p_n(x_n) = p_{n0}\,[\exp(qV/kT) - 1\,], \qquad \Delta n_p(-x_p) = n_{p0}\,[\exp(qV/kT) - 1\,]$$

Step 2: minority-carrier diffusion in the neutral n region. With E = 0 and steady state, the continuity equation reduces to

$$D_p\, d^2\Delta p_n/dx^2 - \Delta p_n/\tau_p = 0$$

For a long region ($\Delta p \to 0$ as $x \to \infty$) the solution is

$$\Delta p_n(x) = \Delta p_n(x_n)\,\exp(-(x - x_n)/L_p), \qquad L_p = \sqrt{(D_p \tau_p)}$$

Step 3: hole diffusion current at $x = x_n$. With no field, the hole current is pure diffusion:

$$J_p(x_n) = -qD_p\, d\Delta p_n/dx\,|_{x_n} = (qD_p/L_p)\,\Delta p_n(x_n) = (qD_p\, p_{n0}/L_p)\,[\exp(qV/kT) - 1]$$

Step 4: electron current at $x = -x_p$. Similarly in the p region:

$$J_n(-x_p) = (qD_n\, n_{p0}/L_n)\,[\exp(qV/kT) - 1]$$

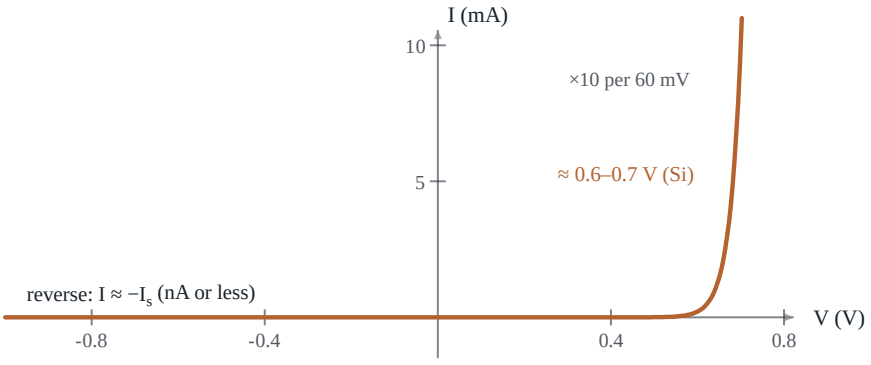
Step 5: total current. With no recombination in the depletion region, J_n is the same at x_n as at $-x_p$, so the total current (constant everywhere) is the sum of the two minority diffusion currents at the depletion edges:

$$J = J_p(x_n) + J_n(-x_p) = [\, qD_p\, p_{n0}/L_p + qD_n\, n_{p0}/L_n\,]\,[\exp(qV/kT) - 1]$$

Using $p_{n0} = n_i^2/N_D$ and $n_{p0} = n_i^2/N_A$:

$$J = J_s\,[\exp(qV/kT) - 1], \qquad J_s = q\, n_i^2\,(D_p/(L_p\, N_D) + D_n/(L_n\, N_A))$$

or, for area A, $\mathbf{I = I_s\,[\exp(qV/kT) - 1]}$ with $I_s = AJ_s$. This is the ideal Shockley diode equation.



Ideal diode characteristic. Forward current grows by a factor of 10 for every 60 mV at 300 K; reverse current saturates at $-I_s$.

Effect of forward bias (V > 0, p side positive):

- The barrier is lowered to $V_{bi} - V$ and the depletion region narrows.
- Minority carriers are injected: the edge densities increase by $\exp(qV/kT)$.
- For $V \gg kT/q$ (above about 0.1 V) the −1 is negligible: $I \approx I_s \exp(qV/kT)$. Current increases by a decade every $2.3kT/q \approx 60$ mV.
- Significant current flows only above the "cut-in" voltage (≈ 0.6–0.7 V in Si, ≈ 0.2–0.3 V in Ge), because I_s is so small.

Effect of reverse bias (V < 0):

- The barrier rises to $V_{bi} + |V|$; the depletion region widens.
- The edge minority densities fall almost to zero ($\exp(qV/kT) \to 0$). Carriers are extracted rather than injected.
- For $|V| \gg kT/q$: $I \approx -I_s$, a small, voltage-independent **reverse saturation current** (Si, about 10^{-15}–10^{-12} A for small diodes) carried by minority carriers generated within a diffusion length of the junction.
- In real diodes, generation in the depletion region adds a current $\propto W \propto \sqrt{|V|}$, and at large |V| avalanche or Zener breakdown occurs.

Numerical example. A Si p^+n diode with $N_D = 10^{16}$ cm^{-3}, $D_p = 10$ cm^2/s, $\tau_p = 1$ µs, area 10^{-4} cm^2: $L_p = \sqrt{(10 \times 10^{-6})} = 3.16\times10^{-3}$ cm; $J_s \approx qn_i^2 D_p/(L_p N_D) = (1.602\times10^{-19})(10^{20})(10)/(3.16\times10^{-3} \times 10^{16}) = 5.1\times10^{-12}$ A/cm^2, so $I_s = 5.1\times10^{-16}$ A. At V = 0.6 V, $I = I_s\, e^{0.6/0.0259} = 5.1\times10^{-16} \times 1.15\times10^{10} \approx 5.9$ µA, and at 0.7 V it is about 280 µA.

Real diodes deviate: recombination in the depletion region at low forward bias ($I \propto e^{qV/2kT}$), high-level injection at high current, series resistance, and generation current and breakdown in reverse. These are summarised by $I = I_s(e^{qV/\eta kT} - 1)$, with ideality factor η between 1 and 2.

Practice questions

These extra questions practise this week's ideas. Work each one on paper before reading the solution; the calculation questions follow the same Question, Given, Find, Solution, Answer layout as the worked examples.

Practice 10.1: built-in potential of a germanium junction

Question. A germanium junction has $N_A = N_D = 1\times10^{16}$ cm^{-3}. Find V_{bi} at 300 K ($n_i = 2.4\times10^{13}$ cm^{-3}) and compare with silicon.

Given: $N_A = N_D = 10^{16}$ cm^{-3}, $n_i = 2.4\times10^{13}$ cm^{-3}, kT/q = 0.0259 V.

Find: V_{bi}.

Solution.

$$V_{bi} = 0.0259 \ln(10^{32}/5.76\times10^{26}) = 0.0259 \ln(1.74\times10^5) = 0.0259 \times 12.06 = 0.31 \text{ V}$$

Answer: $V_{bi} = 0.31$ V, against 0.72 V for silicon with the same doping: the smaller gap gives a smaller barrier.

Practice 10.2: reverse-biased depletion width and field

Question. A silicon p^+n junction has $N_D = 1\times10^{15}$ cm^{-3} and $V_{bi} = 0.7$ V. Find the depletion width and peak field at a reverse bias of 5 V.

Given: $N_D = 10^{15}$ cm^{-3}, $V_{bi} - V = 0.7 + 5 = 5.7$ V, $\varepsilon_s = 1.04\times10^{-12}$ F/cm.

Find: W and E_{max}.

Solution.

$$W = \sqrt{(2\varepsilon_s(V_{bi} - V)/qN_D)} = \sqrt{(2 \times 1.04\times10^{-12} \times 5.7 / (1.6\times10^{-19} \times 10^{15}))} = 2.72\times10^{-4} \text{ cm}$$

$$E_{max} = 2(V_{bi} - V)/W = 2 \times 5.7/2.72\times10^{-4} = 4.2\times10^4 \text{ V/cm}$$

Answer: W = 2.7 µm, $E_{max} = 4.2\times10^4$ V/cm.

Practice 10.3: junction capacitance

Question. Find the junction capacitance of the diode in Practice 10.2 if its area is 1×10^{-3} cm^2.

Given: $W = 2.72\times10^{-4}$ cm, $A = 10^{-3}$ cm^2, $\varepsilon_s = 1.04\times10^{-12}$ F/cm.

Find: C_j.

Solution.

$$C_j = \varepsilon_s A/W = 1.04\times10^{-12} \times 10^{-3} / 2.72\times10^{-4} = 3.8\times10^{-12} \text{ F}$$

Answer: $C_j = 3.8$ pF.

Practice 10.4: saturation current with both sides

Question. A silicon diode has $N_A = 1\times10^{17}$ cm^{-3}, $N_D = 1\times10^{16}$ cm^{-3}, $D_n = 20$ cm^2/s, $L_n = 20$ µm, $D_p = 10$ cm^2/s, $L_p = 10$ µm. Find J_s and the fraction carried by holes.

Given: the values above, $n_i = 10^{10}$ cm^{-3}.

Find: J_s and the hole fraction.

Solution.

$D_p/(L_pN_D) = 10/(10^{-3} \times 10^{16}) = 1.0\times10^{-12}$

$D_n/(L_nN_A) = 20/(2\times10^{-3} \times 10^{17}) = 1.0\times10^{-13}$

$J_s = qn_i^2(1.0\times10^{-12} + 1.0\times10^{-13}) = 1.6\times10^{-19} \times 10^{20} \times 1.1\times10^{-12} = 1.76\times10^{-11}$ A/cm^2

hole fraction $= 1.0/1.1 = 91$ %

Answer: $J_s = 1.8\times10^{-11}$ A/cm^2, 91 % carried by holes injected into the lightly doped n side.

Practice 10.5: temperature and the saturation current

Question. By roughly what factor does I_s of a silicon diode increase when the temperature rises from 300 K to 310 K? (Take $I_s \propto n_i^2$ and ignore other temperature effects.)

Given: $n_i \propto T^{3/2}e^{-E_g/2kT}$, $E_g = 1.12$ eV.

Find: $I_s(310)/I_s(300)$.

Solution.

$n_i(310)/n_i(300) = (310/300)^{3/2} \times \exp[6499 \times (1/300 - 1/310)] = 1.050 \times e^{0.699} = 2.11$

I_s ratio $= 2.11^2 = 4.5$

Answer: about 4.5 times for a 10 K rise: reverse leakage is very temperature-sensitive.

Practice 10.6: forward voltage per decade

Question. How much must the forward voltage of an ideal diode increase to multiply the current by 10 at 300 K? If I = 1 mA at 0.60 V, what is I at 0.66 V?

Given: $I \propto e^{qV/kT}$, $kT/q = 0.0259$ V.

Find: ΔV per decade; I at 0.66 V.

Solution.

$\Delta V = (kT/q) \ln 10 = 0.0259 \times 2.303 = 59.6$ mV

$I(0.66) = 1$ mA $\times e^{0.06/0.0259} = 1 \times 10.1 = 10.1$ mA

Answer: about 60 mV per decade; I ≈ 10 mA at 0.66 V.

Practice 10.7: avalanche or Zener?

Question. A diode breaks down at 3.3 V and its breakdown voltage falls as it warms up. Another breaks down at 50 V and its breakdown voltage rises with temperature. Identify the mechanism in each and explain the temperature behaviour.

Answer. The **3.3 V diode is Zener breakdown**: it is heavily doped, its depletion region is very thin, and electrons tunnel across the gap. Heating shrinks the band gap, making tunnelling easier, so V_{BR} falls. The **50 V diode is avalanche breakdown**: carriers gain energy across a wide depletion region and impact-ionise. Heating increases lattice vibration, so carriers lose more energy to phonons between collisions and need a larger field to ionise, so V_{BR} rises.

Objectives check

Use this table to confirm you have met this week's objectives. Cover the Answer column and answer each check question from memory. If you cannot, go back to the section in the last column before moving on.

#	Objective	Check yourself	Answer	Where it is taught
1	Explain the formation of the depletion region in a p-n junction.	Why is the depletion region empty of mobile carriers?	Carriers diffused across and recombined, leaving only fixed donor and acceptor ions.	Forming the junction
2	Derive and calculate the built-in potential.	Find V_{bi} for $N_A = N_D = 10^{16}$ cm^{-3} in Si.	$0.0259 \ln(10^{12}) = 0.716$ V.	The built-in potential; Example 10.1
3	Calculate the depletion width and electric-field profile.	If $N_A = 100\ N_D$, where does most of the depletion region lie?	On the n side ($x_n = 100\ x_p$).	Depletion approximation; Example 10.2
4	Explain the junction under forward and reverse bias and the law of the junction.	By what factor does forward bias of 0.12 V multiply the injected minority carriers?	$e^{0.12/0.0259} \approx 100$.	The junction under bias
5	Derive and use the Shockley diode equation.	Write the ideal diode equation and its saturation current density.	$I = I_s(e^{qV/kT} - 1)$; $J_s = qn_i^2(D_p/L_pN_D + D_n/L_nN_A)$.	Shockley equation; Example 10.3; Q48
6	Explain breakdown and junction capacitance.	Which breakdown mechanism occurs in very heavily doped junctions?	Zener (tunnelling) breakdown.	Breakdown; junction capacitance

Metal–Semiconductor Interfaces

Plan: Schottky barrier diode physics versus Ohmic contact mechanisms; energy band diagrams under equilibrium and bias.

Learning objectives

By the end of this week you should be able to:

1. Define work function and electron affinity and relate them.
2. Explain how a Schottky barrier forms and calculate Φ_B and V_{bi}.
3. Explain thermionic emission and calculate Schottky diode currents.
4. Draw Schottky band diagrams in equilibrium and under bias.
5. Explain the two ways of making Ohmic contacts.
6. Describe the stability problems of metal contacts.

The big picture

Every device must be wired to the outside world with metal. You might expect a metal touching a semiconductor to behave like a simple wire connection. Sometimes it does: that is an **Ohmic contact**, which lets current through equally well in both directions. But sometimes the contact itself forms a **barrier** and conducts in only one direction, like a diode: a **Schottky contact**.

Which one you get depends on the metal, the semiconductor and its doping. Engineers want Ohmic contacts for wiring, and Schottky contacts when they want a very fast diode. This week explains both using band diagrams, with the same flat-Fermi-level reasoning as the p-n junction (Week 10).

The energies involved

Three quantities describe where the electrons sit, all measured from the **vacuum level** (the energy of an electron just free of the material):

- **Work function Φ_m (metal):** the energy needed to pull an electron from the metal's Fermi level out to the vacuum.
- **Work function Φ_s (semiconductor):** the same for the semiconductor. It depends on doping, because doping moves E_F.
- **Electron affinity χ (semiconductor):** the energy from the **bottom of the conduction band** to the vacuum level. This is a fixed property of the material: Si 4.05 eV, GaAs 4.07 eV, Ge 4.0 eV.

These are linked by

$$\Phi_s = \chi + (E_c - E_F)$$

Metal work functions (eV): Al 4.28, Ti 4.33, Cr 4.5, W 4.55, Au 5.1, Ni 5.15, Pt 5.65.

Example 11.1: work function of doped silicon

Question. Calculate the work function of n-type silicon doped with $N_D = 1\times10^{16}$ cm^{-3} at 300 K. The electron affinity of Si is 4.05 eV.

Given: $\chi = 4.05$ eV, $N_D = 1\times10^{16}$ cm^{-3}, $N_c = 2.8\times10^{19}$ cm^{-3}, $kT = 0.02585$ eV.

Find: Φ_s.

Solution.

Step 1. Position of E_F (Example 5.1): $E_c - E_F = kT\ln(N_c/N_D) = 0.205$ eV

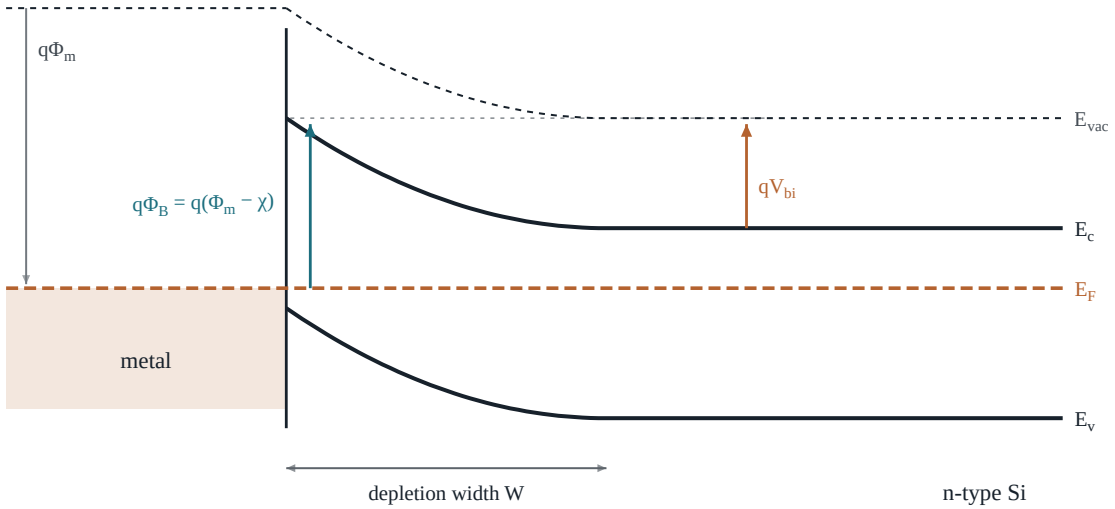
Step 2. $\Phi_s = \chi + (E_c - E_F) = 4.05 + 0.205 = 4.26$ eV

Answer: $\Phi_s = 4.26$ eV.

The Schottky barrier (rectifying contact)

How it forms: metal on n-type with $\Phi_m > \Phi_s$

1. **Before contact**, the semiconductor's Fermi level is **higher** than the metal's (its electrons are less tightly held, since $\Phi_s < \Phi_m$).
2. **On contact**, electrons flow "downhill" from the semiconductor into the metal, until the two Fermi levels line up, just as in a p-n junction.
3. The electrons that left uncover **positive donor ions** in the semiconductor near the interface: a **depletion region**, exactly like the n side of a p^+n junction. A thin sheet of extra electrons sits on the metal surface.
4. The bands in the semiconductor bend **upward** towards the interface.



Metal on n-type silicon ($\Phi_m > \Phi_s$) after contact. An electron in the metal sees the barrier $q\Phi_B$. An electron in the semiconductor sees the smaller built-in barrier qV_{bi}.

There are now **two barriers**, depending on which side the electron starts from:

seen from the metal (barrier height): $\Phi_B = \Phi_m - \chi$ (the ideal Schottky–Mott rule)

seen from the semiconductor (built-in): $V_{bi} = \Phi_m - \Phi_s = \Phi_B - (E_c - E_F)/q$

Example 11.2: ideal Schottky barrier

Question. Gold (Φ_m = 5.1 eV) is deposited on the n-type silicon of Example 11.1. Using the ideal Schottky–Mott model, find the barrier height seen from the metal and the built-in potential seen from the semiconductor. Is the contact rectifying or Ohmic?

Given: Φ_m = 5.1 eV, χ = 4.05 eV, Φ_s = 4.26 eV.

Find: Φ_B, V_{bi}, and the type of contact.

Solution.

$$\Phi_B = \Phi_m - \chi = 5.1 - 4.05 = 1.05 \text{ eV}$$

$$V_{bi} = \Phi_m - \Phi_s = 5.1 - 4.26 = 0.84 \text{ V}$$

$\Phi_m > \Phi_s$ on n-type silicon, so a depletion region forms: the contact is rectifying.

Answer: Φ_B = 1.05 eV, V_{bi} = 0.84 V: a rectifying (Schottky) contact. In practice Fermi-level pinning gives a measured Φ_B of about 0.8 eV for Au on n-Si.

(For p-type semiconductors everything flips: a rectifying contact needs $\Phi_m < \Phi_s$, and the barrier for holes is $\Phi_{Bp} = E_g/q - (\Phi_m - \chi)$.)

Fermi-level pinning (what really happens). Measured barriers depend much less on the metal than the simple rule predicts, because surface states at the interface (Week 9) hold ("pin") the Fermi level near a fixed energy. For example, gold on n-Si gives $\Phi_B \approx 0.8$ eV, not 1.05 eV; on n-GaAs almost every metal gives about 0.8 eV. In calculations, use the measured Φ_B if it is given.

How current flows: thermionic emission

In a Schottky diode the current is carried by **majority carriers** (electrons in n-type) jumping **over** the barrier. This is **thermionic emission**: only electrons with enough thermal energy can get over, so the current depends exponentially on the barrier height.

$$J = J_s \left(e^{qV/\eta kT} - 1 \right), \qquad J_s = A^{**} T^2 e^{-q\Phi_B/kT}$$

Symbol	Meaning
A^{**}	effective Richardson constant (about 110 A/cm^{2}K^2 for n-Si, 8 for n-GaAs)
T	temperature
Φ_B	barrier height
η	ideality factor, 1.0–1.1 for good diodes

The shape is the same as a p-n diode, but J_s is far larger.

Example 11.3: Schottky saturation current

Question. A Schottky diode on n-type silicon has a barrier height of 0.8 eV. Taking A^{**} = 110 A/cm^{2}K^2, calculate the saturation current density at 300 K and the forward voltage needed for 1 A/cm^2.

Given: Φ_B = 0.8 eV, A^{**} = 110 A/cm^{2}K^2, T = 300 K, kT = 0.0259 eV, J = 1 A/cm^2.

Find: J_s, and V at J = 1 A/cm^2.

Solution.

Step 1. $J_s = A^{**}T^2 e^{-q\Phi_B/kT}$:

$$J_s = 110 \times 300^2 \times e^{-0.8/0.0259} = 9.9\times10^6 \times 3.9\times10^{-14} = 3.8\times10^{-7}\ \text{A/cm}^2$$

Step 2. $V = (kT/q)\ln(J/J_s)$ for an ideal diode:

$$V = 0.0259 \times \ln(1/3.8\times10^{-7}) = 0.0259 \times 14.8 = 0.38\ \text{V}$$

Answer: $J_s = 3.8\times10^{-7}$ A/cm^2; 1 A/cm^2 flows at about 0.38 V, compared with about 0.7 V for a p-n diode, because J_s is about 10^5 times larger.

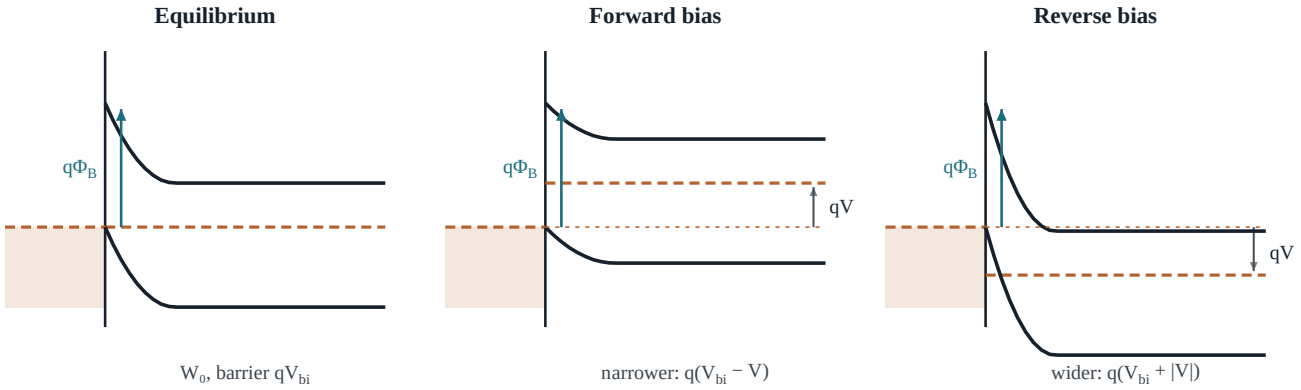
Schottky diode versus p-n diode

	Schottky diode	**p-n junction diode**
Current carried by	Majority carriers only	Minority carriers injected and diffusing
Mechanism	Thermionic emission over the barrier	Diffusion
Stored minority charge	Practically none	$Q = I\tau$
Switching speed	Very fast, no reverse-recovery delay	Slower (recovery time $\approx \tau$)
Turn-on voltage (Si)	0.2–0.4 V	0.6–0.7 V
Reverse leakage	Higher	Lower
Breakdown voltage	Lower	Higher
Uses	Fast rectifiers in power supplies, microwave mixers and detectors, clamps, MESFET gates	General rectification, high voltage, low leakage

Band diagrams under bias

Take the metal on n-type Si. The key rule: **the barrier seen from the metal, $q\Phi_B$, never changes**, because it is fixed by the interface. The applied voltage appears across the depletion region and changes only the barrier seen from the semiconductor.

	Equilibrium (V = 0)	Forward bias (metal +, V > 0)	Reverse bias (metal −, V < 0)
Fermi levels	One flat E_F	Semiconductor E_F raised by qV relative to the metal	Semiconductor E_F lowered by q|V|
Band bending	qV_{bi}	$q(V_{bi} - V)$, smaller	$q(V_{bi} + \|V\|)$, larger
Depletion width	$W_0 = \sqrt{(2\varepsilon_s V_{bi}/qN_D)}$	Narrower	Wider
Electrons semiconductor → metal	Balanced by metal → semiconductor	Increase by $e^{qV/kT}$	Suppressed
Electrons metal → semiconductor	Small, over Φ_B	Unchanged (Φ_B fixed)	Unchanged: this is the saturation current
Net current	0	$J_s(e^{qV/\eta kT} - 1)$, large	$\approx -J_s$, small



The same Schottky contact in equilibrium, forward bias and reverse bias. The barrier from the metal ($q\Phi_B$) stays fixed. Forward bias shrinks the semiconductor-side barrier to $q(V_{bi} - V)$; reverse bias raises it to $q(V_{bi} + |V|)$.

> **In plain words.** The metal side of the hill never changes height. Forward bias lowers the semiconductor side of the hill, so a flood of electrons pours over it into the metal. Reverse bias raises the semiconductor side, stopping that flow, and only the small trickle from the metal side remains.

Ohmic contacts

An **Ohmic contact** has a straight-line, symmetric I-V characteristic (it obeys Ohm's law) and a resistance negligible compared with the device. Current passes freely in both directions.

There are two ways to make one.

1. Choose a low-barrier metal. For n-type, use a metal with $\Phi_m < \Phi_s$. Now electrons flow from the **metal into the semiconductor** on contact, piling up an **accumulation layer** instead of a depletion layer. The bands bend **downward**, and there is no barrier for electrons in either direction. In practice Fermi-level pinning makes suitable metals rare for Si and GaAs, so the second method is used.

2. Dope heavily so that electrons tunnel (the practical method). Dope the semiconductor very heavily just under the contact (n^+ or p^+, above about 10^{19} cm^{-3}). A barrier still exists, but the depletion width $W = \sqrt{(2\varepsilon_s V_{bi}/qN_D)}$ shrinks to a few nanometres. A barrier that thin is no obstacle: electrons **tunnel** straight through it in both directions (quantum tunnelling, called field emission).

> **In plain words.** A wall that is high but paper-thin is no barrier to a quantum particle: it goes straight through. Heavy doping makes the barrier paper-thin.

The quality of an Ohmic contact is measured by its **specific contact resistance** ρ_c ($\Omega\cdot cm^2$), which falls very steeply as the doping increases. Good contacts have $\rho_c \approx 10^{-6}$–10^{-8} $\Omega\cdot cm^2$. Common choices: aluminium or a silicide ($TiSi_2$, $CoSi_2$, NiSi) on n^+ or p^+ silicon; an alloyed Au–Ge–Ni contact on n-GaAs.

Stability of contacts

Contacts must keep working through hot processing steps and years of operation:

- **Aluminium spiking:** silicon dissolves into aluminium and aluminium spikes into the silicon, which can short out a shallow junction. Prevented by adding 1 % Si to the Al, or by a barrier layer (TiN, TiW).
- **Electromigration:** at high current density the flowing electrons gradually push metal atoms along, forming voids and breaks. Reduced by Al–Cu alloys or copper wiring.
- **Interdiffusion and chemical reaction** at high temperature change the barrier height; silicides are used because they form a clean, stable interface.
- **Contamination and native oxide** at the interface raise ρ_c and change Φ_B.

Key points of Week 11

- $\Phi_s = \chi + (E_c - E_F)$. Metal on n-type with $\Phi_m > \Phi_s$ → Schottky barrier; $\Phi_m < \Phi_s$ (or heavy doping) → Ohmic.
- Schottky: depletion region, bands bend up, $\Phi_B = \Phi_m - \chi$ (ideally), $V_{bi} = \Phi_m - \Phi_s$.
- Current by thermionic emission of majority carriers: $J = A^{**}T^2 e^{-q\Phi_B/kT}(e^{qV/kT} - 1)$. Fast, low turn-on voltage, higher leakage.
- Under bias, $q\Phi_B$ is fixed; only the semiconductor-side barrier changes.
- Ohmic contacts: accumulation (low barrier) or tunnelling through a thin barrier made by heavy doping (n^+, p^+).

Examination question solved (Q49)

Q49. Differentiate between a Schottky barrier diode and an Ohmic contact. Explain their energy-band characteristics.

Both are metal–semiconductor junctions. The difference is whether a significant potential barrier to majority-carrier flow exists at the interface.

Schottky barrier diode (rectifying contact).

Condition: n-type semiconductor with $\Phi_m > \Phi_s$ (or p-type with $\Phi_m < \Phi_s$), moderate doping ($\leq 10^{17}$ cm^{-3}).

Energy-band characteristics:

1. Before contact, E_F in the n-semiconductor lies above E_F in the metal by $\Phi_m - \Phi_s$.
2. On contact, electrons flow from semiconductor to metal until the Fermi levels align. A **depletion region** of ionised donors forms in the semiconductor and the bands bend **upward** at the interface.

3. A barrier of height $\mathbf{q\Phi_B = q(\Phi_m - \chi)}$ opposes electron flow from metal to semiconductor; a built-in barrier $\mathbf{qV_{bi} = q(\Phi_m - \Phi_s)}$ opposes flow from semiconductor to metal. (In practice Φ_B is fixed by Fermi-level pinning, e.g. ≈ 0.8 eV for Au on n-Si.)
4. Forward bias (metal positive) lowers the semiconductor-side barrier to $V_{bi} - V$, and electrons are thermionically emitted into the metal: current rises exponentially. Reverse bias raises it to $V_{bi} + |V|$ while Φ_B stays the same, so only a small saturation current flows.

I-V: rectifying, $J = A^{**}T^2 e^{-q\Phi_B/kT}[e^{qV/\eta kT} - 1]$.

Features: majority-carrier device, negligible minority storage, very fast switching, low forward drop (0.2–0.4 V on Si), higher leakage than a p-n diode.

Ohmic contact.

Condition: either n-type with $\Phi_m < \Phi_s$ (p-type with $\Phi_m > \Phi_s$), or, in practice, any metal on a **very heavily doped** (n^+ or p^+, $> 10^{19}$ cm^{-3}) surface layer.

Energy-band characteristics:

1. **Low-barrier case:** electrons flow from the metal into the semiconductor on contact, forming an **accumulation layer**; the bands bend **downward**. There is no barrier for electrons in either direction.
2. **Tunnelling case:** a barrier exists, but heavy doping makes the depletion region only a few nanometres thick ($W \propto 1/\sqrt{N_D}$). Electrons **tunnel** through it freely in both directions (field emission).

I-V: linear and symmetric, $V = IR_c$, with very low specific contact resistance ($\rho_c \approx 10^{-6}$–10^{-8} $\Omega\cdot cm^2$).

Feature	Schottky barrier	Ohmic contact
Work-function condition (n-type)	$\Phi_m > \Phi_s$	$\Phi_m < \Phi_s$ (or heavy doping)
Semiconductor doping	Moderate (10^{15}–10^{17} cm^{-3})	Heavy ($> 10^{19}$ cm^{-3})
Band bending at interface (n-type)	Upward, depletion layer	Downward (accumulation) or very thin barrier
Barrier to majority carriers	Yes, $q\Phi_B$ and qV_{bi}	None, or transparent to tunnelling
Conduction mechanism	Thermionic emission	Free flow or field-emission tunnelling
I-V characteristic	Rectifying, exponential	Linear, symmetric
Resistance	Depends on bias and polarity	Small, constant
Purpose	Fast rectifier, detector, MESFET gate	Connecting devices to the circuit
Examples	Au, Pt, W, PtSi on n-Si; Ti/Au on n-GaAs	Al or silicide on n^+/p^+ Si; Au–Ge–Ni on n^+-GaAs

---

Practice questions

These extra questions practise this week's ideas. Work each one on paper before reading the solution; the calculation questions follow the same Question, Given, Find, Solution, Answer layout as the worked examples.

Practice 11.1: tungsten on n-type silicon

Question. Tungsten ($\Phi_m = 4.55$ eV) is deposited on n-type silicon with $N_D = 1\times10^{15}$ cm^{-3} ($\chi = 4.05$ eV). Using the ideal model, find Φ_B, Φ_s and V_{bi}.

Given: $\Phi_m = 4.55$ eV, $\chi = 4.05$ eV, $N_D = 10^{15}$ cm^{-3}, $N_c = 2.8\times10^{19}$ cm^{-3}.

Find: Φ_B, Φ_s, V_{bi}.

Solution.

$\Phi_B = \Phi_m - \chi = 4.55 - 4.05 = 0.50$ eV

$E_c - E_F = 0.02585 \ln(2.8\times10^{19}/10^{15}) = 0.265$ eV

$\Phi_s = 4.05 + 0.265 = 4.315$ eV

$V_{bi} = \Phi_m - \Phi_s = 4.55 - 4.315 = 0.235$ V

Answer: $\Phi_B = 0.50$ eV, $\Phi_s = 4.32$ eV, $V_{bi} = 0.24$ V.

Practice 11.2: choosing a metal for an Ohmic contact

Question. n-type silicon with $N_D = 1\times10^{16}$ cm^{-3} has $\Phi_s = 4.26$ eV. Using the ideal theory, which of Al (4.28), Ti (4.33), Au (5.1) would give an Ohmic contact? What is done in practice?

Given: $\Phi_s = 4.26$ eV; an Ohmic contact on n-type needs $\Phi_m < \Phi_s$.

Find: a suitable metal.

Solution. All three have $\Phi_m > \Phi_s$ (Al only just, by 0.02 eV), so none meets the ideal Ohmic condition; Au, with the largest Φ_m, would give the strongest barrier.

Answer: None, strictly. In practice the silicon under the contact is doped n^+ (above 10^{19} cm^{-3}) so the barrier becomes thin enough to tunnel through, and Al or a silicide is used.

Practice 11.3: Schottky forward voltage

Question. A Schottky diode on n-Si has $\Phi_B = 0.65$ eV. Find J_s at 300 K ($A^{**} = 110$ A/cm^{2}K^2) and the forward voltage at 10 A/cm^2.

Given: $\Phi_B = 0.65$ eV, $A^{**} = 110$ A/cm^{2}K^2, $T = 300$ K, $J = 10$ A/cm^2.

Find: J_s and V.

Solution.

$J_s = 110 \times 300^2 \times e^{-0.65/0.0259} = 9.9\times10^6 \times 1.26\times10^{-11} = 1.25\times10^{-4}$ A/cm^2

$V = 0.0259 \ln(10/1.25\times10^{-4}) = 0.0259 \times 11.29 = 0.29$ V

Answer: $J_s = 1.25\times10^{-4}$ A/cm^2, $V \approx 0.29$ V.

Practice 11.4: why heavy doping gives tunnelling

Question. Find the zero-bias depletion width under a contact with $V_{bi} = 0.6$ V for $N_D = 1\times10^{16}$ cm^{-3} and for $N_D = 1\times10^{20}$ cm^{-3}. Explain the result.

Given: $V_{bi} = 0.6$ V, $\varepsilon_s = 1.04\times10^{-12}$ F/cm.

Find: W for each doping.

Solution.

$W(10^{16}) = \sqrt{(2 \times 1.04\times10^{-12} \times 0.6 / (1.6\times10^{-19} \times 10^{16}))} = 2.8\times10^{-5}$ cm = 0.28 μm

$W(10^{20}) = W(10^{16})/\sqrt{10^4} = 2.8\times10^{-7}$ cm = 2.8 nm

Answer: 0.28 μm and 2.8 nm. A 2.8 nm barrier is thin enough for electrons to tunnel through, so the heavily doped contact is Ohmic even though the barrier is still there.

Practice 11.5: contact resistance of a small contact

Question. A contact has specific contact resistance $\rho_c = 1\times10^{-6}$ Ω·cm^2 and measures 1 μm × 1 μm. Find its resistance.

Given: $\rho_c = 10^{-6}$ Ω·cm^2, $A = (10^{-4}$ cm$)^2 = 10^{-8}$ cm^2.

Find: R_c.

Solution.

$R_c = \rho_c/A = 10^{-6}/10^{-8} = 100$ Ω

Answer: R_c = 100 Ω, large for a small transistor, which is why modern processes need ρ_c near 10^{-8} Ω·cm^2.

Practice 11.6: an Ohmic contact by accumulation

Question. Describe the band diagram of an ideal Ohmic contact formed by a low work-function metal ($\Phi_m < \Phi_s$) on n-type silicon.

Answer. Before contact the metal's Fermi level is higher than the semiconductor's. On contact, electrons flow from the metal into the semiconductor until the Fermi levels align. The extra electrons form an **accumulation layer** at the interface, and the semiconductor bands bend **downward** towards it. There is no depletion region and no barrier to electrons in either direction, so current flows freely both ways and the I-V characteristic is linear.

Objectives check

Use this table to confirm you have met this week's objectives. Cover the Answer column and answer each check question from memory. If you cannot, go back to the section in the last column before moving on.

#	Objective	Check yourself	Answer	Where it is taught
1	Define work function and electron affinity and relate them.	Write Φ_s in terms of χ.	$\Phi_s = \chi + (E_c - E_F)$.	The energies involved; Example 11.1
2	Explain how a Schottky barrier forms and calculate Φ_B and V_{bi}.	Al (4.28 eV) on n-Si (χ = 4.05 eV): find the ideal barrier height.	$\Phi_B = 4.28 - 4.05 = 0.23$ eV.	The Schottky barrier; Example 11.2
3	Explain thermionic emission and calculate Schottky diode currents.	Why does a Schottky diode switch faster than a p-n diode?	It is a majority-carrier device with no stored minority charge.	Current transport; Example 11.3
4	Draw Schottky band diagrams in equilibrium and under bias.	Which barrier stays constant under bias?	$q\Phi_B$, the barrier seen from the metal.	Band diagrams under bias
5	Explain the two ways of making Ohmic contacts.	Why does heavy doping make a contact Ohmic?	The depletion layer becomes so thin that electrons tunnel through it.	Ohmic contacts
6	Describe the stability problems of metal contacts.	What is electromigration?	Gradual movement of metal atoms by the flowing electrons, forming voids and opens.	Stability of contacts

Interface Characterisation and CAT 2

Plan: methods of characterisation (capacitance–voltage C-V profiling, current–voltage I-V analysis); Continuous Assessment Test 2.

Learning objectives

By the end of this week you should be able to:

1. Describe I-V measurement and extract I_s, η and R_s.
2. Explain how the Schottky barrier height is obtained from I-V data.
3. Explain the principle of C-V measurement.
4. Extract doping and built-in potential from a Mott–Schottky plot.
5. Explain doping profiling and the MOS C-V measurement.

The big picture

After a device is made, how do you know what is inside it? You cannot see the doping or the barrier height. But you can **measure electrically** and compare the results with the theory of Weeks 10–11. Two measurements do most of the work:

- **I-V (current against voltage)** tells you **how current flows**: the saturation current, how ideal the junction is, the series resistance, leakage and breakdown, and for a Schottky diode the barrier height.
- **C-V (capacitance against voltage)** tells you **how charge is arranged**: the doping concentration (and how it changes with depth) and the built-in potential.

The trick in both cases is to plot the data so that the theory predicts a **straight line**, then read the answers from the slope and the intercept.

I-V characterisation

The measurement

A **source-measure unit** (or a power supply with a series resistor and two meters) applies a voltage to the device in steps, from reverse to forward bias, and records the current at each step. Currents from picoamps to amps may need measuring, often on a probe station, sometimes at several temperatures.

Turning the curve into a straight line

For a real diode,

$$I = I_s \left(e^{qV/\eta kT} - 1 \right)$$

where η (the **ideality factor**) is 1 for a perfect diffusion-current diode and up to 2 when recombination in the depletion region dominates. Above about 0.1 V the -1 is negligible, so take the natural log:

$$\ln I = \ln I_s + (q/\eta kT)\, V$$

That is a straight line if you plot **ln I against V** (a semi-log plot):

- the **intercept** at V = 0 gives **ln I_s**, so the saturation current I_s;
- the **slope** is q/(ηkT), so **η = (q/kT) / slope**.

Example 12.1: ideality factor and saturation current from I-V data

Question. A silicon diode carries 0.9 μA at a forward voltage of 0.30 V and 40 μA at 0.40 V. Assuming both points lie on the straight part of the semi-log plot, find the ideality factor and the saturation current (300 K).

Given: I_1 = 0.9 μA at V_1 = 0.30 V; I_2 = 40 μA at V_2 = 0.40 V; kT/q = 0.0259 V.

Find: η and I_s.

Solution.

Step 1. Slope of ln I against V:

$$\text{slope} = \ln(I_2/I_1)/(V_2 - V_1) = \ln(40/0.9)/0.10 = 3.79/0.10 = 37.9\ \text{V}^{-1}$$

Step 2. Ideality factor, slope = q/(ηkT):

$$\eta = 1/(0.0259 \times 37.9) = 1.02$$

Step 3. Extrapolate back to V = 0:

$$I_s = I_1 / e^{\text{slope} \times V_1} = 0.9\times10^{-6} / e^{37.9 \times 0.30} = 0.9\times10^{-6} / 8.7\times10^{4} = 1.0\times10^{-11}\ \text{A}$$

Answer: η = 1.02 (almost ideal diffusion current), **$I_s \approx 1\times10^{-11}$ A.**

What else the I-V curve shows

- **Series resistance R_s:** at high current the curve bends away from the straight line, because part of the voltage is dropped across the resistance of the bulk and contacts: $V = V_{junction} + IR_s$. R_s = (extra voltage)/I.
- **Reverse leakage and breakdown voltage** from the reverse branch.
- **Schottky barrier height** from I_s: $\Phi_B = (kT/q)\ \ln(A{**}AT^2/I_s)$, where A is the diode area. Measuring at several temperatures and plotting $\ln(I_s/T^2)$ against 1/T (an **Arrhenius plot**) gives a straight line of slope $-q\Phi_B/k$.
- **Ohmic contacts:** the I-V between two contacts should be a straight line through the origin, the same in both directions.

Transistors are characterised the same way: MOSFET I_D–V_{GS} curves give the threshold voltage and mobility (Week 14, Practical 4), and BJT curves give the current gain.

C-V characterisation

The measurement

An **LCR meter** applies a small AC voltage (10–50 mV, typically at 1 MHz) on top of a DC bias, and measures the capacitance. The DC bias is stepped through a range of **reverse** voltages.

The theory: a capacitor whose plates move

A reverse-biased one-sided junction (p^+n) or Schottky diode has a depletion region with no mobile carriers, between two conducting regions. It is a **parallel-plate capacitor** whose plate separation is the depletion width W:

$$C = \varepsilon_s A / W$$

From Week 10, $W = \sqrt{(2\varepsilon_s(V_{bi} - V)/qN)}$, where N is the doping on the lightly doped side. **As the reverse voltage increases, W gets bigger and C gets smaller.** Substituting:

$$C = A \sqrt{(q\varepsilon_s N / 2(V_{bi} - V))}$$

Turning it into a straight line: the Mott–Schottky plot

Square both sides and turn upside down:

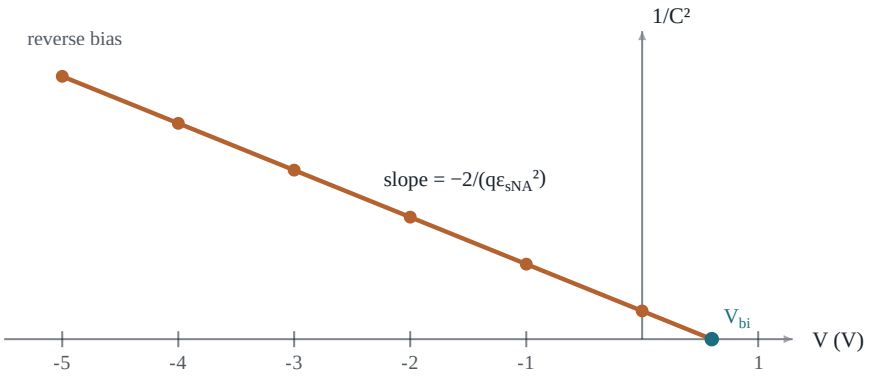
$$1/C^2 = 2(V_{bi} - V) / (q\varepsilon_s N A^2)$$

So if you plot **$1/C^2$ against V**, uniform doping gives a **straight line**:

- **Slope** $= -2/(q\varepsilon_s N A^2)$. Therefore

  $$N = 2 / (q\varepsilon_s A^2 \times |\text{slope}|)$$

- **Intercept** on the voltage axis (where $1/C^2 = 0$) is at $V = V_{bi}$: the **built-in potential**.



Mott–Schottky plot. For uniform doping the points fall on a straight line; the slope gives the doping and the voltage-axis intercept gives V_{bi}.

> **In plain words.** Reverse bias pushes the edge of the depletion region deeper into the semiconductor. How much extra charge is uncovered as the edge moves tells you how many dopant atoms are there. The capacitance measures exactly that.

Example 12.2: doping and built-in potential from a C-V plot

Question. A silicon Schottky diode of area 1×10^{-3} cm^2 gives a straight $1/C^2$ against V plot with slope -1.2×10^{21} F^{-2}V^{-1} that meets the voltage axis at +0.60 V. Find the doping concentration and the built-in potential. Also find the capacitance and depletion width at a reverse bias of 5 V.

Given: $A = 1\times10^{-3}$ cm^2, slope $= -1.2\times10^{21}$ F^{-2}V^{-1}, intercept 0.60 V, $\varepsilon_s = 1.036\times10^{-12}$ F/cm, $q = 1.602\times10^{-19}$ C, $V = -5$ V.

Find: N_D, V_{bi}, and C and W at −5 V.

Solution.

Step 1. Doping from the slope:

$$N_D = 2/(q\varepsilon_s A^2|\text{slope}|) = 2 / (1.602\times10^{-19} \times 1.036\times10^{-12} \times (10^{-3})^2 \times 1.2\times10^{21}) = 2/1.99\times10^{-16} = 1.0\times10^{16}\ \text{cm}^{-3}$$

Step 2. Built-in potential from the intercept: $V_{bi} = 0.60$ V

Step 3. At $V = -5$ V, $V_{bi} - V = 5.6$ V:

$$C = A\sqrt{(q\varepsilon_s N_D / 2(V_{bi} - V))} = 10^{-3} \times \sqrt{(1.66\times10^{-15} / 11.2)} = 1.22\times10^{-11}\ \text{F}$$

$$W = \varepsilon_s A/C = 1.036\times10^{-15} / 1.22\times10^{-11} = 8.5\times10^{-5}\ \text{cm}$$

Answer: $N_D = 1.0\times10^{16}$ cm^{-3}, $V_{bi} = 0.60$ V; at −5 V, C = 12.2 pF and W = 0.85 µm.

Doping profiles

If the doping is **not** uniform, the $1/C^2$ plot curves. The local slope at each bias still gives the doping at the edge of the depletion region, and the capacitance tells you where that edge is:

$$N(W) = 2 / (q\varepsilon_s A^2 \times d(1/C^2)/dV), \quad W = \varepsilon_s A/C$$

Sweeping the bias moves W deeper and maps out the **doping profile** N(x). This only works for one-sided junctions (p^+n, n^+p) or Schottky diodes, where the depletion region is all on one side.

MOS C-V

For a MOS capacitor (Week 14), the C-V curve shows accumulation ($C = C_{ox}$), depletion (C falling) and inversion. It gives the oxide thickness (from $C_{ox} = \varepsilon_{ox}A/t_{ox}$), the flat-band and threshold voltages, the substrate doping, the charge trapped in the oxide, and the density of interface traps. It is the main tool for checking the quality and stability of gate oxides.

Getting accurate C-V results

- Measure only in **reverse** bias (forward bias adds a large diffusion capacitance).
- Subtract stray and probe-pad capacitance (an open-circuit correction).
- Correct for series resistance.
- Keep the AC signal small.
- Choose the frequency so that deep traps do not respond (or so that they do, if you want to measure them).
- Know the device area accurately: N depends on $1/A^2$.

Key points of Week 12

- I-V: plot ln I against V. Intercept → I_s; slope → $\eta = (q/kT)/\text{slope}$; bending at high current → R_s; Arrhenius plot → Φ_B.
- C-V: a reverse-biased junction is a capacitor $C = \varepsilon_s A/W$, with W growing with reverse bias.
- Mott–Schottky: $1/C^2 = 2(V_{bi} - V)/(q\varepsilon_s N A^2)$. Slope → $N = 2/(q\varepsilon_s A^2|\text{slope}|)$; intercept → V_{bi}.
- Non-uniform doping: local slope gives N at depth $W = \varepsilon_s A/C$.
- MOS C-V gives t_{ox}, V_{FB}, V_T, substrate doping and oxide quality.

Examination question solved (Q50)

Q50. Explain the principles of I-V and C-V characterisation of semiconductor devices and describe how C-V measurements can be used to estimate doping concentration.

I-V characterisation: principle. A known DC voltage is applied across the device (swept in steps from reverse to forward bias) and the resulting current is measured with a source-measure unit, usually on a probe station and sometimes at several temperatures. Comparing the measured curve with device theory gives the transport parameters.

For a diode, $I = I_s[\exp(qV/\eta kT) - 1]$. From a semi-log plot of ln I versus V:

- the intercept gives the **saturation current I_s** (and, for Schottky diodes, the **barrier height** $\Phi_B = (kT/q)\ln(AA{*}{*}T^2/I_s)$);
- the slope gives the **ideality factor** η (1 for ideal diffusion current, approaching 2 when depletion-region recombination dominates);
- deviation at high current gives the **series resistance**;
- the reverse branch gives **leakage current** and **breakdown voltage**;
- for contacts, a linear symmetric I-V confirms an **Ohmic** contact and gives contact resistance.

I-V curves of transistors give threshold voltage, transconductance, output resistance, mobility and current gain, and repeated I-V measurement after stress is used to test **stability and reliability**.

C-V characterisation: principle. A reverse-biased p-n junction or Schottky barrier has a depletion region free of carriers between two conducting regions; it behaves like a parallel-plate capacitor with dielectric ε_s and plate separation W:

$$C = \varepsilon_s A / W$$

The DC reverse bias is swept, and the small-signal capacitance is measured with a small AC voltage (typically 1 MHz, a few tens of mV) using an LCR meter. Since W increases with reverse bias, C decreases.

Using C-V to estimate doping. For a one-sided abrupt junction (p^+n or Schottky on n-type) with uniform doping N:

$$W = \sqrt{(\,2\varepsilon_s(V_{bi} - V)/qN\,)} \quad \Rightarrow \quad C = A\sqrt{(\,q\varepsilon_s N / 2(V_{bi} - V)\,)}$$

Rearranging:

$$1/C^2 = 2(V_{bi} - V) / (q\varepsilon_s N A^2)$$

1. Measure C at a series of reverse biases V.
2. Plot **$1/C^2$ against V** (the Mott–Schottky plot). For uniform doping it is a **straight line**.
3. The **slope** is $-2/(q\varepsilon_s N A^2)$, so the doping concentration is **$N = 2/(q\varepsilon_s A^2 \cdot |\text{slope}|)$**.
4. The **intercept** on the V axis gives the built-in potential V_{bi} (and from it the Schottky barrier height).
5. **For non-uniform doping**, use the local slope: $N(W) = 2/(q\varepsilon_s A^2\, d(1/C^2)/dV)$ evaluated at each bias, and the depth $W = \varepsilon_s A/C$. This gives the doping profile N(x) as a function of depth.

Example. A diode of area 10^{-3} cm^2 on Si ($\varepsilon_s = 1.036\times10^{-12}$ F/cm) gives a $1/C^2$ slope magnitude of 1.2×10^{21} F^{-2}V^{-1}:

$$N = 2 / (1.602\times10^{-19} \times 1.036\times10^{-12} \times 10^{-6} \times 1.2\times10^{21}) = 1.0\times10^{16}\ \text{cm}^{-3}$$

Advantages and limits. C-V is non-destructive, fast and needs only a simple diode, but only measures the doping on the lightly doped side, only to depths reachable before breakdown, and with resolution limited by the Debye length.

Accuracy requires a correct area, correction for stray capacitance and series resistance, and a frequency at which deep traps do not respond.

CAT 2 practice test (60 minutes)

Covers Weeks 7–12. Practice questions, not the actual CAT. Use $q = 1.6\times10^{-19}$ C, $kT/q = 0.0259$ V, Si: $n_i = 1\times10^{10}$ cm^{-3}, $\varepsilon_s = 1.04\times10^{-12}$ F/cm.

1. An n-type Si bar ($N_D = 10^{16}$ cm^{-3}, $\mu_n = 1200$ cm^2/V·s) is 1 cm long with 10^{-3} cm^2 cross-section. Find its resistance and the current at 5 V.
2. Find the hole diffusion coefficient in Si if $\mu_p = 450$ cm^2/V·s, and the diffusion length if $\tau_p = 2$ μs.
3. Holes injected into n-Si fall from 10^{14} cm^{-3} at $x = 0$ with $L_p = 30$ μm. Find the hole diffusion current density at $x = 0$ ($D_p = 11.6$ cm^2/s).
4. A Si p^+n junction has $N_D = 5\times10^{15}$ cm^{-3} and $N_A = 10^{18}$ cm^{-3}. Find V_{bi}, the depletion width at zero bias and at −10 V, and the peak field at −10 V.
5. The diode of Q4 has area 10^{-3} cm^2 and $I_s = 1\times10^{-14}$ A. Find the current at 0.6 V and the voltage for 1 mA.
6. A Schottky diode on n-Si has $\Phi_B = 0.7$ eV. Find J_s at 300 K ($A^{**} = 110$ A/cm^{2}K^2) and the forward voltage for $J = 10$ A/cm^2.
7. A $1/C^2$ plot for a one-sided junction of area 2×10^{-3} cm^2 has slope magnitude 4.0×10^{20} F^{-2}V^{-1}. Find the doping.

CAT 2 solutions

1. Resistance of a bar

Question. An n-type Si bar ($N_D = 10^{16}$ cm^{-3}, $\mu_n = 1200$ cm^2/V·s) is 1 cm long with 10^{-3} cm^2 cross-section. Find its resistance and the current at 5 V.

Solution.

$\sigma = qN_D\mu_n = 1.6\times10^{-19} \times 10^{16} \times 1200 = 1.92$ S/cm, $\rho = 0.521$ Ω·cm

$R = \rho L/A = 0.521 \times 1 / 10^{-3} = 521$ Ω

$I = V/R = 5/521 = 9.6$ mA

2. Einstein relation and diffusion length

Question. Find the hole diffusion coefficient in Si if $\mu_p = 450$ cm^2/V·s, and the diffusion length if $\tau_p = 2$ μs.

Solution.

$D_p = (kT/q)\mu_p = 0.0259 \times 450 = 11.7$ cm^2/s

$L_p = \sqrt{(D_p\tau_p)} = \sqrt{(11.7 \times 2\times10^{-6})} = 4.83\times10^{-3}$ cm $= 48$ μm

3. Diffusion current at the injection point

Question. Holes injected into n-Si fall from 10^{14} cm^{-3} at $x = 0$ with $L_p = 30$ μm. Find the hole diffusion current density at $x = 0$ ($D_p = 11.6$ cm^2/s).

Solution.

$\Delta p(x) = \Delta p(0) e^{-x/L_p}$, so dp/dx at $x = 0$ is $-\Delta p(0)/L_p$.

$$J_p = -qD_p\, dp/dx = qD_p \Delta p(0)/L_p = 1.6\times10^{-19} \times 11.6 \times 10^{14} / 3\times10^{-3}$$
$$= 0.062 \text{ A/cm}^2$$

4. Junction parameters

Question. A Si p^+n junction has $N_D = 5\times10^{15}$ cm^{-3} and $N_A = 10^{18}$ cm^{-3}. Find V_{bi}, the depletion width at zero bias and at -10 V, and the peak field at -10 V.

Solution.

$$V_{bi} = 0.0259 \ln(10^{18} \times 5\times10^{15} / 10^{20}) = 0.0259 \ln(5\times10^{13}) = 0.0259 \times 31.54 = 0.817 \text{ V}$$

One-sided ($N_A \gg N_D$): $W = \sqrt{(2\varepsilon_s(V_{bi} - V)/qN_D)}$.

$$V = 0: \quad W = \sqrt{(2 \times 1.04\times10^{-12} \times 0.817 / (1.6\times10^{-19} \times 5\times10^{15}))} = \sqrt{(2.12\times10^{-9})} = 4.6\times10^{-5} \text{ cm} = 0.46 \text{ μm}$$
$$V = -10: \quad W = \sqrt{(2 \times 1.04\times10^{-12} \times 10.817 / 8\times10^{-4})} = \sqrt{(2.81\times10^{-8})} = 1.68\times10^{-4} \text{ cm} = 1.68 \text{ μm}$$
$$E_{max} = 2(V_{bi} - V)/W = 2 \times 10.817 / 1.68\times10^{-4} = 1.29\times10^{5} \text{ V/cm}$$

5. Diode current

Question. The diode of Q4 has area 10^{-3} cm^2 and $I_s = 1\times10^{-14}$ A. Find the current at 0.6 V and the voltage for 1 mA.

Solution.

$$I(0.6 \text{ V}) = 10^{-14} (e^{0.6/0.0259} - 1) = 10^{-14} \times 1.15\times10^{10} = 0.115 \text{ mA}$$
$$V(1 \text{ mA}) = 0.0259 \ln(10^{-3}/10^{-14} + 1) = 0.0259 \times 25.33 = 0.656 \text{ V}$$

6. Schottky diode

Question. A Schottky diode on n-Si has $\Phi_B = 0.7$ eV. Find J_s at 300 K ($A^{**} = 110$ A/cm^{2}K^2) and the forward voltage for $J = 10$ A/cm^2.

Solution.

$$J_s = A^{**}T^2 e^{-q\Phi_B/kT} = 110 \times 9\times10^{4} \times e^{-0.7/0.0259} = 9.9\times10^{6} \times 1.82\times10^{-12} = 1.8\times10^{-5} \text{ A/cm}^2$$
$$V = 0.0259 \ln(10/1.8\times10^{-5}) = 0.0259 \times 13.23 = 0.343 \text{ V}$$

A p-n diode needs about 0.7 V for a similar current density, which is why Schottky diodes are used where a low forward drop matters.

7. Doping from C-V

Question. A $1/C^2$ plot for a one-sided junction of area 2×10^{-3} cm^2 has slope magnitude 4.0×10^{20} F^{-2}V^{-1}. Find the doping.

Solution.

$$N = 2/(q\varepsilon_s A^2 |\text{slope}|) = 2 / (1.6\times10^{-19} \times 1.04\times10^{-12} \times (2\times10^{-3})^2 \times 4.0\times10^{20})$$
$$= 2 / (2.66\times10^{-16}) = 7.5\times10^{15} \text{ cm}^{-3}$$

Practice questions

These extra questions practise this week's ideas. Work each one on paper before reading the solution; the calculation questions follow the same Question, Given, Find, Solution, Answer layout as the worked examples.

Practice 12.1: ideality factor

Question. A diode carries 1 mA at 0.55 V and 10 mA at 0.62 V. Find its ideality factor.

Given: a decade of current over $\Delta V = 0.07$ V, $kT/q = 0.0259$ V.

Find: η.

Solution.

$$\text{slope} = \ln 10 / 0.07 = 32.9\ \text{V}^{-1}$$

$$\eta = 1/(0.0259 \times 32.9) = 1.17$$

Answer: $\eta \approx 1.17$: some depletion-region recombination current is present.

Practice 12.2: series resistance

Question. At 50 mA a diode's measured voltage is 0.80 V, while the straight line of the semi-log plot predicts 0.70 V at that current. Find the series resistance.

Given: $I = 50$ mA, $\Delta V = 0.80 - 0.70 = 0.10$ V.

Find: R_s.

Solution.

$$R_s = \Delta V / I = 0.10/0.05 = 2\ \Omega$$

Answer: $R_s = 2\ \Omega$.

Practice 12.3: barrier height from the saturation current

Question. A Schottky diode of area 1×10^{-4} cm^2 has $I_s = 2\times10^{-9}$ A at 300 K. Find the barrier height ($A^{**} = 110$ A/cm^{2}K^2).

Given: $I_s = 2\times10^{-9}$ A, $A = 10^{-4}$ cm^2, $A^{**} = 110$ A/cm^{2}K^2, $T = 300$ K.

Find: Φ_B.

Solution.

$$\Phi_B = (kT/q)\ln(A^{**}AT^2/I_s) = 0.0259 \times \ln(110 \times 10^{-4} \times 9\times10^4 / 2\times10^{-9})$$

$$= 0.0259 \times \ln(4.95\times10^{11}) = 0.0259 \times 26.93 = 0.70\ \text{eV}$$

Answer: $\Phi_B \approx 0.70$ eV.

Practice 12.4: doping from two C-V points

Question. A one-sided silicon junction of area 1×10^{-3} cm^2 has C = 20 pF at −1 V and 12 pF at −4 V. Assuming uniform doping, find N and V_{bi}.

Given: $C(-1\ \text{V}) = 20$ pF, $C(-4\ \text{V}) = 12$ pF, $A = 10^{-3}$ cm^2, $\varepsilon_s = 1.04\times10^{-12}$ F/cm.

Find: N and V_{bi}.

Solution.

$1/C^2$ at -1 V $= 1/(20\times10^{-12})^2 = 2.50\times10^{21}$ F^{-2}

$1/C^2$ at -4 V $= 1/(12\times10^{-12})^2 = 6.94\times10^{21}$ F^{-2}

$|\text{slope}| = (6.94 - 2.50)\times10^{21} / 3 = 1.48\times10^{21}$ $F^{-2}V^{-1}$

$N = 2/(q\varepsilon_s A^2|\text{slope}|) = 2/(1.6\times10^{-19} \times 1.04\times10^{-12} \times 10^{-6} \times 1.48\times10^{21}) = 8.1\times10^{15}$ cm^{-3}

$1/C^2 = |\text{slope}|(V_{bi} - V)$: $2.50\times10^{21} = 1.48\times10^{21}\,(V_{bi} + 1) \Rightarrow V_{bi} = 0.69$ V

Answer: $N = 8.1\times10^{15}$ cm^{-3}, $V_{bi} = 0.69$ V.

Practice 12.5: oxide thickness from MOS C-V

Question. A MOS capacitor of area 1×10^{-3} cm^2 shows a capacitance of 345 pF in accumulation. Find the oxide thickness.

Given: C = 345 pF, A = 10^{-3} cm^2, $\varepsilon_{ox} = 3.45\times10^{-13}$ F/cm.

Find: t_{ox}.

Solution.

$C_{ox} = C/A = 345\times10^{-12} / 10^{-3} = 3.45\times10^{-7}$ F/cm^2

$t_{ox} = \varepsilon_{ox}/C_{ox} = 3.45\times10^{-13} / 3.45\times10^{-7} = 1\times10^{-6}$ cm

Answer: $t_{ox} = 10$ nm.

Practice 12.6: why one-sided?

Question. Why is C-V profiling done on a p^+n (or n^+p) junction or a Schottky diode rather than a junction with similar doping on both sides?

Answer. The formula $N = 2/(q\varepsilon_s A^2|d(1/C^2)/dV|)$ assumes that the depletion region grows into **one** side only, so that the change in charge measures one doping. In a one-sided junction or Schottky diode, almost all of W lies in the lightly doped side, and the measurement gives that side's doping at depth W. If both sides were similarly doped, W would grow into both at once, and the result would be a mixture (an effective $N_A N_D/(N_A + N_D)$) that cannot be assigned to either side or to a depth.

Objectives check

Use this table to confirm you have met this week's objectives. Cover the Answer column and answer each check question from memory. If you cannot, go back to the section in the last column before moving on.

#	Objective	Check yourself	Answer	Where it is taught
1	Describe I-V measurement and extract I_s, η and R_s.	What plot gives a straight line for a diode, and what does its slope give?	ln I against V; slope = $q/(\eta kT)$.	I-V characterisation; Example 12.1
2	Explain how the Schottky barrier height is obtained from I-V data.	What is plotted in an Arrhenius plot of a Schottky diode?	$\ln(I_s/T^2)$ against $1/T$; slope = $-q\Phi_B/k$.	What else the I-V curve shows
3	Explain the principle of C-V measurement.	Why does the junction capacitance fall with reverse bias?	The depletion width W grows, and $C = \varepsilon_s A/W$.	C-V characterisation
4	Extract doping and built-in potential from a Mott–Schottky plot.	What gives N and what gives V_{bi} in a $1/C^2$ plot?	N from the slope, $N = 2/(q\varepsilon_s A^2$ | slope |); V_{bi} from the voltage intercept.	Mott–Schottky plot; Example 12.2
5	Explain doping profiling and the MOS C-V measurement.	Where is the doping being measured at each bias?	At the depletion edge, depth $W = \varepsilon_s A/C$.	Doping profiles; MOS C-V

Bipolar Junction Transistors (BJTs)

Plan: operational physics; minority-carrier profiles in the active regions; emitter efficiency, base transport factor and current gains (α, β). This page follows your lecturer's Weeks 13–15 note (sections 13.1–13.13) and solves its review questions.

Learning objectives

By the end of this week you should be able to:

1. Explain the structure and operating principle of NPN and PNP transistors.
2. Describe injection, diffusion, recombination and collection in the forward-active region.
3. Sketch and explain the minority-carrier profiles.
4. Define emitter injection efficiency and base transport factor.
5. Derive α and β and relate them to γ and α_T.
6. Solve problems on transistor currents and gains.

The big picture

A transistor is a device in which a **small current or voltage controls a large current**. That is all amplification is.

In a BJT the trick is geometry. Two p-n junctions (Week 10) are built back to back, with a **very thin** middle layer. One junction pushes carriers into the thin layer; the other junction, a fraction of a micrometre away, pulls them out the other side. Because the middle layer is so thin, almost every carrier that goes in comes out at the far side. A small current into the middle layer (the base) is enough to keep this going, while a large current flows straight through. The physics is the minority-carrier diffusion you met in Weeks 7–9.

13.1 Introduction and BJT structure

A **bipolar junction transistor (BJT)** is a three-terminal semiconductor device in which **both electrons and holes** take part in current conduction. That is why it is called "bipolar". The three terminals are the **emitter (E)**, **base (B)** and **collector (C)**. A BJT contains two p-n junctions, arranged either as **NPN** or **PNP**.

Region	Typical doping	Main function
Emitter	Heavily doped (n^+ in an NPN)	Injects majority carriers into the base
Base	Lightly doped and very thin	Controls carrier transport and limits recombination
Collector	Moderately doped	Collects carriers and supports the voltage

Why each choice?

- The **emitter is heavily doped** so that, when its junction is forward biased, it can inject a large concentration of carriers into the base, and so that the base injects very few carriers back into it.
- The **base is very thin** (often well under 1 μm) and lightly doped, so that most injected carriers cross it before they have time to recombine.

- The **collector is moderately doped** and physically larger, so that it can collect the carriers and withstand a relatively high reverse voltage without breaking down.

13.2 BJT biasing and operational physics

For normal **forward-active** operation of an NPN transistor:

- the **emitter–base junction is forward biased**;
- the **collector–base junction is reverse biased**.

What happens, step by step:

1. **Injection.** The forward-biased E–B junction lowers its barrier. Electrons flow from the n-type emitter into the p-type base. In the base they are **minority carriers**.
2. **Diffusion across the base.** There are many injected electrons at the emitter side of the base and almost none at the collector side, so they **diffuse** across. Because the base is thin, most of them get across.
3. **Recombination (a small loss).** A small fraction of the electrons meet holes in the base and recombine.
4. **Collection.** The electrons that reach the collector side are swept into the collector by the strong electric field of the reverse-biased C–B junction. They form most of the collector current.
5. **Base current.** The base current is mainly associated with (a) recombination in the base, which must be replaced by holes flowing in through the base lead, and (b) holes injected from the base into the emitter.

For a **PNP** transistor the carrier polarities are reversed: holes are injected from the p-type emitter into the n-type base and are collected by the p-type collector. All currents and voltages change sign.

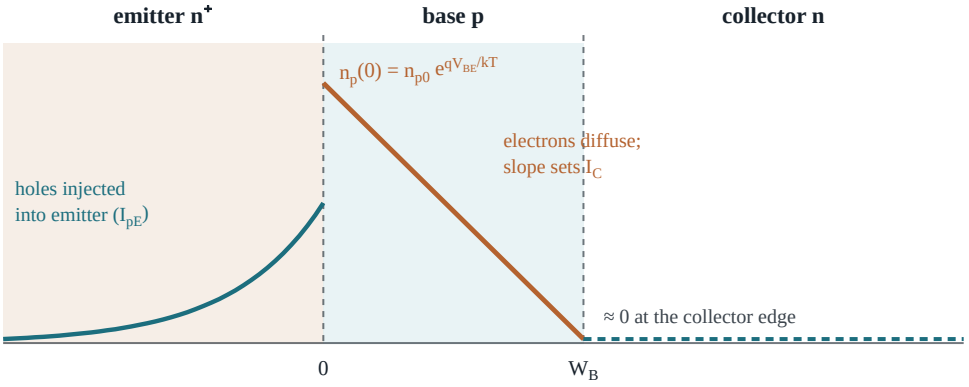
> **In plain words.** The emitter is a crowd pushing through a door (the forward-biased junction) into a narrow corridor (the base). At the other end of the corridor is a slide (the reverse-biased junction) that carries everyone who reaches it away. The corridor is so short that nearly everyone makes it. The base current is the small trickle of people who get lost in the corridor, plus a few who wander back the wrong way.

13.3 Minority-carrier concentration profiles

In a forward-active NPN transistor, electrons are minority carriers in the p-type base. Their concentration is **highest at the emitter side** of the base (because of injection) and **almost zero at the collector side** (because the collector removes them as fast as they arrive). In a thin, uniformly doped base with little recombination, the profile in between is approximately a **straight line**:

$$n_B(x) \approx n_B(0) - [\, n_B(0) - n_B(W_B) \,]\, x/W_B$$

Symbol	Meaning
W_B	neutral base width
x	distance measured from the emitter side of the base
$n_B(0)$	minority-electron concentration at the emitter edge of the base, $n_B(0) = n_{p0}\, e^{qV_{BE}/kT}$
$n_B(W_B)$	minority-electron concentration at the collector edge, ≈ 0



Minority carriers in an NPN transistor in the forward-active mode. Electrons in the base fall almost linearly from $n_B(0)$ to about zero at the collector edge. Holes injected back into the emitter decay away from the junction.

The concentration gradient drives electron **diffusion** through the base. By Fick's law the electron diffusion current density is

$$J_n = q\, D_n\, (dn/dx)$$

where q is the magnitude of the electronic charge and D_n the electron diffusion coefficient (the sign depends on the direction you take as positive). For the straight-line profile, $dn/dx = -n_B(0)/W_B$, so

$$|J_n| = q\, D_n\, n_B(0) / W_B$$

This is the collector current density. It is controlled by V_{BE} through $n_B(0) = n_{p0}\, e^{qV_{BE}/kT}$, and it is larger for a thinner base.

At the collector–base junction, the electric field in the reverse-biased depletion region rapidly sweeps the carriers into the collector. Because the collector only has to collect whatever arrives, the collector current is **almost independent of the collector voltage** in the active region, provided the transistor stays away from breakdown and saturation. This is what makes the BJT behave like a current source.

13.4 Emitter injection efficiency γ

The emitter current is made of two parts:

- $I_{E,n}$: electrons injected from emitter into base (useful: these become the collector current);
- $I_{E,p}$: holes injected from base into emitter (wasted: they only add to base current).

Emitter injection efficiency γ (gamma) measures what fraction of the emitter current is the useful kind:

$$\gamma = I_{E,n} / (I_{E,n} + I_{E,p})$$

A well-designed transistor has γ very close to 1. The electrons injected into the base are limited by the base doping, and the holes injected into the emitter by the emitter doping: the more heavily a side is doped, the fewer carriers the other side can inject into it. This gives

$$\gamma \approx 1 / [\, 1 + (D_E\, N_B\, W_B) / (D_B\, N_E\, W_E)\,]$$

where N_E and N_B are the emitter and base dopings, W_E and W_B their widths, D_E and D_B the minority diffusion coefficients. Making $\mathbf{N_E \gg N_B}$ makes the fraction small and γ close to 1.

Factors that improve emitter efficiency:

- heavy emitter doping relative to base doping;
- appropriate semiconductor material and junction design (for example a wider-gap emitter in a heterojunction transistor);
- proper control of emitter and base widths;
- minimising unwanted carrier injection.

13.5 Base transport factor α_T

Of the electrons injected into the base, a few recombine before reaching the collector. The **base transport factor** α_T is the fraction that make it:

$$\alpha_T = I_{C,n} / I_{E,n}$$

If recombination in the base is negligible, α_T approaches 1. If the base is thick, or the minority-carrier lifetime is short, more carriers recombine and α_T falls. Quantitatively,

$$\alpha_T \approx 1 - W_B^2 / (2L_n^2), \qquad L_n = \sqrt{(D_n \tau_n)}$$

The transport factor depends on base width W_B, the minority-carrier diffusion coefficient D_n and the minority-carrier lifetime τ_n. **A thin base and a long lifetime give α_T close to one.** For example $W_B = 1$ μm and $L_n = 20$ μm give $\alpha_T = 1 - 1/800 = 0.99875$.

13.6 Common-base current gain α

The **common-base current gain** is the ratio of collector current to emitter current:

$$\alpha = I_C / I_E$$

The collector current is slightly less than the emitter current (a little is lost to the base), so α is always less than 1, but for a good transistor it is very close to 1, typically **0.98–0.999**.

Physically, a current must survive two losses to get from emitter to collector: it must be the right kind of carrier (γ) and it must survive the trip (α_T). So

$$\alpha \approx \gamma \, \alpha_T$$

Example 13.1: α from γ and α_T

Question. A transistor has an emitter injection efficiency of 0.995 and a base transport factor of 0.990. Estimate its common-base current gain.

Given: $\gamma = 0.995$, $\alpha_T = 0.990$.

Find: α.

Solution.

$$\alpha \approx \gamma\alpha_T = 0.995 \times 0.990 = 0.98505$$

Answer: α ≈ 0.985.

13.7 Common-emitter current gain β

The **common-emitter current gain** β (also called h_{FE} under specified conditions) is the ratio of collector current to base current:

$$\beta = I_C / I_B$$

Deriving the relation between α and β. Kirchhoff's current law at the transistor: everything that enters at the emitter leaves at the collector or the base,

$$I_E = I_C + I_B$$

Start from α:

$$\alpha = I_C/I_E = I_C/(I_C + I_B)$$

Divide top and bottom by I_B, and use $\beta = I_C/I_B$:

$$\alpha = (I_C/I_B) / (I_C/I_B + 1) = \beta/(\beta + 1)$$

Solve for β: $\alpha(\beta + 1) = \beta \Rightarrow \alpha\beta + \alpha = \beta \Rightarrow \alpha = \beta(1 - \alpha) \Rightarrow$

$$\beta = \alpha / (1 - \alpha)$$

Because α is close to 1, the denominator $1 - \alpha$ is a small number, so **a small change in α produces a large change in β**:

α	**0.98**	**0.99**	**0.995**	**0.999**
$\beta = \alpha/(1 - \alpha)$	49	99	199	999

13.8 BJT current relationships

Quantity	**Expression**	**Meaning**
Emitter current	$I_E = I_C + I_B$	Total transistor current at the emitter
Common-base gain	$\alpha = I_C/I_E$	Collector current relative to emitter current
Common-emitter gain	$\beta = I_C/I_B$	Collector current relative to base current
Relation	$\beta = \alpha/(1 - \alpha)$	Converts α to β
Relation	$\alpha = \beta/(\beta + 1)$	Converts β to α
Physics of α	$\alpha \approx \gamma\alpha_T$	Injection efficiency × transport factor
Collector current	$I_C = I_S\, e^{qV_{BE}/kT}$	Set by the base–emitter voltage

13.9 Worked example 1: finding α

Question. A BJT has an emitter current of 5 mA and a collector current of 4.9 mA. Determine α and the base current.

Given: $I_E = 5$ mA, $I_C = 4.9$ mA. **Find:** α and I_B.

Solution.

$\alpha = I_C/I_E = 4.9/5 = 0.98$

$I_B = I_E - I_C = 5 - 4.9 = 0.1$ mA

$\alpha = 0.98$, $I_B = 0.1$ mA. (And $\beta = 4.9/0.1 = 49$.)

13.10 Worked example 2: finding β from α

Question. A transistor has $\alpha = 0.98$. Determine β. If the base current is 0.2 mA, determine the collector and emitter currents.

Given: $\alpha = 0.98$, $I_B = 0.2$ mA. **Find:** β, I_C, I_E.

Solution.

$\beta = \alpha/(1 - \alpha) = 0.98/0.02 = 49$

$I_C = \beta I_B = 49 \times 0.2$ mA $= 9.8$ mA

$I_E = I_C + I_B = 9.8 + 0.2 = 10.0$ mA

$\beta = 49$, $I_C = 9.8$ mA, $I_E = 10$ mA.

13.11 Worked example 3: injection efficiency and transport factor

Question. An NPN transistor has emitter injection efficiency $\gamma = 0.995$ and base transport factor $\alpha_T = 0.990$. Estimate α and β.

Given: $\gamma = 0.995$, $\alpha_T = 0.990$. **Find:** α and β.

Solution.

$\alpha \approx \gamma\alpha_T = 0.995 \times 0.990 = 0.98505$

$\beta = \alpha/(1 - \alpha) = 0.98505/0.01495 \approx 65.9$

$\alpha \approx 0.985$, $\beta \approx 65.9$. Notice that neither loss is large on its own (0.5 % and 1 %), yet together they hold β down to 66.

Worked example 4: from device design to gain

Question. An NPN silicon transistor has $N_E = 10^{19}$ cm^{-3}, $N_B = 10^{17}$ cm^{-3}, $W_E = 1$ μm, $W_B = 0.5$ μm, D_E (holes in the emitter) $= 2.5$ cm^2/s, D_B (electrons in the base) $= 25$ cm^2/s, base lifetime $\tau_n = 1$ μs and area 10^{-4} cm^2. Calculate the emitter injection efficiency γ, the base transport factor α_T, the current gains α and β, and the collector and base currents at $V_{BE} = 0.65$ V.

Given: the dopings, widths, diffusion coefficients and lifetime above; $n_i = 10^{10}$ cm^{-3}, $kT/q = 0.0259$ V. **Find:** γ, α_T, α, β, I_C and I_B.

Solution.

$\gamma = 1 / [1 + (2.5 \times 10^{17} \times 0.5)/(25 \times 10^{19} \times 1)] = 1/(1 + 5\times10^{-4}) = 0.99950$

$L_n = \sqrt{(25 \times 10^{-6})} = 5\times10^{-3}$ cm $= 50$ μm

$\alpha_T = 1 - 0.5^2/(2 \times 50^2) = 1 - 5\times10^{-5} = 0.99995$

$\alpha = \gamma\alpha_T = 0.99945, \quad \beta = 0.99945/0.00055 \approx 1800$

$I_S = qAD_B n_i^2/(N_B W_B) = (1.6\times10^{-19})(10^{-4})(25)(10^{20})/(10^{17} \times 5\times10^{-5}) = 8\times10^{-15}$ A

$I_C = 8\times10^{-15} \times e^{0.65/0.0259} = 8\times10^{-15} \times 7.9\times10^{10} = 0.63$ mA, $\quad I_B = I_C/\beta \approx 0.35$ μA

Here the emitter efficiency limits β. Real devices have lower β (typically 100–300) because of recombination in the E–B depletion region and other non-ideal effects.

13.12 BJT operating regions

Region	Emitter–base junction	Collector–base junction	Typical use
Cut-off	Reverse biased	Reverse biased	OFF (switching)
Forward active	Forward biased	Reverse biased	Amplification
Saturation	Forward biased	Forward biased	ON (switching)
Reverse active	Reverse biased	Forward biased	Limited, reverse operation

- **Cut-off:** no injection, so essentially no current flows: an open switch.
- **Forward active:** $I_C = \beta I_B$, controlled by the base: an amplifier.
- **Saturation:** both junctions inject; the collector can no longer collect everything independently, $I_C < \beta I_B$, and V_{CE} is small (about 0.2 V): a closed switch.
- **Reverse active:** emitter and collector swap roles; gain is poor because the collector was not designed to inject.

Early effect. In the active region, increasing V_{CE} widens the collector depletion region slightly into the base, making W_B smaller. A thinner base means a steeper profile and a larger I_C, so the output curves slope upward slightly. Extrapolated backwards they meet at $-V_A$, the **Early voltage**.

13.13 Summary of Week 13

- A BJT uses two p-n junctions and three semiconductor regions (emitter, base, collector).
- In forward-active operation the E–B junction is forward biased and the C–B junction reverse biased.
- The thin base lets most injected minority carriers reach the collector; their profile across the base is nearly a straight line.
- Emitter injection efficiency γ measures the desired carrier injection relative to the total emitter current.
- Base transport factor α_T measures the fraction of injected carriers that cross the base without recombining.
- $\alpha = I_C/I_E \approx \gamma\alpha_T$ is the common-base gain; $\beta = I_C/I_B$ is the common-emitter gain.
- The key relationship is $\boldsymbol{\beta = \alpha/(1 - \alpha)}$.

Review questions, Week 13, solved

1. Define a BJT and explain why it is called a bipolar device.

A bipolar junction transistor is a three-terminal semiconductor device consisting of two p-n junctions formed by three alternately doped regions, emitter, base and collector, arranged as NPN or PNP. A small base current (or base–emitter voltage) controls a much larger collector current, so it can amplify and switch. It is called **bipolar** because current is carried by both kinds of carrier: in an NPN, electrons carry the main emitter-to-collector current while holes carry the base current (recombination in the base and injection into the emitter). In contrast, a MOSFET is unipolar: its channel current is carried by one carrier type only.

2. Draw and label an NPN transistor and indicate the direction of conventional current and electron flow.

Draw three blocks side by side: **n^+ emitter | p base | n collector**, with terminal leads E, B and C. Mark the E–B junction "forward biased" (base positive with respect to emitter, $V_{BE} \approx 0.7$ V) and the C–B junction "reverse biased" (collector positive with respect to base).

- **Electron flow:** from the emitter, through the thin base, into the collector (left to right). A small number recombine with holes in the base; the holes they use up flow in through the base lead.
- **Conventional current:** opposite to electron flow. I_C flows **into** the collector, I_B flows **into** the base, and $I_E = I_C + I_B$ flows **out of** the emitter. In the circuit symbol, the arrow on the emitter points **outwards** for an NPN, showing the direction of conventional emitter current.

3. Explain the operation of a BJT in the forward-active region.

With the E–B junction forward biased, the emitter barrier is lowered and the heavily doped n^+ emitter injects a large number of electrons into the p-type base, where they become minority carriers with concentration $n_B(0) = n_{p0e}{}^{qV_{BE}/kT}$ at the emitter edge. The reverse-biased collector holds the electron concentration at the collector edge of the base at almost zero. The resulting nearly linear concentration gradient makes the electrons diffuse across the thin base. Since the base is much thinner than the diffusion length, only a small fraction recombine. On reaching the C–B depletion region, the electrons are swept into the collector by its electric field, forming the collector current $I_C = \alpha I_E$. The base supplies holes to replace those lost in recombination and those injected into the emitter, which is the small base current $I_B = I_C/\beta$. Because I_C depends on V_{BE} but hardly on V_{CB}, the transistor acts as a controlled current source, which allows amplification.

4. What is emitter injection efficiency? Why should it be close to unity?

Emitter injection efficiency $\gamma = I_{E,n}/(I_{E,n} + I_{E,p})$ is the fraction of the emitter current carried by carriers injected from the emitter into the base (electrons in an NPN), rather than by carriers injected from the base back into the emitter (holes). It should be close to unity because only the injected electrons can reach the collector and form useful output current; the back-injected holes simply add to the base current without contributing to I_C. Since $\alpha \approx \gamma\alpha_T$ and $\beta = \alpha/(1 - \alpha)$, even a small reduction of γ below 1 sharply reduces the current gain. γ is made close to 1 by doping the emitter much more heavily than the base.

5. Define the base transport factor and state two factors that affect it.

The base transport factor $\alpha_T = I_{C,n}/I_{E,n}$ is the fraction of minority carriers injected into the base that cross it without recombining and reach the collector. It is approximately $\alpha_T \approx 1 - W_B{}^2/(2L_n{}^2)$. Factors that affect it (any two):

1. **Base width W_B**: a thinner base gives fewer recombinations and a higher α_T.
2. **Minority-carrier lifetime τ_n** in the base: a longer lifetime (fewer traps, cleaner material) increases L_n and α_T.
3. **Minority-carrier diffusion coefficient D_n**: faster diffusion means less time in the base and higher α_T.
4. **Base doping**: heavier doping shortens lifetime and lowers mobility, reducing α_T.

6. Derive β = α/(1 − α) from the definitions of α and β.

Definitions: $\alpha = I_C/I_E$ and $\beta = I_C/I_B$. By Kirchhoff's current law, $I_E = I_C + I_B$.

$$\alpha = I_C/I_E = I_C/(I_C + I_B)$$

Dividing numerator and denominator by I_B:

$$\alpha = (I_C/I_B)/(I_C/I_B + 1) = \beta/(\beta + 1)$$

Rearranging: $\alpha(\beta + 1) = \beta \Rightarrow \alpha\beta + \alpha = \beta \Rightarrow \alpha = \beta - \alpha\beta = \beta(1 - \alpha)$, so

$$\beta = \alpha/(1 - \alpha)$$

7. A transistor has I_E = 12 mA and I_C = 11.76 mA. Calculate α, I_B and β.

$\alpha = I_C/I_E = 11.76/12 = 0.98$

$I_B = I_E - I_C = 12 - 11.76 = 0.24$ mA

$\beta = I_C/I_B = 11.76/0.24 = 49$

Check: $\alpha/(1 - \alpha) = 0.98/0.02 = 49$. ✓

8. Explain the difference between cut-off, active and saturation regions.

	Cut-off	**Forward active**	**Saturation**
E–B junction	Reverse (or zero) biased	Forward biased	Forward biased
C–B junction	Reverse biased	Reverse biased	Forward biased
Currents	$I_B \approx 0$, $I_C \approx 0$ (leakage only)	$I_C = \beta I_B$	$I_C < \beta I_B$; I_C set by the external circuit
V_{CE}	Large (≈ supply voltage)	Intermediate	Small, V_{CE}(sat) ≈ 0.1–0.3 V
Behaves as	Open switch (OFF)	Current-controlled current source	Closed switch (ON)
Use	Digital OFF state	Amplifiers	Digital ON state

In cut-off there is no injection, so no carriers cross the base. In the active region the collector collects all carriers that arrive, independently of its voltage. In saturation the collector junction is also forward biased and starts injecting carriers back into the base, so the collector current can no longer rise with base current; the transistor is fully ON.

Practice questions

These extra questions practise this week's ideas. Work each one on paper before reading the solution; the calculation questions follow the same Question, Given, Find, Solution, Answer layout as the worked examples.

Practice 13.1: currents from β

Question. A transistor has $\beta = 80$ and $I_C = 4$ mA. Find I_B, I_E and α.

Given: $\beta = 80$, $I_C = 4$ mA.

Find: I_B, I_E, α.

Solution.

$I_B = I_C/\beta = 4/80 = 0.05$ mA $= 50$ µA

$I_E = I_C + I_B = 4.05$ mA

$\alpha = \beta/(\beta + 1) = 80/81 = 0.988$

Answer: $I_B = 50$ µA, $I_E = 4.05$ mA, $\alpha = 0.988$.

Practice 13.2: from doping to gain

Question. An NPN transistor has $N_E = 5\times10^{18}$ cm^{-3}, $N_B = 5\times10^{16}$ cm^{-3}, equal emitter and base widths, $D_E = 3$ cm^2/s and $D_B = 20$ cm^2/s. Its base transport factor is 0.995. Find γ, α and β.

Given: N_E, N_B, D_E, D_B above, $W_E = W_B$, $\alpha_T = 0.995$.

Find: γ, α, β.

Solution.

$D_E N_B W_B/(D_B N_E W_E) = (3 \times 5\times10^{16})/(20 \times 5\times10^{18}) = 1.5\times10^{-3}$

$\gamma = 1/(1 + 1.5\times10^{-3}) = 0.9985$

$\alpha = \gamma\alpha_T = 0.9985 \times 0.995 = 0.9935$

$\beta = \alpha/(1 - \alpha) = 0.9935/0.0065 = 153$

Answer: $\gamma = 0.9985$, $\alpha = 0.9935$, $\beta \approx 153$.

Practice 13.3: base transport factor

Question. A BJT has a base width of 1 µm and an electron diffusion length in the base of 10 µm. Find α_T. What base width would give $\alpha_T = 0.999$?

Given: $W_B = 1$ µm, $L_n = 10$ µm.

Find: α_T, and W_B for $\alpha_T = 0.999$.

Solution.

$\alpha_T = 1 - W_B^2/(2L_n^2) = 1 - 1/(2 \times 100) = 0.995$

For 0.999: $W_B^2/(2L_n^2) = 0.001 \Rightarrow W_B = L_n\sqrt{0.002} = 10 \times 0.0447 = 0.45$ µm

Answer: $\alpha_T = 0.995$; a 0.45 µm base gives 0.999.

Practice 13.4: collector current and V_{BE}

Question. A transistor has $I_S = 1\times10^{-15}$ A. Find I_C at $V_{BE} = 0.70$ V, and the V_{BE} needed for $I_C = 5$ mA.

Given: $I_S = 10^{-15}$ A, $kT/q = 0.0259$ V.

Find: $I_C(0.70\text{ V})$ and $V_{BE}(5\text{ mA})$.

Solution.

$I_C = I_S\, e^{V_{BE}/0.0259} = 10^{-15} \times e^{27.03} = 10^{-15} \times 5.5\times10^{11} = 0.55$ mA

$V_{BE} = 0.0259 \ln(5\times10^{-3}/10^{-15}) = 0.0259 \times 29.24 = 0.757$ V

Answer: I_C = 0.55 mA at 0.70 V; V_{BE} = 0.76 V for 5 mA.

Practice 13.5: the Early effect

Question. A transistor has an Early voltage of 100 V and I_C = 1 mA at V_{CE} = 5 V. Find its output resistance and I_C at V_{CE} = 10 V (same base drive).

Given: V_A = 100 V, I_C(5 V) = 1 mA.

Find: r_o and I_C(10 V).

Solution.

$r_o = V_A/I_C = 100/10^{-3} = 100$ kΩ

$I_C(10\text{ V}) = I_C(5\text{ V}) \times (1 + 10/100)/(1 + 5/100) = 1 \times 1.10/1.05 = 1.048$ mA

Answer: r_o = 100 kΩ; I_C rises to 1.05 mA.

Practice 13.6: base transit time

Question. An NPN transistor has a 0.2 μm base and D_n = 20 cm^2/s. Find the base transit time and the upper limit it places on f_T.

Given: $W_B = 2\times10^{-5}$ cm, D_n = 20 cm^2/s.

Find: τ_B and f_T.

Solution.

$\tau_B = W_B^2/(2D_n) = (2\times10^{-5})^2/40 = 1\times10^{-11}$ s

$f_T \lesssim 1/(2\pi\tau_B) = 1.6\times10^{10}$ Hz

Answer: τ_B = 10 ps, f_T up to about 16 GHz. Thin bases make fast transistors.

Practice 13.7: why the reverse-active gain is poor

Question. Explain why a BJT has a much lower current gain in the reverse-active mode.

Answer. In reverse-active mode the collector acts as the emitter. But the collector is only moderately (lightly) doped and much larger, while the base is more heavily doped than it. The injection efficiency is therefore poor (the base injects many holes back into the collector), and the collector-side geometry collects carriers badly. The result is a reverse β of typically only 1–5.

Objectives check

Use this table to confirm you have met this week's objectives. Cover the Answer column and answer each check question from memory. If you cannot, go back to the section in the last column before moving on.

#	Objective	Check yourself	Answer	Where it is taught
1	Explain the structure and operating principle of NPN and PNP transistors.	Why is the base made very thin?	So that injected carriers cross it before they recombine.	13.1–13.2
2	Describe injection, diffusion, recombination and collection in the forward-active region.	Which junction is forward biased in forward-active operation?	The emitter–base junction (collector–base is reverse biased).	13.2
3	Sketch and explain the minority-carrier profiles.	What shape is the electron profile in a thin base, and what does its slope set?	Almost a straight line; its slope sets the collector current.	13.3
4	Define emitter injection efficiency and base transport factor.	How is γ made close to 1?	By doping the emitter much more heavily than the base.	13.4–13.5
5	Derive α and β and relate them to γ and α_T.	Write β in terms of α.	$\beta = \alpha/(1 - \alpha)$.	13.6–13.8
6	Solve problems on transistor currents and gains.	$\beta = 100$, $I_C = 5$ mA: find I_B and I_E.	$I_B = 50$ μA, $I_E = 5.05$ mA.	Worked examples 1–4; review Q7

Field-Effect Transistors (MOSFETs)

Plan: ideal MOS capacitor physics (accumulation, depletion, inversion); MOSFET threshold voltage; linear and saturation operating regions. This page follows your lecturer's note (sections 14.1–14.17) and solves its review questions 9–17.

Learning objectives

By the end of this week you should be able to:

1. Describe the structure of an ideal MOS capacitor and calculate C_{ox}.
2. Explain accumulation, depletion and inversion.
3. Explain how a MOS capacitor becomes a MOSFET, and enhancement versus depletion mode.
4. Define the threshold voltage and calculate it.
5. Determine the operating region and calculate drain current.
6. Compare BJTs and MOSFETs.

The big picture

A BJT is controlled by a current. A MOSFET is controlled by a **voltage**, and the control works through an electric field, without any current flowing into the control terminal.

The idea: put a metal plate (the **gate**) just above the silicon surface, separated by a very thin insulator. Put a positive voltage on the plate, and it attracts electrons to the silicon surface just beneath it, the same way a charged comb attracts bits of paper. With enough voltage, those electrons form a thin conducting sheet, a **channel**, connecting two contacts (source and drain). Remove the voltage and the channel disappears. The gate switches the channel on and off, and nothing flows into the gate. That is why billions of MOSFETs can sit on one chip without melting it.

14.1 Introduction to MOSFETs

A **metal-oxide-semiconductor field-effect transistor (MOSFET)** is a voltage-controlled semiconductor device. The **gate voltage** controls the conductivity of a channel between the **source** and **drain** terminals. The gate is separated from the semiconductor by a thin insulating oxide layer, so ideally the DC gate current is essentially zero.

The basic MOS structure is:

> gate (metal or polysilicon) / oxide (SiO_2 insulator) / semiconductor substrate

In an **n-channel** MOSFET the substrate is normally **p-type**, and an **electron inversion layer** forms when the gate voltage is sufficiently positive. In a **p-channel** MOSFET the polarities are reversed: n-type substrate, hole inversion layer, negative gate voltage.

14.2 The ideal MOS capacitor

A MOS capacitor can be viewed as a **parallel-plate capacitor**: one plate is the gate, the other is the semiconductor surface, and the oxide is the dielectric between them. The gate voltage creates an electric field through the oxide and

changes the distribution of charge near the semiconductor surface.

$$C_{ox} = \varepsilon_{ox} / t_{ox}$$

Symbol	Meaning
C_{ox}	oxide capacitance per unit area (F/cm^2)
ε_{ox}	oxide permittivity, $\varepsilon_{ox} = \varepsilon_r \varepsilon_0$ with $\varepsilon_r \approx 3.9$ for SiO_2: $\varepsilon_{ox} = 3.45\times10^{-13}$ F/cm
t_{ox}	oxide thickness

A **thinner oxide** gives a larger capacitance per unit area, so the same gate voltage induces more charge: stronger electrostatic control of the channel. For $t_{ox} = 10$ nm $= 10^{-6}$ cm, $C_{ox} = 3.45\times10^{-13}/10^{-6} = 3.45\times10^{-7}$ F/cm^2.

"**Ideal**" means: no charge trapped in the oxide, no trap states at the interface, no difference between the gate and semiconductor work functions, and an oxide that conducts no current at all.

For the p-type substrate we also need the **bulk Fermi potential** φ_F, which is how far the Fermi level lies below the intrinsic level in the bulk (from Week 5, $E_i - E_F = kT \ln(N_A/n_i)$):

$$\varphi_F = (kT/q) \ln(N_A/n_i)$$

For $N_A = 10^{17}$ cm^{-3}: $\varphi_F = 0.0259 \ln(10^7) = 0.417$ V.

14.3 Accumulation

Take a p-type silicon substrate and apply a **negative** gate voltage relative to it. The negative gate attracts the positive majority carriers, **holes**, towards the semiconductor–oxide interface. The surface now contains more holes than it does in equilibrium. This is **accumulation**.

- p-type substrate + negative gate voltage → hole accumulation.
- The surface majority-carrier concentration increases.
- The surface becomes more strongly p-type.
- In the band diagram the bands bend **upward** at the surface (E_v moves closer to E_F).

14.4 Depletion

Apply a **small positive** gate voltage. Holes are now **repelled** from the surface. The region near the oxide is depleted of mobile majority carriers. What remains is the fixed negative charge of the **ionised acceptor atoms**, which cannot move. This is the **depletion region**.

- p-type substrate + moderately positive gate voltage → depletion.
- Mobile holes are pushed away from the surface.
- A negative space-charge region of ionised acceptors remains, of width W_d.
- The bands bend **downward** at the surface by the surface potential ψ_s.

The depletion width grows with the band bending:

$$W_d = \sqrt{(2\varepsilon_{si} \psi_s / qN_A)}$$

14.5 Inversion

As the positive gate voltage is increased further, the surface potential becomes positive enough to attract **electrons** (the minority carriers of the p-type substrate) towards the surface. When the electron concentration at the surface becomes large enough that the surface behaves as **n-type**, the semiconductor is said to be **inverted**. This is **inversion**.

For an ideal p-type substrate, **strong inversion** is conventionally defined as the point where the surface is as strongly n-type as the bulk is p-type. That happens when the band bending is twice the bulk Fermi potential:

$$\psi_s \approx 2\varphi_F$$

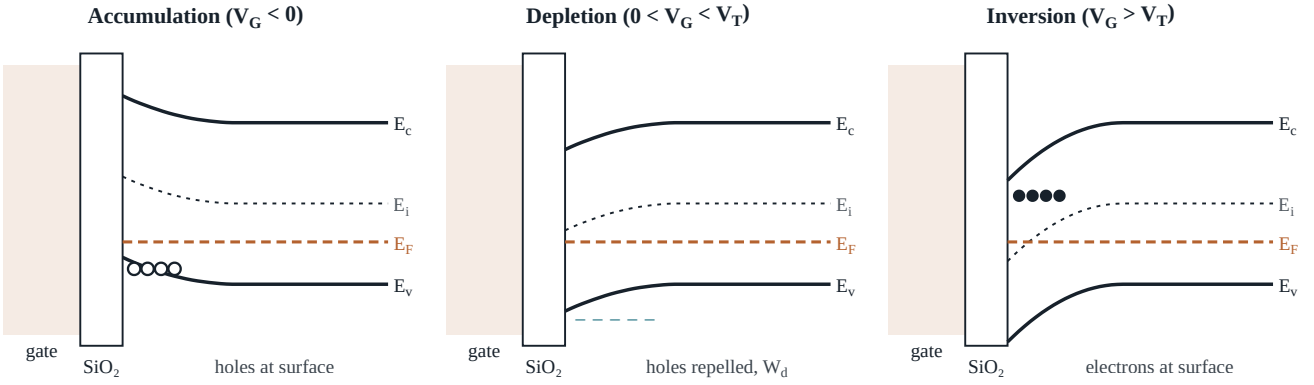
Why $2\varphi_F$? In the bulk, E_F is φ_F below E_i. Bending the bands down by φ_F brings E_i down to E_F (the surface is intrinsic). Bending by another φ_F puts E_i φ_F *below* E_F, which is exactly the mirror image of the bulk: n-type to the same degree.

Once strong inversion is reached, the depletion width stops growing at its maximum

$$W_{dmax} = \sqrt{(4\varepsilon_{si} \varphi_F / qN_A)}$$

and any further gate voltage simply adds electrons to the inversion layer. **This electron layer is the channel of the MOSFET.**

The exact sign conventions for φ_F and ψ_s differ between textbooks. The physical idea is what matters: the gate field first **depletes** the surface, then attracts minority carriers until an **inversion layer** forms.



Band diagrams of an ideal MOS capacitor on p-type silicon. Accumulation: bands bend up and holes gather at the surface. Depletion: bands bend down and holes are pushed away, leaving negative acceptor ions. Inversion: the bands bend so far that the surface becomes n-type.

14.6 From MOS capacitor to MOSFET

In an **n-channel enhancement MOSFET**, two heavily doped **n^+ regions** form the **source** and **drain** inside a p-type substrate. The gate lies above the region between them, which has length L (channel length) and width W.

- With V_{GS} below threshold there is no channel. Source and drain are two n^+ islands in p-type silicon, i.e. two back-to-back p-n junctions: no current flows.
- With a sufficiently positive gate-to-source voltage, an electron inversion channel forms and connects source to drain.
- Applying a drain-to-source voltage V_{DS} then makes electrons **drift** from source to drain. Conventional drain current flows from drain to source.

The gate voltage controls whether the channel exists and how strongly it conducts. This is the fundamental reason the MOSFET is a **field-effect** device.

Enhancement and depletion mode. In an **enhancement-mode** device (the normal kind) there is no channel at $V_{GS} = 0$; a gate voltage above V_T must be applied to create (enhance) one. A **depletion-mode** device is built with a channel already present at $V_{GS} = 0$ (by doping the surface n-type); a negative gate voltage is needed to remove (deplete) it, so its V_T is negative.

14.7 Threshold voltage

The **threshold voltage** V_T is the gate-to-source voltage at which a strong inversion channel forms and significant conduction begins in an enhancement-mode MOSFET.

A commonly used long-channel expression for an ideal nMOS is:

$$V_T = V_{FB} + 2\varphi_F + \gamma\sqrt{(2\varphi_F)}$$

with the **body-effect coefficient**

$$\gamma = \sqrt{(2q\varepsilon_{si} N_A)} / C_{ox}$$

Symbol	Meaning
V_{FB}	flat-band voltage: the gate voltage that makes the bands flat; zero for an ideal MOS, $V_{FB} = \Phi_{ms} - Q_f/C_{ox}$ in a real one
$2\varphi_F$	the band bending needed for strong inversion
$\gamma\sqrt{(2\varphi_F)}$	the voltage dropped across the oxide to support the depletion charge, $\lvert Q_{dmax}\rvert/C_{ox}$
N_A	substrate acceptor concentration
ε_{si}	silicon permittivity, $11.7\varepsilon_0 = 1.04\times10^{-12}$ F/cm

Reading the formula. The gate voltage has three jobs: get to flat bands (V_{FB}), bend the bands by $2\varphi_F$, and hold up the charge of the depletion region across the oxide. **Heavier substrate doping** raises both φ_F and the depletion charge, so V_T rises. A **thinner oxide** (larger C_{ox}) lowers V_T.

(This γ is not the same as the emitter injection efficiency γ of Week 13; the same Greek letter is used for different things.)

For simplified ideal MOSFET problems, the threshold voltage may simply be given. Always use the convention and parameters supplied in the question.

Example 14.1: calculating the threshold voltage

Question. An n-channel MOSFET is built on p-type silicon with $N_A = 1\times10^{17}$ cm^{-3}, a 10 nm SiO_2 gate oxide and an n^+ polysilicon gate, for which $V_{FB} = -0.98$ V. There is no oxide charge. Calculate the threshold voltage at 300 K.

Given: $N_A = 1\times10^{17}$ cm^{-3}, $t_{ox} = 10$ nm $= 1\times10^{-6}$ cm, $V_{FB} = -0.98$ V, $\varepsilon_{ox} = 3.45\times10^{-13}$ F/cm, $\varepsilon_{si} = 1.04\times10^{-12}$ F/cm, $n_i = 10^{10}$ cm^{-3}, $kT/q = 0.0259$ V.

Find: $V_T = V_{FB} + 2\varphi_F + \gamma\sqrt{(2\varphi_F)}$.

Solution.

Step 1. Bulk Fermi potential:

$$\varphi_F = 0.0259 \ln(10^{17}/10^{10}) = 0.417 \text{ V}, \quad 2\varphi_F = 0.834 \text{ V}$$

Step 2. Oxide capacitance:

$C_{ox} = 3.45\times10^{-13} / 10^{-6} = 3.45\times10^{-7}$ F/cm^2

Step 3. Body-effect coefficient:

$\gamma = \sqrt{(2 \times 1.6\times10^{-19} \times 1.04\times10^{-12} \times 10^{17})} / 3.45\times10^{-7} = 1.824\times10^{-7} / 3.45\times10^{-7} = 0.529$ V$^{1/2}$

$\gamma\sqrt{(2\varphi_F)} = 0.529 \times \sqrt{0.834} = 0.529 \times 0.913 = 0.483$ V

Step 4. Add the three terms:

$V_T = -0.98 + 0.834 + 0.483 = 0.34$ V

Answer: $V_T \approx 0.34$ V.

14.8 MOSFET regions of operation

For a long-channel enhancement nMOS transistor, the region depends on V_{GS} compared with V_T, and on V_{DS} compared with $V_{GS} - V_T$ (called the **overdrive voltage**).

Condition	Operating region	Physical condition
$V_{GS} < V_T$	Cut-off	No strong inversion channel
$V_{GS} > V_T$ and $V_{DS} < V_{GS} - V_T$	Linear (triode)	Channel exists along the entire length
$V_{GS} > V_T$ and $V_{DS} \geq V_{GS} - V_T$	Saturation	Channel pinches off near the drain

14.9 Cut-off region

When $V_{GS} \leq V_T$, an enhancement nMOS has no strong inversion channel. Ideally the drain current is zero, apart from small leakage:

$I_D \approx 0$

14.10 Linear (triode) region

When $V_{GS} > V_T$ and V_{DS} is smaller than $V_{GS} - V_T$, the channel exists all the way from source to drain. The long-channel drain current is:

$I_D = \mu_n C_{ox} (W/L) [(V_{GS} - V_T) V_{DS} - V_{DS}^2/2]$

Symbol	Meaning
μ_n	electron mobility in the channel
C_{ox}	oxide capacitance per unit area
W, L	channel width and length

Where it comes from (in words). The amount of electron charge in the channel per unit area is C_{ox} × (how far the gate is above threshold). At the source end that is $C_{ox}(V_{GS} - V_T)$; at the drain end the channel is at V_{DS}, so the gate is only $V_{GS} - V_{DS}$ above it and the charge is $C_{ox}(V_{GS} - V_T - V_{DS})$. The current is (charge per area) × (width) × (drift velocity μ × field). Adding this up along the channel gives the formula; the $-V_{DS}^2/2$ term is the effect of the channel thinning towards the drain.

For **small V_{DS}** the V_{DS}^2 term is negligible, and

$$I_D \approx \mu_n C_{ox}(W/L)(V_{GS} - V_T)\, V_{DS}$$

The current is proportional to V_{DS}: the transistor behaves as a **voltage-controlled resistor** whose resistance is set by the gate.

14.11 Saturation region

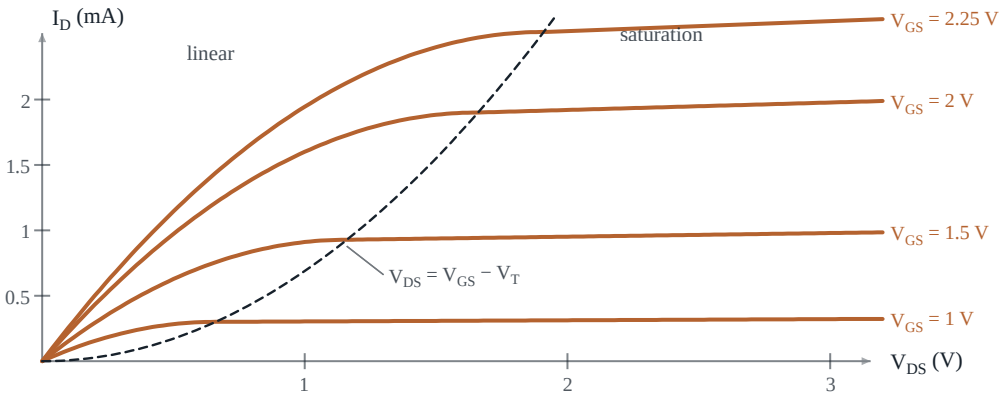
As V_{DS} is increased, the channel becomes progressively thinner near the drain. When

$$V_{DS} = V_{GS} - V_T$$

the channel charge at the drain end falls to zero: this is the **onset of pinch-off**. For larger V_{DS} the device is in **saturation**. The extra voltage is dropped across a short pinched-off region near the drain, and the current stops rising. Ignoring channel-length modulation:

$$I_D = \tfrac{1}{2}\,\mu_n\, C_{ox}\,(W/L)\,(V_{GS} - V_T)^2$$

This is one of the most important MOSFET equations for examination calculations. In saturation the current depends on V_{GS} but (ideally) not on V_{DS}: the MOSFET is a **voltage-controlled current source**, which is what an amplifier needs.



Output characteristics of an n-channel MOSFET. Each curve rises (linear region) until $V_{DS} = V_{GS} - V_T$ (dashed line), then flattens (saturation).

14.12 Transconductance

In saturation the small-signal **transconductance** is

$$g_m = \partial I_D / \partial V_{GS}$$

For the ideal long-channel saturation equation:

$$g_m = \mu_n C_{ox}\,(W/L)(V_{GS} - V_T) = 2I_D/(V_{GS} - V_T)$$

Transconductance measures how strongly the drain current responds to a small change in gate voltage (unit: siemens, A/V). It plays the role for a MOSFET that β plays for a BJT.

14.13 Worked example 1: determining the operating region

Question. An nMOS transistor has $V_T = 1$ V, $V_{GS} = 3$ V and $V_{DS} = 1$ V. Determine its operating region.

Given: $V_T = 1$ V, $V_{GS} = 3$ V, $V_{DS} = 1$ V. **Find:** the region of operation.

Solution.

$V_{GS} > V_T$ (3 > 1), so a channel exists.

$V_{GS} - V_T = 3 - 1 = 2$ V

$V_{DS} = 1$ V < 2 V

The transistor is in the **linear (triode) region**.

14.14 Worked example 2: saturation current

Question. An nMOS transistor has $\mu_n C_{ox}(W/L) = 2$ mA/V^2, $V_T = 1$ V and $V_{GS} = 4$ V. Find the saturation drain current.

Given: $k = \mu_n C_{ox}(W/L) = 2$ mA/V^2, $V_T = 1$ V, $V_{GS} = 4$ V. **Find:** I_D(sat).

Solution.

$I_D = \tfrac{1}{2} \times 2 \times (4 - 1)^2$ mA $= 1 \times 9 = 9$ mA

$I_D = 9$ mA.

14.15 Worked example 3: linear-region current

Question. An nMOS transistor has $\mu_n C_{ox}(W/L) = 1$ mA/V^2, $V_T = 1$ V, $V_{GS} = 3$ V and $V_{DS} = 0.5$ V. Calculate the drain current.

Given: $k = 1$ mA/V^2, $V_T = 1$ V, $V_{GS} = 3$ V, $V_{DS} = 0.5$ V. **Find:** I_D.

Solution.

First check the region: $V_{GS} - V_T = 2$ V > $V_{DS} = 0.5$ V, so it is linear.

$I_D = 1 \times [\,(3 - 1)(0.5) - (0.5)^2/2\,]$ mA $= [1 - 0.125]$ mA $= 0.875$ mA

$I_D = 0.875$ mA.

14.16 Comparison of BJT and MOSFET

Feature	BJT	MOSFET
Control	Mainly current-controlled (base current)	Voltage-controlled (gate voltage)
Input current	Requires base current	Ideally zero DC gate current
Carriers	Electrons and holes (bipolar)	Majority-carrier conduction in the channel (unipolar)
Input impedance	Moderate	Very high
Typical use	Amplification, switching, analogue	Digital logic, switching, amplification, power
Key gain parameter	β or α	g_m

14.17 Summary of Week 14

- An ideal MOS capacitor has three electrostatic conditions: accumulation, depletion and inversion.

- On a p-type substrate: negative gate → accumulation; moderate positive gate → depletion; sufficiently positive gate → inversion.
- The MOSFET uses the inversion layer as a controllable channel between source and drain.
- Threshold voltage is the gate voltage at the onset of strong inversion, $V_T = V_{FB} + 2\varphi_F + \gamma\sqrt{(2\varphi_F)}$.
- For an nMOS enhancement device: $V_{GS} < V_T$ gives cut-off; $V_{GS} > V_T$ and $V_{DS} < V_{GS} - V_T$ gives the linear region; $V_{GS} > V_T$ and $V_{DS} \geq V_{GS} - V_T$ gives saturation.
- Linear: $I_D = \mu C_{ox}(W/L)[(V_{GS} - V_T)V_{DS} - V_{DS}^2/2]$. Saturation: $I_D = \tfrac{1}{2}\mu C_{ox}(W/L)(V_{GS} - V_T)^2$.

Review questions, Week 14, solved

9. What is a MOS capacitor? Draw and label its basic structure.

A MOS (metal-oxide-semiconductor) capacitor is a three-layer structure in which a conducting gate electrode is separated from a semiconductor substrate by a thin insulating oxide. It acts as a parallel-plate capacitor whose capacitance per unit area is $C_{ox} = \varepsilon_{ox}/t_{ox}$ in accumulation; the gate voltage controls the charge at the semiconductor surface.

Sketch (top to bottom): a gate contact (metal or n^+ polysilicon) labelled **G**; a thin layer of **SiO_2** with thickness **t_{ox}** marked; the **p-type Si substrate** (doping N_A); a metal **substrate (body) contact** at the bottom labelled **B**. Mark V_G applied between G and B.

10. Explain accumulation, depletion and inversion for a p-type silicon substrate.

- **Accumulation ($V_G < 0$):** the negative gate attracts the majority holes to the oxide–silicon interface. The surface becomes more strongly p-type; the bands bend upward.
- **Depletion (small $V_G > 0$):** the positive gate repels holes from the surface, leaving a layer of fixed negative acceptor ions of width W_d. There are almost no mobile carriers in this layer; the bands bend downward.
- **Inversion ($V_G > V_T$):** the downward band bending becomes large enough ($\psi_s \approx 2\varphi_F$) to attract minority electrons to the surface in large numbers. The surface becomes n-type: an electron inversion layer forms, which is the MOSFET channel. The depletion width stops growing at W_{dmax}.

11. Why is an oxide layer used between the gate and the semiconductor?

1. It **insulates** the gate from the channel, so the gate draws essentially no DC current. This gives the MOSFET its very high input impedance and low power consumption.
2. It lets the gate control the surface through the **electric field** only, as the dielectric of a capacitor. A thin oxide gives a large C_{ox} and strong control.
3. SiO_2 forms a very **clean, stable interface** with silicon (few interface traps), so the surface carriers have good mobility and V_T is stable.
4. It **passivates** and protects the silicon surface.

12. Define threshold voltage.

The threshold voltage V_T is the gate-to-source voltage at which the semiconductor surface reaches strong inversion (surface potential $\psi_s \approx 2\varphi_F$), so that a conducting inversion channel forms between source and drain and significant drain current begins to flow. For an ideal nMOS, $V_T = V_{FB} + 2\varphi_F + \gamma\sqrt{(2\varphi_F)}$, with $\gamma = \sqrt{(2q\varepsilon_{si}N_A)}/C_{ox}$.

13. State the conditions for cut-off, linear and saturation operation of an nMOS transistor.

- **Cut-off:** $V_{GS} < V_T$. No channel, $I_D \approx 0$.
- **Linear (triode):** $V_{GS} > V_T$ and $V_{DS} < V_{GS} - V_T$. $I_D = \mu_n C_{ox}(W/L)[(V_{GS} - V_T)V_{DS} - V_{DS}^2/2]$.
- **Saturation:** $V_{GS} > V_T$ and $V_{DS} \geq V_{GS} - V_T$. $I_D = \tfrac{1}{2}\mu_n C_{ox}(W/L)(V_{GS} - V_T)^2$.

14. Derive the condition $V_{DS} \geq V_{GS} - V_T$ for saturation.

Physical argument. An inversion channel exists at a point in the channel only where the gate is more than V_T above the local channel voltage. At the source end the channel voltage is 0, so the gate–channel voltage is V_{GS}. At the drain end the channel voltage is V_{DS}, so the gate–channel voltage is $V_{GD} = V_{GS} - V_{DS}$. The channel at the drain end disappears (pinches off) when this falls to threshold:

$$V_{GS} - V_{DS} = V_T \quad \Rightarrow \quad V_{DS} = V_{GS} - V_T$$

For any V_{DS} larger than this, the drain end stays pinched off and the device is saturated. Hence saturation requires $V_{DS} \geq V_{GS} - V_T$.

Mathematical argument. The linear-region equation $I_D = k[(V_{GS} - V_T)V_{DS} - V_{DS}^2/2]$ is a parabola in V_{DS}. Its maximum is where

$$dI_D/dV_{DS} = k[(V_{GS} - V_T) - V_{DS}] = 0 \quad \Rightarrow \quad V_{DS} = V_{GS} - V_T$$

Beyond this point the equation would predict a falling current, which does not happen physically; instead the current stays at its maximum value, $I_D = \tfrac{1}{2}k(V_{GS} - V_T)^2$. So saturation begins at $V_{DS} = V_{GS} - V_T$.

15. Calculate the saturation current when $\mu_n C_{ox}(W/L) = 1.5$ mA/V^2, $V_{GS} = 5$ V and $V_T = 2$ V.

$$I_D = \tfrac{1}{2}\,\mu_n C_{ox}(W/L)(V_{GS} - V_T)^2 = \tfrac{1}{2} \times 1.5 \times (5 - 2)^2 = 0.75 \times 9 = 6.75 \text{ mA}$$

$I_D = 6.75$ mA.

16. Calculate the linear-region current for $\mu_n C_{ox}(W/L) = 2$ mA/V^2, $V_{GS} = 4$ V, $V_T = 1$ V and $V_{DS} = 0.8$ V.

Check the region: $V_{GS} - V_T = 3$ V $> V_{DS} = 0.8$ V, so linear.

$$I_D = 2 \times [\,(4 - 1)(0.8) - (0.8)^2/2\,] = 2 \times [2.4 - 0.32] = 2 \times 2.08 = 4.16 \text{ mA}$$

$I_D = 4.16$ mA.

17. Compare a BJT and a MOSFET in terms of input impedance and control mechanism.

- **Control mechanism.** A BJT is current-controlled: the collector current is set by the base current ($I_C = \beta I_B$), or equivalently by the base–emitter voltage through carrier injection across a forward-biased junction. A MOSFET is voltage-controlled: the gate voltage creates an electric field across the oxide that induces the channel charge, with no carrier injection from the gate.
- **Input impedance.** The BJT's base–emitter junction is forward biased and must be supplied with base current, so its input impedance is moderate (typically kΩ). The MOSFET's gate is insulated by the oxide and draws essentially no DC current, so its input impedance is extremely high (10^{12} Ω or more), limited only by the gate capacitance at high frequency.

Practice questions

These extra questions practise this week's ideas. Work each one on paper before reading the solution; the calculation questions follow the same Question, Given, Find, Solution, Answer layout as the worked examples.

Practice 14.1: oxide capacitance of a thin gate

Question. Find C_{ox} for a 3 nm SiO_2 gate oxide.

Given: $t_{ox} = 3\times10^{-7}$ cm, $\varepsilon_{ox} = 3.45\times10^{-13}$ F/cm.

Find: C_{ox}.

Solution.

$$C_{ox} = \varepsilon_{ox}/t_{ox} = 3.45\times10^{-13} / 3\times10^{-7} = 1.15\times10^{-6}\ \text{F/cm}^2$$

Answer: $C_{ox} = 1.15\ \mu\text{F/cm}^2$.

Practice 14.2: bulk potential and maximum depletion width

Question. For p-type silicon with $N_A = 5\times10^{16}$ cm^{-3}, find φ_F and the maximum depletion width at strong inversion.

Given: $N_A = 5\times10^{16}$ cm^{-3}, $n_i = 10^{10}$ cm^{-3}, $\varepsilon_{si} = 1.04\times10^{-12}$ F/cm.

Find: φ_F and W_{dmax}.

Solution.

$$\varphi_F = 0.0259 \ln(5\times10^{6}) = 0.400\ \text{V}$$

$$W_{dmax} = \sqrt{(4\varepsilon_{si}\varphi_F/qN_A)} = \sqrt{(4 \times 1.04\times10^{-12} \times 0.400 / (1.6\times10^{-19} \times 5\times10^{16}))} = 1.44\times10^{-5}\ \text{cm}$$

Answer: $\varphi_F = 0.40$ V, $W_{dmax} = 0.144\ \mu$m.

Practice 14.3: threshold voltage of an ideal MOSFET

Question. Using the substrate of Practice 14.2 with a 20 nm oxide and an ideal gate ($V_{FB} = 0$), find the threshold voltage.

Given: $\varphi_F = 0.400$ V, $W_{dmax} = 1.44\times10^{-5}$ cm, $t_{ox} = 2\times10^{-6}$ cm.

Find: $V_T = V_{FB} + 2\varphi_F + qN_{AWdmax}/C_{ox}$.

Solution.

$$C_{ox} = 3.45\times10^{-13}/2\times10^{-6} = 1.725\times10^{-7}\ \text{F/cm}^2$$

$$qN_{AWdmax} = 1.6\times10^{-19} \times 5\times10^{16} \times 1.44\times10^{-5} = 1.15\times10^{-7}\ \text{C/cm}^2$$

$$V_T = 0 + 0.80 + 1.15\times10^{-7}/1.725\times10^{-7} = 0.80 + 0.67 = 1.47\ \text{V}$$

Answer: $V_T \approx 1.47$ V. A thinner oxide or a real n^+ poly gate (negative V_{FB}) would lower it.

Practice 14.4: current in both regions

Question. An nMOS has $\mu_n C_{ox}(W/L) = 0.5$ mA/V^2 and $V_T = 0.8$ V, with $V_{GS} = 2.0$ V. Find I_D at $V_{DS} = 0.4$ V and at $V_{DS} = 2.0$ V.

Given: $k = 0.5$ mA/V^2, $V_T = 0.8$ V, $V_{GS} = 2.0$ V (overdrive 1.2 V).

Find: I_D at each V_{DS}.

Solution.

$V_{DS} = 0.4\ \text{V} < 1.2\ \text{V} \rightarrow$ linear: $I_D = 0.5 \times [1.2 \times 0.4 - 0.4^2/2] = 0.5 \times 0.40 = 0.20\ \text{mA}$

$V_{DS} = 2.0\ \text{V} \geq 1.2\ \text{V} \rightarrow$ saturation: $I_D = \tfrac{1}{2} \times 0.5 \times 1.2^2 = 0.36\ \text{mA}$

Answer: 0.20 mA (linear) and 0.36 mA (saturation).

Practice 14.5: transconductance

Question. Find the transconductance of the transistor of Practice 14.4 in saturation.

Given: $I_D = 0.36\ \text{mA}$, $V_{GS} - V_T = 1.2\ \text{V}$.

Find: g_m.

Solution.

$g_m = 2I_D/(V_{GS} - V_T) = 2 \times 0.36/1.2 = 0.60\ \text{mS}$

(check: $g_m = k(V_{GS} - V_T) = 0.5 \times 1.2 = 0.60\ \text{mS}$)

Answer: $g_m = 0.60$ mS.

Practice 14.6: body effect

Question. A MOSFET has $\gamma = 0.5\ \text{V}^{1/2}$ and $2\varphi_F = 0.8\ \text{V}$. By how much does its threshold voltage rise when the source is 2 V above the body?

Given: $\gamma = 0.5\ \text{V}^{1/2}$, $2\varphi_F = 0.8\ \text{V}$, $V_{SB} = 2\ \text{V}$.

Find: ΔV_T.

Solution.

$\Delta V_T = \gamma(\sqrt{(2\varphi_F + V_{SB})} - \sqrt{(2\varphi_F)}) = 0.5 \times (\sqrt{2.8} - \sqrt{0.8}) = 0.5 \times (1.673 - 0.894) = 0.39\ \text{V}$

Answer: V_T rises by 0.39 V.

Practice 14.7: sizing a transistor

Question. A MOSFET must carry 1 mA in saturation with an overdrive of 0.5 V. If $\mu_n C_{ox} = 200\ \mu\text{A/V}^2$, what W/L is needed?

Given: $I_D = 1\ \text{mA}$, $V_{GS} - V_T = 0.5\ \text{V}$, $\mu_n C_{ox} = 200 \times 10^{-6}\ \text{A/V}^2$.

Find: W/L.

Solution.

$W/L = 2I_D / (\mu_n C_{ox} (V_{GS} - V_T)^2) = 2 \times 10^{-3} / (200 \times 10^{-6} \times 0.25) = 40$

Answer: W/L = 40.

Practice 14.8: channel-length modulation

Question. Why does a real MOSFET's saturation current rise slightly with V_{DS}?

Answer. Beyond pinch-off, increasing V_{DS} moves the pinch-off point slightly towards the source, so the effective channel length becomes $L - \Delta L$. Since $I_D \propto 1/L$, the current rises a little. This is **channel-length modulation**, written $I_D = I_D(\text{sat})(1 + \lambda V_{DS})$. It is the MOSFET's equivalent of the BJT's Early effect and gives a finite output resistance $r_o \approx 1/(\lambda I_D)$.

Objectives check

Use this table to confirm you have met this week's objectives. Cover the Answer column and answer each check question from memory. If you cannot, go back to the section in the last column before moving on.

#	Objective	Check yourself	Answer	Where it is taught
1	Describe the structure of an ideal MOS capacitor and calculate C_{ox}.	Find C_{ox} for a 5 nm SiO_2 layer.	$3.45\times10^{-13}/5\times10^{-7} = 6.9\times10^{-7}$ F/cm^2.	14.2
2	Explain accumulation, depletion and inversion.	On p-type Si, what forms at the surface in inversion?	A layer of electrons (the surface becomes n-type).	14.3–14.5
3	Explain how a MOS capacitor becomes a MOSFET, and enhancement versus depletion mode.	Does an enhancement-mode nMOS conduct at $V_{GS} = 0$?	No: a gate voltage above V_T is needed to create the channel.	14.6
4	Define the threshold voltage and calculate it.	What three contributions make up V_T?	V_{FB}, the band bending $2\varphi_F$, and the depletion-charge term $\gamma\sqrt{(2\varphi_F)}$.	14.7; Example 14.1
5	Determine the operating region and calculate drain current.	$V_T = 1$ V, $V_{GS} = 2$ V, $V_{DS} = 3$ V: which region?	Saturation ($V_{DS} \geq V_{GS} - V_T = 1$ V).	14.8–14.15
6	Compare BJTs and MOSFETs.	Which has the higher input impedance, and why?	The MOSFET: its gate is insulated by the oxide and draws no DC current.	14.16; review Q17

Revision and Final Exam Preparation

Plan: overview of the entire curriculum, problem-solving clinic and past-question review. The sections follow the Week 15 headings in your lecturer's note: course overview, formula sheet, problem-solving clinic, past-question style exercises, final examination checklist and comprehensive revision questions.

Learning objectives

By the end of this week you should be able to:

1. Summarise the whole course and how its ideas connect.
2. Recall and apply the key formulas.
3. Answer definitions, derivations and calculations in examination style.
4. Solve mixed past-question style problems.
5. Check readiness for the final examination.

15.1 Course overview

The whole course tells one story: **how do we make, control and use the free charges in a crystal?**

Module	Weeks	The question it answers	The one idea to carry into the exam
1. Material physics and growth	1–3	What is a semiconductor, and how do we make and dope it?	Conductivity is controllable because of a moderate band gap and doping. Crystals are grown pure by CZ and epitaxy, and doped by diffusion (erfc or Gaussian) or implantation (Gaussian at R_p).
2. Bands and statistics	4–6	How many electrons and holes are there?	$n = N_{ce}^{-(E_c-E_F)/kT}$, $p = N_{ve}^{-(E_F-E_v)/kT}$, $np = n_i^2$. E_F moves 60 mV per decade of doping, and back to mid-gap when hot.
3. Transport and injection	7–9	How do the carriers move, and how long do extra ones last?	$J = qn\mu E + qD\,dn/dx$, $D/\mu = kT/q$. Excess minority carriers recombine with lifetime τ and spread over $L = \sqrt{(D\tau)}$.
4. Junctions and interfaces	10–12	What happens where two materials meet?	$V_{bi} = (kT/q)\ln(N_A N_D/n_i^2)$; $I = I_s(e^{qV/kT} - 1)$; Schottky $\Phi_B = \Phi_m - \chi$; $1/C^2$ against V gives the doping.
5. Devices	13–15	How do transistors work?	BJT: $\beta = \alpha/(1-\alpha)$, $\alpha = \gamma\alpha_T$. MOSFET: $V_T = V_{FB} + 2\varphi_F + \gamma\sqrt{(2\varphi_F)}$; $I_D(\text{sat}) = \tfrac{1}{2}\mu C_{ox}(W/L)(V_{GS} - V_T)^2$.

How the ideas connect. Doping (Week 3) sets n and p (Week 5). n and p set conductivity (Week 7). Joining p and n regions creates a built-in field (Week 10). Forward bias injects minority carriers that diffuse and recombine (Weeks 7–9), which gives the diode equation. Two junctions close together give the BJT (Week 13); a field-induced inversion layer gives the MOSFET (Week 14).

15.2 Formula sheet

Physical constants

Quantity	Symbol	Value
Electronic charge	q	1.602×10^{-19} C
Boltzmann constant	k	1.381×10^{-23} J/K = 8.617×10^{-5} eV/K
Thermal energy at 300 K	kT	0.0259 eV
Thermal voltage at 300 K	kT/q	0.0259 V (25.9 mV)
Planck constant	h	6.626×10^{-34} J·s
Reduced Planck constant	ħ	1.055×10^{-34} J·s
Free electron mass	m_0	9.109×10^{-31} kg
Permittivity of free space	ε_0	8.854×10^{-14} F/cm
Speed of light	c	3×10^{10} cm/s
Electron-volt	1 eV	1.602×10^{-19} J
Photon wavelength	λ	1.24/E (μm, with E in eV)

Material data at 300 K

	Si	Ge	GaAs
Crystal	Diamond	Diamond	Zinc blende
Lattice constant a (Å)	5.43	5.65	5.65
Atoms/cm^3	5.0×10^{22}	4.4×10^{22}	4.4×10^{22}
E_g (eV)	1.12 (indirect)	0.66 (indirect)	1.42 (direct)
n_i (cm^{-3})	1×10^{10}	2.4×10^{13}	2×10^{6}
N_c (cm^{-3})	2.8×10^{19}	1.04×10^{19}	4.7×10^{17}
N_v (cm^{-3})	1.04×10^{19}	6.0×10^{18}	7.0×10^{18}
μ_n (cm^2/V·s)	1350	3900	8500
μ_p (cm^2/V·s)	480	1900	400
D_n, D_p (cm^2/s)	35, 12.4	101, 49	220, 10
ε_r	11.7	16.0	12.9
χ (eV)	4.05	4.0	4.07
v_{sat} (cm/s)	1×10^{7}	6×10^{6}	≈ 1×10^{7} (peak 2×10^{7})
Melting point (°C)	1414	937	1238

Crystals and Miller indices

Miller indices: intercepts → reciprocals → clear fractions → (h k l)

$d_{hkl} = a / \sqrt{(h^2 + k^2 + l^2)}$ (cubic)

Diamond: 8 atoms/cell, coordination 4, $r_{nn} = (\sqrt{3}/4)a$, APF = 0.34

SC 1, BCC 2, FCC 4 atoms/cell; APF 0.52, 0.68, 0.74

Growth and doping

$SiO_2 + 2C \rightarrow Si + 2CO$; $Si + 3HCl \rightarrow SiHCl_3 + H_2$; $SiHCl_3 + H_2 \rightarrow Si + 3HCl$

Segregation: $k_0 = C_s/C_l$; $C_s = k_0 C_0 (1 - f)^{k_0 - 1}$

Fick 1: $F = -D\,\partial C/\partial x$; Fick 2: $\partial C/\partial t = D\,\partial^2 C/\partial x^2$; $D = D_0\, e^{-E_a/kT}$

Constant source: $C = C_s\, \mathrm{erfc}(x/2\sqrt{(Dt)})$; $Q = (2/\sqrt{\pi})C_s\sqrt{(Dt)}$

Limited source: $C = (Q/\sqrt{(\pi Dt)})\exp(-x^2/4Dt)$

Implant: $N(x) = (\Phi/\sqrt{(2\pi)}\Delta R_p)\exp(-(x - R_p)^2/2\Delta R_p^{\,2})$; $N_p \approx 0.4\Phi/\Delta R_p$; $\Phi = It/qA$

Bands and statistics

Schrödinger: $-(\hbar^2/2m)\, d^2\psi/dx^2 + V\psi = E\psi$

Kronig–Penney: $P\,\sin(\alpha a)/(\alpha a) + \cos(\alpha a) = \cos(ka)$

$1/m^* = (1/\hbar^2)\, d^2E/dk^2$

$f(E) = 1/(1 + e^{(E - E_F)/kT})$

$g_c(E) = (4\pi(2m_n^*)^{3/2}/h^3)\sqrt{(E - E_c)}$

$n = N_c\, e^{-(E_c - E_F)/kT}$; $p = N_v\, e^{-(E_F - E_v)/kT}$

$N_c = 2(2\pi m_n^* kT/h^2)^{3/2}$

$n_i = \sqrt{(N_c N_v)}\, e^{-E_g/2kT}$; $np = n_i^2$

$n = n_i\, e^{(E_F - E_i)/kT}$; $p = n_i\, e^{(E_i - E_F)/kT}$

$n = (N_D - N_A)/2 + \sqrt{(((N_D - N_A)/2)^2 + n_i^{\,2})}$

$E_F - E_i = kT\ln(N_D/n_i)$; $E_i - E_F = kT\ln(N_A/n_i)$

Transport

$v_d = \mu E$; $\mu = q\tau_m/m^*$; $1/\mu = 1/\mu_L + 1/\mu_I$

$J_{drift} = q(n\mu_n + p\mu_p)E$; $\sigma = q(n\mu_n + p\mu_p)$; $\rho = 1/\sigma$

$J_n = qn\mu_n E + qD_n\, dn/dx$; $J_p = qp\mu_p E - qD_p\, dp/dx$

Einstein: $D/\mu = kT/q$

$v_d = \mu E/[1 + (\mu E/v_{sat})^{\beta}]^{1/\beta}$

Recombination and continuity

SRH: $U = (np - n_i^2)/[\tau_p(n + n_1) + \tau_n(p + p_1)]$; $\tau = 1/(\sigma v_{th} N_t)$

Low-level, n-type: $U = \Delta p/\tau_p$; $\Delta p(t) = \Delta p(0)e^{-t/\tau_p}$

$\partial \Delta p/\partial t = D_p\, \partial^2 \Delta p/\partial x^2 - \mu_p E\, \partial \Delta p/\partial x + G_L - \Delta p/\tau_p$

$L = \sqrt{(D\tau)}$; $\Delta p(x) = \Delta p(0)e^{-x/L_p}$

Surface: $U_s = S\Delta p_s$; $S = \sigma v_{th} N_{st}$; $1/\tau_{eff} = 1/\tau_b + 2S/W$

Junctions and contacts

$V_{bi} = (kT/q)\ln(N_A N_D/n_i^2)$

$W = \sqrt{((2\varepsilon_s/q)(V_{bi} - V)(1/N_A + 1/N_D))}$; $N_A x_p = N_D x_n$; $E_{max} = qN_D x_n/\varepsilon_s$

$I = I_s(e^{qV/kT} - 1)$; $J_s = qn_i^2(D_p/(L_p N_D) + D_n/(L_n N_A))$

$C_j = \varepsilon_s A/W$; $1/C^2 = 2(V_{bi} - V)/(q\varepsilon_s N A^2)$; $N = 2/(q\varepsilon_s A^2 |d(1/C^2)/dV|)$

Schottky: $\Phi_B = \Phi_m - \chi$; $V_{bi} = \Phi_m - \Phi_s$; $J_s = A{**}T^2 e^{-q\Phi_B/kT}$

Ohmic (tunnelling): $N_D \gtrsim 10^{19}\ \text{cm}^{-3}$

Devices

BJT: $I_C = I_S\, e^{qV_{BE}/kT}$, $I_S = qAD_B n_i^2/(N_B W_B)$

$\gamma = 1/[1 + D_E N_B W_B/(D_B N_E W_E)]$, $\alpha_T \approx 1 - W_B^2/2L_n^2$

$\alpha = \gamma\alpha_T$, $\beta = \alpha/(1 - \alpha)$, $I_E = I_C + I_B$, $\tau_B = W_B^2/2D_B$

MOS: $C_{ox} = \varepsilon_{ox}/t_{ox}$, $\varphi_F = (kT/q)\ln(N_A/n_i)$, $W_{dmax} = \sqrt{(4\varepsilon_s \varphi_F/qN_A)}$

$V_T = V_{FB} + 2\varphi_F + \sqrt{(4q\varepsilon_s N_A \varphi_F)}/C_{ox}$, $V_{FB} = \Phi_{ms} - Q_f/C_{ox}$

Linear: $I_D = \mu C_{ox}(W/L)[(V_{GS} - V_T)V_{DS} - V_{DS}^2/2]$

Saturation: $I_D = \tfrac{1}{2}\mu C_{ox}(W/L)(V_{GS} - V_T)^2$, $g_m = \mu C_{ox}(W/L)(V_{GS} - V_T)$

Rules of thumb

- Each decade of doping moves E_F by 60 mV; each 60 mV of forward bias raises diode current tenfold (300 K).
- n_i(Si) roughly doubles every 8 °C.
- Doping $10^{16}\ \text{cm}^{-3}$ in Si = 1 dopant per 5×10^6 Si atoms; $\rho \approx 0.5\ \Omega\cdot$cm (n-type), 1.4 Ω·cm (p-type).
- Diffusion length in good Si: tens to hundreds of μm.

15.3 Problem-solving clinic

How to answer each type of question

1. **Definitions** (2–3 marks). One clear sentence, then an example, then the formula if there is one. *"The Fermi level is the energy at which the probability of occupation is ½; in n-type Si it lies near E_c; $E_F - E_i = kT\ln(N_D/n_i)$."*
2. **"Explain" and "describe"** questions want the **mechanism**, not just a list. Use the pattern: what happens physically → the equation → the consequence.
3. **Derivations.** List the assumptions first, number the steps, and box the final result. Examiners give marks for the assumptions.

4. **Calculations.** Write the formula, substitute **with units**, give the answer to 2–3 significant figures with units. Then sanity-check the size of the answer.
5. **Comparisons.** Use a table with three to six rows.
6. **Diagrams** earn marks: band diagrams, carrier profiles, the CZ puller, MOS band bending, BJT and MOSFET cross-sections, I-V curves.

Unit traps that cost marks

Trap	Fix
Mixing μm and cm	1 μm = 10^{-4} cm. Convert depths and widths to cm before using cm^2/s, cm^{-3}
Mixing eV and J	kT = 0.02585 eV at 300 K; only convert to joules (× 1.6×10^{-19}) when multiplying by something in SI
Forgetting the factor ½	$I_D(sat) = ½\mu C_{ox}(W/L)(V_{GS} - V_T)^2$
Using the wrong form of diffusion length	Use the form the question gives: √(Dt), √(2Dt) or 2√(Dt)
Natural log against $\log_{10}$	All the semiconductor formulas use ln. $\ln(10^n) = 2.303n$
Dose units	Dose is per cm^2 (atoms/cm^2); concentration is per cm^3

Sanity checks

Quantity	Typical size
Carrier concentrations	$10^4 – 10^{20}$ cm^{-3}
n_i (Si, 300 K)	10^{10} cm^{-3}
$E_F - E_i$	0.2 – 0.5 eV
Diffusion coefficients (dopants in a furnace)	$10^{-15} – 10^{-12}$ cm^2/s
Carrier diffusion coefficients	1 – 100 cm^2/s
Built-in voltage	0.5 – 1.0 V
Depletion width	0.01 – 10 μm
BJT β	20 – 500
MOSFET V_T	0.2 – 2 V

15.4 Past-question style exercises (with solutions)

E1. Find the Miller indices of a plane with intercepts 4a, 2b and ∞c.

Reciprocals 1/4, 1/2, 0. Multiply by 4: **(1 2 0)**. The plane is parallel to the z-axis.

E2. Constant-source diffusion with $C_s = 5\times10^{19}$ cm^{-3}, $D = 2\times10^{-13}$ cm^2/s, $t = 2$ h. Find the concentration at 0.3 μm.

$t = 7200$ s, $\sqrt{(Dt)} = \sqrt{(2\times10^{-13} \times 7200)} = \sqrt{(1.44\times10^{-9})} = 3.79\times10^{-5}$ cm

$u = x/2\sqrt{(Dt)} = 3\times10^{-5} / (2 \times 3.79\times10^{-5}) = 0.395$

$\text{erfc}(0.395) = 0.576$

$C = 5\times10^{19} \times 0.576 = 2.88\times10^{19}$ cm^{-3}

E3. Silicon is doped with $N_A = 2\times10^{16}$ cm^{-3} at 300 K ($n_i = 10^{10}$ cm^{-3}). Find p, n and the Fermi level relative to E_i.

$p \approx N_A = 2\times10^{16}$ cm^{-3}

$n = n_i^2/p = 10^{20}/(2\times10^{16}) = 5\times10^{3}$ cm^{-3}

$E_i - E_F = kT \ln(p/n_i) = 0.02585 \ln(2\times10^{6}) = 0.02585 \times 14.51 = 0.375$ eV

E_F lies 0.375 eV below E_i.

E4. Find the resistivity of n-type Si with $N_D = 10^{15}$ cm^{-3} and $\mu_n = 1350$ cm^2/V·s.

$\sigma = qN_D\mu_n = 1.6\times10^{-19} \times 10^{15} \times 1350 = 0.216$ S/cm

$\rho = 1/\sigma = 4.63$ Ω·cm

E5. Holes in n-Si have $\mu_p = 450$ cm^2/V·s and lifetime 5 μs. Find D_p and the diffusion length L_p.

$D_p = (kT/q)\mu_p = 0.02585 \times 450 = 11.6$ cm^2/s

$L_p = \sqrt{(D_p\tau_p)} = \sqrt{(11.6 \times 5\times10^{-6})} = 7.6\times10^{-3}$ cm $= 76$ μm

E6. A Si p$^+$n junction has $N_A = 10^{17}$ cm^{-3} and $N_D = 10^{15}$ cm^{-3}. Find V_{bi} and the zero-bias depletion width.

$V_{bi} = (kT/q) \ln(N_AN_D/n_i^2) = 0.02585 \ln(10^{32}/10^{20}) = 0.02585 \times 27.63 = 0.714$ V

One-sided junction, so the depletion region is almost all on the lightly doped n side:

$W \approx \sqrt{(2\varepsilon_s V_{bi}/qN_D)} = \sqrt{(2 \times 1.04\times10^{-12} \times 0.714 / (1.6\times10^{-19} \times 10^{15}))} = 9.6\times10^{-5}$ cm $= 0.96$ μm

E7. A diode has $I_s = 2\times10^{-14}$ A. Find the current at V = 0.65 V and 300 K.

$I = I_s(e^{qV/kT} - 1) = 2\times10^{-14} \times e^{0.65/0.02585} = 2\times10^{-14} \times 8.3\times10^{10} = 1.66$ mA

E8. A BJT has $\beta = 150$ and $I_B = 20$ μA. Find I_C, I_E and α.

$I_C = \beta I_B = 150 \times 20$ μA $= 3.00$ mA

$I_E = I_C + I_B = 3.02$ mA

$\alpha = \beta/(\beta + 1) = 150/151 = 0.9934$

E9. An nMOS has $\mu_n C_{ox}(W/L) = 0.8$ mA/V^2 and $V_T = 0.7$ V, with $V_{GS} = 2$ V. Find I_D for $V_{DS} = 0.5$ V and for $V_{DS} = 3$ V.

$V_{GS} - V_T = 1.3$ V.

$V_{DS} = 0.5$ V < 1.3 V $\rightarrow$ linear: $I_D = 0.8 \times [1.3 \times 0.5 - 0.5^2/2] = 0.8 \times 0.525 = 0.42$ mA

$V_{DS} = 3$ V ≥ 1.3 V $\rightarrow$ saturation: $I_D = \tfrac{1}{2} \times 0.8 \times 1.3^2 = 0.4 \times 1.69 = 0.676$ mA

15.5 Final examination checklist

Before the exam, can you:

- [Week 1] compare conductors, semiconductors and insulators; Si, Ge and GaAs; find Miller indices including ∞ and negative intercepts?
- [Week 2] describe MG-Si → EG-Si, the CZ method with its advantages and limitations, and compare VPE, LPE, MBE?
- [Week 3] state Fick's laws; use the erfc and Gaussian profiles; calculate junction depth, implant peak and dose?
- [Week 4] state the Schrödinger equation and explain the Kronig–Penney model; compare direct and indirect gaps; derive n_i?
- [Week 5] use the Fermi–Dirac function; calculate n, p and E_F for any doping; explain the three temperature regions?
- [Week 7] calculate drift current and conductivity; explain scattering; state and use the Einstein relation?
- [Week 8] explain velocity saturation, hot carriers, SRH recombination and lifetime?
- [Week 9] write the continuity equation and solve it for steady injection; explain surface recombination velocity?
- [Week 10] explain the depletion region; calculate V_{bi} and W; derive the Shockley equation?
- [Week 11] draw Schottky and Ohmic band diagrams in equilibrium and under bias?
- [Week 12] extract doping and V_{bi} from a $1/C^2$ plot; extract I_s and η from an I-V plot?
- [Week 13] explain BJT operation and minority-carrier profiles; use γ, α_T, α, β?
- [Week 14] explain accumulation, depletion, inversion; use V_T; calculate I_D in each region?

On the day: bring a scientific calculator (with e^x and ln), a ruler for diagrams, and spare pens. Read every question first, start with the one you know best, and leave 10 minutes to check units.

15.6 Comprehensive revision questions

Work through these without notes, then check against the week shown. Full worked answers for the 50 PGD examination questions are on the next page (the answer key) and in each week.

1. Define a semiconductor and explain why its conductivity rises with temperature. (Week 1)
2. Explain covalent bonding in silicon and electron–hole pair generation. (Week 1)
3. Find the Miller indices of planes with intercepts (a) a, 2b, 3c; (b) 2a, 3b, ∞c. (Week 1)
4. Describe the conversion of MG-Si to EG-Si and the Czochralski method. (Week 2)
5. Compare VPE, LPE and MBE. (Week 2)
6. State Fick's first and second laws and the two diffusion profiles. (Week 3)

7. Compare diffusion and ion implantation. Find the junction depth of an implant with $N_p = 10^{18}$ cm^{-3}, $R_p = 0.2$ μm, $\Delta R_p = 0.04$ μm into $N_B = 10^{16}$ cm^{-3}. (Week 3; answer $x_j = 0.2 + 0.04\sqrt{(2 \ln 100)} = 0.321$ μm)
8. Explain the Kronig–Penney model and how it gives energy bands. (Week 4)
9. Derive $n_i = \sqrt{(N_c N_v)} e^{-E_g/2kT}$. (Week 4)
10. Write the Fermi–Dirac function and describe it at 0 K and 300 K. (Week 5)
11. Find n, p and E_F for Si with $N_D = 5\times10^{15}$ cm^{-3}. (Week 5; $n = 5\times10^{15}$, $p = 2\times10^{4}$ cm^{-3}, $E_F - E_i = 0.339$ eV)
12. Derive the drift current density and the Einstein relation. (Week 7)
13. Explain SRH recombination and define minority-carrier lifetime. (Week 8)
14. Derive the minority-carrier continuity equation and solve it for steady injection into a long bar. (Week 9)
15. Explain the formation of the depletion region; derive the Shockley diode equation. (Week 10)
16. Distinguish Schottky and Ohmic contacts with band diagrams. (Week 11)
17. Explain how C-V measurements give the doping concentration. (Week 12)
18. Explain BJT operation and derive $\beta = \alpha/(1 - \alpha)$. (Week 13)
19. Explain accumulation, depletion and inversion; derive the saturation condition of a MOSFET. (Week 14)
20. Compare a BJT and a MOSFET. (Week 14)

Practice questions

These extra questions practise this week's ideas. Work each one on paper before reading the solution; the calculation questions follow the same Question, Given, Find, Solution, Answer layout as the worked examples.

Practice 15.1: one sample, every property

Question. n-type silicon has $N_D = 2\times10^{16}$ cm^{-3}, $\mu_n = 1200$ cm^2/V·s, $\mu_p = 420$ cm^2/V·s and $\tau_p = 2$ μs. At 300 K find (a) n and p, (b) $E_F - E_i$, (c) the conductivity, (d) D_p and (e) the hole diffusion length.

Given: $N_D = 2\times10^{16}$ cm^{-3}, $n_i = 10^{10}$ cm^{-3}, $\mu_n = 1200$, $\mu_p = 420$ cm^2/V·s, $\tau_p = 2\times10^{-6}$ s.

Find: n, p, $E_F - E_i$, σ, D_p, L_p.

Solution.

(a) $n = 2\times10^{16}$ cm^{-3}, $p = 10^{20}/2\times10^{16} = 5\times10^{3}$ cm^{-3}

(b) $E_F - E_i = 0.0259 \ln(2\times10^{6}) = 0.376$ eV

(c) $\sigma = qn\mu_n = 1.6\times10^{-19} \times 2\times10^{16} \times 1200 = 3.84$ S/cm

(d) $D_p = 0.0259 \times 420 = 10.9$ cm^2/s

(e) $L_p = \sqrt{(10.9 \times 2\times10^{-6})} = 4.66\times10^{-3}$ cm $= 46.6$ μm

Answer: $n = 2\times10^{16}$, $p = 5\times10^{3}$ cm^{-3}; $E_F = E_i + 0.376$ eV; $\sigma = 3.84$ S/cm; $D_p = 10.9$ cm^2/s; $L_p = 47$ μm.

Practice 15.2: a diode made from that sample

Question. A p^+n diode ($N_A = 1\times10^{19}$ cm^{-3}, area 1×10^{-3} cm^2) is made on the silicon of Practice 15.1. Find V_{bi}, the zero-bias depletion width, I_s and the current at 0.65 V.

Given: $N_A = 10^{19}$, $N_D = 2\times10^{16}$ cm^{-3}, $D_p = 10.9$ cm^2/s, $L_p = 4.66\times10^{-3}$ cm, $A = 10^{-3}$ cm^2.

Find: V_{bi}, W, I_s, I(0.65 V).

Solution.

$V_{bi} = 0.0259 \ln(10^{19} \times 2\times10^{16}/10^{20}) = 0.0259 \ln(2\times10^{15}) = 0.913$ V

$W \approx \sqrt{(2\varepsilon_s V_{bi}/qN_D)} = \sqrt{(2 \times 1.04\times10^{-12} \times 0.913/(1.6\times10^{-19} \times 2\times10^{16}))} = 2.44\times10^{-5}$ cm $= 0.24$ μm

$J_s \approx qn_i^2 D_p/(L_p N_D) = 1.6\times10^{-19} \times 10^{20} \times 10.9/(4.66\times10^{-3} \times 2\times10^{16}) = 1.87\times10^{-12}$ A/cm^2

$I_s = 1.87\times10^{-15}$ A

$I(0.65\text{ V}) = 1.87\times10^{-15} \times e^{0.65/0.0259} = 1.87\times10^{-15} \times 7.9\times10^{10} = 0.15$ mA

Answer: $V_{bi} = 0.91$ V, $W = 0.24$ μm, $I_s = 1.9\times10^{-15}$ A, $I(0.65\text{ V}) = 0.15$ mA.

Practice 15.3: a complete MOSFET calculation

Question. An nMOS has $N_A = 1\times10^{17}$ cm^{-3}, $t_{ox} = 5$ nm, $V_{FB} = -0.9$ V, $\mu_n = 350$ cm^2/V·s and $W/L = 20$. Find V_T and the saturation current at $V_{GS} = 1.2$ V.

Given: $N_A = 10^{17}$ cm^{-3}, $t_{ox} = 5\times10^{-7}$ cm, $V_{FB} = -0.9$ V, $\mu_n = 350$ cm^2/V·s, $W/L = 20$.

Find: V_T and $I_D(\text{sat})$.

Solution.

$C_{ox} = 3.45\times10^{-13}/5\times10^{-7} = 6.9\times10^{-7}$ F/cm^2

$\varphi_F = 0.417$ V, $2\varphi_F = 0.834$ V

$\gamma = \sqrt{(2q\varepsilon_{si}N_A)}/C_{ox} = 1.824\times10^{-7}/6.9\times10^{-7} = 0.264$ V$^{1/2}$, $\gamma\sqrt{(2\varphi_F)} = 0.241$ V

$V_T = -0.9 + 0.834 + 0.241 = 0.18$ V

$k = \mu_n C_{ox}(W/L) = 350 \times 6.9\times10^{-7} \times 20 = 4.83$ mA/V^2

$I_D(\text{sat}) = \tfrac{1}{2} \times 4.83 \times (1.2 - 0.18)^2 = 2.5$ mA

Answer: $V_T \approx 0.18$ V, $I_D(\text{sat}) \approx 2.5$ mA.

Practice 15.4: essay practice

Question. In about one page, explain how a p-n junction diode rectifies, starting from the formation of the depletion region and ending with the Shockley equation. (Use the structure of Week 10 and Q47–Q48.)

Answer outline. (1) Diffusion of holes and electrons across the junction uncovers fixed ions: the depletion region and its field. (2) Equilibrium: drift balances diffusion, E_F is flat, $V_{bi} = (kT/q)\ln(N_A N_D/n_i^2)$. (3) Forward bias lowers the barrier; minority carriers are injected (law of the junction, $p_n = p_{n0}e^{qV/kT}$). (4) They diffuse and recombine over L_p, L_n; the diffusion currents at the depletion edges add to give $I = I_s(e^{qV/kT} - 1)$. (5) Reverse bias raises the barrier: only the tiny $-I_s$ flows. Hence rectification. Include the band diagram and the I-V sketch.

Objectives check

Use this table to confirm you have met this week's objectives. Cover the Answer column and answer each check question from memory. If you cannot, go back to the section in the last column before moving on.

#	Objective	Check yourself	Answer	Where it is taught
1	Summarise the whole course and how its ideas connect.	Trace the chain from doping to the diode equation in one sentence.	Doping sets n and p; a p-n junction forms a barrier; forward bias injects minority carriers that diffuse and recombine, giving $I = I_s(e^{qV/kT} - 1)$.	15.1
2	Recall and apply the key formulas.	Write the mass-action law, the Einstein relation and the saturation MOSFET current.	$np = n_i^2$; $D/\mu = kT/q$; $I_D = \tfrac{1}{2}\mu C_{ox}(W/L)(V_{GS} - V_T)^2$.	15.2
3	Answer definitions, derivations and calculations in examination style.	What should a derivation answer begin with?	A list of the assumptions.	15.3
4	Solve mixed past-question style problems.	Find V_{bi} for $N_A = 10^{17}$ and $N_D = 10^{15}$ cm^{-3} in Si.	0.714 V.	15.4, E6
5	Check readiness for the final examination.	Can you tick every line of the checklist in 15.5?	If not, revise the week named on that line.	15.5–15.6

Answer Key — All 50 Examination Questions

The key point of every PGD examination answer in one or two lines. Cover the right-hand column, say the answer aloud, then check. The full worked answer is in the week shown in brackets.

Material physics and semiconductor growth (Q1–Q13)

Q	Key answer
1 (Week 1)	Conductivity between metal and insulator, controllable by doping, T, light. Metal: no gap, σ falls with T. Insulator: gap > 4 eV. Semiconductor: gap ≈ 1 eV, two carriers, σ rises with T.
2 (Week 1)	Elemental: one element, Group IV, covalent, e.g. **Si, Ge**. Compound: two or more elements (III–V, II–VI), tunable and often direct gap, e.g. **GaAs, InP**.
3 (Week 1)	Abundant and cheap; excellent native SiO_2; 1.12 eV gap gives low leakage to 150 °C; large perfect crystals; strong, good thermal conductor; well-understood doping.
4 (Week 1)	Gaps 1.12 / 0.66 / 1.42 eV; indirect / indirect / direct; μ_n 1350 / 3900 / 8500. Si for ICs and power; Ge for IR and SiGe HBTs; GaAs for RF, LEDs, lasers.
5 (Week 1)	Lattice = periodic array of points; crystal = lattice + basis. Structure sets bands and gap, mobility, anisotropy, doping sites, epitaxy, cleavage and defects.
6 (Week 1)	Smallest integers proportional to reciprocals of intercepts. Steps: intercepts → reciprocals → clear fractions → (h k l). Infinite intercept → 0; negative → bar.
7 (Week 1)	2, 3, 1 → 1/2, 1/3, 1 → × 6 → **(3 2 6)**. In a cubic crystal d = a/7.
8 (Week 1)	Covalent: shared pairs, directional. Ionic: electron transfer, Coulomb attraction. Metallic: electron sea. **Si is covalent**: four sp^3 bonds, tetrahedral, 109.5°.
9 (Week 2)	SiO_2 + 2C → MG-Si (98 %); Si + 3HCl → $SiHCl_3$ (300 °C); fractional distillation; Siemens CVD $SiHCl_3 + H_2$ → Si + 3HCl (1100 °C) → EG-Si (9N–11N) polysilicon; then CZ/FZ.
10 (Week 2)	Seed dipped into melt at 1414 °C and slowly pulled (≈ 1 mm/min) with counter-rotation; melt freezes epitaxially on seed; Dash neck removes dislocations; shoulder, body, tail.
11 (Week 2)	Advantages: large diameter, dislocation-free, controlled doping, low cost, oxygen strengthens and getters. Limitations: O and C contamination from crucible (limits resistivity), axial doping variation from segregation.
12 (Week 2)	VPE: gas reacts on hot substrate ($SiCl_4 + 2H_2$ → Si + 4HCl). LPE: precipitation from cooling saturated melt. MBE: atomic beams in UHV with shutters, monolayer control.
13 (Week 2)	Temperature VPE 1000–1250 °C > LPE 700–900 °C > MBE 500–650 °C. Precision MBE (atomic) > VPE > LPE. Uses: VPE production Si/MOCVD LEDs; LPE LEDs, thick layers; MBE quantum wells, HEMTs.

Impurity control and doping (Q14–Q21)

Q	Key answer
14 (Week 3)	Controlled addition of impurities (10^{14}–10^{20} cm^{-3}). Needed because intrinsic σ is too low and T-sensitive; sets carrier type, makes junctions, sets V_{bi}, V_T, β, contacts.
15 (Week 3)	n-type: Group V donors, electrons majority, E_F near E_c, $n \approx N_D$. p-type: Group III acceptors, holes majority, E_F near E_v, $p \approx N_A$.
16 (Week 3)	Donor: one extra valence electron, gives it to CB, becomes +, level near E_c (P, As, Sb). Acceptor: one fewer, takes an electron creating a hole, becomes −, level near E_v (B, Al, Ga, In).
17 (Week 3)	Thermally activated jumps down a gradient. Vacancy (substitutional, slow: B, P, As), interstitial (fast: Au, Cu, Fe), interstitialcy. $D = D_0 e^{-E_a/kT}$. Predeposition + drive-in.
18 (Week 3)	First law $F = -D\partial C/\partial x$ (flux ∝ gradient). Second law $\partial C/\partial t = D\partial^2 C/\partial x^2$ (from continuity), gives the profile's time evolution.
19 (Week 3)	Constant source → $C = 10^{20}\ \mathrm{erfc}(x/2\sqrt{(Dt)})$ cm^{-3}. Maximum (10^{20}) at the surface, falling monotonically with depth; penetrates as √t; dose $Q = (2/\sqrt{\pi})C_s\sqrt{(Dt)}$.
20 (Week 3)	Ionised, mass-selected dopant ions accelerated (keV–MeV) into the wafer; Gaussian profile at R_p; anneal. Advantages: precise dose, independent depth, low temperature, resist masks, shallow and buried profiles.
21 (Week 3)	Diffusion: high T, surface-peaked erfc/Gaussian, coupled dose/depth, no damage, cheap batch. Implant: room T, peak at R_p, ±1 % dose, damage needs anneal, channelling, costly, standard for VLSI.

Energy bands and carrier statistics (Q22–Q34)

Q	Key answer
22 (Week 4)	Wave nature; fixed nuclei (Born–Oppenheimer); perfect periodic potential; independent electrons in average potential; Bloch functions; Pauli principle; effective mass; Fermi–Dirac statistics.
23 (Week 4)	$-(\hbar^2/2m)d^2\psi/dx^2 + V\psi = E\psi$. Kinetic-energy term + potential-energy term = total energy E; $\|\psi\|^2$ is probability density; E eigenvalues give quantisation.
24 (Week 4)	Periodic square wells and barriers; Bloch condition gives $P\sin(\alpha a)/\alpha a + \cos \alpha a = \cos ka$. Only energies with LHS between ±1 are allowed → bands and gaps; edges at $k = n\pi/a$.
25 (Week 4)	VB: highest full band at 0 K (bond electrons, holes). CB: lowest empty band (free electrons). Gap $E_g = E_c - E_v$: no states.
26 (Week 4)	Direct: CB min and VB max at same k, photon only, efficient light (GaAs). Indirect: different k, phonon needed, poor emitter (Si, Ge). $\lambda = 1.24/E_g$ μm.
27 (Week 4)	Pure, $n = p = n_i$. $n = N_{ce}^{-(E_c-E_F)/kT}$, $p = N_{ve}^{-(E_F-E_v)/kT}$; multiply: $\mathbf{n_i = \sqrt{(N_cN_v)}\, e^{-E_g/2kT}} \propto T^{3/2}e^{-E_g/2kT}$.
28 (Week 4)	Intrinsic: thermal only, $n = p = n_i$, strongly T-dependent. Extrinsic: from dopants, $n \neq p$, $n \approx N_D$ or $p \approx N_A$, nearly T-independent. Linked by $np = n_i^2$.
29 (Week 4)	$n = p$ and $np = n_i^2$ ⇒ $\mathbf{n = p = 1.0\times10^{10}\ cm^{-3}}$.
30 (Week 5)	$f(E) = 1/(1 + e^{(E-E_F)/kT})$: occupation probability of a state; ½ at E_F. Gives n and p, defines E_F, constant E_F in equilibrium sets junction physics.
31 (Week 5)	At 0 K f is a step: 1 below E_F, 0 above. VB full, CB empty, pure semiconductor is an insulator.
32 (Week 5)	g(E) = states per unit energy per unit volume, $g_c \propto (m_n*)^{3/2}\sqrt{(E - E_c)}$. $n = \int g\, f\, dE$; gives N_c, N_v; zero in gap.
33 (Week 5)	n-type: E_F rises toward E_c, $E_F - E_i = kT \ln(N_D/n_i)$. p-type: falls toward E_v. 60 mV per decade; degenerate at very high doping; returns to E_i at high T.
34 (Week 4)	$n \approx N_D = \mathbf{5\times10^{16}\ cm^{-3}}$. ($p = 2\times10^3\ cm^{-3}$; $E_F - E_i = 0.40$ eV in Si.)

Carrier transport and injection (Q35–Q46)

Q	Key answer
35 (Week 7)	$\mu = v_d/E = q\tau/m*$. Depends on effective mass, temperature (lattice $T^{-3/2}$, impurity $T^{3/2}$), doping, field, defects, surfaces, orientation, strain.
36 (Week 7)	Lattice (acoustic, optical, polar phonons), ionised impurity, neutral impurity, carrier–carrier, defects, surface roughness, alloy, intervalley. Matthiessen $1/\mu = \Sigma 1/\mu_i$.
37 (Week 7)	$v_{dn} = -\mu_n E$, $v_{dp} = \mu_p E$; $J = \rho v$: $J_n = qn\mu_n E$, $J_p = qp\mu_p E$; $\mathbf{J = q(n\mu_n + p\mu_p)E}$, $\sigma = q(n\mu_n + p\mu_p)$.
38 (Week 7)	$J = qn\mu E = 1.602\times10^{-19} \times 2\times10^{16} \times 1350 \times 10 =$ **43.3 A/cm^2**.
39 (Week 7)	Current from carriers moving down a concentration gradient by random motion. Fick: $F = -D\, dC/dx$. $J_n = qD_n\, dn/dx$, $J_p = -qD_p\, dp/dx$.
40 (Week 7)	$\mathbf{D/\mu = kT/q}$ (0.0259 V). From zero equilibrium current in a non-uniform sample. Si: $D_n = 35$, $D_p = 12.4$ cm^2/s.
41 (Week 8)	High field: optical-phonon emission caps v_d at $v_{sat} \approx 10^7$ cm/s. Hot carriers ($T_e \gg T$): impact ionisation, avalanche, oxide injection and V_T drift, substrate current; LDD as cure.
42 (Week 8)	Generation creates e–h pairs (thermal, optical, impact); recombination annihilates them (radiative, SRH, Auger, surface). Equal in equilibrium; net $U = \Delta p/\tau$.
43 (Week 8)	Recombination via mid-gap trap: capture/emission of electrons and holes. $U = (np - n_i^2)/[\tau_p(n + n_1) + \tau_n(p + p_1)]$; maximum for $E_t \approx E_i$; $\tau = 1/\sigma v_{th} N_t$.
44 (Week 8)	Average time an excess minority carrier survives: $\Delta p(t) = \Delta p(0)e^{-t/\tau}$. Sets $L = \sqrt{(D\tau)}$, diode leakage, switching speed, BJT gain, solar-cell efficiency.
45 (Week 9)	$\partial p/\partial t = -(1/q)\partial J_p/\partial x + G - R$; minority form $\partial \Delta p/\partial t = D\partial^2 \Delta p/\partial x^2 - \Delta p/\tau + G$. Used for diodes (exponential profile), BJTs, lifetime and Haynes–Shockley, solar cells.
46 (Week 9)	S defined by $U_s = S\Delta p_s$ (cm/s), $S = \sigma v_{th} N_{st}$. Dangling-bond surface states are SRH centres: $1/\tau_{eff} = 1/\tau_b + 2S/W$. Passivate with SiO_2, SiN_x, Al_2O_3.

Junctions, interfaces and characterisation (Q47–Q50)

Q	Key answer
47 (Week 10)	Diffusion of carriers uncovers fixed ions → depletion region and field; drift balances diffusion; flat E_F. $\mathbf{V_{bi} = (kT/q)\ln(N_A N_D/n_i^2)}$; $W = \sqrt{(2\varepsilon_s V_{bi}(N_A + N_D)/qN_A N_D)}$.
48 (Week 10)	Assumptions; law of junction $p_n = p_{n0e}^{qV/kT}$; diffusion equation → exponential decay over L_p; currents at edges; $\mathbf{I = I_s(e^{qV/kT} - 1)}$, $J_s = qn_i^2(D_p/L_p N_D + D_n/L_n N_A)$. Forward: exponential rise; reverse: $-I_s$.
49 (Week 11)	Schottky: $\Phi_m > \Phi_s$ (n-type), upward bending, depletion, barrier $\Phi_B = \Phi_m - \chi$, rectifying, thermionic, fast. Ohmic: $\Phi_m < \Phi_s$ or heavy doping, accumulation or tunnelling, linear I-V.
50 (Week 12)	I-V: ln I vs V gives I_s, η, R_s, breakdown, Φ_B. C-V: $C = \varepsilon_s A/W$; $\mathbf{1/C^2 = 2(V_{bi} - V)/(q\varepsilon_s N A^2)}$; slope gives $N = 2/(q\varepsilon_s A^2 \cdot \text{slope})$, intercept gives V_{bi}; local slope gives profile N(W).

REVISION & PRACTICALS

Laboratory and Practical Component

The course plan requires at least four practical or simulation experiments (3 hours/week). Each practical below gives the aim, theory, procedure, a results table and the analysis. **The readings in the tables are example values** that show what correct results look like; replace them with your own measurements.

Practical 1 — Simulation of carrier transport: Fermi-level shifts in Python

Aim. Model how the Fermi level of silicon moves with temperature and with doping, and find the temperature at which a doped sample becomes intrinsic.

Theory (Weeks 4–5).

$n_i(T) = \sqrt{(N_c N_v)}\, e^{-E_g(T)/2kT}$, $N_c, N_v \propto T^{3/2}$, $E_g(T) = 1.17 - 4.73\times10^{-4}T^2/(T + 636)$ eV

$n = N_D/2 + \sqrt{((N_D/2)^2 + n_i^2)}$ (complete ionisation)

$E_F - E_i = kT \ln(n/n_i)$

Procedure.

1. Run the script below (Python 3 with NumPy and Matplotlib; the same steps work in MATLAB).
2. Plot $E_F - E_i$ against T from 200 K to 700 K for $N_D = 10^{14}$, 10^{15}, 10^{16} and 10^{17} cm^{-3}.
3. Plot $E_F - E_i$ against N_D at 300 K from 10^{14} to 10^{18} cm^{-3} and measure the slope per decade.
4. For each doping, record the temperature at which $n_i = N_D$.

```
import numpy as np
import matplotlib.pyplot as plt

k = 8.617e-5                        # Boltzmann constant, eV/K
Nc300, Nv300 = 2.8e19, 1.04e19      # effective densities of states at 300 K, cm^-3

def Eg(T):                          # silicon band gap, eV (Varshni)
    return 1.17 - 4.73e-4 * T**2 / (T + 636)

def ni(T):
    Nc = Nc300 * (T / 300)**1.5
    Nv = Nv300 * (T / 300)**1.5
    raw = np.sqrt(Nc * Nv) * np.exp(-Eg(T) / (2 * k * T))
    return raw * 1e10 / ni_raw300  # scale so n_i(300 K) = 1e10 cm^-3

ni_raw300 = np.sqrt(Nc300 * Nv300) * np.exp(-Eg(300) / (2 * k * 300))

def EF_minus_Ei(T, ND):
    n = ND / 2 + np.sqrt((ND / 2)**2 + ni(T)**2)
    return k * T * np.log(n / ni(T))

T = np.linspace(200, 700, 501)
for ND in (1e14, 1e15, 1e16, 1e17):
    plt.plot(T, EF_minus_Ei(T, ND), label=f"N_D = {ND:.0e} cm^-3")
    Tint = T[np.argmin(abs(ni(T) - ND))]
```

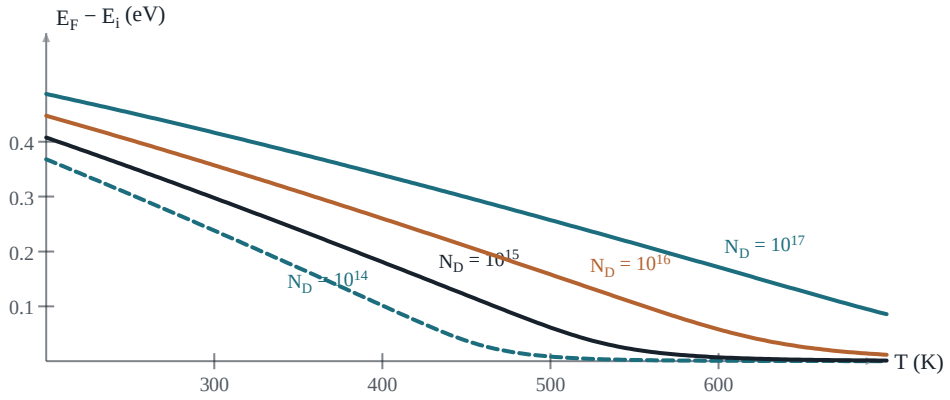
```
    print(f"N_D = {ND:.0e}: E_F - E_i at 300 K = {EF_minus_Ei(300, ND):.3f} eV, "
          f"n_i = N_D at about {Tint:.0f} K")
plt.xlabel("T (K)"); plt.ylabel("E_F - E_i (eV)"); plt.legend(); plt.grid()
plt.show()

slope = (EF_minus_Ei(300, 1e17) - EF_minus_Ei(300, 1e16))
print(f"Shift per decade of doping at 300 K: {1000 * slope:.1f} mV")
```

Results (output of the script):

N_D (cm^{-3})	$E_F - E_i$ at 300 K (eV)	Temperature where $n_i = N_D$
10^{14}	0.238	≈ 470 K
10^{15}	0.298	≈ 550 K
10^{16}	0.357	≈ 650 K
10^{17}	0.417	above 700 K

Shift per decade of doping at 300 K: 59.5 mV (= kT ln 10).



Output of Practical 1: Fermi level against temperature for silicon. Each curve falls towards zero (mid-gap) once n_i approaches N_D.

Discussion points. (1) Why does the curve flatten near zero at high T? (2) Why does heavier doping raise the maximum operating temperature? (3) What would change for germanium (E_g = 0.66 eV)? Germanium reaches $n_i = 10^{15}$ cm^{-3} at only about 400 K, which is why Ge devices fail at modest temperatures.

Practical 2 — I-V characterisation of a p-n junction and a Schottky diode

Aim. Measure and plot the forward and reverse characteristics of a silicon p-n diode (e.g. 1N4148) and a Schottky diode (e.g. 1N5817 or BAT85); extract the saturation current, ideality factor and series resistance, and compare the two.

Theory (Weeks 10–12). $I = I_s[\exp(qV/\eta kT) - 1]$. Above about 0.1 V, $\ln I = \ln I_s + qV/(\eta kT)$, so a semi-log plot is a straight line with slope $q/(\eta kT)$ and intercept $\ln I_s$.

Apparatus. DC supply (0–10 V) or source-measure unit, series resistor (1 kΩ for forward, none for reverse), two digital multimeters, breadboard, diodes.

Procedure.

1. Connect supply → resistor → diode (anode to +). Measure the diode voltage V and the current I (from the resistor voltage, or an ammeter).

2. **Forward:** increase the supply so that V rises from 0 to 0.8 V (p-n) or 0 to 0.5 V (Schottky) in 0.05 V steps. Do not exceed the rated current.
3. **Reverse:** reverse the diode, remove the resistor (or use 100 kΩ) and step V from 0 to −10 V, reading the small reverse current (µA range for Schottky, nA range for p-n).
4. Plot I against V on linear axes, and ln I (or $\log_{10}$ I) against V for the forward branch.

Example readings (forward):

V (V)	I, p-n diode	I, Schottky diode
0.15	0.003 µA	0.016 mA
0.20	0.02 µA	0.10 mA
0.25	0.13 µA	0.63 mA
0.30	0.9 µA	4.0 mA
0.35	5.7 µA	25 mA
0.40	40 µA	(beyond rating)
0.50	1.7 mA	—
0.60	45 mA (bending: series resistance)	—

Analysis.

- From the straight part of the p-n semi-log plot: between 0.3 V and 0.4 V, I rises from 0.9 µA to 40 µA, a factor of 44.4, so ln(44.4) = 3.79 per 0.1 V, slope = 37.9 V^{-1}. Then $\eta = 1/(0.0259 \times 37.9) = 1.02$. Extrapolating to V = 0: $I_s = 0.9\ \mu A / e^{3.79\times 3} \approx 1\times10^{-11}$ A.
- Schottky diode: between 0.25 V and 0.35 V the current rises 40 times per 0.1 V, giving $\eta \approx 1.05$ and $I_s \approx 0.1\ mA / e^{0.20/0.0272} \approx 6\times10^{-8}$ A. It reaches a given current at about 0.25–0.3 V lower voltage than the p-n diode; I_s is several orders of magnitude larger, and the reverse current is much larger (µA instead of nA).
- At high current the curves bend away from the straight line: the extra voltage is IR_s. $R_s = \Delta V/I$ where ΔV is the deviation from the extrapolated line.
- Schottky barrier height: $\Phi_B = (kT/q)\ \ln(A{**}AT^2/I_s)$, using the diode area A from the datasheet.

Discussion. Explain the lower turn-on voltage and higher leakage of the Schottky diode in terms of majority-carrier thermionic emission over Φ_B versus minority-carrier diffusion (Weeks 10–11).

Practical 3 — C-V profiling: doping and built-in potential

Aim. Measure the capacitance of a reverse-biased one-sided junction (a varactor such as BB112, or a Schottky diode) against bias, and extract the doping concentration and the built-in potential.

Theory (Week 12).

$$C = A\sqrt{\left(q\varepsilon_s N / 2(V_{bi} - V)\right)} \quad \Rightarrow \quad 1/C^2 = 2(V_{bi} - V)/(q\varepsilon_s N A^2)$$

The plot of $1/C^2$ against V is a straight line: the slope gives N, the voltage intercept gives V_{bi}.

Apparatus. LCR meter with DC bias (1 MHz, 20–50 mV AC level), or a capacitance bridge; DC supply; shielded test fixture.

Procedure.

1. Null the fixture with the device removed (open-circuit correction) to remove stray capacitance.
2. Connect the diode in reverse bias. Record C at V = 0, −1, −2, … −5 V (or the device's rated range).
3. Plot C against V and $1/C^2$ against V. Fit a straight line to $1/C^2$.
4. Take the area A from the datasheet (or measure the contact diameter).

Example readings (A = 1.0×10^{-3} cm^2, Si):

V (V)	C (pF)	$1/C^2$ (F^{-2})
0	37.2	7.2×10^{20}
−1	22.8	1.92×10^{21}
−2	17.9	3.12×10^{21}
−3	15.2	4.33×10^{21}
−4	13.4	5.57×10^{21}
−5	12.2	6.72×10^{21}

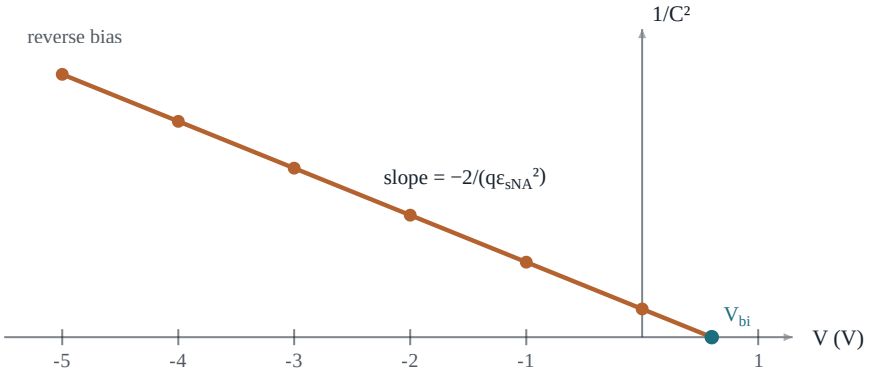
Analysis.

Slope: $(6.72\times10^{21} - 7.2\times10^{20})/(-5 - 0) = -1.20\times10^{21}$ $F^{-2}V^{-1}$

$N = 2/(q\varepsilon_s A^2|\text{slope}|) = 2/(1.602\times10^{-19} \times 1.036\times10^{-12} \times 10^{-6} \times 1.20\times10^{21}) = 1.0\times10^{16}$ cm^{-3}

Intercept: $1/C^2 = 0$ at $V = 7.2\times10^{20}/1.20\times10^{21} = +0.60$ V $\Rightarrow$ $V_{bi} = 0.60$ V

Depletion depth at −5 V: $W = \varepsilon_s A/C = 1.036\times10^{-15}/12.2\times10^{-12} = 0.85$ µm



Mott–Schottky plot of the example readings. A straight line means uniform doping; curvature would reveal a doping profile N(W).

Discussion. Why must the junction be one-sided? What would curvature in $1/C^2$ tell you? Why measure in reverse bias only?

Practical 4 — MOSFET parameter extraction: threshold voltage and channel mobility

Aim. Measure the transfer and output characteristics of an n-channel enhancement MOSFET (e.g. a CD4007 array transistor or 2N7000) and extract V_{th}, μ_n and the transconductance.

Theory (Week 14). In the linear region at small V_{DS}:

$$I_D \approx \mu_n C_{ox}(W/L)[(V_{GS} - V_T)V_{DS} - V_{DS}^2/2]$$

so I_D against V_{GS} is a straight line of slope $g_m = \mu_n C_{ox}(W/L)V_{DS}$ that meets $I_D = 0$ at $V_{GS} = V_T + V_{DS}/2$. In saturation, $\sqrt{I_D} = \sqrt{(\mu_n C_{ox} W/2L)}(V_{GS} - V_T)$.

Apparatus. Two DC supplies (or a semiconductor parameter analyser), ammeter, voltmeters. C_{ox}, W and L from the datasheet or test structure.

Procedure.

1. **Transfer curve (linear):** set $V_{DS} = 0.1$ V. Step V_{GS} from 0 to 2.5 V in 0.25 V steps; record I_D.
2. **Transfer curve (saturation):** set $V_{DS} = V_{GS}$ (gate tied to drain) and repeat; plot $\sqrt{I_D}$ against V_{GS}.
3. **Output curves:** for $V_{GS} = 1.0$, 1.5, 2.0 V, sweep V_{DS} from 0 to 3 V; record I_D.
4. Find V_T by linear extrapolation at the point of maximum slope; find μ_n from the slope.

Example readings (linear region, $V_{DS} = 0.1$ V; device with $C_{ox} = 3.45\times10^{-7}$ F/cm^2, W/L = 10):

V_{GS} (V)	0.5	1.0	1.5	2.0	2.5
I_D (μA)	15	84	153	222	291

Analysis.

$g_m = \Delta I_D/\Delta V_{GS} = (291 - 84)\ \mu A / 1.5\ V = 1.38\times10^{-4}$ S

Line through (1.0 V, 84 μA): $I_D = 0$ at $V_{GS} = 1.0 - 84/138 = 0.39$ V

$V_T = 0.39 - V_{DS}/2 = 0.39 - 0.05 = 0.34$ V

$\mu_n = g_m L / (W C_{ox} V_{DS}) = 1.38\times10^{-4} / (10 \times 3.45\times10^{-7} \times 0.1) = 400$ cm^2/V·s

The channel mobility (400) is about a third of bulk electron mobility in Si (1350) because of surface scattering at the Si/SiO_2 interface (Week 7). At high V_{GS} the measured slope falls further (mobility degradation by the vertical field), so the extraction uses the maximum-slope point.

Discussion. Compare V_T from the linear and saturation methods. Why does mobility degrade at high gate voltage? How would a thicker oxide change V_T and g_m?

Report format

For each practical: title and aim; brief theory with the equations used; circuit diagram; apparatus; procedure; tables of readings with units; graphs (labelled axes, units, fitted lines); calculations; results with uncertainties; discussion of sources of error (lead resistance, stray capacitance, self-heating, meter resolution); conclusion.